

Processor Programming Reference (PPR) for AMD Family 19h Model 11h, Revision B2 Processors Volume 8 of 9

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15 System Hub (SYSHUB)

15.1 Overview

The System Hub is a crossbar switch used to connect the high speed IO logic into the system. This allows the high speed IO devices to quickly and efficiently access the memory controllers on the system while allowing efficient priority control preventing any high speed interface from dominating the memory controller.

The following sections describe the System Hub and IO capabilities.

15.2 SATA

15.2.1 Definitions

Table 267: Link Definitions

Term	Description
HBA	Host Bus Adapter.
PXCMD	Port x Command and Status. X indicates port number. Refer to registers SYSHUB::SATA::PORT::SATA_AHCI_P_CMD.

15.2.2 SATA Functional Overview

The following sections describe the register programming requirements for proper operation of the integrated SATA for the processor.

15.2.2.1 SATA Main Features

Supported features:

- Gen I (1.5Gbps), Gen II (3.0Gbps), and Gen III (6.0Gbps) rates are supported
- Spread spectrum clocking is supported on receive sides
- AHCI 1.3.1 compliant
- 64 bit addressing
- NCQ (Native Command Queuing)
- SNotification
- Mechanical presence switch
- Staggered spin-up
- Aggressive link power management
- Command list override
- Command completion coalescing
- Port multiplier
- FIS (Frame Information Structure) based Switching (FBS)
- AHCI mode only
- Supported capabilities defined in PXCMD register
- Automatic partial to slumber transitions

SATA II supported features:

- Port Multiplier; supports up to 15 devices per port
- NCQ (Native Command Queuing) – up to 32 command queues for each device
- Auto-Active in DMA Setup FIS
- Asynchronous signal recovery
- Asynchronous Notification
- SATA power management features
 - SATA Link Power management - support both partial and slumber modes
 - Support both host initiated and device initiated LPM
 - Support Asynchronous Notification
 - Unused SATA ports can be disabled for lower power state
 - Device Sleep

Miscellaneous features:

- 4 sata instances
- SGPIO
- UBM supported
-
- Hot Plug; 100ms timer to ping external device (except after power-down to PARTIAL/SLUMBER modes)
- Host and device initiated interface power management
- Independent bus master channel for each port allows supporting simultaneous bus master transfers on all ports
- 64-bit addressing

15.2.2.2 Address Mapping

There are 4 SATA tiles: sata0, sata1, sata2, sata3.

Table 268: SATA0 Address Mapping

Registers Space	Sub-Category	SMN Base Address	Size
SATA Controller registers			
	Sata_ahci - SATA AHCI registers (Bar5)	0x03101000 ~ 0x031017FF	2K

Table 269: SATA1 Address Mapping

Registers Space	Sub-Category	SMN Base Address	Size
SATA Controller registers			
	Sata_ahci - SATA AHCI registers (Bar5)	0x03201000 ~ 0x032017FF	2K

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Table 270: SATA2 Address Mapping

Registers Space	Sub-Category	SMN Base Address	Size
SATA Controller registers			
	Sata_ahci - SATA AHCI registers (Bar5)	0x03301000 ~ 0x033017FF	2K

Table 271: SATA3 Address Mapping

Registers Space	Sub-Category	SMN Base Address	Size
SATA Controller registers			
	Sata_ahci - SATA AHCI registers (Bar5)	0x03401000 ~ 0x034017FF	2K

15.2.3 Registers

15.2.3.1 SATA Advanced Host Controller Interface (AHCI) Registers

Table 272: List of Acronyms and Terms used in SATA AHCI

Term	Definition
AHCI	Advanced Host Controller Interface.
BOHC	BIOS/OS Handoff Control and Status.
CCC	Command Completion Coalescing.
DRQ	Data Transfer Request.
FIS	Frame Information Structure.
PIO	Programmed IO.
PRD	Physical Region Descriptor.
SAF-TE	SCSI Accessed Fault-Tolerant Enclosure.
SATA	Serial AT (Advanced Technology) Attachment.
SES	SCSI-3 Enclosure Services Command Set.

SATA[0...3]AHCIx00000000 (SYSHUB::SATA::AHCI::SATA_AHCI_GHC_CAP)

Read-only. Reset: F737_FF87h.

This register indicates basic capabilities of the HBA to driver software.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx00000000; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description												
31	GHC_CAP_S64A. Read-only. Reset: 1. 0=These bits are read-only and treated as 0s by HBA. This bit is RO. 1=HBA will make the 32-bit upper bits of the port DMA Descriptor, the PRD Base, and each PRD entry read/write. Supports 64-bit Addressing. Indicates whether HBA can access 64-bit data structures.												
30	GHC_CAP_SNCQ. Read-only. Reset: 1. 0=Native command queueing is not supported and software should not issue any native command. 1=HBA can handle DMA Setup FISes. Indicates whether HBA supports Serial ATA native command queueing.												
29	GHC_CAP_SSNTF. Read-only. Reset: 1. 0=HBA does not support the PxSNTF (SNotification) register and its associated functionality. 1=HBA supports the PxSNTF (SNotification) register and its associated functionality. Supports SNotification Register.												
28	GHC_CAP_SMPS. Read-only. Reset: 1. 0=This function is not supported. This value is loaded by the BIOS prior to OS initialization. 1=HBA supports mechanical presence switches on its ports for use in hot plug operations. Supports Mechanical Presence Switch.												
27	GHC_CAP_SSS. Read-only. Reset: 0. 0=This function is not supported. This value is loaded by the BIOS prior to OS initialization. 1=HBA supports staggered spin-up on its ports, for use in balancing power spikes. Supports Staggered Spin-up.												
26	GHC_CAP_SALP. Read-only. Reset: 1. 0=This function is not supported and software will treat PxCMD.ALPE and PxCMD.ASP bits as reserved. 1=HBA can support auto-generating link requests to the Partial or Slumber states when there are no commands to process. Supports Aggressive Link Power Management.												
25	GHC_CAP_SAL. Read-only. Reset: 1. 0=This function is not supported. This bit is RO. 1=HBA supports a single activity indication output pin. This pin can be connected to an LED on the platform to indicate device activity on any drive. Supports Activity LED.												
24	GHC_CAP_SCLO. Read-only. Reset: 1. 0=HBA is not capable of clearing the BSY and DRQ bits in the status register in order to issue a software reset if these bits are still set from a previous operation. This bit is RO. 1=HBA supports the PxCMD.CLO bit and its associated function. Supports Command List Override.												
23:20	GHC_CAP_ISS. Read-only. Reset: 3h. Interface Speed Support. Indicates the maximum speed HBA can support on its ports. These encodings match the system software programmable PxSCTL.DET.SPD field. Values are: ValidValues:												
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>Reserved.</td></tr> <tr> <td>1h</td><td>Gen1 (1.5 Gbps).</td></tr> <tr> <td>2h</td><td>Gen 1 (1.5 Gbps) and Gen 2 (3 Gbps).</td></tr> <tr> <td>3h</td><td>Gen1 (1.5 Gbps) and Gen 2 (3 Gbps) and Gen 3 (6 Gbps).</td></tr> <tr> <td>Fh-4h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	Reserved.	1h	Gen1 (1.5 Gbps).	2h	Gen 1 (1.5 Gbps) and Gen 2 (3 Gbps).	3h	Gen1 (1.5 Gbps) and Gen 2 (3 Gbps) and Gen 3 (6 Gbps).	Fh-4h	Reserved.
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3h	Gen1 (1.5 Gbps) and Gen 2 (3 Gbps) and Gen 3 (6 Gbps).												
Fh-4h	Reserved.												
19	GHC_CAP_SNZO. Read-only. Reset: 0. Reserved.												
18	GHC_CAP_SAM. Read-only. Reset: 1. 0=Indicates that in addition to the native AHCI mechanism (via ABAR), the SATA controller implements a legacy, task-file based register interface such as SFF-8038i. 1=Indicates the SATA controller does not implement a legacy, task-file based register interface. This bit is RO. Supports AHCI mode only. The SATA controller may optionally support AHCI access mechanisms only.												
17	GHC_CAP_SPM. Read-only. Reset: 1. 0=When cleared to 0, a port multiplier is not supported, and a port multiplier may not be attached to this HBA. 1=When set, a port multiplier using command-based switching is supported. Supports Port Multiplier. Indicates whether HBA can support a port multiplier.												
16	GHC_CAP_FBSS. Read-only. Reset: 1. 0=Indicates HBA does not support FIS-based switching. AHCI 1.0 and 1.1 HBAs will have this bit cleared to 0. 1=Indicates HBA supports Port Multiplier FIS-based switching. FIS-based Switching Supported.												

15	GHC_CAP_PMD . Read-only. Reset: 1. 0=HBA only supports single DRQ block data transfers for the PIO command protocol. 1=HBA supports multiple DRQ block data transfers for the PIO command protocol. 0 This bit is RO as 1. PIO Multiple DRQ Block.																
14	GHC_CAP_SSC . Read-only. Reset: 1. 0=Software must not allow HBA to initiate transitions to the Slumber state via aggressive link power management nor the PxCMD.ICC field in each port, and the PxSCTL.IPM field in each port must be programmed to disallow device initiated Slumber requests. 1=HBA and device initiated Slumber requests can be supported. Slumber State Capable.																
13	GHC_CAP_PSC . Read-only. Reset: 1. 0=Software must not allow HBA to initiate transitions to the Partial state via aggressive link power management nor the PxCMD.ICC field in each port, and the PxSCTL.IPM field in each port must be programmed to disallow device initiated Partial requests. 1=HBA and device initiated Partial requests can be supported. Partial State Capable.																
12:8	GHC_CAP_NCS . Read-only. Reset: 1Fh. Number of Command Slots. 0-based value indicate the number of command slots per port supported by this HBA. A minimum of 1 and a maximum of 32 slots per port can be supported. The same number of command slots is available on each implemented port. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>1 command slot per port supported.</td></tr> <tr> <td>01h</td><td>2 command slots per port supported.</td></tr> <tr> <td>02h</td><td>3 command slots per port supported.</td></tr> <tr> <td>03h</td><td>4 command slots per port supported.</td></tr> <tr> <td>1Dh-04h</td><td>5 to 30 command slots per port supported.</td></tr> <tr> <td>1Eh</td><td>31 command slots per port supported.</td></tr> <tr> <td>1Fh</td><td>32 command slots per port supported.</td></tr> </table>	Value	Description	00h	1 command slot per port supported.	01h	2 command slots per port supported.	02h	3 command slots per port supported.	03h	4 command slots per port supported.	1Dh-04h	5 to 30 command slots per port supported.	1Eh	31 command slots per port supported.	1Fh	32 command slots per port supported.
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03h	4 command slots per port supported.																
1Dh-04h	5 to 30 command slots per port supported.																
1Eh	31 command slots per port supported.																
1Fh	32 command slots per port supported.																
7	GHC_CAP_CCCS . Read-only. Reset: 1. 0=Indicates that HBA does not support command completion coalescing and the CCC_CTL and CCC_PORTS global HBA registers are not implemented. 1=Indicates that HBA supports command completion coalescing. When command completion coalescing is supported, HBA has implemented the CCC_CTL and the CCC_PORTS global HBA registers. Command Completion Coalescing Supported.																
6	GHC_CAP_EMS . Read-only. Reset: 0. 0=Indicates that HBA does not support enclosure management and the EM_LOC and EM_CTL global HBA registers are not implemented. 1=Indicates that HBA supports enclosure management. When enclosure management is supported, HBA has implemented the EM_LOC and EM_CTL global HBA registers. Enclosure Management Supported.																
5	GHC_CAP_SXS . Read-only. Reset: 0. 0=Indicates that HBA has no Serial ATA ports that have a signal-only connector externally accessible. 1=Indicates that HBA has one or more Serial ATA ports that have a signal-only connector (i.e. power is not part of that connector) that is externally accessible. If this bit is set to 1, software may refer to the PxCMD.ESP bit to determine whether a specific port has its signal connector externally accessible as a signal-only connector. Supports External SATA.																
4:0	GHC_CAP_NP . Read-only. Reset: 07h. Number of Ports. 0-based value indicating the maximum number of ports supported by the HBA silicon. A maximum of 32 ports can be supported. A value of 0h, indicating one port, is the minimum requirement. Note that the number of ports indicated in this field may be more than the number of ports indicated in the GHC.PI register. The default value of this register is 8 ports since the HBA is in AHCI mode after power-on reset. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>A value of 0h, indicating one port, is the minimum requirement.</td></tr> <tr> <td>01h</td><td>2 ports supported.</td></tr> <tr> <td>1Eh-02h</td><td>3 to 31 ports supported.</td></tr> <tr> <td>1Fh</td><td>32 ports supported.</td></tr> </table>	Value	Description	00h	A value of 0h, indicating one port, is the minimum requirement.	01h	2 ports supported.	1Eh-02h	3 to 31 ports supported.	1Fh	32 ports supported.						
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1Eh-02h	3 to 31 ports supported.																
1Fh	32 ports supported.																

SATA[0...3]AHCIx00000004 (SYSHUB::SATA::AHCI::SATA_AHCI_GHC_GHC)

Reset: 8000_0000h.

This register controls various global actions of HBA.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx00000004; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description
31	GHC_GHC_AE. Read-write. Reset: 1. 0=When cleared, software shall only communicate with the HBA using legacy mechanisms. 1=When set, indicates that communication to HBA shall be via AHCI mechanisms. AHCI Enable. When set, indicates that communication to HBA shall be via AHCI mechanisms. This can be used by an HBA that supports both legacy mechanisms (such as SFF-8038i) and AHCI to know when HBA is running under an AHCI driver. When set, software shall only communicate with the HBA using AHCI. This bit is RO as CAP.SAM=1.
30:3	GHC_GHC_Reserved. Read-only. Reset: 000_0000h. Reserved.
2	GHC_GHC_MRSM. Read-only. Reset: 0. 0=When this bit is cleared to 0, HBA has not reverted to single MSI mode. 1=When this bit is set to 1, single MSI mode operation is in use and software is responsible for clearing bits in the IS register to clear interrupts. MSI Revert to Single Message. When set to 1 by hardware, indicates that HBA requested more than one MSI vector but has reverted to using the first vector only. When this bit is cleared to 0, HBA has not reverted to single MSI mode (i.e., hardware is already in single MSI mode, software has allocated the number of messages requested, or hardware is sharing interrupt vectors if MC.MME < MC.MMC). HBA may revert to single MSI mode when the number of vectors allocated by the host is less than the number requested. This bit shall only be set to 1 when the following conditions hold: 1) MC.MSIE = 1 (MSI is enabled). 2) MC.MMC > 0 (multiple messages requested). 3) MC.MME > 0 (more than one message allocated). 4) MC.MME != MC.MMC (messages allocated not equal to number requested). When this bit is set to 1, single MSI mode operation is in use and software is responsible for clearing bits in the IS register to clear interrupts. This bit shall be cleared to 0 by hardware when any of the four conditions stated is false. This bit is also cleared to 0 when MC.MSIE = 1 and MC.MME = 0h. In this case, hardware has been programmed to use single MSI mode, and is not reverting to that mode. Read only.
1	GHC_GHC_IE. Read-write. Reset: 0. 0=When cleared (reset default), all interrupt sources from all ports are disabled. 1=When set, interrupts are enabled. Interrupt Enable. This global bit enables interrupts from the HBA.
0	GHC_GHC_HR. Read,Write-1-only. Reset: 0. 0=A write of 0 has no effect on this bit. 1=When set by software, this bit causes an internal reset of the HBA. When HBA has performed the reset action, it shall reset this bit to 0. HBA Reset. All state machines that relate to data transfers and queuing shall return to an idle condition, and all ports shall be re-initialized via COMRESET (if staggered spin-up is not supported). If staggered spin-up is supported, then it is software's responsibility to spin-up each port after the reset has completed.

SATA[0...3]AHCIx00000008 (SYSHUB::SATA::AHCI::SATA_AHCI_GHC_IS)

Read,Write-1-to-clear. Reset: 0000_0000h.

This register indicates which of the ports within the controller have an interrupt pending and require service.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx00000008; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description
31:0	GHC_IS. Read,Write-1-to-clear. Reset: 0000_0000h. Interrupt Pending Status. If set, indicates that the corresponding port has an interrupt pending. Software can use this information to determine which ports require service after an interrupt. The IPS[x] bit is only defined for ports that are implemented or for the command completion coalescing interrupt defined by CCC_CTL.INT. All other bits are reserved. A write of 1 will clear these bits. Bit[0]: DHRS. Bit[1]: PSS. Bit[2]: DSS. Bit[3]: SDBS. Bit[4]: UFS. Bit[5]: DPS. Bit[6]: PCS. Bit[7]: DMPS. Bit[21:8]: Reserved. Bit[22]: PRCS. Bit[23]: IPMS. Bit[24]: OFS. Bit[25]: Reserved. Bit[26]: INFS. Bit[27]: IFS. Bit[28]: HBDS. Bit[29]: HBFS. Bit[30]: TFES. Bit[31]: CPDS.

SATA[0...3]AHCIx0000000C (SYSHUB::SATA::AHCI::SATA_AHCI_GHC_PI)

Reset: 0000_00FFh.

This register indicates which ports are exposed by the HBA. It is loaded by the BIOS. It indicates which ports that HBA supports are available for software to use. For example, on an HBA that supports 3 ports as indicated in CAP.NP, only ports 0 and 1 could be available, with port 2 being unavailable. Software must not read or write to registers within unavailable ports. The intent of this register is to allow system vendors to build platforms that support less than the full number of ports implemented on the HBA silicon.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx0000000C; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description
31:8	GHC_PI_Reserved_31_8. Read-only. Reset: 00_0000h. Reserved.
7:0	GHC_PI. Read-write. Reset: FFh. Port Implemented. If a bit is set to 1, the corresponding port is available for software to use. If a bit is cleared to 0, the port is not available for software to use. The maximum number of bits set to 1 shall not exceed CAP.NP + 1, although the number of bits set in this register may be fewer than CAP.NP + 1. At least one bit should be set to 1. This register indicates which ports are exposed by the HBA. It is loaded by the BIOS. It indicates which ports that HBA supports are available for software to use. For example, on an HBA that supports 3 ports as indicated in CAP.NP, only ports 0 and 1 could be available, with port 2 being unavailable. Software must not read or write to registers within unavailable ports.

SATA[0...3]AHCIx00000010 (SYSHUB::SATA::AHCI::SATA_AHCI_GHC_VS)

Read-write. Reset: 0001_0301h.

This register indicates the major and minor version of the AHCI specification that the HBA implementation supports. The upper two bytes represent the major version number, and the lower two bytes represent the minor version number. Example: Version 3.12 would be represented as 00030102h. Three versions of the specification are valid: 0.95, 1.0, 1.1, 1.2, 1.3, and 1.3.1.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx00000010; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description
31:16	GHC_VS_MJR. Read-write. Reset: 0001h. Major Version Number. Indicates the major version is 1.
15:0	GHC_VS_MNR. Read-write. Reset: 0301h. Minor Version Number. Indicates the minor version is 31.

SATA[0...3]AHCIx00000014 (SYSHUB::SATA::AHCI::SATA_AHCI_GHC_CCC_CTL)

Reset: 0001_0140h.

The Command Completion Coalescing Control register is used to configure the command completion coalescing feature for the entire HBA. Implementation Note: HBA state variables (examples include hCccComplete and hCccTimer) are used to describe the required externally visible behavior. Implementations are not required to have internal state values that directly correspond to these variables.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx00000014; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description												
31:16	GHC_CCC_CTL_TV. Read-write. Reset: 0001h. Timeout Value. The timeout value is specified in 1 millisecond intervals. The timer accuracy shall be within 5%. hCccTimer is loaded with this timeout value. hCccTimer is only decremented when commands are outstanding on selected ports. HBA will signal a CCC interrupt when hCccTimer has decremented to 0. hCccTimer is reset to the timeout value on the assertion of each CCC interrupt. A timeout value of 0 is reserved.												
15:8	GHC_CCC_CTL_CC. Read-write. Reset: 01h. Command Completions. Specifies the number of command completions that are necessary to cause a CCC interrupt. HBA has an internal command completion counter, hCccComplete. hCccComplete is incremented by one each time a selected port has a command completion. When hCccComplete is equal to the command completions value, a CCC interrupt is signaled. The internal command completion counter is reset to 0 on the assertion of each CCC interrupt. A value of 0 for this field shall disable CCC interrupts being generated based on the number of commands completed, i.e. CCC interrupts are only generated based on the timer in this case. ValidValues: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>00h</td><td>A value of 0 for this field shall disable CCC interrupts being generated based on the number of commands completed.</td></tr> <tr> <td>01h</td><td>1 command completions are necessary to cause a CCC interrupt.</td></tr> <tr> <td>02h</td><td>2 command completions are necessary to cause a CCC interrupt.</td></tr> <tr> <td>03h</td><td>3 command completions are necessary to cause a CCC interrupt.</td></tr> <tr> <td>FFh-04h</td><td>4 to 255 command completions are necessary to cause a CCC interrupt.</td></tr> </table>	Value	Description	00h	A value of 0 for this field shall disable CCC interrupts being generated based on the number of commands completed.	01h	1 command completions are necessary to cause a CCC interrupt.	02h	2 command completions are necessary to cause a CCC interrupt.	03h	3 command completions are necessary to cause a CCC interrupt.	FFh-04h	4 to 255 command completions are necessary to cause a CCC interrupt.
Value	Description												
00h	A value of 0 for this field shall disable CCC interrupts being generated based on the number of commands completed.												
01h	1 command completions are necessary to cause a CCC interrupt.												
02h	2 command completions are necessary to cause a CCC interrupt.												
03h	3 command completions are necessary to cause a CCC interrupt.												
FFh-04h	4 to 255 command completions are necessary to cause a CCC interrupt.												
7:3	GHC_CCC_CTL_INT. Read-only. Reset: 08h. CCC Interrupt. Specifies the interrupt used by the CCC feature. This interrupt must be marked as unused in the Ports Implemented register (ABAR + 0Ch) by having the corresponding bit being set to 0. Thus, the CCC interrupt corresponds to the interrupt for an unimplemented port on the controller. When a CCC interrupt occurs, the corresponding bit of ABAR + 08h [31:0] (Interrupt Status. Interrupt Pending Status) shall be asserted to 1. This field also specifies the interrupt vector used for MSI. Note: 1) MS driver will not support CCC interrupt. 2) AMD in-house driver will support CCC interrupt but we require AMD driver to put the CCC_CTRL.INT to bit n+1 of PI[n:0] if all the PI[n:0] are all 1s or Empty PI[n:0] slot. So as to save system resources. For example, (i) For port 0 ~ 7 all implemented within PI, CCC_CTL.INT must be set to 08h. (ii) For port 0 and port 7 implemented within PI, CCC_CTL.INT can be set to any value of 1~6 but not >=8h.												
2:1	GHC_CCC_CTL_Reserved. Read-only. Reset: 0h. Reserved.												
0	GHC_CCC_CTL_En. Read-write. Reset: 0. 0=When cleared to 0, the command completion coalescing feature is disabled and no CCC interrupts are generated. 1=When set to 1, the command completion coalescing feature is enabled and CCC interrupts may be generated based on timeout or command completion conditions. CCC_CTL Enable.												

SATA[0...3]AHCIx00000018 (SYSHUB::SATA::AHCI::SATA_AHCI_GHC_CCC_PORTS)

Read-write. Reset: 0000_0000h.

The command completion coalescing ports register is used to specify the ports that are coalesced as part of the [CCC](#) feature when CCC_CTL.EN = 1.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx00000018; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description
31:0	GHC_CCC_PRT. Read-write. Reset: 0000_0000h. Ports. This register is bit significant. Each bit corresponds to a particular port, where bit 0 corresponds to port 0. If a bit is set to 1, the corresponding port is part of the command completion coalescing feature. If a bit is cleared to 0, the port is not part of the command completion coalescing feature. Bits set to 1 in this register must also have the corresponding bit set to 1 in the Ports Implemented register (ABAR + 0Ch). An updated value for this field shall take effect within one timer increment (1 millisecond).

SATA[0...3]AHCIx0000001C (SYSHUB::SATA::AHCI::SATA_AHCI_GHC_EM_LOC)

Read-only. Reset: 0000_0000h.

Enclosure Management Location

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx0000001C; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description
31:16	GHC_EM_LOC_OFST. Read-only. Reset: 0000h. Buffer Size. Specifies the size of the transmit message buffer area in Dwords. If both transmit and receive buffers are supported, then the transmit buffer begins at ABAR[EM_LOC.OFST*4] and the receive buffer directly follows it. If both transmit and receive buffers are supported, both buffers are of the size indicated in the Buffer Size field. A value of 0 is invalid.
15:0	GHC_EM_LOC_SZ. Read-only. Reset: 0000h. Buffer Size. Specifies the size of the transmit message buffer area in Dwords. If both transmit and receive buffers are supported, then the transmit buffer begins at ABAR[EM_LOC.OFST*4] and the receive buffer directly follows it. If both transmit and receive buffers are supported, both buffers are of the size indicated in the Buffer Size field. A value of 0 is invalid.

SATA[0...3]AHCIx00000020 (SYSHUB::SATA::AHCI::SATA_AHCI_GHC_EM_CTL)

Reset: 0000_0000h.

Enclosure Management Control

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx00000020; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description
31:28	GHC_EM_CTL_Reserved4. Read-only. Reset: 0h. Reserved.
27	GHC_EM_CTL_ATTR_PM. Read-only. Reset: 0. 0=HBA does not support enclosure management messages for devices attached via a Port Multiplier. When cleared to 0, software should use the Serial ATA enclosure management bridge that is built into many Port Multipliers for enclosure services with these devices. 1=HBA supports enclosure management messages for devices attached via a Port Multiplier. Enclosure Management Port Multiplier Support.
26	GHC_EM_CTL_ATTR_ALHD. Read-only. Reset: 0. Activity LED Hardware Driven. If set to 1, HBA drives the activity LED for the LED message type in hardware and does not utilize software settings for this LED. HBA does not begin transmitting the hardware based activity signal until after software has written CTL.TM=1 after a reset condition.
25	GHC_EM_CTL_ATTR_XMT. Read-only. Reset: 0. 0=HBA supports transmitting and receiving messages. 1=HBA only supports transmitting messages and does not support receiving messages. Enclosure Management Transmit Only.
24	GHC_EM_CTL_ATTR_SMB. Read-only. Reset: 0. 0=There are separate receive and transmit buffers so that unsolicited messages could be supported. 1=HBA has one message buffer that is shared for messages to transmit and messages received. In this case, unsolicited received messages are not supported and it is software's responsibility to manage access to this buffer. Single Message Buffer.
23:20	GHC_EM_CTL_Reserved3. Read-only. Reset: 0h. Reserved.
19	GHC_EM_CTL_SUPP_SGPIO. Read-only. Reset: 0. SGPIO Enclosure Management Messages. If set to 1, HBA supports the SGPIO register interface message type.
18	GHC_EM_CTL_SUPP_SES2. Read-only. Reset: 0. SES-2 Enclosure Management Messages. If set to 1, HBA supports the SES-2 message type.
17	GHC_EM_CTL_SUPP_SAFTE. Read-only. Reset: 0. SAF-TE Enclosure Management Messages. If set to 1, HBA supports the SAF-TE message type.
16	GHC_EM_CTL_SUPP_LED. Read-only. Reset: 0. LED Message Types. If set to 1, HBA supports the LED message type.
15:10	GHC_EM_CTL_Reserved2. Read-only. Reset: 00h. Reserved.
9	GHC_EM_CTL_RST. Read,Write-1-only. Reset: 0. 0=A write of 0 to this bit has no effect. 1=HBA shall reset all enclosure management message logic and the attached enclosure processor (if applicable) and take all appropriate reset actions to ensure messages can be transmitted/received after the reset. After HBA completes the reset operation, HBA shall set the value to 0. Enclosure Management Reset.
8	GHC_EM_CTL_TM. Read,Write-1-only. Reset: 0. 0=A write of 0 to this bit by software shall have no effect. Software shall not change the contents of the message buffer while CTL.TM is set to 1. 1=HBA shall transmit the message contained in the message buffer. When the message is completely sent, HBA shall clear this bit to 0. Enclosure Management Transmit Message.
7:1	GHC_EM_CTL_Reserved1. Read-only. Reset: 00h. Reserved.
0	GHC_EM_CTL_STS_MR. Read-only. Reset: 0. 0=A write of 0 to this bit by software shall have no effect. 1=HBA sets this bit to a 1 when a message is completely received into the message buffer. When software detects this bit is a 1, software should read the message and perform any actions necessary. When software finishes reading the message in the buffer, software writes a 1 to this bit in order to clear it. Enclosure Management Message Received.

SATA[0...3]AHCIx00000024 (SYSHUB::SATA::AHCI::SATA_AHCI_HBA_CAP_EXT)

Read-only. Reset: 0000_0010h.

HBA Capabilities Extended - This register indicates capabilities of the HBA to driver software.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx00000024; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description
31:6	GHC_CAP_EXT_Reserved1. Read-only. Reset: 000_0000h. Reserved.
5	GHC_CAP2_DESO. Read-only. Reset: 0. 0=HBA may enter DevSleep from any link state (Active, Partial, or Slumber). 1=HBA shall ignore software directed entry to DevSleep via PxCMD.ICC unless PxSSTS.IPM = 6h. DevSleep Entrance from Slumber Only. This field specifies that HBA shall only assert the DEVSLP signal if the interface is in Slumber.
4	GHC_CAP2_SADM. Read-only. Reset: 1. 0=This function is not supported and software shall treat the PxDEVSLP.ADSE field as reserved. 1=HBA supports hardware assertion of the DEVSLP signal after the idle timeout expires. Supports Aggressive Device Sleep Management.
3	GHC_CAP2_SDS. Read-only. Reset: 0. 0=DEVSLP is not supported and software shall not set PxCMD.ICC to 8h. 1=HBA supports the Device Sleep feature. Supports Device Sleep.
2:1	GHC_CAP_EXT_Reserved. Read-only. Reset: 0h. Reserved.
0	GHC_CAP_EXT_BOH. Read-only. Reset: 0. 0=HBA does not support the BIOS/OS handoff mechanism. It indicates that HBA does not support BIOS/OS handoff and the BOHC global HBA register is not implemented. 1=HBA supports the BIOS/OS handoff mechanism. When BIOS/OS handoff is supported, HBA has implemented the BOHC global HBA register (Offset:28h). BIOS/OS Handoff.

SATA[0...3]AHCIx00000028 (SYSHUB::SATA::AHCI::SATA_AHCI_BOHC)

Reset: 0000_0000h.

BIOS/OS Handoff Control and Status - This register controls various global actions of the HBA. This register is not affected by HBA reset.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]AHCIx00000028; SATA[3:0]AHCI=03[4:1]0_1000h

Bits	Description
31:5	BOHC_Reserved. Read-only. Reset: 000_0000h. Reserved.
4	BOHC_BB. Read-write. Reset: 0. BIOS Busy. This bit is used by the BIOS to indicate that it is busy cleaning up for ownership change.
3	BOHC_OOC. Read,Write-1-to-clear. Reset: 0. 0=A write of 0 to this bit has no effect. 1=Clear this bit. OS Ownership Change. This bit is set to 1 when the OOS bit transits from 0 to 1.
2	BOHC_SOOE. Read-write. Reset: 0. SMI on OS Ownership Change Enable. This bit, when set to 1, enables an SMI when the OOC bit has been set to 1.
1	BOHC_OOS. Read-write. Reset: 0. OS Owned Semaphore. The system software sets this bit to request ownership of the HBA controller. Ownership is obtained when this bit returns 1 and the BOS bit returns 0.
0	BOHC_BOS. Read-write. Reset: 0. 0=In response to a request for ownership of the HBA by system software via OOS. 1=Establish ownership of the HBA controller. BIOS Owned Semaphore.

15.2.3.2 SATA Port Registers

Table 273: List of Acronyms and Terms used in SATA PORT

Term	Definition
AHCI	Advanced Host Controller Interface.
D2H	Device to HBA.
FIS	Frame Information Structure.
LBA	Logical Block Addressing.
PIO	Programmed IO Commands.
PRD	Physical Region Descriptor.

SATA0PORT0x00000000...SATA3PORT7x00000000 (SYSHUB::SATA::PORT::SATA_AHCI_P_CLB)

Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000000; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000000; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000000; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000000; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:10	P_CLB . Read-write. Reset: 00_0000h. Indicates the 32-bit base physical address for the command list for this port.
9:0	P_CLB_Reserved . Read-only. Reset: 000h. Reserved.

SATA0PORT0x00000004...SATA3PORT7x00000004 (SYSHUB::SATA::PORT::SATA_AHCI_P_CLBU)

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000004; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000004; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000004; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000004; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:0	P_CLBU . Read-write. Reset: 0000_0000h. Indicates the upper 32-bits for the command list base physical address for this port.

SATA0PORT0x00000008...SATA3PORT7x00000008 (SYSHUB::SATA::PORT::SATA_AHCI_P_FB)

Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000008; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000008; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000008; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000008; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:8	P_FB . Read-write. Reset: 00_0000h. Indicates the 32-bit base physical address for received FISes.
7:0	P_FB_Reserved . Read-only. Reset: 00h. Reserved.

SATA0PORT0x0000000C...SATA3PORT7x0000000C (SYSHUB::SATA::PORT::SATA_AHCI_P_FBU)

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x0000000C; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x0000000C; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x0000000C; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x0000000C; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:0	P_FBU . Read-write. Reset: 0000_0000h. Indicates the upper 32-bits for the received FIS base physical address for this port.

SATA0PORT0x00000010...SATA3PORT7x00000010 (SYSHUB::SATA::PORT::SATA_AHCI_P_IS)

Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000010; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000010; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000010; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000010; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31	P_IS_CPDS. Read,Write-1-to-clear. Reset: 0. When set, a device status has changed as detected by the cold presence detect logic.
30	P_IS_TFES. Read,Write-1-to-clear. Reset: 0. This bit is set whenever the status register is updated by the device and the error bit (bit 0) is set.
29	P_IS_HBFS. Read,Write-1-to-clear. Reset: 0. Indicates that HBA encountered a host bus error that it cannot recover from, such as a bad software pointer.
28	P_IS_HBDS. Read,Write-1-to-clear. Reset: 0. Indicates that HBA encountered a data error (uncorrectable ECC / parity) when reading from or writing to system memory.
27	P_IS_IFS. Read,Write-1-to-clear. Reset: 0. Indicates that HBA encountered an error on the Serial ATA interface, which caused the transfer to stop.
26	P_IS_INFS. Read,Write-1-to-clear. Reset: 0. Indicates that HBA encountered an error on the Serial ATA interface and was able to continue operation.
25	P_IS_Reserved2. Read-only. Reset: 0. Reserved.
24	P_IS_OFS. Read,Write-1-to-clear. Reset: 0. Indicates that HBA received more bytes from a device than was specified in the PRD table for the command.
23	P_IS_IPMS. Read,Write-1-to-clear. Reset: 0. Indicates that HBA received a FIS from a device whose Port Multiplier field did not match what was expected.
22	P_IS_PRCs. Read-only. Reset: 0. When set to 1 indicates the internal PhyRdy signal changed state. To clear this bit, software must clear POSERR.DIAG.N to 0.
21:8	P_IS_Reserved1. Read-only. Reset: 0000h. Reserved.
7	P_IS_DMPS. Read,Write-1-to-clear. Reset: 0. When set, indicates that a mechanical presence switch attached to this port has been opened or closed, which may lead to a change in the connection state of the device.
6	P_IS_PCS. Read-only. Reset: 0. 0=No change in Current Connect Status. 1=Change in Current Connect Status.
5	P_IS_DPS. Read,Write-1-to-clear. Reset: 0. When the bit is set, a PRD with the 1-bit set has transferred all of its data.
4	P_IS_UFS. Read-only. Reset: 0. When the bit is set, an unknown FIS was received with the 1-bit set and has been copied into system memory.
3	P_IS_SDBS. Read,Write-1-to-clear. Reset: 0. When the bit is set, a Set Device Bits FIS has been received with the 1-bit set and has been copied into system memory.
2	P_IS_DSS. Read,Write-1-to-clear. Reset: 0. When the bit is set, a DMA Setup FIS has been received with the 1-bit set and has been copied into system memory.
1	P_IS_PSS. Read,Write-1-to-clear. Reset: 0. When the bit is set, A PIO Setup FIS has been received with the 1-bit set, and it has been copied into system memory, and the data related to that FIS has been transferred.
0	P_IS_DHRS. Read,Write-1-to-clear. Reset: 0. When the bit is set, a D2H Register FIS has been received with the 1-bit set, and has been copied into system memory.

SATA0PORT0x00000014...SATA3PORT7x00000014 (SYSHUB::SATA::PORT::SATA_AHCI_P_IE)

Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000014; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000014; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000014; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000014; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31	P_IE_CPDE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0S.CPDS are set.
30	P_IE_TFEE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0S.TFES are set.
29	P_IE_HBFE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.HBFS are set.
28	P_IE_HBDE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.HBDS are set.
27	P_IE_IFE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.IFS are set.
26	P_IE_INFE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.INFS are set.
25	P_IE_Reserved2. Read-only. Reset: 0. Reserved.
24	P_IE_OFE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.OFS are set.
23	P_IE_IPME. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.IPMS are set.
22	P_IE_PRCE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.PRCS are set to 1.
21:9	P_IE_Reserved1. Read-only. Reset: 0000h. Reserved.
8	P_IE_ZPDE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.ZPDS are set.
7	P_IE_DMPE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.DMPS are set.
6	P_IE_PCE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.PCS are set.
5	P_IE_DPE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, HBA will generate an interrupt if both GHC.IE and P0IS.DPS are set.
4	P_IE_UFE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, if both GHC.IE and P0IS.UFS are set to 1, HBA will generate an interrupt.
3	P_IE_SDBE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, if both GHC.IE and P0IS.SDBS are set, HBA will generate an interrupt.
2	P_IE_DSE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, if both GHC.IE and P0IS.DSS are set, HBA will generate an interrupt.
1	P_IE_PSE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, if both GHC.IE and P0IS.PSS are set, HBA will generate an interrupt.
0	P_IE_DHRE. Read-write. Reset: 0. 0=Interrupt is disabled. 1=Interrupt is enabled. When set, and if both GHC.IE and P0IS.DHRS are set, HBA will generate an interrupt.

SATA0PORT0x00000018...SATA3PORT7x00000018 (SYSHUB::SATA::PORT::SATA_AHCI_P_CMD)

Reset: 0040_2006h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000018; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000018; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000018; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000018; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description**31:28 P_CMD_ICC.** Read-write. Reset: 0h. This field is used to control power management states of the interface.**ValidValues:**

Value	Description
0h	0000: No-Op/Idle: When software read this value, it indicates HBA is ready to accept a new interface control command, although the transition to the previously selected state may not have occurred yet.
1h	0001: Active: This shall cause HBA to request a transition of the interface into the active state.
2h	0010: Partial: This shall cause HBA to request a transition of the interface into partial state. The SATA device may reject the request and interface shall remain in current state.
3h	0011: Slumber: This shall cause HBA to request a transition of the interface into slumber state. The SATA device may reject the request and interface shall remain in current state.
4h	0100: Reserved.
5h	0101: Reserved.
6h	0110: Reserved.
7h	1000: DevSleep: This shall cause HBA to request to assert the DEVSLP signal associated with the port. HBA shall ignore the Device Sleep Idle Timeout value. Software shall only request DevSleep when the interface is in an idle state.
Fh-8h	Reserved.

27 P_CMD_ASP. Read-write. Reset: 0. 0=When cleared, and ALPE is set, HBA shall aggressively enter the Partial state when it clears the PxCI register and the PxSACT register is cleared or when it clears the PxSACT register and PxCI is cleared. 1=When set to 1, and ALPE is set, HBA shall aggressively enter the Slumber state when it clears the PxCI register and the PxSACT register is cleared or when it clears the PxSACT register and PxCI is cleared. If CAP.SALP(ABAR+00h[26]) is cleared to 0, software shall treat this bit as reserved.

26 P_CMD_ALPE. Read-write. Reset: 0. 0=When cleared to 0, HBA shall not aggressively enter a lower link power state (Partial or Slumber). 1=When set to 1, HBA shall aggressively enter a lower link power state (Partial or Slumber) based upon the setting of the ASP bit. When set to 1, HBA shall aggressively enter a lower link power state (Partial or Slumber) based upon the setting of the [ASP](#) bit.

25 P_CMD_DLAЕ. Read-write. Reset: 0. 0=When cleared to 0, HBA shall only drive the LED pin active for commands if POCMD.ATAPI is set to 0. 1=When set to 1, HBA shall drive the LED pin active for commands regardless of the state of POCMD.ATAPI (ABAR + port offset + 18h[24]). When set to 1, HBA shall drive the LED pin active for commands regardless of the state of POCMD.ATAPI.

24 P_CMD_ATAPI. Read-write. Reset: 0. 0=This connected device is not an ATAPI device. 1=When set to 1, this connected device is an ATAPI device. When set to 1, the connected device is an ATAPI device.

23 P_CMD_Reserved2. Read-only. Reset: 0. Reserved.

22 P_CMD_FBSCP. Read-only. Reset: 1. 0=When cleared to 0, it indicates that this port does not support FIS-based switching. 1=When set to 1, it indicates that this port supports Port Multiplier FIS-based switching. This bit may only be set to 1 if both CAP.SPM (ABAR + 00h[17]) and CAP.FBSS (ABAR + 00h[16]) are set to 1.

21 P_CMD_ESP. Read-only. Reset: 0. 0=When cleared to 0, it indicates that this port's signal connector is not externally accessible on a signal only connector. ESP is mutually exclusive with the HPCP bit in this register. 1=When set to 1, it indicates that this port's signal connector is externally accessible on a signal only connector. CAP.SXS (ABAR + 00h[5]) shall be set to 1. ESP is mutually exclusive with the HPCP bit in this register.

20 P_CMD_CPD. Read-only. Reset: 0. 0=When cleared to 0, the platform does not support cold presence detection on this port. 1=When set to 1, the platform supports cold presence detection on this port. If set to 1, the platform supports cold presence detection on this port.

19	P_CMD_MPSP. Read-only. Reset: 0. 0=If cleared to 0, the platform does not support a mechanical presence switch attached to this port. 1=If set to 1, the platform supports an mechanical presence switch attached to this port. If set to 1, the platform supports an mechanical presence switch attached to this port.
18	P_CMD_HPCP. Read-only. Reset: 0. 0=When cleared to 0, it indicates that this port's signal and power connectors are not externally accessible via a joint signal and power connector. 1=When set to 1, it indicates that this port's signal and power connectors are externally accessible via a joint signal and power connector for blindmate device hot plug. HPCP is mutually exclusive with the ESP bit in this register.
17	P_CMD_PMA. Read-write. Reset: 0. 0=When cleared to 0 by software, a Port Multiplier is not attached to the HBA for this port. 1=When set to 1 by software, a Port Multiplier is attached to the HBA for this port. This bit is read/write for HBAs that support a Port Multiplier. Software is responsible for detecting whether a Port Multiplier is present.
16	P_CMD_CPS. Read-only. Reset: 0. 0=If cleared to 0, HBA detects via cold presence that there is no device attached to this port. 1=If set to 1, HBA detects via cold presence that a device is attached to this port. The CPS bit reports whether a device is currently detected on this port via cold presence detection.
15	P_CMD_CR. Read-write. Reset: 0. When this bit is set, the command list DMA engine for the port is running.
14	P_CMD_FR. Read-write. Reset: 0. When set, the FIS Receive DMA engine for the port is running.
13	P_CMD_MPSS. Read-only. Reset: 1. The MPSS bit reports the state of a mechanical presence switch attached to this port.
12:8	P_CMD_CCS. Read-only. Reset: 00h. This field is valid when P_CMD_ST (bit 0) is set to 1 and shall be set to the command slot value of the command that is currently being issued by the HBA.
7:5	P_CMD_Reserved1. Read-only. Reset: 0h. Reserved.
4	P_CMD_FRE. Read-write. Reset: 0. 0=When cleared, received FISes are not accepted by the HBA, except for the first D2H register FIS after the initialization sequence. And there are no FISes posted to the FIS receiving area. 1=When set, HBA may post received FISes into the FIS receive area pointed to by PxFB (and for 64-bit HBAs, PxFBU). When set, HBA may post received FISes into the FIS receive area pointed to by PxFB (and for 64-bit HBAs, PxFBU).
3	P_CMD_CLO. Read-write. Reset: 0. 0=Setting this bit to 0 when PxTFD.STS.BSY and PxTFD.STS.DRQ have been cleared to 0. A write of 0 to this register shall have no effect. 1=Setting this bit to 1 causes P0TFD.STS.BSY and P0TFD.STS.DRQ to be cleared to 0. Setting this bit to 1 causes P0TFD.STS.BSY and P0TFD.STS.DRQ to be cleared to 0.
2	P_CMD_POD. Read-write. Reset: 1. This bit is read/write for HBAs that support cold presence detection on this port as indicated by P0CMD.CPD set to 1. This bit reads 1 only for HBAs that do not support cold presence detect.
1	P_CMD_SUD. Read-write. Reset: 1. This bit is read/write for HBAs that support staggered spin-up via CAP.SSS. This bit reads 1 only for HBAs that do not support staggered spin-up.
0	P_CMD_ST. Read-write. Reset: 0. When set, HBA may process the command list.

SATA0PORT0x0000001C...SATA3PORT7x0000001C (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED1)

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x0000001C; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x0000001C; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x0000001C; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x0000001C; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:0 **P_RESERVED1.** Read-write. Reset: 0000_0000h. Reserved.

SATA0PORT0x00000020...SATA3PORT7x00000020 (SYSHUB::SATA::PORT::SATA_AHCI_P_TFD)

Read-only. Reset: 0000_007Fh.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000020; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000020; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000020; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000020; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:16 **P_TFD_Reserved.** Read-only. Reset: 0000h. Reserved.15:8 **P_TFD_ERR.** Read-only. Reset: 00h. Contains the newest copy of the task file error register.7:0 **P_TFD_STS.** Read-only. Reset: 7Fh. Contains the newest copy of the task file status register.**SATA0PORT0x00000024...SATA3PORT7x00000024 (SYSHUB::SATA::PORT::SATA_AHCI_P_SIG)**

Read-only. Reset: FFFF_FFFFh.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000024; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000024; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000024; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000024; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:0 **P_SIG.** Read-only. Reset: FFFF_FFFFh. Contains the signature received from a device on the first D2H Register FIS.

SATA0PORT0x00000028...SATA3PORT7x00000028 (SYSHUB::SATA::PORT::SATA_AHCI_P_SSTS)

Read-only. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000028; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000028; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000028; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000028; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description**31:12 P_SSTS_Reserved.** Read-only. Reset: 0_0000h. Reserved.**11:8 P_SSTS_IPM.** Read-only. Reset: 0h. Indicates the current interface state. Read only.**ValidValues:**

Value	Description
0h	0000: Device not present or communication not established.
1h	0001: Interface in Active state.
2h	0010: Interface in Partial power management state.
3h	0110: Interface in Slumber power management state.
4h	1000: Interface in DevSleep power management state.
Fh-5h	Reserved.

7:4 P_SSTS_SPD. Read-only. Reset: 0h. Indicates the negotiated interface communication speed. Read only.**ValidValues:**

Value	Description
0h	0000: Device not present or communication not established.
1h	0001: Generation 1 communication rate negotiated.
2h	0010: Generation 2 communication rate negotiated.
3h	0011: Generation 3 communication rate negotiated.
Fh-4h	Reserved.

3:0 P_SSTS_DET. Read-only. Reset: 0h. Indicates the interface device detection and Phy state. Read only.**ValidValues:**

Value	Description
0h	0000: No device detected and PHY communication not established.
1h	0001: Device presence detected but PHY communication not established.
2h	0011: Device presence detected and PHY communication established.
3h	0100: PHY in offline mode as a result of the interface being disabled or running in a BIST loopback mode.
Fh-4h	Reserved.

SATA0PORT0x0000002C...SATA3PORT7x0000002C (SYSHUB::SATA::PORT::SATA_AHCI_P_SCTL)

Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x0000002C; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x0000002C; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x0000002C; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x0000002C; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description																				
31:20	P_SCTL_Reserved. Read-only. Reset: 000h. Reserved.																				
19:16	P_SCTL_PMP. Read-only. Reset: 0h. This field is not used by AHCI. Read only.																				
15:12	P_SCTL_SPM. Read-only. Reset: 0h. This field is not used by AHCI. Read only.																				
11:8	P_SCTL_IPM. Read-write. Reset: 0h. Indicates which power states HBA is allowed to transition to.																				
	ValidValues:																				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>0000: No interface restrictions.</td></tr> <tr> <td>1h</td><td>0001: Transitions to the Partial state are disabled.</td></tr> <tr> <td>2h</td><td>0010: Transitions to the Slumber state are disabled.</td></tr> <tr> <td>3h</td><td>0011: Transitions to both Partial and Slumber states are disabled.</td></tr> <tr> <td>4h</td><td>0100: Transitions to the DevSleep power management state are disabled.</td></tr> <tr> <td>5h</td><td>0101: Transitions to the Partial and DevSleep power management states are disabled.</td></tr> <tr> <td>6h</td><td>0110: Transitions to the Slumber and DevSleep power management states are disabled.</td></tr> <tr> <td>7h</td><td>0111: Transitions to the Partial, Slumber and DevSleep power management states are disabled.</td></tr> <tr> <td>Fh-8h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	0000: No interface restrictions.	1h	0001: Transitions to the Partial state are disabled.	2h	0010: Transitions to the Slumber state are disabled.	3h	0011: Transitions to both Partial and Slumber states are disabled.	4h	0100: Transitions to the DevSleep power management state are disabled.	5h	0101: Transitions to the Partial and DevSleep power management states are disabled.	6h	0110: Transitions to the Slumber and DevSleep power management states are disabled.	7h	0111: Transitions to the Partial, Slumber and DevSleep power management states are disabled.	Fh-8h	Reserved.
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Fh-8h	Reserved.																				
7:4	P_SCTL_SPD. Read-write. Reset: 0h. Indicates the highest allowable speed of the interface.																				
	ValidValues:																				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>0000: Reserved.</td></tr> <tr> <td>1h</td><td>0001: Gen 1 (1.5 Gbps).</td></tr> <tr> <td>2h</td><td>0010: Gen 2 (3 Gbps).</td></tr> <tr> <td>3h</td><td>0011: Gen 3 (6 Gbps).</td></tr> <tr> <td>Fh-4h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	0000: Reserved.	1h	0001: Gen 1 (1.5 Gbps).	2h	0010: Gen 2 (3 Gbps).	3h	0011: Gen 3 (6 Gbps).	Fh-4h	Reserved.								
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3h	0011: Gen 3 (6 Gbps).																				
Fh-4h	Reserved.																				
3:0	P_SCTL_DET. Read-write. Reset: 0h. Controls the HBA's device detection and interface initialization.																				
	ValidValues:																				
	<table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>0000: No device detection or initialization action requested.</td></tr> <tr> <td>1h</td><td>0001: Perform interface communication initialization sequence to establish communication.</td></tr> <tr> <td>2h</td><td>0100: Disable the Serial ATA interface and put PHY in offline mode.</td></tr> <tr> <td>Fh-3h</td><td>Reserved.</td></tr> </table>	Value	Description	0h	0000: No device detection or initialization action requested.	1h	0001: Perform interface communication initialization sequence to establish communication.	2h	0100: Disable the Serial ATA interface and put PHY in offline mode.	Fh-3h	Reserved.										
Value	Description																				
0h	0000: No device detection or initialization action requested.																				
1h	0001: Perform interface communication initialization sequence to establish communication.																				
2h	0100: Disable the Serial ATA interface and put PHY in offline mode.																				
Fh-3h	Reserved.																				

SATA0PORT0x00000030...SATA3PORT7x00000030 (SYSHUB::SATA::PORT::SATA_AHCI_P_SERR)

Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000030; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000030; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000030; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000030; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:27	P_SERR_DIAG_Reserved. Read-only. Reset: 00h. Reserved.
26	P_SERR_DIAG_X. Read,Write-1-to-clear. Reset: 0. Exchanged (X)
25	P_SERR_DIAG_F. Read-only. Reset: 0. Unknown FIS Type (F)
24	P_SERR_DIAG_T. Read-only. Reset: 0. Transport state transition error (T)
23	P_SERR_DIAG_S. Read-only. Reset: 0. Link Sequence Error (S)
22	P_SERR_DIAG_H. Read-only. Reset: 0. Handshake Error (H)
21	P_SERR_DIAG_C. Read-only. Reset: 0. CRC Error (C)
20	P_SERR_DIAG_D. Read-only. Reset: 0. Disparity Error (D)
19	P_SERR_DIAG_B. Read-only. Reset: 0. 10B to 8B Decode Error (B)
18	P_SERR_DIAG_W. Read-only. Reset: 0. Comm Wake (W)
17	P_SERR_DIAG_I. Read-only. Reset: 0. Phy Internal Error (I)
16	P_SERR_DIAG_N. Read,Write-1-to-clear. Reset: 0. PhyRdy Change (N)
15:12	P_SERR_ERR_Reserved2. Read-only. Reset: 0h. Reserved.
11	P_SERR_ERR_E. Read,Write-1-to-clear. Reset: 0. Internal Error (E)
10	P_SERR_ERR_P. Read-only. Reset: 0. Protocol Error (P)
9	P_SERR_ERR_C. Read-only. Reset: 0. Persistent Communication or Data Integrity Error (C)
8	P_SERR_ERR_T. Read-only. Reset: 0. Transient Data Integrity Error (T)
7:2	P_SERR_ERR_Reserved1. Read-only. Reset: 00h. Reserved.
1	P_SERR_ERR_M. Read-only. Reset: 0. Recovered Communications Error (M)
0	P_SERR_ERR_I. Read-only. Reset: 0. Recovered Data Integrity Error (I)

SATA0PORT0x00000034...SATA3PORT7x00000034 (SYSHUB::SATA::PORT::SATA_AHCI_P_SACT)

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000034; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000034; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000034; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000034; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:0	P_SACT. Read-write. Reset: 0000_0000h. This field is bit significant. Each bit corresponds to the TAG and command slot of a native queued command.

SATA0PORT0x00000038...SATA3PORT7x00000038 (SYSHUB::SATA::PORT::SATA_AHCI_P_CI)

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000038; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000038; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000038; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000038; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:0	P_CI. Read-write. Reset: 0000_0000h. This field is bit significant. Each bit corresponds to a command slot

SATA0PORT0x0000003C...SATA3PORT7x0000003C (SYSHUB::SATA::PORT::SATA_AHCI_P_NTF)

Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x0000003C; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x0000003C; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x0000003C; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x0000003C; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:16 **P_NTF_Reserved**. Read-only. Reset: 0000h. Reserved.15:0 **P_NTF_PMN**. Read,Write-1-to-clear. Reset: 0000h. This field indicates whether a particular device with the corresponding PM Port number issued a Set Device Bits FIS to the host with the Notification bit set.**SATA0PORT0x00000040...SATA3PORT7x00000040 (SYSHUB::SATA::PORT::SATA_AHCI_P_FBSS)**

Reset: 0000_2000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000040; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000040; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000040; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000040; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:20 **P_FBSS_Reserved2**. Read-only. Reset: 000h. Reserved.19:16 **P_FBSS_DWE**. Read-only. Reset: 0h. Set by hardware to the value of the Port Multiplier port number of the device that experienced a fatal error condition. This field is only valid when POFBS.SDE = 1.15:12 **P_FBSS_ADO**. Read-only. Reset: 2h. This register exposes the number of active devices that the FIS-based switching implementation has been optimized for.11:8 **P_FBSS_DEV**. Read-write. Reset: 0h. Set by software to the Port Multiplier port value of the next command to issue.7:3 **P_FBSS_Reserved1**. Read-only. Reset: 00h. Reserved.2 **P_FBSS_SDE**. Read-only. Reset: 0. 0=When cleared to 0 and a fatal error condition has occurred. The error applies to the entire port and PxCMD.ST shall be cleared to 0 by software. This bit is cleared on PxFBS.DEC being set to 1 or on PxCMD.ST being cleared to 0. 1=When set to 1 and a fatal error condition has occurred, hardware believes the error is localized to one device such that software is first error recovery step should be to utilize the POFBS.DEC functionality. When set to 1 and a fatal error condition has occurred, hardware believes the error is localized to one device such that software is first error recovery step should be to utilize the POFBS.DEC functionality.1 **P_FBSS_DEC**. Read-write. Reset: 0. 0=A write of 0 to this bit by software shall have no effect. 1=When set to 1 by software, HBA shall clear the device-specific error condition and HBA shall flush any commands outstanding for the device that experienced the error. When hardware has completed error recovery actions, hardware shall clear the bit to 0. When set to 1 by software, HBA shall clear the device-specific error condition and HBA shall flush any commands outstanding for the device that experienced the error.0 **P_FBSS_En**. Read-write. Reset: 0. 0=When cleared to 0, FIS-based switching is not being used. 1=When set to 1, a Port Multiplier is attached and HBA shall use FIS-based switching to communicate with it. FIS-based switching enable bit

SATA0PORT0x00000044...SATA3PORT7x00000044 (SYSHUB::SATA::PORT::SATA_AHCI_P_DEVSLP)

Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000044; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000044; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000044; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000044; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:29	P_DEVSLP_Reserved. Read-only. Reset: 0h. Reserved.
28:25	P_DEVSLP_DM. Read-only. Reset: 0h. 0-based value that specifies the DITO multiplier that HBA applies to the specified DITO value, effectively extending the range of DITO from 1ms to 16383ms.
24:15	P_DEVSLP_DITO. Read-write. Reset: 000h. This field specifies the amount of the time (in approximate 1ms granularity) that HBA shall wait before driving the DEVSLP signal.
14:10	P_DEVSLP_MDAT. Read-write. Reset: 00h. This field specifies the minimum amount of time (in 1ms granularity) that HBA must assert the DEVSLP signal before it may be de-asserted.
9:2	P_DEVSLP_DETO. Read-write. Reset: 00h. This field specifies the maximum duration (in approximate 1ms granularity) from DEVSLP de-assertion until the device is ready to accept OOB .
1	P_DEVSLP_DSP. Read-only. Reset: 0. 0=The platform does not support DEVSLP on this port. 1=The platform supports DEVSLP on this port. This bit may only be set to 1 if CAP2.SDS is set to 1. DSP is mutually exclusive with the PxCMD.HPCP bit and PxCMD.ESP bit.
0	P_DEVSLP_ADSE. Read-write. Reset: 0. 0=When set to 0 and CAP2.DESO = 1, HBA shall assert the DEVSLP signal after the port has been idle (PxCI = 0h and PxSACT = 0h) for the amount of time specified by PxDEVSLP.DITO and the interface is in Slumber (PxSSTS.IPM = 6h).. 1=When set to 1, HBA shall assert the DEVSLP signal after the port has been idle (PxCI = 0h and PxSACT = 0h) for the amount of time specified by the PxDEVSLP.DITO register. This bit is read/write for HBAs that support aggressive DEVSLP management (CAP2.SADM = '1'). This bit is read-only for HBAs that do not support aggressive Device Sleep management (CAP2.SADM = 0) or CAP2.SDS is cleared to 0.

SATA0PORT0x00000048...SATA3PORT7x00000048 (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED3)

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000048; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000048; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000048; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000048; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:0	P_RESERVED3. Read-write. Reset: 0000_0000h. Reserved.

SATA0PORT0x0000004C...SATA3PORT7x0000004C (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED4)

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x0000004C; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x0000004C; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x0000004C; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x0000004C; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:0	P_RESERVED4. Read-write. Reset: 0000_0000h. Reserved.

SATA0PORT0x00000050...SATA3PORT7x00000050 (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED5)

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000050; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000050; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000050; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000050; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits	Description
31:0	P_RESERVED5. Read-write. Reset: 0000_0000h. Reserved.

SATA0PORT0x00000054...SATA3PORT7x00000054 (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED6)

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000054; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000054; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000054; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000054; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:0 **P_RESERVED6.** Read-write. Reset: 0000_0000h. Reserved.**SATA0PORT0x00000058...SATA3PORT7x00000058 (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED7)**

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000058; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000058; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000058; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000058; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:0 **P_RESERVED7.** Read-write. Reset: 0000_0000h. Reserved.**SATA0PORT0x0000005C...SATA3PORT7x0000005C (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED8)**

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x0000005C; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x0000005C; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x0000005C; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x0000005C; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:0 **P_RESERVED8.** Read-write. Reset: 0000_0000h. Reserved.**SATA0PORT0x00000060...SATA3PORT7x00000060 (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED9)**

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000060; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000060; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000060; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000060; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:0 **P_RESERVED9.** Read-write. Reset: 0000_0000h. Reserved.**SATA0PORT0x00000064...SATA3PORT7x00000064 (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED10)**

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000064; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000064; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000064; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000064; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:0 **P_RESERVED10.** Read-write. Reset: 0000_0000h. Reserved.**SATA0PORT0x00000068...SATA3PORT7x00000068 (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED11)**

Read-write. Reset: 0000_0000h.

_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x00000068; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h

_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x00000068; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h

_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x00000068; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h

_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x00000068; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h

Bits Description31:0 **P_RESERVED11.** Read-write. Reset: 0000_0000h. Reserved.

SATA0PORT0x0000006C...SATA3PORT7x0000006C (SYSHUB::SATA::PORT::SATA_AHCI_P_RESERVED12)

Read-write. Reset: 0000_0000h.	
_instSATA0_port[7:0]_aliasSMN; SATA0PORT[7:0]x0000006C; SATA0PORT[7:0]=0310_1[48,40,38,30,28,20,18,10]0h	
_instSATA1_port[7:0]_aliasSMN; SATA1PORT[7:0]x0000006C; SATA1PORT[7:0]=0320_1[48,40,38,30,28,20,18,10]0h	
_instSATA2_port[7:0]_aliasSMN; SATA2PORT[7:0]x0000006C; SATA2PORT[7:0]=0330_1[48,40,38,30,28,20,18,10]0h	
_instSATA3_port[7:0]_aliasSMN; SATA3PORT[7:0]x0000006C; SATA3PORT[7:0]=0340_1[48,40,38,30,28,20,18,10]0h	
Bits	Description
31:0	P_RESERVED12. Read-write. Reset: 0000_0000h. Reserved.

15.2.3.3 SATA Serial General Purpose IO (SGPIO) Registers

Table 274: List of Acronyms and Terms used in SATA SGPIO

Term	Definition
SGPIO	Serial General Purpose IO.

SATA[0...3]SGPIOx00000000 (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_MESS_HEADER)

Reset: 0000_0000h.	
_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx00000000; SATA[3:0]SGPIO=03[4:1]0_1500h	
Bits	Description
31:28	SGPIO_MEASSAGE_HEADER_Reserved2. Read-only. Reset: 0h. Specifies the type of the message.
27:24	SGPIO_MEASSAGE_HEADER_MTYPE. Read-write. Reset: 0h. Specifies the type of the message.
23:16	SGPIO_MEASSAGE_HEADER_DSIZE. Read-write. Reset: 00h. Specifies the data size in bytes.
15:8	SGPIO_MEASSAGE_HEADER_MSIZ. Read-write. Reset: 00h. Specifies the size of the message in bytes. The message size does not include the one Dword header. A value of '0' is invalid.
7:0	SGPIO_MEASSAGE_HEADER_Reserved. Read-only. Reset: 00h. Reserved.

SATA[0...3]SGPIOx00000004 (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_CFG)

Read-write. Reset: 0000_0000h.	
_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx00000004; SATA[3:0]SGPIO=03[4:1]0_1500h	
Bits	Description
31:24	SGPIO_REG_INDEX. Read-write. Reset: 00h. Specifies the index of the first register in the bank to write. Currently supported index range is: REG_TYPE=00h: 0 to 1, REG_TYPE=03h: 0, REG_TYPE=04h: 0 to 7, REG_TYPE=0Ch: 0. It is software is responsibility to make sure the index of which the first register it attempts to access is valid and supported, as hardware does not provide a mechanism to return a status to software indicating that the current operation fails. An out-of-range index for a specific type of register will be treated as do not care and none of the data for such index will be written.
23:16	SGPIO_REG_TYPE. Read-write. Reset: 00h. Specifies the bank of registers to write. Currently supported Register Types are: 00h: SGPIO Configuration, 03h: Transmit, 04h: General Purpose Transmit, C0h: AMD Configuration, All other values are reserved. It is software is responsibility to make sure the type of register that it attempts to access is valid and supported, as hardware does not provide a mechanism to return a status to software indicating that the current operation fails. This field has a second functionality when combined with ABAR + F4h[1:0]. It serves as the MUX select to read out the designated internal SGPIO registers to locations starting from (ABAR + EMB.OFST*4 + 0Ch).
15:8	SGPIO_FUNCTION. Read-write. Reset: 00h. To be compatible with existing SGPIO-aware software, this field shall be set to 82h when performing any command issued to SGPIO targets.
7:0	SGPIO_FRAME_TYPE. Read-write. Reset: 00h. To be compatible with existing SGPIO-aware software, this field shall be set to 40h when performing any command issued to SGPIO targets.

SATA[0...3]SGPIOx00000008 (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_REG_CNT)

Reset: 0000_0000h.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx00000008; SATA[3:0]SGPIO=03[4:1]0_1500h

Bits	Description
31:8	SGPIO_RESERVED0. Read-only. Reset: 00_0000h. Reserved.
7:0	SGPIO_REG_COUNT. Read-write. Reset: 00h. Specifies the number of registers starting with the specified index to write. Currently supported register count range is: REG_TYPE=00h: 1 to 2, REG_TYPE=03h: 1, REG_TYPE=04h: 1 to 8, REG_TYPE=0Ch: 1. It is software is responsibility to make sure the register count is valid and supported, as hardware does not provide a mechanism to return a status to software indicating that the current operation fails. An out-of-range register count for a specific type of register will be treated as do not care and none of the data for such register count will be written.

SATA[0...3]SGPIOx0000000C (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_WR_DATA0)

Read-write. Reset: 0000_0000h.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx0000000C; SATA[3:0]SGPIO=03[4:1]0_1500h

Bits	Description
31:0	SGPIO_WR_DATA0. Read-write. Reset: 0000_0000h. (ABAR + EMB.OFST * 4 + 0Ch) to (ABAR + EMB.OFST * 4 + 2Bh) houses 256 bits of data registers. Once Register Type, Index, and Count are specified, the values in this field will be written to the internal copies of SGPIO register.

SATA[0...3]SGPIOx00000010 (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_WR_DATA1)

Read-write. Reset: 0000_0000h.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx00000010; SATA[3:0]SGPIO=03[4:1]0_1500h

Bits	Description
31:0	SGPIO_WR_DATA1. Read-write. Reset: 0000_0000h. (ABAR + EMB.OFST * 4 + 0Ch) to (ABAR + EMB.OFST * 4 + 2Bh) houses 256 bits of data registers. Once Register Type, Index, and Count are specified, the values in this field will be written to the internal copies of SGPIO register.

SATA[0...3]SGPIOx00000014 (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_WR_DATA2)

Read-write. Reset: 0000_0000h.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx00000014; SATA[3:0]SGPIO=03[4:1]0_1500h

Bits	Description
31:0	SGPIO_WR_DATA2. Read-write. Reset: 0000_0000h. (ABAR + EMB.OFST * 4 + 0Ch) to (ABAR + EMB.OFST * 4 + 2Bh) houses 256 bits of data registers. Once Register Type, Index, and Count are specified, the values in this field will be written to the internal copies of SGPIO register.

SATA[0...3]SGPIOx00000018 (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_WR_DATA3)

Read-write. Reset: 0000_0000h.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx00000018; SATA[3:0]SGPIO=03[4:1]0_1500h

Bits	Description
31:0	SGPIO_WR_DATA3. Read-write. Reset: 0000_0000h. (ABAR + EMB.OFST * 4 + 0Ch) to (ABAR + EMB.OFST * 4 + 2Bh) houses 256 bits of data registers. Once Register Type, Index, and Count are specified, the values in this field will be written to the internal copies of SGPIO register.

SATA[0...3]SGPIOx0000001C (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_WR_DATA4)

Read-write. Reset: 0000_0000h.

_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx0000001C; SATA[3:0]SGPIO=03[4:1]0_1500h

Bits	Description
31:0	SGPIO_WR_DATA4. Read-write. Reset: 0000_0000h. (ABAR + EMB.OFST * 4 + 0Ch) to (ABAR + EMB.OFST * 4 + 2Bh) houses 256 bits of data registers. Once Register Type, Index, and Count are specified, the values in this field will be written to the internal copies of SGPIO register.

SATA[0...3]SGPIOx00000020 (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_WR_DATA5)

Read-write. Reset: 0000_0000h.

`_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx00000020; SATA[3:0]SGPIO=03[4:1]0_1500h`

Bits	Description
31:0	SGPIO_WR_DATA5. Read-write. Reset: 0000_0000h. (ABAR + EMB.OFST * 4 + 0Ch) to (ABAR + EMB.OFST * 4 + 2Bh) houses 256 bits of data registers. Once Register Type, Index, and Count are specified, the values in this field will be written to the internal copies of SGPIO register.

SATA[0...3]SGPIOx00000024 (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_WR_DATA6)

Read-write. Reset: 0000_0000h.

`_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx00000024; SATA[3:0]SGPIO=03[4:1]0_1500h`

Bits	Description
31:0	SGPIO_WR_DATA6. Read-write. Reset: 0000_0000h. (ABAR + EMB.OFST * 4 + 0Ch) to (ABAR + EMB.OFST * 4 + 2Bh) houses 256 bits of data registers. Once Register Type, Index, and Count are specified, the values in this field will be written to the internal copies of SGPIO register.

SATA[0...3]SGPIOx00000028 (SYSHUB::SATA::SGPIO::SATA_AHCI_SGPIO_WR_DATA7)

Read-write. Reset: 0000_0000h.

`_inst[SATA[3:0]]_aliasSMN; SATA[3:0]SGPIOx00000028; SATA[3:0]SGPIO=03[4:1]0_1500h`

Bits	Description
31:0	SGPIO_WR_DATA7. Read-write. Reset: 0000_0000h. (ABAR + EMB.OFST * 4 + 0Ch) to (ABAR + EMB.OFST * 4 + 2Bh) houses 256 bits of data registers. Once Register Type, Index, and Count are specified, the values in this field will be written to the internal copies of SGPIO register.

15.3 USB

This section describes the USB3.1 host controller.

15.3.1 USB Functional Overview

This document includes the BIOS requirements for USB.

USB has two duplicated USB HOST Controllers. Each controller USB0 has 2 USB3.1 SuperSpeedPlus Type-A PHYs and 2 USB2.0 only PHYs.

15.3.1.1 USB Address mapping

There are two NBIOs in one DIE. Each NBIO has three nBIFs: nbif0, nbif1 and nbif2. USB is located in nbif0.dev0.func4.

Each USB container is allocated a 2MB address space on the [SMN](#).

Table 275: USB0/1 Address Mapping

	NBIO Information	SMN address of USB CFG Space
USB0	NBIO1-nBIF0: DEV0.F4	0x16C0_0000
USB1	NBIO0-nBIF0: DEV0.F4	0x16E0_0000

Each USB container's 2MB space is partitioned into five major register groups as shown in the table below. The Start Address and End Address are relative to the container's Base Address.

USB IP register space covers 2 groups of registers, PHY control/parameter registers and host container registers. The

index ranges are listed in below table. A short summary is below:

1. PHY control/parameter register index range is 0x0010_0000 to 0x0017_FFFF (512KB).
2. USB S5 container register index range is 0x0018_0000 to 0x001D_FFFF (384KB).
3. USB S0 container register index range is 0x001E_0000 to 0x001F_FFFF (128KB)

Table 276: USB Address Mapping

Register Group	Address Range	Size	Power domain	Description
Host Controller	0000_0000h ~ 000F_FFFFh	1MB	S0 domain	Host Controller Register
PHY	0010_0000h ~ 0010_FFFFh	64KB	S5 domain	Reserved
	0011_0000h ~ 0011_7FFFh	32KB		2.0 PHY Internal Registers
	0011_8000h ~ 0011_FFFFh	32KB		2.0 PHY Parameter Control Registers
	0012_0000h ~ 0015_FFFFh	256KB		3.1 PHY internal Registers
	0016_0000h ~ 0016_FFFFh	64KB		3.1 PHY Parameter Control Registers
	0017_0000h ~ 0017_FFFFh	64KB		Reserved
S5 Container	0018_0000h ~ 001D_FFFFh	384KB	S5 domain	USB S5 Container Register
S0 Container	001E_0000h ~ 001E_FFFFh	64KB	S0 domain	Reserved
	001F_0000h ~ 001F_00FFh	48KB		MCA configuration registers
	001F_0100h ~ 001F_012Fh			S0 logger (ECC logger)
	001F_0130h ~ 001F_BFFFh			Reserved for future logger
	001F_C000h ~ 001F_DFFFh	8K		Reserved
MCA bank	001F_E000h ~ 001F_FFFFh	8K	S0 domain	Reserved for 32 MCA banks

15.3.1.1.1 PHY Register Group

512KB of the USB address map is allocated to the PHY Internal Control registers (registers inside the PHY) and PHY Parameter Control registers (registers outside the PHY which control the PHY input pins). The PHY group register map is sub-divided three ways according to the PHY type:

- USB2.0 PHY
- USB3.1 PHY

15.3.1.1.1.1 USB2.0 PHY

The USB2.0 PHY Internal Control registers and Parameter Control register space address format is specified in the following table. PHY Internal Control registers are registers that are part of the PHY macro. PHY Parameter Control registers are registers outside the PHY macro that control PHY input pins.

Table 277: USB 2.0 PHY

Address Bits	Field
1:0	Reserved.
9:2	Register Number Allocates space for up to 256 registers per type.
13:10	PHY/Lane Number Allocates space for up to 16 PHYs and 16-lanes.
15:14	Register Type 00: PHY Common Register 01: PHY Per-Lane Register 10: PHY Common Parameter Register 11: PHY Per-Lane Parameter Register

18:16	PHY Type 001: USB2.0 PHY Registers, including PHY Internal and Parameter Control registers
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The resulting USB2.0 PHY address map is:

Table 278: USB 2.0 PHY address map

Type	Start Address	End Address	Description
USB 2.0	0011_0000h	0011_03FFh	PHY 0 Common Registers
	0011_0400h	0011_07FFh	PHY 1 Common Registers
	...	...	...
	0011_3C00h	0011_3FFFh	PHY 15 Common Registers
	0011_4000h	0011_43FFh	PHY Lane 0 Registers
	0011_4400h	0011_47FFh	PHY Lane 1 Registers
	...	...	...
	0011_7C00h	0011_7FFFh	PHY Lane 15 Registers
	0011_8000h	0011_83FFh	PHY 0 Common Parameter Control Registers
	0011_8400h	0011_87FFh	PHY 1 Common Parameter Control Registers
	...	...	...
	0011_BC00h	0011_BFFFh	PHY 15 Common Parameter Control Registers
	0011_C000h	0011_C3FFh	Lane 0 Parameter Control Registers
	0011_C400h	0011_C7FFh	Lane 1 Parameter Control Registers
	...	...	...
	0011_FC00h	0011_FFFFh	Lane 15 Parameter Control Registers

15.3.1.1.1.2 USB3.1 PHY and USB 3.1 /DP PHY

The USB3.1 PHY Internal Control registers and Parameter Control register space address format is specified in the following table. PHY Internal Control registers are registers that are part of the PHY macro. PHY Parameter Control registers are registers outside the PHY macro that control PHY input pins.

Table 279: USB 3.1 Address Mapping

Address Bits	Field
1:0	Reserved.
15:2	Register Number
18:16	PHY Type 010~101: USB3.1 PHY or USB3.1/DP PHY Internal Control Registers 101~111: USB3.1 PHY or USB3.1/DP PHY Parameter Control Registers

The resulting USB3.1 PHY address map is:

Table 280: USB PHY or USB3.1/DP PHY address map

Type	Start Address	End Address	Description
USB3.1	0012_0000h	0015_FFFFh	PHY Internal Control Registers

15.3.2 USB Programming Sequences

15.3.2.1 Programming Architecture

The majority of USB register programming sequences are implemented via [MP1](#) firmware. The primary role of the BIOS is to signal MP1 to execute appropriate Reset or Power State programming sequence.

15.3.2.1.1 BIOS-to-MP1 Messages

The following messages are defined for the BIOS to communicate with the [MP1](#):

- **USB_COLD_RESET_START**
 - Signal MP1 to execute Cold Reset, System Reset, S4/S5 Exit sequence.
 - May need to use a parameter field to separate USB0 and USB1.
- **USB_S3_ENTRY**
 - Signal MP1 to execute S3 Entry sequence.
 - May need to use a parameter field to separate USB0 and USB1.
- **USB_S3_EXIT**
 - Signal MP1 to execute S3 Exit sequence.
 - May need to use a parameter field to separate USB0 and USB1.
- **USB_CONFIG**
 - Information for platform dependent and independent registers to be programmed by MP1 firmware in USB sequences.
 - Parameter fields are used for host, group, and various configuration related information. Refer to 15.3.2.1.2 [USB Register Groups] and 15.3.2.1.3 [USB_CONFIG Messages]

15.3.2.1.2 USB Register Groups

15.3.2.1.2.1 Group Registers

USB registers fall into 5 categories that are differentiated by when they need to be programmed.

- Group 1: USB Container Registers (Group1A, both dependent and independent) and USB31 PHY Parameter Registers (Group1B, both dependent and independent).
 - Refer to 15.3.2.3.1 [Group 1 (USB Container registers and USB31 PHY Parameter registers)]
 - Programmed before reset is de-asserted to PHY and Host Controller.
 - Programmed per cold-boot, warm-reset, and S4/S5 exit.
- Group 2: USB31 PHY Internal Registers.
 - Refer to 15.3.2.3.2 [Group 2 (USB31 PHY Internal Registers)]
 - Programmed after USB31 PHY initialization is done and is ready to use.
 - Programmed per cold-boot, warm-reset, S4/S5 exit, and link-reset.
- Group 3: USB IP Core Registers
 - Refer to 15.3.2.3.3 [Group 3 (IP Core Registers)]
 - Programmed after reset is de-asserted to PHY and Host Controller.
 - Programmed per cold-boot, warm-reset, S4/S5 exit, link-reset, S3 exit, and D3 exit.
- Group 4: USB31 PHY Internal Registers.
 - Refer to 15.3.2.3.4 [Group 4 (USB31 PHY Internal Registers)]
 - Programmed after reset is de-asserted to PHY.
 - Programmed after USB31 PHY SRAM firmware loading.
 - Programmed per cold-boot, warm-reset, S4/S5 exit, and link-reset.
- Group 5: USB20 PHY Internal Registers.

- Refer to 15.3.2.3.5 [Group 5 (USB20 PHY Internal Registers)]
- Programmed after reset is de-asserted to PHY.
- Programmed per cold-boot, warm-reset, S4/S5 exit, and link-reset.

15.3.2.1.2.2 Registers Programming Sequences

USB registers fall into 2 categories that are differentiated by how they are programmed.

1. Platform-Independent Registers: [MP1](#) firmware directly program the registers when different sequence stages.
2. Platform-Dependent Registers: MP1 firmware co-works with BIOS message to cover the programming. Detailed information can refer to 15.3.2.1.3 [USB_CONFIG Messages]

15.3.2.1.3 USB_CONFIG Messages

USB_CONFIG messages are defined to pass the programming information of the 4 groups of platform-dependent registers from BIOS to [MP1](#) firmware.

On reception of USB_CONFIG messages, MP1 firmware does not program USB registers immediately. Instead, MP1 firmware needs to save the message contents to a local location (such as MP1 SRAM) as the USB registers are programmed in different stages of USB sequences. USB_CONFIG messages may also include functional sub-sequence and workaround sub-sequence flags as all the sub-sequences are executed inside USB sequences. MP1 firmware needs to keep a count for all USB_CONFIG messages received so to avoid missing any register programming.

For the Group registers, MP1 firmware needs to store Group 2, Group 3, Group 4 and Group 5 values/contents to a powered location (such as system DRAM) as the registers need to be reprogrammed in system low-power cycles where BIOS is not available and can't pass the register information to firmware. The system low-power cycles include link-resets cycles, where USB is reset, and S3/S4/S5 cycles, where USB is put to D3 by OS.

MP1 firmware shall be implemented in a generic way that all USB registers passed through the messages are supported.

15.3.2.1.3.1 USB_CONFIG Message Format

The software messages have 5 [DW](#) parameters. USB_CONFIG message uses 4 DW of the parameters. The message formats are defined in following tables.

Table 281: DW Message Parameters

Double Word	Function	Descriptions
DW0	Operand0	Operand0 for associated opcode (DW3[31:24]). Currently the field is defined only for opcode 0x01 (register modification).
DW1	Operand1	Operand1 for associated opcode (DW3[31:24]). Currently the field is defined only for opcode 0x01 (register modification).
DW2	Operand2	Operand2 for associated opcode (DW3[31:24]). Currently the field is defined only for opcode 0x01 (register modification).
DW3	Opcode (bit[31:24]) + Host Select (bit[23:20]) + Reserved (bit[19:16]) + Parameter (bit[15:0])	Opcode (bits[31:24]) defines the operations associated with the message. Host Select (bits[23:20]) indicates the host number. Reserved (bits[19:16]) is for future use and should be 0. Parameter (bits[15:0]) defines other information associated with the opcode.

		See below Table 282 [Opcode List] for opcodes currently defined.
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Table 282: Opcode List

DW3 Opcode, Bit[31:24]	DW3 Parameter, Bit[15:0]	Descriptions
0x00	N/A	No OP
0x01	Group Number Code	Register modification with group number. The group number for the register modification listed in 15.3.2.1.2 [USB Register Groups] is defined here. The register index, data, and mask are defined in other DW of the message. The modification needs to be done using read-modify-write. See Table 283 [DW0~DW2 and Parameter/DW[15:0] for Opcode 0x01, Register Modification].
0x02~0xBF	N/A	Reserved.
0xC0	Sub-sequence Enable Code	Sub-sequence enables. The opcode enables specific sub-sequences within the main sequences and bit[15:0] is used to identify different sub-sequences. One sub-sequence is currently defined using parameter 0x01 for PHY SRAM loading sequence for A0. See Table 284 [DW0~DW2 for Opcode 0xC0 and Parameter/DW3[15:0] 0x01/0x02, Sub-sequence Enable]. DW0 ~ DW2 (operand0 ~ operand2) may be used as necessary.
0xC1~0xCF	N/A	Reserved.
0xD0	USB3 Feature Enable Code for Models 10h-1Fh except Model 18h	This opcode is used to enable or disable USB3 feature. See Table 285 [DW0~DW2 for Opcode 0xD0/0xF5 and Parameter/DW3[15:0] USB3 feature Enable].
0xD1~0xF4	N/A	Reserved.
0xF5	USB3 Feature Enable Code for Model 18h	This opcode is used to enable or disable USB3 feature. See Table 285 [DW0~DW2 for Opcode 0xD0/0xF5 and Parameter/DW3[15:0] USB3 feature Enable].
0xF6~0xFF	N/A	Reserved.
Note: Bit[23:20] is Host Select for all opcodes.		

Table 283: DW0~DW2 and Parameter/DW[15:0] for Opcode 0x01, Register Modification

Double Word	Function	Descriptions
DW0	Register Offset	32-bit offset of the register to be accessed. The register is accessed through SMN USB base address + USB register offset.
DW1	Write Bit Enable (Mask)	32-bit bit-level enables for write data. 1'b0: no modification to the corresponding register bit position. 1'b1: Corresponding Write Data bit to be written to the same register bit position
DW2	Write Data	Write data, filtered by the bit-level enables (DW2), to be updated to the register indexed by the register offset (DW0).
DW3[15:0]	Group Number Code	Bit[3:0] indicates the group number. Bit[15:4] is reserved.

Table 284: DW0~DW2 for Opcode 0xC0 and Parameter/DW3[15:0] 0x01/0x02, Sub-sequence Enable

Double Word	Function	Descriptions
DW0	N/A	Reserved.
DW1	N/A	Reserved.
DW2	N/A	Reserved.
DW3[15:0]	0x02: Standalone-PHY SRAM Loading	Opcode 0xC0 and parameter 0x02 enable USB31 standalone-PHY SRAM loading sequence defined in 15.3.2.4.2.1 [USB31 Standalone-PHY SRAM Firmware Loading Sub-sequence] with bit[23:20] indicating the USB controller.

Table 285: DW0~DW2 for Opcode 0xD0/0xF5 and Parameter/DW3[15:0] USB3 feature Enable

Double Word	Function	Descriptions
DW0	N/A	Reserved.
DW1	N/A	Reserved.
DW2	N/A	Reserved.
DW3[15:0]	USB3 feature enable/disable	When opcode (DW3[31:24]) is 0xD0/0xF5, DW3[15:0] is used to enable/disable DbgSC feature for different controller. DW3[0]: DbgSC feature enable/disable. • 1: Enable • 0: Disable (default) DW3[23:1]: reserved.

15.3.2.2 Reset and Power State Transition Sequences

There are six primary reset and power-state transition sequences:

- Cold Reset, System Reset, S4/S5 Exit
- D3 Entry
- D3 Exit
- S3/S4/S5 Entry
 - Includes Isolation Enable sequence
- S3 Exit
 - Includes Isolation Disable sequence
- Link Reset

15.3.2.3 Group Register Configuration

15.3.2.3.1 Group 1 (USB Container registers and USB31 PHY Parameter registers)

Since the PHY related programming will be skipped and USB31PHY CR reset will be gated when pipe is switched, we separate Group 1 to 2 groups: Group1A and Group1B.

15.3.2.3.1.1 Group 1A Platform-Independent Registers

Table 286: Group 1A (Platform-Independent Registers)

Step #	Applies To		Register	
	Write Data		Write Mask	
	Description			
1	USB0 USB1	USB31::USBCONTAINERREGCNTR0::INTERRUPT_CONTROL_CNTR0		
	0000_0100		0000_0100	
	Enable legacy SMI support by setting USB31::USBCONTAINERREGCNTR0::INTERRUPT_CONTROL_CNTR0[USB_LEGACY_SMI_EN]			
3	USB0 USB1	USB31::USBCONTAINERREGCNTR0::SYSBUS_BLOCK_CONTROL_CNTR0		
	0012_0000		0012_0000	
	Set SYSBUS_BLOCK_WATERMARK_LO to 18us (USB31::USBCONTAINERREGCNTR0::SYSBUS_BLOCK_CONTROL_CNTR0[SYSBUS_BLOCK_WATERMARK_0])			
6	USB0 USB1	USB31::USBCONTAINERREGCNTR0::DBGSC_PIPE_SWITCH_CTL_CNTR0		
	0000_0004		0000_0004	
	Enable DbgSC PHY reset mask by setting bit[2], USB_DBGSC_PSWITCH_PHY_RESET_MASK			

15.3.2.3.1.2 Group 1A Platform-Dependent Registers

Table 287: Group 1A (Platform-Dependent Registers)

Step #	Applies To		Register	
	Write Data		Write Mask	
	Description			
1	USB0 USB1	USB31::USBCONTAINERREGCNTR0::OVERCURRENT_MAP_0_CNTR0		
	0000_0000		0000_FFFF	
	Enable and route Overcurrent pin 0 to HC0 Port0 (USB2.0), Port1 (USB2.0), Port2 (USB3.1), Port3 (USB3.1)			
2	USB0 USB1	USB31::USBCONTAINERREGCNTR0::PORT_CONTROL_CNTR0		
	0003_0100		0003_0100	
	Bit[8], OC_POLARITY_CONTROL (Note: the programming value is the same as default value) Bit[31:16], USB31_PORT_FORCE_GEN1_EN			
3	USB0 USB1	USB31::USBCONTAINERREGCNTR0::DEVICE_REMOVABLE_CNTR0		
	0000_0000		0003_0003	
	Bit[15:0], USB2_PORT_DEVICE_REMOVABLE Bit[31:16], SS_PORT_DEVICE_REMOVABLE Note: the programming value is the same as default value			

Note:

- Please refer 15.3.2.9.3 [Overcurrent Mapping] before setting the mapping control. If need to map the selected port to Overcurrent pin 0, the selected port's Port Overcurrent Select should be set to 0. For example,
 - To map Port0 to Overcurrent pin 0, set bit[3:0] of register USB31::USBCONTAINERREGCNTR0::OVERCURRENT_MAP_0_CNTR0 to 0x0
 - To map Port1 to Overcurrent pin 0, set bit[7:4] of register USB31::USBCONTAINERREGCNTR0::OVERCURRENT_MAP_0_CNTR0 to 0x0
- USB2_PORT_DISABLE and USB3_PORT_DISABLE can only be programmed by [ASP](#), thus not included in Table 287 [Group 1A (Platform-Dependent Registers)].

15.3.2.3.1.3 Group 1B Platform-Independent Registers

Table 288: Group 1B (Platform-Independent Registers)

Step #	Applies To	Register	
		Write Data	Write Mask
		Description	
1	USB0 USB1	USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARAC TL0_CNTR0	
		0000_0001	0000_0001
		Bit[0], ext_ctrl_sel	
2	USB0 USB1	USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARAC TL16_CNTR0	
		0010_0000	0010_0000
		Bit[20], pipe_rx_cdr_legacy_en	

15.3.2.3.1.4 Group 1B Platform-Dependent Registers

Table 289: Group 1B (Platform-Dependent Registers)

Step #	Applies To	Register	
		Write Data	Write Mask
		Description	
1	USB0 USB1	USB31::USB20LANEPARAMCTLREGCNTR0PORT::USB_20LANEPARA CTL0_CNTR0	
		3223_3001	F3BF_7007
		Bit[2:0], COMPDISTUNE0 is set to 0x1 Bit[14:12], SQRXTUNE0 is set to 0x3 Bit[19:16], TXFSLSTUNE0 is set to 0x3 Bit[21:20], TXPREEMPAMPTUNE0 is set to 0x2 Bit[23], TXPREEMPPULSETUNE0 is set to 0 Bit[25:24], TXRISETUNE0 is set to 0x2 Bit[31:28], TXVREFTUNE0 is set to 0x3 Note: the programming value is the same as default value	
2	USB0 USB1	USB31::USB20LANEPARAMCTLREGCNTR0PORT::USB_20LANEPARA CTL1_CNTR0	
		0000_000B	0000_000F
		Bit[1:0], TXHSXVTUNE0 is set to 0x3 Bit[3:2], TXRESTUNE0 is set to 0x2 Note: the programming value is the same as default value	

3	USB0 USB1	USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL10_CNTR0	
		0000_0000	0000_0707
	Bit[2:0], rx_term_ctrl Bit[10:8], tx_term_ctrl		

15.3.2.3.2 Group 2 (USB31 PHY Internal Registers)

15.3.2.3.2.1 Group 2 Platform-Independent Registers

Currently, there is no programming requirement for Group 2 Platform-Independent registers.
Reserved for future use.

15.3.2.3.3 Group 3 (IP Core Registers)

15.3.2.3.3.1 Group 3 Platform-Independent Registers

Table 290: Group 3 (Platform-Independent Registers)

Step #	Applies To	Register	
		Write Data	Write Mask
		Description	
1	USB0 USB1	FCHUSBDWCUSB31CNTR0::GSYSBLKWINCTRL	
		2000_0000	2000_0000
		Bit[29], erst_prefetching_en.	
2	USB0 USB1	FCHUSBDWCUSB31CNTR0::GUCTL	
		0000_0000	0004_0000
		Bit[18], EN_EXTD_TBC_CAP	
3	USB0 USB1	FCHUSBDWCUSB31CNTR0::GUCTL2	
		3000_0000	FE00_0000
		Bit[31:25], PERIODIC_UF_THR	
4	USB0 USB1	FCHUSBDWCUSB31CNTR0::GUCTL2	
		0014_0000	01FC_0000
		Bit[24:18], ASYNC_UF_THR	
5	USB0 USB1	FCHUSBDWCUSB31CNTR0::GSMACCTL	
		0000_0004	0000_0004
		Bit[2], host_mask_nump0	
6	USB0 USB1	FCHUSBDWCUSB31CNTR0::GUCTL3	
		0000_0000	0020_0000
		Bit[21], BLOCK_CONCURRENT_IN_CTRL_XFERS	

15.3.2.3.3.2 Group 3 Platform-Dependent Registers

Currently, there is no programming requirement for Group 3 Platform-Dependent registers.
Reserved for future use.

15.3.2.3.4 Group 4 (USB31 PHY Internal Registers)

Currently, there is no programming requirement for Group 4 Platform-Independent registers and Group 4 Platform-Dependent registers.
Reserved for future use.

15.3.2.3.5 Group 5 (USB20 PHY Internal Registers)**15.3.2.3.5.1 Group 5 Platform-Dependent Registers**

Currently, there is no programming requirement for Group 5 Platform-Dependent register.
Reserved for future use.

15.3.2.4 BIOS Menu Options and Sub-sequences**15.3.2.4.1 BIOS Menu Options**

The platform configuration features described in 15.3.2.8 [Note on S-State Entry/Exit] can be enabled/disabled or configured through BIOS menu options. The chapter lists the features to be covered using BIOS menu options. All register updates from user inputs are sent to [MP1](#) firmware using USB_CONFIG messages.

The features to be included in BIOS menu options are:

- SuperSpeed Link Speed Limit
- Over-current Control
- Battery Charger Enable

15.3.2.4.2 Sub-Sequences

Currently following sub-sequences, ComboPHY SRAM firmware loading sequence and ComboPHY display (DP) programming sequence are defined.

15.3.2.4.2.1 USB31 Standalone-PHY SRAM Firmware Loading Sub-sequence

The sub-sequence is required and is enabled through USB_CONFIG message with opcode 0xC0, parameter 0x01 and host_number indicating USB0 or USB1. Refer from Table 281 [DW Message Parameters] to Table 284 [DW0~DW2 for Opcode 0xC0 and Parameter/DW3[15:0] 0x01/0x02, Sub-sequence Enable] for USB_CONFIG message formats.

The SRAM loading sub-sequence needs to be executed every time after the PHY is reset through phy_reset.

1. Program USB31::USBCONTAINERREGCNTR0::PORT_CONTROL_CNTR0[USB31_PORT_SEL] to 4'h0.
2. [MP1](#) firmware communicates with [ASP](#) firmware to start PHY SRAM loading sequence.

3. On reception of MP1 firmware request for PHY firmware loading, ASP firmware retrieves PHY firmware binary code and writes the firmware code to PHY SRAM through APB interface.
 - a) USB APB offset range for USB31 type-C PHY SRAM is 0x0015_0000 ~ 0x0015_7FFF.
 - b) The SRAM locations can be written only one offset at a time through APB interface with no burst.

15.3.2.5 Note on USB31 Combo/Standalone PHY Programming

Both USB31 combo PHY and USB31 standalone PHY have the entire 16-bit address range (64kB) used. After conversion, the range is 256kB, almost the whole space assigned for USB31 PHYs.

To cover multiple USB31 PHY designs a register select field from a USB container is assigned for USB31 PHY selections as detailed below. The number of lanes inside the PHYs do not have effect on the PHY selection (although the select field is improperly named as USB31_PORT_SEL).

- USB31::USBCONTAINERREGCNTR0::PORT_CONTROL_CNTR0[USB31_PORT_SEL]
 - USB31_PORT_SEL = 0 -> PHY0
 - USB31_PORT_SEL = 1 -> PHY1
 - USB31_PORT_SEL = 2 -> PHY2

And so on, up to 16 PHYs.

Whenever software needs to program a register in a different PHY from the one currently selected by USB31_PORT_SEL, software must set the field to a value corresponding to the PHY.

Once USB31_PORT_SEL is set for a PHY, all registers in the same PHY can be programmed in a batch without the need to touch the select field until another PHY needs to be selected. It may be preferred to group the register from the same PHY together.

15.3.2.6 Note on RSMU Interrupt from USB

USB has 8 RSMU power-management interrupts for use in controlling power management sequences. Currently only intr_1 is used and needs to be enabled.

Table 291: RSMU_PWRMGMT_INTR_ENABLE

RSMU_PWRMGMT_INTR_ENABLE_Px - RW – 32 bits		
Field Name	Bits	Description
RESERVED (RO)	31:16	Reserved bits.
ENABLE (RW)	15:0	Enable of Px (x=0/1/2) interrupt. Each bit is mapped to one interrupt signal. Bit0 map to interrupt input 0 and so no. Access Control: Only MP1/ASP

15.3.2.7 Note on Current Power Status for USB

Please note that the read value of
 USB31::USBCONTAINERREGCNTR0::CURRENT_POWER_STATE_CNTR0[CURRENT_POWER_STATE] bits
 reflects the current power state of USB host.

When enter into D3 cold state, don't start to remove the power until
 USB31::USBCONTAINERREGCNTR0::CURRENT_POWER_STATE_CNTR0[CURRENT_POWER_STATE] =2'b11
 (D3).

When exit from D3 state, don't start the host transaction until
 USB31::USBCONTAINERREGCNTR0::CURRENT_POWER_STATE_CNTR0[CURRENT_POWER_STATE] =2'b00
 (D0).

Table 292: Current Power State

Current Power State - RO - 32 bits - 0x0018_0108			
Field Name	Bits	Default	Description
Current_power_state	9:8	00b	The current power state of the USB core. When equal to '3', the USB core is in D3 state. When equal to '0', the USB core is in D0 State.

15.3.2.8 Note on S-State Entry/Exit

The implementation of the host controller is compliant to the xHCI standard with regard to saving state and powering down as well as powering up and restoring state. As such it is compatible with standard xHCI drivers. There are however a number of implementation specific programming steps for the BIOS to indicate S-State entry and exit to the USB controller so that it can properly support Wake/PME. Note that the steps described below are already part of the S-State Entry and Exit sequences specified in previous sections, this section is purely informative.

- S-State Entry
- Use standard software steps to save xHC state (xHCI 1.1 Specification).
 - Enable desired wake conditions via PORTSC (WCE, WDE, WOE).
- Use standard PCI Power Management interface to enable PME and place xHC in D3.
- BIOS sends USB_S3_ENTRY message to [MP1](#) to trigger MP1 sequence for USB S3 entry.
 - This informs USB controller that system will be entering low-power state in which USB should not be reset.
- MP1 needs to wait for acknowledgment (USB_RSMU_PME_TO_ACK through USB_RSMU_pwrmtgt_intr_1) before informing BIOS that USB is ready for system S3 entry.
- After MP1 firmware returns for USB_S3_ENTRY message, BIOS waits T_UsbIsoOn for USB to complete the internal S3 entry sequence.
- S-State Exit
 - BIOS sends USB_S3_EXIT message to MP1 to trigger MP1 sequence for USB S3 exit.
 - MP1 then executes USB S3 Exit sequence.
 - Use standard software steps to restore xHC state (xHCI 1.1 Specification, Section 4.23.2).

15.3.2.9 Platform Configuration Features

This section describes a number of common platform configuration features. The programming of these features would generally occur during the Platform-Dependent Register Configuration of the Reset and Power State Transition

Sequences.

15.3.2.9.1 Number of Ports

This feature allows a platform to expose fewer ports to software than are physically present in the USB host controller. This configuration is implemented via the USB2_NUM_PORTS and USB3_NUM_PORTS fields of the PORT_CONTROL_CNTR register (USB31::USBCONTAINERREGCNTR0::PORT_CONTROL_CNTR0). Valid values of USB2_NUM_PORTS is from 1 to the actual number of USB2 ports implemented in the host controller. Valid values of USB3_NUM_PORTS is from 1 to the actual number of USB3 ports implemented in the host controller. xHCI HCSPARAMS1.MaxPorts = USB2_NUM_PORTS + USB3_NUM_PORTS.

15.3.2.9.2 Port Disable

This feature allows a platform to disable ports such that the ports do not report connect/disconnect events, and always remain in the disabled state. This feature is configured via USB31::USBCONTAINERREGCNTR0::PORT_DISABLE_WRITE_ONCE_CNTR0[USB2_PORT_DISABLE_WO] and USB31::USBCONTAINERREGCNTR0::PORT_DISABLE_WRITE_ONCE_CNTR0[USB3_PORT_DISABLE_WO]. Each of these registers is a bit-map where each bit corresponds to a port.

Table 293: PORT_DISABLE_WRITE_ONCE_CNTR0

Port Disable Write Once - RWO - 32 bits - 0x001D_BC00			
Field Name	Bits	Default	Description
USB2_PORT_DISABLE	15:0	0000h	USB2 Port Disable (write-once) When set USB2_PORT_DISABLE[n] to 1, USB2 port<n> is disabled. Only writing the value '1' has any effect. Writing a '0' has no effect. Fields can be cleared by cold boot reset and system reset.
USB3_PORT_DISABLE	31:16	0000h	SS Port Disable (write-once). When set USB3_PORT_DISABLE[n] to 1, USB3 port<n> is disabled. Only writing the value '1' has any effect. Writing a '0' has no effect. Fields can be cleared by cold boot reset and system reset.

15.3.2.9.3 Overcurrent Mapping

This feature allows a platform to map SOC overcurrent pins to USB ports. Overcurrent mapping is configured via the USB31::USBCONTAINERREGCNTR0::OVERCURRENT_MAP_0_CNTR0 and USB31::USBCONTAINERREGCNTR0::OVERCURRENT_MAP_1_CNTR0 registers. A four-bit field is allocated for each USB port in the SOC. For an SOC with M USB2 ports and N USB3 ports, USB2 ports are numbered 0 to M and USB3 ports are numbered M+1 to M+N. For overcurrent mapping purposes, port numbers are not affected by any change to USB31::USBCONTAINERREGCNTR0::PORT_CONTROL_CNTR0[USB2_NUM_PORTS] or USB31::USBCONTAINERREGCNTR0::PORT_CONTROL_CNTR0[USB3_NUM_PORTS]. PORTx_OVERCURRENT_SELECT = Fh maps no overcurrent pin to Port x, effectively disabling overcurrent detection

for Port x.

Table 294: Overcurrent Map 0

Overcurrent Map 0 - RW - 32 bits - 0x0018_0120			
Field Name	Bits	Default	Description
Port 0 Overcurrent Select	3:0	Fh	Overcurrent input mapped to Port 0
Port 1 Overcurrent Select	7:4	Fh	Overcurrent input mapped to Port 1
Port 2 Overcurrent Select	11:8	Fh	Overcurrent input mapped to Port 2
Port 3 Overcurrent Select	15:12	Fh	Overcurrent input mapped to Port 3
Port 4 Overcurrent Select	19:16	Fh	Overcurrent input mapped to Port 4
Port 5 Overcurrent Select	23:20	Fh	Overcurrent input mapped to Port 5
Port 6 Overcurrent Select	27:24	Fh	Overcurrent input mapped to Port 6
Port 7 Overcurrent Select	31:28	Fh	Overcurrent input mapped to Port 7.

Note:

- The number of overcurrent inputs is related with port number, and it is a feature and will vary for different implementations of the IP. Up to 15 Overcurrent inputs are supported.
 - If Overcurrent_Select for a port is set to 0h, map the selected port overcurrent input from overcurrent pin 0.
 - If Overcurrent_Select for a port is set to 1h, map the selected port overcurrent input from overcurrent pin 1.
 - If Overcurrent_Select for a port is set to Fh, Overcurrent detection is disabled for that port.
- If Overcurrent_Select for a port is set to to value > the number of Overcurrent inputs, Overcurrent detection is disabled for that port.
 - In this project, if Overcurrent_Select for a port is set to 4 ~ 15, Overcurrent detection is disabled for that port.

Table 295: Overcurrent Map1

Overcurrent Map 1 - RW - 32 bits - 0x0018_0124			
Field Name	Bits	Default	Description
Port 8 Overcurrent Select	3:0	Fh	Overcurrent input mapped to Port 8
Port 9 Overcurrent Select	7:4	Fh	Overcurrent input mapped to Port 9
Port 10 Overcurrent Select	11:8	Fh	Overcurrent input mapped to Port 10
Port 11 Overcurrent Select	15:12	Fh	Overcurrent input mapped to Port 11
Port 12 Overcurrent Select	19:16	Fh	Overcurrent input mapped to Port 12
Port 13 Overcurrent Select	23:20	Fh	Overcurrent input mapped to Port 13
Port 14 Overcurrent Select	27:24	Fh	Overcurrent input mapped to Port 14
Port 15 Overcurrent Select	31:28	Fh	Overcurrent input mapped to Port 15

Note:

- The number of overcurrent inputs is related with port number, and it is a feature and will vary for different implementations of the IP. Up to 15 Overcurrent inputs are supported.
 - If Overcurrent_Select for a port is set to 0h, map the selected port overcurrent input from overcurrent pin 0.
 - If Overcurrent_Select for a port is set to 1h, map the selected port overcurrent input from overcurrent pin 1.
 - If Overcurrent_Select for a port is set to Fh, Overcurrent detection is disabled for that port.
- If Overcurrent_Select for a port is set to to value > the number of Overcurrent inputs, Overcurrent detection is disabled for that port.

1. In this project, if Overcurrent_Select for a port is set to 4 ~ 15, Overcurrent detection is disabled for that port.

15.3.2.9.4 SuperSpeed Link Speed Limit

This feature allows a platform to limit a SuperSpeed Gen2 capable port to Gen1 operation. This feature is configured via the PORT_CONTROL_CNTR register. To enable the Link Speed Limit feature on Port P:

- Set PORT_CONTROL_CNTR0.USB31_PORT_FORCE_GEN1_EN[P+16]

15.3.2.10 PORTSC Port Speed

The xHCI specification PORTSC register Port Speed field identifies the speed of the connected USB device. The xHCI 1.1 Specification defines Default USB Speed ID Values for Full-speed, Low-speed, High-speed, and SuperSpeed. However it does not define the Default USB Speed ID Value for SuperSpeedPlus. The Default USB Speed ID Value for SuperSpeedPlus is 5, as defined in the table below:

Table 296: Port Speed

Default Speed ID Value	Definition	Bit Rate	Protocol
1	Full-speed	12 Mb/s	USB 2.0
2	Low-speed	1.5Mb/s	USB 2.0
3	High-speed	480 Mb/s	USB 2.0
4	SuperSpeed	5 Gb/s	USB 3.1
5	SuperSpeedPlus	10 Gb/s	USB 3.1

15.3.2.11 USB3 Compliance Mode

The USB3 specification defines that a downstream port shall disable its transition path from Polling LFPS to Compliance Mode upon PowerOn Reset or Warm Reset. The method to direct a USB3 downstream port to enable the transition path from Polling, LFPS to Compliance is defined in the xHCI specification. If the port is in the Disconnected state, a write to PORTSC with PORTSC[PLS]=10d (Compliance Mode) enables the transition to Compliance Mode.

15.3.2.12 Protocol Speed ID

The xHCI v1.1 specification defines bits[15:14] of the Protocol Speed ID ([PSI](#)) Dword as RsvdP.USB defines PSI[15:14] as the Link Protocol (LP) field:

Table 297: Protocol Speed

31: 16	15: 14	13: 9	8	7: 6	5: 4	3: 0
Protocol Speed ID Mantissa	LP	RsvdP	PFD	PLT	PSIE	PSIV

If the xHCI Protocol Extended Capability Revision Major = 02h, then LP shall be set to 0. In this case the link protocol (LS, FS, or HS) depends on the reported link speed.

If the xHCI Protocol Extended Capability Revision Major = 03h, then the LP field identifies the link-level protocol supported by ports associated with the PSI Dword: LP = 0 for SuperSpeed and LP = 1 for SuperSpeedPlus.

15.3.3 Registers

15.3.3.1 USB 3.1 – USB Container Register

USBCONTAINER[0...1]x0000001C

(USB31::USBCONTAINERREGCNTR0::SMU_ASSIST_POWER_STATE_CONTROL_CNTR0)

Read-write. Reset: 0000_0000h.

SMU assist Power State Control register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x0000001C; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:9	SMU_ASSIST_POWER_STATE_RESERVED_31_9. Read-write. Reset: 00_0000h. Reserved
8	SMU_FW_ASSIST_D_STATE. Read-write. Reset: 0. Description: The mode bit to select the source of USB power state setting 1: USB detect power state setting through SMU_CTL_Power_State[1:0] that is controlled by SMU FW. 0: USB detect power state setting through USB shadowed PMCSR.PowerState that is updated by nBIF.
7:2	SMU_ASSIST_POWER_STATE_RESERVED_7_2. Read-write. Reset: 00h. Reserved
1:0	SMU_CTL_POWER_STATE. Read-write. Reset: 0h. Description: SMU FW controlled Power State. 00b: D0 11b: D3 If software attempts to write an unsupported state to this field, the data is discarded and no state change occurs.

USBCONTAINER[0...1]x00000024 (USB31::USBCONTAINERREGCNTR0::RAM_CLR_Control_CNTR0)

Ram clear Control register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000024; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:9	RAM_CLR_Control_RESERVED_31_9. Read-write. Reset: 00_0000h. Reserved
8	CLR_RAM_DONE. Read,Write-1-to-clear. Description: Write-One-Clear bit The status bit of clearing IP core RAM0/1/2 It is set by hardware when all RAM0/1/2 cleared. It is cleared by SW when writing 1 to it.
7:1	RAM_CLR_Control_RESERVED_7_1. Read-write. Reset: 00h. Reserved
0	CLR_RAM_TRIGGER. Read-write. Description: Trigger bit of clearing IP core RAM0/1/2 1: When SW writes 1 to it, it will be set to 1, and hardware starts to clear IP core RAM0/1/2. 0: After SW writes 1 to it, SW should not write 0 to clear it before CLR_RAM_DONE is set to 1. After SW writes 1 to trigger RAM clear operation, and read back 1 from CLR_RAM_DONE, SW can write 0 to clear this bit.

USBCONTAINER[0...1]x00000028 (USB31::USBCONTAINERREGCNTR0::RAM_CLR_Enable_CNTR0)

Read-write. Reset: 0000_0000h.

Ram clear Enable register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000028; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:2	RAM_CLR_Enable_RESERVED_31_2. Read-write. Reset: 0000_0000h. Reserved
1	RAM_CLK_OVERRIDE_DISABLE. Read-write. Reset: 0. Description: Ram Clk override disable bit 0: Enable Ram Clk override. When it is 1'b0, the ram_clk for memory modules are clocked by ungated ram clk 1: Disable Ram Clk override. When it is 1'b1, the ram_clk for memory modules are always clocked by gated clks.
0	CLR_RAM_EN. Read-write. Reset: 0. Description: Ram clear operation enable bit 0: Disable ram clear operation by hardware. When it is 1'b0, SW can't(hardware gated to ignore SW writing 1) write 1 to CLR_RAM_TRIGGER(0x16D8_0024[0]), and RAM IF forced from IP core, but not from the new added ram clear logic. 1: Enable ram clear operation by hardware. When it is 1'b1, SW can write 1 to CLR_RAM_TRIGGER(0x16D8_0024[0]) to trigger the RAM clear operation.

USBCONTAINER[0...1]x00000100 (USB31::USBCONTAINERREGCNTR0::RESET_CNTR0)

Read-write. Reset: 0000_0009h.

RESET Control register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000100; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:16	USB_SOFT_PORT_RESET . Read-write. Reset: 0000h. Description: Soft PORT_RESET, per-port reset control. 1: Reset
15:11	RESET_RESERVED_15_11 . Read-write. Reset: 00h. Reserved
10	USB_DEBUG_PHY31_RESET . Read-write. Reset: 0. USB3.1 PHY Reset for Debugging
9	USB_DEBUG_PHY30_RESET . Read-write. Reset: 0. USB3.0 PHY Reset for Debugging
8	USB_DEBUG_PHY20_RESET . Read-write. Reset: 0. USB2.0 PHY Reset for Debugging
7:4	RESET_RESERVED_7_4 . Read-write. Reset: 0h. Reserved
3	USB_SOFT_RAM_CLR_RESET . Read-write. Reset: 1. Description: Soft Ram Clear Reset. 1: Reset
2	USB_SOFT_PHY_RESET . Read-write. Reset: 0. Description: Soft Phy Reset. 1: Reset
1	USB_SOFT_CONTROLLER_RESET . Read-write. Reset: 0. Description: Soft Host Controller Reset. 1: Reset
0	USB_S5_RESET . Read-write. Reset: 1. Description: Global Soft Reset Asserted (1) when a> Cold boot reset/System reset is active Or b> programming this bit to 1 De-asserted (0) when a> programming this bit to 0 1: Reset IP Controller, PHY and other S5 logic except container registers.

USBCONTAINER[0...1]x00000104 (USB31::USBCONTAINERREGCNTR0::PGFSM_OVERRIDE_CNTR0)

Read-write. Reset: 0000_0033h.

PGFSM OVERRIDE Control register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000104; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:8	PGFSM_OVERRIDE_RESERVED_31_8. Read-write. Reset: 00_0000h. Reserved
7	PGFSM_OVERRIDE_DAUGHTER_SLEEP. Read-write. Reset: 0. PGFSM Daughter-Sleep Override
6	PGFSM_OVERRIDE_MOTHER_SLEEP. Read-write. Reset: 0. PGFSM Mother-Sleep Override
5	PGFSM_OVERRIDE_SCAN_ISO. Read-write. Reset: 1. Description: PGFSM Scan Isolation Override. 0: Isolate
4	PGFSM_OVERRIDE_ISO. Read-write. Reset: 1. Description: PGFSM Isolation Override. 0: Isolate
3	PGFSM_OVERRIDE_SD. Read-write. Reset: 0. Description: PGFSM SD Override. 1: Memory Shutdown
2	PGFSM_OVERRIDE_DS. Read-write. Reset: 0. Description: PGFSM DS Override. 1: Memory DeepSleep
1	PGFSM_OVERRIDE_HARDRESETB. Read-write. Reset: 1. Description: PGFSM Hard ResetB Override. 0: Reset
0	PGFSM_OVERRIDE_COLDRESETB. Read-write. Reset: 1. Description: PGFSM Cold ResetB Override. 0: Reset

USBCONTAINER[0...1]x00000108**(USB31::USBCONTAINERREGCNTR0::CURRENT_POWER_STATE_CNTR0)**

Reset: 0000_0000h.

Current Power State Status register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000108; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:10	CURRENT_POWER_STATE_RESERVED_31_10. Read-write. Reset: 00_0000h. Reserved
9:8	CURRENT_POWER_STATE. Read-only. Reset: 0h. Description: The current power state of the ip core. When equal to '3', the USB core is in D3 state. When equal to '0', the USB core is in D0 State.
7:6	CURRENT_POWER_STATE_RESERVED_7_6. Read-write. Reset: 0h. Reserved
5:4	CURRENT_POWER_STATE_U3PMU. Read-only. Reset: 0h. Description: The current power state of U3PMU. When equal to '3', the U3PMU is controlling the PHY, and when equal to '0', the core is controlling the PHY. When both inputs from the PMUs are equal to '3', the core is prepared to lose Vcc. When both inputs are equal to '0', software may access the CSRs.
3:2	CURRENT_POWER_STATE_RESERVED_3_2. Read-write. Reset: 0h. Reserved
1:0	CURRENT_POWER_STATE_U2PMU. Read-only. Reset: 0h. Description: The current power state of U2PMU. When equal to '3', the U2PMU is controlling the PHY, and when equal to '0', the core is controlling the PHY. When both inputs from the PMUs are equal to '3', the core is prepared to lose Vcc. When both inputs are equal to '0', software may access the CSRs.

**USBCONTAINER[0...1]x0000010C
(USB31::USBCONTAINERREGCNTR0::HOST_POWER_CLOCK_CNTR_CNTR0)**

Read-write. Reset: 0000_0000h.

Host Power and Clock Control register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x0000010C; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:7	Reserved.
6	USB_WAKE_RO_PCMLRX_CTRL. Read-write. Reset: 0. Description: RO switch and PCMLRX enable control Take effect only when bit[7] is 0. When set to 0 (default), selects CGPLL ref_clk as PHY ref_clk and enables PCMLRX. When set to 1, selects RO clock as PHY ref_clk and disables PCMLRX. Bit[6] is the switch control bit (not a chicken bit) and needs to be set/cleared in real-time sequence when bit[7] is 0.
5:0	Reserved.

USBCONTAINER[0...1]x00000114 (USB31::USBCONTAINERREGCNTR0::DEVICE_REMOVABLE_CNTR0)

Read-write. Reset: 0000_0000h.

Device Removable Control register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000114; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:16	SS_PORT_DEVICE_REMOVABLE. Read-write. Reset: 0000h. Description: SS Port Device Removable. Indicates if the device attached to a downstream port (SS port) is permanently attached or not.
15:0	USB2_PORT_DEVICE_REMOVABLE. Read-write. Reset: 0000h. Description: USB2 Port Device Removable. Indicates if the device attached to a downstream port (usb2.0 port) is permanently attached or not.

**USBCONTAINER[0...1]x00000118
(USB31::USBCONTAINERREGCNTR0::INTERRUPT_CONTROL_CNTR0)**

Read-write. Reset: 0000_0000h.

Interrupt Control register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000118; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:13	INTERRUPT_RESERVED_31_13. Read-write. Reset: 0_0000h. Reserved
12	Reserved.
11:9	INTERRUPT_RESERVED_11_9. Read-write. Reset: 0h. Reserved
8	USB_LEGACY_SMI_EN. Read-write. Reset: 0. Description: Enable bit to send out legacy SMI . 0: Disable 1: Enable
7:5	INTERRUPT_RESERVED_7_5. Read-write. Reset: 0h. Reserved
4	USB_HSE_TO_SMI_EN. Read-write. Reset: 0. Description: Enable bit to send out SMI when detect host system error and USBCMD.HSEE is set. 0: Disable 1: Enable
3:1	INTERRUPT_RESERVED_3_1. Read-write. Reset: 0h. Reserved
0	USB_HSE_TO_NMI_EN. Read-write. Reset: 0. Description: Enable bit to send out NMI when detect host system error and USBCMD.HSEE is set. 0: Disable 1: Enable

**USBCONTAINER[0...1]x0000011C
(USB31::USBCONTAINERREG_CNTR0::DBGSC_PIPE_SWITCH_CTRL_CNTR0)**

Read-write. Reset: 0000_0000h.

DBGSC Pipe Switch Control Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x0000011C; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:23	DBGSC_PIPE_SWITCH_CTRL_RESERVED_31_23. Read-write. Reset: 000h. Reserved
22	USB_DBGSC_SSP_CONN_DET_TXDETRX_DIS. Read-write. Reset: 0. Description: Control bit to disable the PIPE.TxDetectRx condition for SuperSpeed Port connect detection 1: PIPE.TxDetectRx won't trigger DBGSC SuperSpeed Port connect detection 0: PIPE.TxDetectRx will trigger DBGSC SuperSpeed Port connect detection
21	USB_DBGSC_SSP_CONN_DET_PD_3_DIS. Read-write. Reset: 0. Description: Control bit to disable the PIPE.Powerdown==3 condition for SuperSpeed Port connect detection 1: PIPE.Powerdown==3 won't trigger DBGSC SuperSpeed Port connect detection 0: PIPE.Powerdown==3 will trigger DBGSC SuperSpeed Port connect detection
20	USB_DBGSC_SSP_CONN_DET_RESET_DIS. Read-write. Reset: 0. Description: Control bit to disable the PIPE.RESET condition for SuperSpeed Port connect detection 1: PIPE.RESET won't trigger DBGSC SuperSpeed Port connect detection 0: PIPE.RESET will trigger DBGSC SuperSpeed Port connect detection
19:18	DBGSC_PIPE_SWITCH_CTRL_RESERVED_19_18. Read-write. Reset: 0h. Reserved
17	USB_DBGSC_U2_CONN_DET_SUSPEND_DIS. Read-write. Reset: 0. Description: Control bit to disable the SUSPENDM condition for USB2 port connect detection 1: SUSPENDM won't trigger DBGSC USB2 connect detection 0: SUSPENDM will trigger DBGSC USB2 connect detection
16	USB_DBGSC_U2_CONN_DET_PORTRST_DIS. Read-write. Reset: 0. Description: Control bit to disable the PORTRST condition for USB2 port connect detection 1: PORTRST won't trigger DBGSC USB2 connect detection 0: PORTRST will trigger DBGSC USB2 connect detection
15:10	USB_DBGSC_DBUS_MUX1_SEL_CTRL. Read-write. Reset: 00h. dbgsc pswitch debugbus mux1 selection control
9:4	USB_DBGSC_DBUS_MUX0_SEL_CTRL. Read-write. Reset: 00h. dbgsc pswitch debugbus mux0 selection control
3	DBGSC_PIPE_SWITCH_CTRL_RESERVED_3. Read-write. Reset: 0. Reserved
2	USB_DBGSC_PSWITCH_PHY_RESET_MASK. Read-write. Reset: 0. Description: Dbgsc pipe switch phy reset mask bit 1: mask the phy reset once the pipe is switched to dbgsc HOST
1	USB_DBGSC_PSWITCH_DISABLE. Read-write. Reset: 0. Description: Dbgsc pipe switch disable bit 1: Reset
0	USB_DBGSC_PSWITCH_SOFT_RESET. Read-write. Reset: 0. Description: Dbgsc pipe switch soft reset 1: Reset

USBCONTAINER[0...1]x00000120**(USB31::USBCONTAINERREGCNTR0::OVERCURRENT_MAP_0_CNTR0)**

Read-write. Reset: FFFF_FFFFh.

Overcurrent Map 0 register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000120; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:28	PORT7_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 7
27:24	PORT6_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 6
23:20	PORT5_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 5
19:16	PORT4_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 4
15:12	PORT3_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 3
11:8	PORT2_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 2
7:4	PORT1_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 1
3:0	PORT0_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 0

USBCONTAINER[0...1]x00000124**(USB31::USBCONTAINERREGCNTR0::OVERCURRENT_MAP_1_CNTR0)**

Read-write. Reset: FFFF_FFFFh.

Overcurrent Map 1 register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000124; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:28	PORT15_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 15
27:24	PORT14_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 14
23:20	PORT13_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 13
19:16	PORT12_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 12
15:12	PORT11_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 11
11:8	PORT10_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 10
7:4	PORT9_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 9
3:0	PORT8_OVERCURRENT_SELECT . Read-write. Reset: Fh. Overcurrent input mapped to Port 8

USBCONTAINER[0...1]x00000130 (USB31::USBCONTAINERREGCNTR0::PORT_CONTROL_CNTR0)

Read-write. Reset: 0000_0922h.

Port Number Control register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000130; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:16	USB31_PORT_FORCE_GEN1_EN. Read-write. Reset: 0000h. USB31 PORT FORCE GEN1 EN
15:12	USB31_PORT_SEL. Read-write. Reset: 0h. Description: Selecting USB3.1 PHY ports. The field is used to select USB3.1 PHYs on CR interface accesses. Totally 16 PHYs can be supported although we only have 1 now. The field may also be used for PHY selection in other use cases if necessary.
11	USB31_CR_IF_SEL. Read-write. Reset: 1. Description: Selecting USB3.1 PHY CR interface. 0: selects JTAG path to access USB3.1 PHY registers. 1: selects CR IF to access USB3.1 PHY registers.
10	USB31_CR_IF_TIMEOUT_DISABLE. Read-write. Reset: 0. Description: USB31 PHY CR IF Timeout control bit 0: Enable timeout (default) 1: Disable timeout
9	PPCTRL_ENABLE. Read-write. Reset: 0. Description: PPC (port power control) enable Controls PPC bit of xHCI HCCPARAMS1 register for host port power control indication. 0: PPC = 0 1: PPC = 1
8	OC_POLARITY_CONTROL. Read-write. Reset: 1. Description: Over-current input polarity configure 0: OC pin is high when OC occurs 1: OC pin is low when OC occurs (default).
7:4	USB3_NUM_PORTS. Read-write. Reset: 2h. Description: Number of SS Ports. Controls the 'Number of Ports' field of the xHCI HCPARAMS1 register. 'Number of Ports' = USB2_NUM_PORTS + USB3_NUM_PORTS. 0h is not a valid value. Default is the fch_usb_u3_port_number feature. DRD0(USB0): 1, HC1(USB1): 2, DRD1(USB2): 1
3:0	USB2_NUM_PORTS. Read-write. Reset: 2h. Description: Number of USB2 Ports. Controls the 'Number of Ports' field of the xHCI HCPARAMS1 register. 'Number of Ports' = USB2_NUM_PORTS + USB3_NUM_PORTS. 0h is not a valid value. Default is the fch_usb_u2_port_number feature. DRD0(USB0): 1, HC1(USB1): 6, DRD1(USB2): 1

USBCONTAINER[0...1]x00000140**(USB31::USBCONTAINERREGCNTR0::MEM_ECC_SMI_STATUS_CNTR0)**

Read-write. Reset: 0000_0000h.

Memory ECC [SMI](#) Interrupt Status register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000140; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:1	MEM_ECC_SMI_STATUS_RESERVED_31_1. Read-write. Reset: 0000_0000h. Reserved
0	Reserved.

USBCONTAINER[0...1]x00000154**(USB31::USBCONTAINERREGCNTR0::SYSBUS_BLOCK_CONTROL_CNTR0)**

Read-write. Reset: 0000_0000h.

Sysbus Block Control register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000154; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:24	SYSBUS_BLOCK_WATERMARK_1. Read-write. Reset: 00h. sysbus_block_watermark_time_high[7:0]
23:16	SYSBUS_BLOCK_WATERMARK_0. Read-write. Reset: 00h. sysbus_block_watermark_time_low[7:0]
15:5	SYSBUS_BLOCK_CONTROL_RESERVED_15_5. Read-write. Reset: 000h. Reserved
4	SYSBUS_BLOCK_WATERMARK_0_SEL. Read-write. Reset: 0. when it's asserted, watermark_0 will be driven out ok_to_block_sysbus[0], otherwise, watermark_1 will be driven out
3	BLOCK_SYSBUS_OVRD_EN_HI. Read-write. Reset: 0. Enable for override to ok_to_block_sysbus[1]
2	BLOCK_SYSBUS_OVRD_EN_LO. Read-write. Reset: 0. Enable for override to ok_to_block_sysbus[0]
1	OK_TO_BLOCK_SYSBUS_OVRD_HI. Read-write. Reset: 0. Override value for ok_to_block_sysbus[1]
0	OK_TO_BLOCK_SYSBUS_OVRD_LO. Read-write. Reset: 0. Override value for ok_to_block_sysbus[0]

USBCONTAINER[0...1]x00000248**(USB31::USBCONTAINERREGCNTR0::RSMU_MGCG_USB_OVERRIDE_CNTR0)**

Read-write. Reset: 0000_000Dh.

IP override control for RSMU MGCG ENABLE signal

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000248; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:4	RSMU_MGCG_USB_OVERRIDE_RESERVED_31_4. Read-write. Reset: 000_0000h. Reserved
3	RSMU_MGCG_USB_override_val_1. Read-write. Reset: 1. Description: Override value for RSMU MGCG ENABLE bit.For u_gate_TESTCLK_r0 0: Override RSMU MGCG Enable to 0. 1: Override RSMU MGCG Enable to 1. (default)
2	RSMU_MGCG_USB_override_en_1. Read-write. Reset: 1. Description: Override enable bit for RSMU MGCG ENABLE bit.For u_gate_TESTCLK_r0 0: Do no override RSMU MGCG Enable bit. 1: Override RSMU MGCG Enable bit.(default)
1	RSMU_MGCG_USB_override_val_0. Read-write. Reset: 0. Description: Override value for RSMU MGCG ENABLE bit.For u_gate_CLK_Core_Bus_Clk_AXI_up 0: Override RSMU MGCG Enable to 0. (default) 1: Override RSMU MGCG Enable to 1.
0	RSMU_MGCG_USB_override_en_0. Read-write. Reset: 1. Description: Override enable bit for RSMU MGCG ENABLE bit.For u_gate_CLK_Core_Bus_Clk_AXI_up 0: Do no override RSMU MGCG Enable bit. 1: Override RSMU MGCG Enable bit.(default)

USBCONTAINER[0...1]x00000280 (USB31::USBCONTAINERREGCNTR0::BCConfig1_CNTR0)

Read-write. Reset: 000A_0000h.

Battery Charging Control 1 Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000280; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31	BCCL_SOFT_RESET. Read-write. Reset: 0. Description: BCCL soft reset 0: BCCL reset is HW controlled. 1: BCCL is reset.
30:20	BCCONFIG_1_RESERVED_30_20. Read-write. Reset: 000h. Reserved
19	PULLDOWN_EN. Read-write. Reset: 1. Description: When set, DPPULLDOWN_OUT and DMPULLDOWN_OUT are asserted all the time. This enables PHY to operate in BC. This setting affects all ports with CDP_ENABLE=1.
18	SUSPENDM_CONTROL_EN. Read-write. Reset: 0. Description: When set, BCCL asserts SUSPENDM_OUT to PHY upon power up. Control of SUSPENDM_OUT is disabled once LINESTATE is not 2'b00(USB connection found) and host is out of reset
17	SUSPENDM_CHECK_EN. Read-write. Reset: 1. Description: When set, BC operation may only start if SUSPENDM is asserted. Otherwise, SUSPENDM signal is not checked for enabling/disabling BC operation
16	PRIMARY_DET_MODE. Read-write. Reset: 0. Description: Primary Detection Mode 1: Primary detection based on connection. CDP drives Vdm_src after a disconnection, and disable Vdm_src after a connection is found. 0: Primary detection based on handshake. When there is no connection, and Vlgc>DP>Vdat_ref, CDP drives Vdm_src asynchronously. CDP stops driving Vdm_src when Vdat_ref>DP or device connects(DP>Vlgc)
15:0	CDP_ENABLE. Read-write. Reset: 0000h. Description: CDP Enable (per port) 1: CDP Enabled 0: CDP Disabled Bit 0 refers to port0, bit 1 refers to port 1 and so on

USBCONTAINER[0...1]x00000284 (USB31::USBCONTAINERREGCNTR0::BCConfig2_CNTR0)

Reset: 0000_0064h.

Battery Charging Control 2 Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000284; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:16	BCCONFIG_2_RESERVED_31_16. Read-write. Reset: 0000h. Reserved
15:0	BCCL_CLK_FREQ. Read-only. Reset: 0064h. Description: Defines the frequency of BCCL_CLK.(MHz) BCCL_CLK_PERIOD = 1/BCCL_CLK_FREQ

USBCONTAINER[0...1]x00000288 (USB31::USBCONTAINERREGCNTR0::BCConfig3_CNTR0)

Read-write. Reset: 00FA_00FAh.

Battery Charging Control 3 Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000288; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:16	CONNECT_DURATION . Read-write. Reset: 00FAh. Description: Defines the length of time(in unit of BCCL_CLK_PERIOD) for validating connection. This duration should be at least 2.5us(see table 7.2 of USB 2.0 spec).
15:0	DISCONNECT_DURATION . Read-write. Reset: 00FAh. Description: Defines the length of time(in unit of BCCL_CLK_PERIOD) for validating disconnection. This duration should be at least 2.5us(see table 7.2 of USB 2.0 spec).

USBCONTAINER[0...1]x0000028C (USB31::USBCONTAINERREGCNTR0::BCConfig4_CNTR0)

Read-write. Reset: 004F_004Fh.

Battery Charging Control 4 Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x0000028C; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:16	CHGDET_DEBOUNCE_TIME . Read-write. Reset: 004Fh. Defines the length of time(in unit of 256*BCCL_CLK_DIVIDER) in asserting VDATSRCENB/VDATDETENB after DPPULLDOWN_OUT/DMPULLDOWN_OUT are asserted
15:0	VDAT_ASSERTION_DELAY . Read-write. Reset: 004Fh. Defines the length of time(in unit of 256*BCCL_CLK_PERIOD) in debouncing CHGDET

USBCONTAINER[0...1]x00000290 (USB31::USBCONTAINERREGCNTR0::BCSpare0_CNTR0)

Read-write. Reset: 0000_0000h.

Battery Charging Spare 0 Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000290; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:0	BCSPARE0_RESERVED_31_0 . Read-write. Reset: 0000_0000h. Reserved

USBCONTAINER[0...1]x00000294 (USB31::USBCONTAINERREGCNTR0::BCSpare1_CNTR0)

Read-write. Reset: 0000_0000h.

Battery Charging Spare 1 Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000294; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:0	BCSPARE1_RESERVED_31_0 . Read-write. Reset: 0000_0000h. Reserved

USBCONTAINER[0...1]x00000298 (USB31::USBCONTAINERREGCNTR0::BCIntrCtrl_CNTR0)

Read-write. Reset: 0000_0000h.

Battery Charging Interrupt Control Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00000298; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:3	BCINTRCTRL_RESERVED_31_3. Read-write. Reset: 0000_0000h. Reserved
2:1	BCCL_INTR_ROUTE. Read-write. Reset: 0h. Description: BCCL interrupt routing select 00: RSMU 01: SMI 10: Reserved 11: Reserved
0	BCCL_INTR_ENABLE. Read-write. Reset: 0. Description: BCCL interrupt enable 0: Disable 1: Enable When enabled, BCCL interrupt will be routed according to BCCL_INTR_ROUTE

USBCONTAINER[0...1]x0000029C (USB31::USBCONTAINERREGCNTR0::BCINTEn_CNTR0)

Read-write. Reset: 0000_0000h.

Battery Charging Interrupt Enable Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x0000029C; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:16	BCINTEN_RESERVED_31_16. Read-write. Reset: 0000h. Reserved
15:0	PRIMARY_DET_INT_EN. Read-write. Reset: 0000h. Description: Primary Detect Interrupt Enable (per port) 1: Primary_Det_INT is asserted when BCStatus.PRIMARY_DET_STS Bit 0 refers to port0, bit 1 refers to port1 and so on

USBCONTAINER[0...1]x000002A0 (USB31::USBCONTAINERREGCNTR0::BCStatus_P0_CNTR0)

Battery Charging Status Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x000002A0; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:4	BCSTATUS_RESERVED_31_4. Read-write. Reset: 000_0000h. Reserved
3	CHGDET. Read-only. Reset: 0. Description: CHGDET Status This field reflects the current status of CHGDET
2	FSVMINUS. Read-only. Reset: 0. Description: FSVMINUS Status This field reflects the current status of FSVMINUS
1	FSVPLUS. Read-only. Reset: 0. Description: FSVPLUS Status This field reflects the current status of FSVPLUS
0	PRIMARY_DET_STS. Read,Write-1-to-clear. Description: primary Detection Status This field reflects Primary Detection signaling of Vlgc>DP>Vdat_ref was once received from device. Once asserted, this signal is kept at 1, until software writes 1 to clear

USBCONTAINER[0...1]x000002A4 (USB31::USBCONTAINERREGCNTR0::BCStatus_P1_CNTR0)**Battery Charging Status Register**

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x000002A4; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:4	BCSTATUS_RESERVED_31_4. Read-write. Reset: 000_0000h. Reserved
3	CHGDET. Read-only. Reset: 0. Description: CHGDET Status This field reflects the current status of CHGDET
2	FSVMINUS. Read-only. Reset: 0. Description: FSVMINUS Status This field reflects the current status of FSVMINUS
1	FSVPLUS. Read-only. Reset: 0. Description: FSVPLUS Status This field reflects the current status of FSVPLUS
0	PRIMARY_DET_STS. Read,Write-1-to-clear. Description: primary Detection Status This field reflects Primary Detection signaling of Vlgc>DP>Vdat_ref was once received from device. Once asserted, this signal is kept at 1, until software writes 1 to clear

USBCONTAINER[0...1]x0005BC00**(USB31::USBCONTAINERREGCNTR0::PORT_DISABLE_WRITE_ONCE_CNTR0)**

Read-write.

Port Disable Write Once Control register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x0005BC00; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:16	USB3_PORT_DISABLE_WO. Read-write. Description: SS Port Disable (RW1S). if some platform set the test_powerdown_ssp=1, we should set USB3_PORT_DISABLE_WO[16] to 1. When set USB3_PORT_DISABLE[n] to 1, USB3 port<n> is disabled. Only writing the value '1' has any effect. Writing a '0' has no effect. Fields can be cleared by cold boot reset and system reset. Note: Bit[16] should not be asserted to disable eSS Port0.
15:0	USB2_PORT_DISABLE_WO. Read-write. Description: USB2 Port Disable (RW1S). When set USB2_PORT_DISABLE[n] to 1, USB2 port<n> is disabled. Only writing the value '1' has any effect. Writing a '0' has no effect. Fields can be cleared by cold boot reset and system reset. Note: Bit[0] should not be asserted to disable USB2 Port0.

15.3.3.2 USB 3.1 – Controller Register

USB[0...1]x000000 (FCHUSBDCUSB31CNTR0::CAPLENGTH)

Read-only.

Capability Registers Length

Host Controller Operational Registers = Base address + CAPLENGTH

where CAPLENGTH is `DWC_USB31_HOST_CAP_REG_LEN whose default value is 20h.

_instUSB0_aliasHOST; USB0x000000; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000000; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	HCIVERSION. Read-only. Reset: 0110h. HC Interface Version Number (HCIVERSION)
15:8	Reserved.
7:0	CAPLENGTH. Read-only. Reset: 20h. Capability Registers Length (CAPLENGTH)

USB[0...1]x000004 (FCHUSBDCUSB31CNTR0::HCSPARAMS1)

Read-only.

Structural Parameters 1 Register

For register definitions, refer to the xHCI specification.

_instUSB0_aliasHOST; USB0x000004; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000004; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	MAXPORTS. Read-only. Reset: 04h. Description: Number of Ports (MaxPorts) - Number of ports implemented is defined by the parameter (`DWC_USB31_HOST_NUM_U2_ROOT_PORTS + `DWC_USB31_HOST_NUM_U3_ROOT_PORTS) - Number of ports enabled is controlled by the controller input signals host_num_u2_port[3:0]+host_num_u3_port[3:0] Note: In USB 2.0-only mode, the host_num_u3_port signal is zero.
23:19	Reserved.
18:8	MAXINTRS. Read-only. Reset: 008h. Description: Number of Interrupters (MaxIntrs) Defined by the configurable parameter `DWC_USB31_HOST_NUM_INTERRUPTER_SUPT
7:0	MAXSLOTS. Read-only. Reset: 40h. Description: Number of device slots (MaxSlots) Defined by configurable parameter `DWC_USB31_NUM_DEVICE_SUPT

USB[0...1]x000008 (FCHUSBDCUSB31CNTR0::HCSPARAMS2)

Read-only.

Structural Parameters 2 Register

For register definitions, refer to the xHCI specification.

_instUSB0_aliasHOST; USB0x000008; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000008; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:27	MAXSCRATCHPADBUFS. Read-only. Reset: 02h. Description: Max Scratchpad Bufs Lo The value is calculated based on chosen configuration parameter values. Possible values are 1-4.
26	SPR. Read-only. Reset: 1. Scratchpad Restore (SPR)
25:21	MAXSCRATCHPADBUFS_HI. Read-only. Reset: 00h. Description: Max Scratchpad Bufs HI The controller automatically updates this field.
20:8	Reserved.
7:4	ERSTMAX. Read-only. Reset: Fh. Event Ring Segment Table Max (ERST Max)
3:0	IST. Read-only. Reset: 1h. Isochronous Scheduling Threshold (IST)

USB[0...1]x00000C (FCHUSBDCUSB31CNTR0::HCSPARAMS3)

Read-only.

Structural Parameters 3 Register

For register definitions, refer to the xHCI specification.

_instUSB0_aliasHOST; USB0x00000C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00000C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	U2_DEVICE_EXIT_LAT. Read-only. Reset: 0007h. U2 Device Exit Latency
15:8	Reserved.
7:0	U1_DEVICE_EXIT_LAT. Read-only. Reset: 0Ah. U1 Device Exit Latency

USB[0...1]x000010 (FCHUSBWCUSB31CNTR0::HCCPARAMS1)

Reset: 0118_FFCDh.

Capability Parameters 1 Register

For register definitions, refer to the xHCI specification.

_instUSB0_aliasHOST; USB0x000010; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x000010; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	XECP. Read-only. Reset: 0118h. Description: xHCI Extended Capabilities Pointer (xECP) Based on configuration, controller automatically updates it. The value of this pointer was changed in Release 1.80a_lp02. Prior to this release, the extended Capability registers were mapped to after the Doorbell registers. With this change, these registers are now mapped to after the Operational registers.
15:12	MAXPSASIZE. Read-only. Reset: Fh. Description: Maximum Primary Stream Array Size (MaxPSASize) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
11	CFC. Read-only. Reset: 1. Contiguous Frame ID Capability (CFC)
10	SEC. Read-only. Reset: 1. Description: Stopped EDLTA Capability (SEC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
9	SPC. Read-only. Reset: 1. Description: Short Packet Capability (SPC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
8	PAE. Read-only. Reset: 1. Description: Parse All Event Data (PAE) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7	NSS. Read-only. Reset: 1. Description: No Secondary SID Support (NSS) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
6	LTC. Read-only. Reset: 1. Description: Latency Tolerance Messaging Capability (LTC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

5	LHRC. Read-only. Reset: 0. Description: Light HC Reset Capability For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
4	PIND. Read-only. Reset: 0. Description: Port Indicators (PIND) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3	PPC. Read-only, Volatile. Reset: 1. Description: Port Power Control For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
2	CSZ. Read-only. Reset: 1. Description: Context Size (CSZ) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
1	BNC. Read-only. Reset: 0. Description: Bandwidth Negotiation Capability (BNC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	AC64. Read-only. Reset: 1. Description: 64-bit Addressing Capability (AC64) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000014 (FCHUSBWCUSB31CNTR0::DBOFF)

Read-only.

Doorbell Offset Register

For register definitions, refer to the xHCI specification.

The value of this register is 8192. So the DoorBell Registers Start on a 4K Boundary.

This value was changed in Release 1.80a_lp02 to make it aligned to a Page Boundary. Before the 1.80a_lp02 release, the value was not on a Page boundary.

_instUSB0_aliasHOST; USB0x000014; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000014; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	DOORBELL_ARRAY_OFFSET. Read-only. Reset: 0000_0800h. Description: Doorbell Array Offset - RO Based on configuration, controller automatically updates it. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
1:0	Reserved.

USB[0...1]x000018 (FCHUSBWCUSB31CNTR0::RTSOFF)

Read-only.

Runtime Register Space Offset Register

Base address + 0x18

where RTSOFF is `DWC_USB31_HC_RUNTIME_REGISTER_SPACE_OFFSET. This parameter is defined as 4096. So the Run time registers start on a fresh 4K Page.

This value was changed in Release 1.80a_lp02 to make it aligned to a Page Boundary. Before the 1.80a_lp02 release, the value was not on a Page boundary.

_instUSB0_aliasHOST; USB0x000018; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000018; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:5	RUNTIME_REG_SPACE_OFFSET. Read-only. Reset: 000_0080h. Description: Runtime Register Space Offset Based on configuration, controller automatically updates it. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
4:0	Reserved.

USB[0...1]x00001C (FCHUSBDCUSB31CNTR0::HCCPARAMS2)

Read-only. Reset: 0000_00FFh.

Host Controller Capability Parameters 2

For register definitions, refer to the xHCI specification.

_instUSB0_aliasHOST; USB0x00001C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00001C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:8	Reserved.
7	ETC_TSC. Read-only. Reset: 1. Extended TBC TRB Status Capability-ETC_TSC
6	ETC. Read-only. Reset: 1. ETC
5	CIC. Read-only. Reset: 1. Description: Configuration Information Capability (CIC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
4	LEC. Read-only. Reset: 1. Description: Large ESIT Payload Capability (LEC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3	CTC. Read-only. Reset: 1. Description: Compliance Transition Capability (CTC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
2	FSC. Read-only. Reset: 1. Description: Force Save Context Capability (FSC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
1	CMC. Read-only. Reset: 1. Description: Configure Endpoint Command Max Exit Latency Too Large Capability (CMC) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	U3C. Read-only. Reset: 1. Description: U3 Entry Capability (U3C) For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000020 (FCHUSBWCUSB31CNTR0::USBCMD)

USB Command Register

For a description of this standard USB register field see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x000020; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000020; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:15	Reserved.
14	ETE. Read-write,Reserved. Reset: 0. Description: Extended TBC Enable For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
13	CME. Read-write. Reset: 0. Description: CEM Enable For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
12	Reserved.
11	EU3S. Read-write. Reset: 0. Description: EU3S For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
10	EWE. Read-write. Reset: 0. Description: EWE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
9	CRS. Read-write,Reserved. Reset: 0. Description: Controller Restore State This command is similar to the USBCMD.CRS bit in host mode and initiates the restore process. When software sets this bit to '1', the controller immediately sets DSTS.RSS to '1'. When the controller has finished the restore process, it sets DSTS.RSS to '0'. Note: When read, this field always returns '0'.
8	CSS. Read-write,Reserved. Reset: 0. Description: Controller Save State This command is similar to the USBCMD.CSS bit in host mode and initiates the save process. When software sets this bit to '1', the controller immediately sets DSTS.SSS to '1'. When the controller has finished the save process, it sets DSTS.SSS to '0'. Note: When read, this field always returns '0'.
7	LHCRST. Read-write. Reset: 0.

	Description: Light Host Controller Reset For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1. The following bits reset the internal logic of the host controller. Under soft reset, some CSR accesses may fail (Timeout). - HCRST - LHCRST
6:4	Reserved.
3	HSEE. Read-write. Reset: 0. Description: HSEE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
2	INTE. Read-write. Reset: 0. Description: INTE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
1	HCRST. Read-write. Reset: 0. Description: HCRST The following bits reset the internal logic of the host controller. Under soft reset, some CSR accesses may fail (Timeout). - HCRST - LHCRST
0	R_S. Read-write,Reserved. Reset: 0. Description: R_S For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000024 (FCHUSBWCUSB31CNTR0::USBSTS)**USB Status Register Bit Definitions**

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x000024; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000024; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:13	Reserved.
12	HCE. Read-only,Volatile. Reset: 0. Description: Host Controller Error (HCE) - RO Default = 0. '0' = No internal xHC error conditions exist and '1' = Internal xHC error condition. This flag must be set to indicate that an internal error condition has been detected which requires software to reset and reinitialize the xHC.
11	CNR. Read-only,Volatile. Reset: 1. Description: Controller Not Ready (CNR) - RO Default = '1'. '0' = Ready and '1' = Not Ready. Software must not write to the Doorbell or Operational register of the xHC, other than the USBSTS register, until CNR = '0'. This flag is set by the xHC after a Chip Hardware Reset and cleared when the xHC is ready to begin accepting register writes. This flag remains cleared ('0') until the next Chip Hardware Reset.
10	SRE. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: Save/Restore Error This bit is currently not supported.
9	RSS. Read-only,Volatile. Reset: 0. Description: Restore State Status This bit is similar to the USBSTS.RSS in host mode. When the controller has finished the restore process, it completes the command by setting DSTS.RSS to '0'.
8	SSS. Read-only,Volatile. Reset: 0. Description: Save State Status This bit is similar to the USBSTS.SSS in host mode. When the controller has finished the save process, it completes the command by setting DSTS.SSS to '0'.
7:5	Reserved.
4	PCD. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0.

	Description: Port Change Detect For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1..
3	EINT. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: Event Interrupt For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1..
2	HSE. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: HSE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1..
1	Reserved.
0	HCH. Read-only,Volatile. Reset: 1. Description: HCH For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1..

USB[0...1]x000028 (FCHUSBDCUSB31CNTR0::PAGESIZE)

Read-only.

Page Size Register Bit Definitions

Use this register to enable or disable the reporting of specific USB Device Notification Transaction Packets being received.

A Notification Enable (Nx, where x = 0 to 15) flag is defined for each of the 16 possible device notification types. If a flag is set for a specific notification type, a Device Notification Event is generated when the respective notification packet is received. After reset, all notifications are disabled.

This register is written as a Dword. Byte writes produce undefined results.

_instUSB0_aliasHOST; USB0x000028; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000028; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	Reserved.
15:0	PAGE_SIZE. Read-only. Reset: 0001h. PAGE_SIZE

USB[0...1]x000034 (FCHUSBWCUSB31CNTR0::DNCTRL)

Device Notification Register Bit Definitions

For a description of this standard USB register field see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x000034; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000034; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	Reserved.
15:0	N0_N15. Read-write. Reset: 0000h.
	Description: N0_N15
	For a description of this standard USB register field see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000038 (FCHUSBWDCUSB31CNTR0::CRCR_LO)

Reset: 0000_0000h.

CRCR_LO

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x000038; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000038; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:6	CMD_RING_PNTR. Read-write,Volatile. Reset: 000_0000h. Description: CMD_RING_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
5:4	Reserved.
3	CRR. Read-only. Reset: 0. Description: CRR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
2	CA. Read-write,Read,Write-1-only,Volatile. Reset: 0. Description: CA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
1	CS. Read-write,Read,Write-1-only,Volatile. Reset: 0. Description: CS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	RCS. Read-write,Volatile. Reset: 0. Description: RCS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00003C (FCHUSBWDCUSB31CNTR0::CRCR_HI)

Read-write,Volatile. Reset: 0000_0000h.

CRCR_HI

_instUSB0_aliasHOST; USB0x00003C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00003C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:0	CMD_RING_PNTR. Read-write,Volatile. Reset: 0000_0000h. Description: COMMAND_RING_POINTER Reading this field always returns '0'. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000050 (FCHUSBDCUSB31CNTR0::DCBAAP_LO)**DCBAAP_LO**

_instUSB0_aliasHOST; USB0x000050; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000050; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:6	DEVICE_CONTEXT_BAAP. Read-write. Reset: 000_0000h. Description: DEVICE_CONTEXT_BAAP For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
5:0	Reserved.

USB[0...1]x000054 (FCHUSBDCUSB31CNTR0::DCBAAP_HI)

Read-write. Reset: 0000_0000h.

DCBAAP_HI

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x000054; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000054; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	DEVICE_CONTEXT_BAAP. Read-write. Reset: 0000_0000h. DEVICE_CONTEXT_BAAP

USB[0...1]x000058 (FCHUSBDCUSB31CNTR0::CONFIG)**Configure Register Bit Definitions**

This register is in the Aux Power well. It is only reset by platform hardware during a cold reset or in response to a Host Controller Reset (HCRST).

_instUSB0_aliasHOST; USB0x000058; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000058; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:10	Reserved.
9	CIE. Read-write, Volatile. Reset: 0. Description: Configuration Information Enable For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
8	U3E. Read-write, Volatile. Reset: 0. Description: U3 Entry Enable For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7:0	MAXSLOTSN. Read-write, Volatile. Reset: 00h. Description: MAXSLOTSN For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000420 (FCHUSBWCUSB31CNTR0::PORTSC_20_REGS0_PORTSC_20)

Reset: 0000_02A0h.

Port Status and Control Register Bit Definitions

The PORTSC Register Access fails (Timeout) if the UTMI/ULPI clock is not running or one of the following bits is asserted.

- PR
- ORC

_instUSB0_aliasHOST; USB0x000420; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000420; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	Reserved.
30	DR. Read-only,Volatile. Reset: 0. Description: Reset Value For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
29:28	Reserved.
27	WOE. Read-write. Reset: 0. Description: WOE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
26	WDE. Read-write. Reset: 0. Description: WDE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
25	WCE. Read-write. Reset: 0. Description: WCE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
24	CAS. Read-only. Reset: 0. Description: Cold Attach Status For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
23	Reserved.
22	PLC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PLC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
21	PRC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PRC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
20	OCC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0.

	Description: OCC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
19	Reserved.
18	PEC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PEC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
17	CSC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: CSC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
16	LWS. Read-write,Reserved. Reset: 0. Description: LWS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:14	PIC. Read-write,Volatile. Reset: 0h. Description: PIC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
13:10	PORTSPEED. Read-only. Reset: 0h. Description: PORTSPEED For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
9	PP. Read-write,Volatile. Reset: 1. Description: PP For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
8:5	PLS. Read-write,Volatile. Reset: 5h. Description: PLS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
4	PR. Read-write,Read,Write-1-only,Volatile. Reset: 0. PR
3	OCA. Read-only,Volatile. Reset: 0. Description: OCA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
2	Reserved.
1	PED. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0.

	Description: PED For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	CCS. Read-only, Volatile. Reset: 0. Description: CCS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000424 (FCHUSBWCUSB31CNTR0::PORTSC_20_REGS0_PORTPMSC_20)

Reset: 0000_0000h.

USB3 Port Power Management Status and Control Register Bit Definitions

This register is in the Aux Power well. It is only reset by platform hardware during a cold reset or in response to a Host Controller Reset (HCRST).

_instUSB0_aliasHOST; USB0x000424; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000424; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:28	PRTTSTCTRL. Read-write. Reset: 0h. Description: Port Test Control (Test Mode) When this field is 0, the port is NOT operating in a test mode. A non-zero value indicates that it is operating in testmode and the specific test mode is indicated by the specific value. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
27:17	Reserved.
16	HLE. Read-write. Reset: 0. Description: Hardware LPM Enable If this bit is set to 1 , then hardware controlled LPM shall be enabled for this port. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	L1DSLOT. Read-write. Reset: 00h. Description: L1DSLOT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7:4	BESL. Read-write. Reset: 0h. Description: Best Effort Service Latency Default = '0'. System software sets this field to indicate to the recipient device how long the xHC will drive resume if it (the xHC) initiates an exit from L1. The BESL value is encoded as follows: Value Description 0h 125us. (default) 1h 150us. 2h 200us. Fh 10 ms as per the xHCI specification Table 13. The value of 0000b is interpreted as 50us. Each incrementing value up adds 75us. to the previous value. Based on this rule, the maximum value resume drive time is at encoding value 1111b which represents BESL=10ms and Resume duration of 9.950ms. Note that the BESL field is used by both software and hardware controlled LPM. Refer to section 4.23.5.1.1 for more information on BESL use. Refer to Section 4.1 of the USB2 LPM spec for more information on the use of the BESL field.
3	RWE. Read-write. Reset: 0.

	Description: Remote Wake Enable System software sets this flag to enable or disable the device for remote wake from L1. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
	L1S. Read-only. Reset: 0h. Description: L1 Status (L1S) - RO. Default = 0. This field is used by software to determine whether an L1-based suspend request (LMP transaction) was successful, specifically: Value Meaning 0 Invalid - This field is ignored by software. - 1 Success - Port successfully transitioned to L1 (ACK) - 2 Not Yet - Device is unable to enter L1 at this time (NYET) - 3 Not Supported -Device does not support L1 transitions (STALL) - 4 Timeout/Error -Device failed to respond to the LPM Transaction or an error occurred - 5-7 Reserved The value of this field is only valid when the port resides in the L0 or L1 state (PLS = '0' or '2'). Refer to section 4.23.5.1.1 for more information.

USB[0...1]x000428 (FCHUSBDCUSB31CNTR0::PORTSC_20_REGS0_PORTLI_20)

Read-only. Reset: 0000_0000h.	
Port Link Info Register	
_instUSB0_aliasHOST; USB0x000428; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}	
_instUSB1_aliasHOST; USB1x000428; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}	
Bits	Description
31:0	Reserved.

USB[0...1]x00042C (FCHUSBWCUSB31CNTR0::PORTSC_20_REGS0_PORHLPMC_20)

Reset: 0000_0000h.

USB2 Port Hardware LPM Control Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x00042C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00042C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:14 Reserved.

13:10 **BESLD.** Read-write. Reset: 0h.**Description:** Best Effort Service Latency Deep (BESLD)

The BESL value is encoded as follows: Value Description 0h 125us. (default) 1h 150us. 2h 200us. Fh 10 ms as per the xHCI specification Table 13. The value of 0000b is interpreted as 50us. Each incrementing value up adds 75us. to the previous value.

Based on this rule, the maximum value resume drive time is at encoding value 1111b which represents BESL=10ms and Resume duration of 9.950ms.

Note that the BESL field is used by both software and hardware controlled LPM.

Refer to section 4.23.5.1.1 for more information on BESL use. Refer to Section 4.1 of the USB2 LPM spec for more information on the use of the BESL field.

9:2 **L1_TIMEOUT.** Read-write. Reset: 00h.**Description:** PORHLPMC_20 L1_TIMEOUT.

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

1:0 **HIRDM.** Read-write. Reset: 0h.**Description:** Host Initiated Resume Duration Mode (HIRDM)

- RWS. Default = 0h.

Indicates which BESL value must be used.

The following are permissible values:

Value Description

0 Initiate L1 using BESL only on timeout.

(default) 1 Initiate L1 using BESLD on timeout.

If rejected by device, initiate L1 using BESL.

3-2 Reserved.

USB[0...1]x000430 (FCHUSBWCUSB31CNTR0::PORTSC_20_REGS1_PORTSC_20)

Reset: 0000_02A0h.

Port Status and Control Register Bit Definitions

The PORTSC Register Access fails (Timeout) if the UTMI/ULPI clock is not running or one of the following bits is asserted.

- PR
- ORC

_instUSB0_aliasHOST; USB0x000430; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000430; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	Reserved.
30	DR. Read-only,Volatile. Reset: 0. Description: Reset Value For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
29:28	Reserved.
27	WOE. Read-write. Reset: 0. Description: WOE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
26	WDE. Read-write. Reset: 0. Description: WDE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
25	WCE. Read-write. Reset: 0. Description: WCE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
24	CAS. Read-only. Reset: 0. Description: Cold Attach Status For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
23	Reserved.
22	PLC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PLC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
21	PRC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PRC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
20	OCC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0.

	Description: OCC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
19	Reserved.
18	PEC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PEC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
17	CSC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: CSC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
16	LWS. Read-write,Reserved. Reset: 0. Description: LWS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:14	PIC. Read-write,Volatile. Reset: 0h. Description: PIC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
13:10	PORTSPEED. Read-only. Reset: 0h. Description: PORTSPEED For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
9	PP. Read-write,Volatile. Reset: 1. Description: PP For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
8:5	PLS. Read-write,Volatile. Reset: 5h. Description: PLS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
4	PR. Read-write,Read,Write-1-only,Volatile. Reset: 0. PR
3	OCA. Read-only,Volatile. Reset: 0. Description: OCA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
2	Reserved.
1	PED. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0.

	Description: PED For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	CCS. Read-only, Volatile. Reset: 0. Description: CCS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000434 (FCHUSBWCUSB31CNTR0::PORTSC_20_REGS1_PORTPMSC_20)

Reset: 0000_0000h.

USB3 Port Power Management Status and Control Register Bit Definitions

This register is in the Aux Power well. It is only reset by platform hardware during a cold reset or in response to a Host Controller Reset (HCRST).

_instUSB0_aliasHOST; USB0x000434; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000434; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:28	PRTTSTCTRL. Read-write. Reset: 0h. Description: Port Test Control (Test Mode) When this field is 0, the port is NOT operating in a test mode. A non-zero value indicates that it is operating in testmode and the specific test mode is indicated by the specific value. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
27:17	Reserved.
16	HLE. Read-write. Reset: 0. Description: Hardware LPM Enable If this bit is set to 1, then hardware controlled LPM shall be enabled for this port. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	L1DSLOT. Read-write. Reset: 00h. Description: L1DSLOT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7:4	BESL. Read-write. Reset: 0h. Description: Best Effort Service Latency Default = '0'. System software sets this field to indicate to the recipient device how long the xHC will drive resume if it (the xHC) initiates an exit from L1. The BESL value is encoded as follows: Value Description 0h 125us. (default) 1h 150us. 2h 200us. Fh 10 ms as per the xHCI specification Table 13. The value of 0000b is interpreted as 50us. Each incrementing value up adds 75us. to the previous value. Based on this rule, the maximum value resume drive time is at encoding value 1111b which represents BESL=10ms and Resume duration of 9.950ms. Note that the BESL field is used by both software and hardware controlled LPM. Refer to section 4.23.5.1.1 for more information on BESL use. Refer to Section 4.1 of the USB2 LPM spec for more information on the use of the BESL field.
3	RWE. Read-write. Reset: 0.

	Description: Remote Wake Enable System software sets this flag to enable or disable the device for remote wake from L1. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
2:0	L1S. Read-only. Reset: 0h. Description: L1 Status (L1S) - RO. Default = 0. This field is used by software to determine whether an L1-based suspend request (LMP transaction) was successful, specifically: Value Meaning 0 Invalid - This field is ignored by software. 1 Success - Port successfully transitioned to L1 (ACK) 2 Not Yet - Device is unable to enter L1 at this time (NYET) 3 Not Supported -Device does not support L1 transitions (STALL) 4 Timeout/Error -Device failed to respond to the LPM Transaction or an error occurred 5-7 Reserved The value of this field is only valid when the port resides in the L0 or L1 state (PLS = '0' or '2'). Refer to section 4.23.5.1.1 for more information.

USB[0...1]x000438 (FCHUSBDCUSB31CNTR0::PORTSC_20_REGS1_PORTLI_20)

	Read-only. Reset: 0000_0000h.				
	Port Link Info Register				
	_instUSB0_aliasHOST; USB0x000438; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}				
	_instUSB1_aliasHOST; USB1x000438; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}				
	<table border="1"> <thead> <tr> <th>Bits</th><th>Description</th></tr> </thead> <tbody> <tr> <td>31:0</td><td>Reserved.</td></tr> </tbody> </table>	Bits	Description	31:0	Reserved.
Bits	Description				
31:0	Reserved.				

USB[0...1]x00043C (FCHUSBWCUSB31CNTR0::PORTSC_20_REGS1_PORThLPMC_20)

Reset: 0000_0000h.

USB2 Port Hardware LPM Control Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x00043C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00043C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:14	Reserved.
13:10	BESLD. Read-write. Reset: 0h. Description: Best Effort Service Latency Deep (BESLD) The BESL value is encoded as follows: Value Description 0h 125us. (default) 1h 150us. 2h 200us. Fh 10 ms as per the xHCI specification Table 13. The value of 0000b is interpreted as 50us. Each incrementing value up adds 75us. to the previous value. Based on this rule, the maximum value resume drive time is at encoding value 1111b which represents BESL=10ms and Resume duration of 9.950ms. Note that the BESL field is used by both software and hardware controlled LPM. Refer to section 4.23.5.1.1 for more information on BESL use. Refer to Section 4.1 of the USB2 LPM spec for more information on the use of the BESL field.
9:2	L1_TIMEOUT. Read-write. Reset: 00h. Description: PORThLPMC_20 L1_TIMEOUT. For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
1:0	HIRDM. Read-write. Reset: 0h. Description: Host Initiated Resume Duration Mode (HIRDM) - RWS. Default = 0h. Indicates which BESL value must be used. The following are permissible values: Value Description 0 Initiate L1 using BESL only on timeout. (default) 1 Initiate L1 using BESLD on timeout. If rejected by device, initiate L1 using BESL. 3-2 Reserved.

USB[0...1]x000440 (FCHUSBWCUSB31CNTR0::PORTSC_30_REGS0_PORTSC_30)

Reset: 0000_02A0h.

Port Status and Control Register Bit Definitions

The PORTSC Register Access fails (Timeout) if the UTMI/ULPI clock is not running or one of the following bits is asserted.

- PR
- ORC
- WPR

_instUSB0_aliasHOST; USB0x000440; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000440; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	WPR. Read-write,Reserved. Reset: 0. Description: Reset Value For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1. Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.
30	DR. Read-only. Reset: 0. Description: Reset Value For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
29:28	Reserved.
27	WOE. Read-write. Reset: 0. Description: WOE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
26	WDE. Read-write. Reset: 0. Description: WDE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
25	WCE. Read-write. Reset: 0. Description: WCE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
24	CAS. Read-only. Reset: 0. Description: Cold Attach Status For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
23	CEC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0.

	Description: CEC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
22	PLC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PLC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
21	PRC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PRC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1. Programming this field with random data will cause side effect. Bit Bash register testing is not recommended.
20	OCC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: OCC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
19	WRC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: WRC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
18	PEC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PEC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
17	CSC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: CSC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
16	LWS. Read-write,Reserved. Reset: 0. Description: LWS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:14	PIC. Read-write,Reserved,Volatile. Reset: 0h. Description: PIC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
13:10	PORTSPEED. Read-only. Reset: 0h.

	Description: PORTSPEED Note: PORTSPEED for Enhanced SuperSpeed 10Gbps is 5 For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
9	PP. Read-write,Reserved,Volatile. Reset: 1. Description: PP For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
8:5	PLS. Read-write,Volatile. Reset: 5h. Description: PLS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
4	PR. Read-write,Read,Write-1-only,Volatile. Reset: 0. Description: PR Programming this field with random data will cause side effect. Bit Bash register testing is not recommended.
3	OCA. Read-only,Volatile. Reset: 0. Description: OCA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
2	Reserved.
1	PED. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PED For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	CCS. Read-only,Volatile. Reset: 0. Description: CCS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000444 (FCHUSBWCUSB31CNTR0::PORTSC_30_REGS0_PORTPMSC_30)

Reset: 0000_0000h.

USB3 Port Power Management Status and Control Register Bit Definitions

This register is in the Aux Power well. It is only reset by platform hardware during a cold reset or in response to a Host Controller Reset (HCRST).

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x000444; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000444; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:17	Reserved.
16	FLA. Read-write,Volatile. Reset: 0. Description: FLA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	U2_TIMEOUT. Read-write,Volatile. Reset: 00h. Description: U2_TIMEOUT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7:0	U1_TIMEOUT. Read-write,Volatile. Reset: 00h. Description: U1_TIMEOUT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000448 (FCHUSBDCUSB31CNTR0::PORTSC_30_REGS0_PORTLI_30)

Read-only. Reset: 0000_0000h.

Port Link Info Register

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x000448; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000448; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:24 Reserved.

23:20 **TLC.** Read-only. Reset: 0h.**Description:** Tx Lane Count (TLC)

Currently this field supports value 0

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

19:16 **RLC.** Read-only. Reset: 0h.**Description:** Rx Lane Count (RLC)

Currently this field supports value 0

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

15:0 **LINK_ERROR_COUNT.** Read-only. Reset: 0000h.**Description:** LINK_ERROR_COUNT

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00044C (FCHUSBDCUSB31CNTR0::PORTSC_30_REGS0_PORHLPMC_30)

Read-only. Reset: 0000_0000h.

USB2 Port Hardware LPM Control Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x00044C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00044C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:0 Reserved.

USB[0...1]x000450 (FCHUSBWCUSB31CNTR0::PORTSC_30_REGS1_PORTSC_30)

Reset: 0000_02A0h.

Port Status and Control Register Bit Definitions

The PORTSC Register Access fails (Timeout) if the UTMI/ULPI clock is not running or one of the following bits is asserted.

- PR
- ORC
- WPR

_instUSB0_aliasHOST; USB0x000450; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000450; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	WPR. Read-write,Reserved. Reset: 0. Description: Reset Value For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1. Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.
30	DR. Read-only. Reset: 0. Description: Reset Value For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
29:28	Reserved.
27	WOE. Read-write. Reset: 0. Description: WOE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
26	WDE. Read-write. Reset: 0. Description: WDE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
25	WCE. Read-write. Reset: 0. Description: WCE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
24	CAS. Read-only. Reset: 0. Description: Cold Attach Status For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
23	CEC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0.

	Description: CEC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
22	PLC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PLC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
21	PRC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PRC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1. Programming this field with random data will cause side effect. Bit Bash register testing is not recommended.
20	OCC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: OCC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
19	WRC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: WRC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
18	PEC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PEC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
17	CSC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: CSC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
16	LWS. Read-write,Reserved. Reset: 0. Description: LWS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:14	PIC. Read-write,Reserved,Volatile. Reset: 0h. Description: PIC For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
13:10	PORTSPEED. Read-only. Reset: 0h.

	Description: PORTSPEED Note: PORTSPEED for Enhanced SuperSpeed 10Gbps is 5 For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
9	PP. Read-write,Reserved,Volatile. Reset: 1. Description: PP For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
8:5	PLS. Read-write,Volatile. Reset: 5h. Description: PLS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
4	PR. Read-write,Read,Write-1-only,Volatile. Reset: 0. Description: PR Programming this field with random data will cause side effect. Bit Bash register testing is not recommended.
3	OCA. Read-only,Volatile. Reset: 0. Description: OCA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
2	Reserved.
1	PED. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: PED For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	CCS. Read-only,Volatile. Reset: 0. Description: CCS For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000454 (FCHUSBWCUSB31CNTR0::PORTSC_30_REGS1_PORTPMSC_30)

Reset: 0000_0000h.

USB3 Port Power Management Status and Control Register Bit Definitions

This register is in the Aux Power well. It is only reset by platform hardware during a cold reset or in response to a Host Controller Reset (HCRST).

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x000454; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000454; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:17	Reserved.
16	FLA. Read-write,Volatile. Reset: 0. Description: FLA For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	U2_TIMEOUT. Read-write,Volatile. Reset: 00h. Description: U2_TIMEOUT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7:0	U1_TIMEOUT. Read-write,Volatile. Reset: 00h. Description: U1_TIMEOUT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000458 (FCHUSBDCUSB31CNTR0::PORTSC_30_REGS1_PORTLI_30)

Read-only. Reset: 0000_0000h.

Port Link Info Register

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x000458; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000458; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:24 Reserved.

23:20 **TLC.** Read-only. Reset: 0h.**Description:** Tx Lane Count (TLC)

Currently this field supports value 0

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

19:16 **RLC.** Read-only. Reset: 0h.**Description:** Rx Lane Count (RLC)

Currently this field supports value 0

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

15:0 **LINK_ERROR_COUNT.** Read-only. Reset: 0000h.**Description:** LINK_ERROR_COUNT

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00045C (FCHUSBDCUSB31CNTR0::PORTSC_30_REGS1_PORTHLCPMC_30)

Read-only. Reset: 0000_0000h.

USB2 Port Hardware LPM Control Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x00045C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00045C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:0 Reserved.

USB[0...1]x000460 (FCHUSB0WCUSB31CNTR0::USBLEGSUP)

USBLEGSUP	
_instUSB0_aliasHOST; USB0x000460; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}	
_instUSB1_aliasHOST; USB1x000460; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}	
Bits	Description
31:25	Reserved.
24	HC_OS_OWNED. Read-write. Reset: 0. Description: HC_OS_OWNED SEMAPHORE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
23:17	Reserved.
16	HC_BIOS_OWNED. Read-write. Reset: 0. Description: HC_BIOS_OWNED SEMAPHORE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	NEXT_CAPABILITY_POINTER. Read-only. Reset: 04h. Description: NEXT_CAPABILITY_POINTER For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7:0	CAPABILITY_ID. Read-only. Reset: 01h. Description: CAPABILITY_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000464 (FCHUSBWCUSB31CNTR0::USBLEGCTLSTS)

USBLEGCTLSTS	
_instUSB0_aliasHOST; USB0x000464; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}	
_instUSB1_aliasHOST; USB1x000464; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}	
Bits	Description
31	SMI_ON_BAR. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: SMI_ON_BAR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
30	SMI_ON_PCI. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: SMI_ON_PCI COMMAND For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
29	SMI_ON_OS. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: SMI_ON_OS OWNERSHIP CHANGE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
28:21	Reserved.
20	SMI_ON_HOST. Read-only. Reset: 0. Description: SMI_ON_HOST SYSTEM ERROR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
19:17	Reserved.
16	SMI_ON_EVENT. Read-only. Reset: 0. Description: SMI_ON_EVENT INTERRUPT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15	SMI_ON_BAR_E. Read-write. Reset: 0. Description: SMI_ON_BAR ENABLE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
14	SMI_ON_PCI_E. Read-write. Reset: 0. Description: SMI_ON_PCI COMMAND ENABLE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
13	SMI_ON_OS_E. Read-write. Reset: 0. Description: SMI_ON_OS OWNERSHIP ENABLE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
12:5	Reserved.
4	SMI_ON_HOST_E. Read-write. Reset: 0.

	Description: SMI_ON_HOST SYSTEM ERROR ENABLE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3:1	Reserved.
0	USB_SMI_ENABLE. Read-write. Reset: 0. Description: USB_SMI_ENABLE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000470 (FCHUSBDCUSB31CNTR0::SUPTPRT2_DW0)

Read-only. Reset: 0200_0402h.	
SUPTPRT2_DW0	
_instUSB0_aliasHOST; USB0x000470; USB0={NBIFEPFNCFG:: BASE_ADDR_2 _dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG:: BASE_ADDR_1 _dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}	
_instUSB1_aliasHOST; USB1x000470; USB1={NBIFEPFNCFG:: BASE_ADDR_2 _dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG:: BASE_ADDR_1 _dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}	
Bits	Description
31:24	MAJOR_REVISION. Read-only. Reset: 02h. MAJOR_REVISION
23:16	MINOR_REVISION. Read-only. Reset: 00h. MINOR_REVISION
15:8	NEXT_CAPABILITY_POINTER. Read-only. Reset: 04h. NEXT_CAPABILITY_POINTER
7:0	CAPABILITY_ID. Read-only. Reset: 02h. Description: CAPABILITY_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000474 (FCHUSBDCUSB31CNTR0::SUPTPRT2_DW1)

Read-only. Reset: 2042_5355h.	
Register SUPTPRT2_DW1	
_instUSB0_aliasHOST; USB0x000474; USB0={NBIFEPFNCFG:: BASE_ADDR_2 _dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG:: BASE_ADDR_1 _dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}	
_instUSB1_aliasHOST; USB1x000474; USB1={NBIFEPFNCFG:: BASE_ADDR_2 _dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG:: BASE_ADDR_1 _dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}	
Bits	Description
31:0	NAME_STRING. Read-only. Reset: 2042_5355h. Description: NAME_STRING For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000478 (FCHUSBWCUSB31CNTR0::SUPTPRT2_DW2)

Reset: 0018_0201h.

xHCI Supported Protocol Capability_ Data Word 2

For a description of other register fields, see section 7.2 of the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x000478; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000478; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:28	PSIC. Read-only. Reset: 0h. PSIC
27:25	MHD. Read-only. Reset: 0h. Description: Hub Depth For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
24:21	Reserved.
20	BLC. Read-only. Reset: 1. Description: BESL LPM Capability. When this bit is set to: - 1: The ports described by this xHCI Supported Protocol Capability applies BESL timing to the BESL and BESLD fields of the PORTPMSC and PORTHLPMC registers. - 0: The ports described by this xHCI Supported Protocol Capability applies BESL timing to the BESL and BESLD fields of the PORTPMSC and PORTHLPMC registers.
19	HLC. Read-only. Reset: 1. Description: Compatible Port Offset. Compatible Port Count
18	IHI. Read-only. Reset: 0. IHI
17	HSO. Read-only. Reset: 0. HSO
16	L1C. Read-only. Reset: 0. Description: L1C For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	COMPATIBLE_PORT_COUNT. Read-only, Volatile. Reset: 02h. Description: COMPATIBLE_PORT_COUNT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7:0	COMPATIBLE_PORT_OFFSET. Read-only. Reset: 01h. Description: COMPATIBLE_PORT_OFFSET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00047C (FCHUSBDCUSB31CNTR0::SUPTPRT2_DW3)

Read-only.

Register SUPTPRT2_DW3

_instUSB0_aliasHOST; USB0x00047C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00047C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description

31:5 Reserved.

4:0 **PROTCL_SLT_TY**. Read-only. Reset: 00h. Protocol Slot Type**USB[0...1]x000480 (FCHUSBDCUSB31CNTR0::SUPT_30_REGS0_SUPTPRT3_DW0)**

Reset: 0310_0802h.

Register SUPTPRT3_DW0

_instUSB0_aliasHOST; USB0x000480; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000480; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description31:24 **MAJOR_REVISION**. Read-only. Reset: 03h.**Description:** MAJOR_REVISION

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

23:16 **MINOR_REVISION**. Read-only. Reset: 10h.**Description:** MINOR_REVISION

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

15:8 **NEXT_CAPABILITY_POINTER**. Read-only,Volatile. Reset: 08h.**Description:** NEXT_CAPABILITY_POINTER

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

7:0 **CAPABILITY_ID**. Read-only. Reset: 02h.**Description:** CAPABILITY_ID

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000484 (FCHUSBDCUSB31CNTR0::SUPT_30_REGS0_SUPTPRT3_DW1)

Read-only. Reset: 2042_5355h.

Register SUPTPRT3_DW1

_instUSB0_aliasHOST; USB0x000484; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000484; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description31:0 **NAME_STRING**. Read-only. Reset: 2042_5355h.**Description:** NAME_STRING

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x000488 (FCHUSBDWCUSB31CNTR0::SUPT_30_REGS0_SUPTPRT3_DW2)

Reset: 2000_0103h.

SUPTPRT3_DW2

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Due to side-effects this register field COMPATIBLE_PORT_OFFSET is not recommended for Bit-Bash testing.

_instUSB0_aliasHOST; USB0x000488; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000488; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:28	PSIC . Read-only. Reset: 2h. PSIC
27:16	Reserved.
15:8	COMPATIBLE_PORT_COUNT . Read-only, Volatile. Reset: 01h. Description: COMPATIBLE_PORT_COUNT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7:0	COMPATIBLE_PORT_OFFSET . Read-only, Volatile. Reset: 03h. Description: COMPATIBLE_PORT_OFFSET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00048C (FCHUSBDWCUSB31CNTR0::SUPT_30_REGS0_SUPTPRT3_DW3)

Read-only. Reset: 0000_0000h.

SUPTPRT3_DW3

_instUSB0_aliasHOST; USB0x00048C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00048C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:5	Reserved.
4:0	PROTCL_SLT_TY . Read-only. Reset: 00h. Protocol Slot Type

USB[0...1]x000490 (FCHUSBDWCUSB31CNTR0::SUPT_30_REGS0_SUPTPRT3_DW4)

Read-only. Reset: 0005_0134h.

SUPTPRT3_DW4

_instUSB0_aliasHOST; USB0x000490; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x000490; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	PSIM . Read-only. Reset: 0005h. PSIM
15:14	LP . Read-only. Reset: 0h. Link Protocol
13:9	Reserved.
8	PFD . Read-only. Reset: 1. PSI full-duplex
7:6	PLT . Read-only. Reset: 0h. PSI type
5:4	PSIE . Read-only. Reset: 3h. Protocol speed ID exponent
3:0	PSIV . Read-only. Reset: 4h. Protocol speed ID value

USB[0...1]x000494 (FCHUSBDCUSB31CNTR0::SUPT_30_REGS0_SUPTPRT3_DW5)

Reset: 000A_4135h.

SUPTPRT3_DW5

_instUSB0_aliasHOST; USB0x000494; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000494; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	PSIM. Read-only, Volatile. Reset: 000Ah. protocol speed mantissa
15:14	LP. Read-only. Reset: 1h. Link Protocol
13:9	Reserved.
8	PFD. Read-only. Reset: 1. PSI full-duplex
7:6	PLT. Read-only. Reset: 0h. PSI type
5:4	PSIE. Read-only. Reset: 3h. Protocol speed ID exponent
3:0	PSIV. Read-only. Reset: 5h. PSIV

USB[0...1]x000498 (FCHUSBDCUSB31CNTR0::SUPT_30_REGS0_SUPTPRT3_DW6)

Read-only. Reset: 0000_0000h.

SUPTPRT3_DW6

_instUSB0_aliasHOST; USB0x000498; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x000498; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	Reserved.

USB[0...1]x00049C (FCHUSBDCUSB31CNTR0::SUPT_30_REGS0_SUPTPRT3_DW7)

Read-only. Reset: 0000_0000h.

SUPTPRT3_DW7

_instUSB0_aliasHOST; USB0x00049C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00049C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	Reserved.

USB[0...1]x0004A0 (FCHUSBDCUSB31CNTR0::SUPT_30_REGS1_SUPTPRT3_DW0)

Reset: 0310_0802h.

Register SUPTPRT3_DW0

_instUSB0_aliasHOST; USB0x0004A0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004A0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	MAJOR_REVISION. Read-only. Reset: 03h. Description: MAJOR_REVISION For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
23:16	MINOR_REVISION. Read-only. Reset: 10h. Description: MINOR_REVISION For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	NEXT_CAPABILITY_POINTER. Read-only, Volatile. Reset: 08h. Description: NEXT_CAPABILITY_POINTER For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7:0	CAPABILITY_ID. Read-only. Reset: 02h. Description: CAPABILITY_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0004A4 (FCHUSBDCUSB31CNTR0::SUPT_30_REGS1_SUPTPRT3_DW1)

Read-only. Reset: 2042_5355h.

Register SUPTPRT3_DW1

_instUSB0_aliasHOST; USB0x0004A4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004A4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	NAME_STRING. Read-only. Reset: 2042_5355h. Description: NAME_STRING For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0004A8 (FCHUSBDCUSB31CNTR0::SUPT_30_REGS1_SUPTPRT3_DW2)

Reset: 2000_0103h.

SUPTPRT3_DW2

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Due to side-effects this register field COMPATIBLE_PORT_OFFSET is not recommended for Bit-Bash testing.

_instUSB0_aliasHOST; USB0x0004A8; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0004A8; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:28	PSIC . Read-only. Reset: 2h. PSIC
27:16	Reserved.
15:8	COMPATIBLE_PORT_COUNT . Read-only, Volatile. Reset: 01h. Description: COMPATIBLE_PORT_COUNT For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
7:0	COMPATIBLE_PORT_OFFSET . Read-only, Volatile. Reset: 03h. Description: COMPATIBLE_PORT_OFFSET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0004AC (FCHUSBDCUSB31CNTR0::SUPT_30_REGS1_SUPTPRT3_DW3)

Read-only. Reset: 0000_0000h.

SUPTPRT3_DW3

_instUSB0_aliasHOST; USB0x0004AC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0004AC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:5	Reserved.
4:0	PROTCL_SLT_TY . Read-only. Reset: 00h. Protocol Slot Type

USB[0...1]x0004B0 (FCHUSBDCUSB31CNTR0::SUPT_30_REGS1_SUPTPRT3_DW4)

Read-only. Reset: 0005_0134h.

SUPTPRT3_DW4

_instUSB0_aliasHOST; USB0x0004B0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0004B0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	PSIM . Read-only. Reset: 0005h. PSIM
15:14	LP . Read-only. Reset: 0h. Link Protocol
13:9	Reserved.
8	PFD . Read-only. Reset: 1. PSI full-duplex
7:6	PLT . Read-only. Reset: 0h. PSI type
5:4	PSIE . Read-only. Reset: 3h. Protocol speed ID exponent
3:0	PSIV . Read-only. Reset: 4h. Protocol speed ID value

USB[0...1]x0004B4 (FCHUSBDWCUSB31CNTR0::SUPT_30_REGS1_SUPTPRT3_DW5)

Reset: 000A_4135h.

SUPTPRT3_DW5

_instUSB0_aliasHOST; USB0x0004B4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004B4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	PSIM. Read-only, Volatile. Reset: 000Ah. protocol speed mantissa
15:14	LP. Read-only. Reset: 1h. Link Protocol
13:9	Reserved.
8	PFD. Read-only. Reset: 1. PSI full-duplex
7:6	PLT. Read-only. Reset: 0h. PSI type
5:4	PSIE. Read-only. Reset: 3h. Protocol speed ID exponent
3:0	PSIV. Read-only. Reset: 5h. PSIV

USB[0...1]x0004B8 (FCHUSBDWCUSB31CNTR0::SUPT_30_REGS1_SUPTPRT3_DW6)

Read-only. Reset: 0000_0000h.

SUPTPRT3_DW6

_instUSB0_aliasHOST; USB0x0004B8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004B8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	Reserved.

USB[0...1]x0004BC (FCHUSBDWCUSB31CNTR0::SUPT_30_REGS1_SUPTPRT3_DW7)

Read-only. Reset: 0000_0000h.

SUPTPRT3_DW7

_instUSB0_aliasHOST; USB0x0004BC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004BC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	Reserved.

USB[0...1]x0004C0 (FCHUSBDWCUSB31CNTR0::DCID)

Read-only.

DCID

_instUSB0_aliasHOST; USB0x0004C0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004C0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:21	Reserved.
20:16	DCERSTMAX. Read-only. Reset: 0Fh. DCERSTMAX
15:8	NEXT_CAPABILITY_POINTER. Read-only. Reset: 00h. NEXT_CAPABILITY_POINTER
7:0	CAPABILITY_ID. Read-only. Reset: 0Ah. CAPABILITY_ID

USB[0...1]x0004C4 (FCHUSBWDCUSB31CNTR0::DCDB)

Register DCDB

_instUSB0_aliasHOST; USB0x0004C4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004C4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	Reserved.
15:8	DBTARGET . Read-write,Reserved. Reset: 00h. DBTARGET
7:0	Reserved.

USB[0...1]x0004C8 (FCHUSBWDCUSB31CNTR0::DCERSTSZ)

DCERSTSZ

_instUSB0_aliasHOST; USB0x0004C8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004C8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	Reserved.
15:0	ERS_TABLE_SIZE . Read-write. Reset: 0000h. ERS_TABLE_SIZE

USB[0...1]x0004CC (FCHUSBWDCUSB31CNTR0::RSVD0)

Read-only. Reset: 0000_0000h.

Reserved-0

_instUSB0_aliasHOST; USB0x0004CC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004CC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	Reserved.

USB[0...1]x0004D0 (FCHUSBWDCUSB31CNTR0::DCERSTBA_LO)

Read-write.

DCERSTBA_LO

_instUSB0_aliasHOST; USB0x0004D0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004D0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:4	ERS_TABLE_BAR . Read-write. Reset: 000_0000h. ERS_TABLE_BAR
3:0	Reserved.

USB[0...1]x0004D4 (FCHUSBWDCUSB31CNTR0::DCERSTBA_HI)

Read-write. Reset: 0000_0000h.

Register DCERSTBA_HI

_instUSB0_aliasHOST; USB0x0004D4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004D4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERS_TABLE_BAR . Read-write. Reset: 0000_0000h. ERS_TABLE_BAR

USB[0...1]x0004D8 (FCHUSBDCUSB31CNTR0::DCERDP_LO)

Read-write.

DCERDP_LO

_instUSB0_aliasHOST; USB0x0004D8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004D8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:4	DEQUEUE_POINTER. Read-write. Reset: 000_0000h. DEQUEUE_POINTER
3	Reserved.
2:0	DESI. Read-write. Reset: 0h. DESI

USB[0...1]x0004DC (FCHUSBDCUSB31CNTR0::DCERDP_HI)

Read-write. Reset: 0000_0000h.

DCERDP_HI

_instUSB0_aliasHOST; USB0x0004DC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004DC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	DEQUEUE_POINTER. Read-write. Reset: 0000_0000h. DEQUEUE_POINTER

USB[0...1]x0004E0 (FCHUSBDCUSB31CNTR0::DCCTRL)

DCCTRL

_instUSB0_aliasHOST; USB0x0004E0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004E0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	DCE. Read-write. Reset: 0. DCE
30:24	DEVICE_ADDRESS. Read-only. Reset: 00h. DEVICE_ADDRESS
23:16	DEBUG_MAX_BURST_SIZE. Read-only. Reset: 0Fh. DEBUG_PROTOCOL
15:5	Reserved.
4	DRC. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. DRC
3	HIT. Read-write,Read,Write-1-only,Volatile. Reset: 0. HIT
2	HOT. Read-write,Read,Write-1-only,Volatile. Reset: 0. HOT
1	LSE. Read-write. Reset: 0. LSE
0	DCR. Read-only. Reset: 0. DCR

USB[0...1]x0004E4 (FCHUSBDCUSB31CNTR0::DCST)

Read-only.

DCST

_instUSB0_aliasHOST; USB0x0004E4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004E4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	DEBUG_PORT_NUMBER. Read-only. Reset: 00h. DEBUG_PORT_NUMBER
23:2	Reserved.
1	SBR. Read-only. Reset: 0. SBR
0	ER. Read-only. Reset: 0. ER

USB[0...1]x0004E8 (FCHUSBDCUSB31CNTR0::DCPORTSC)**Register DCPORTSC**

_instUSB0_aliasHOST; USB0x0004E8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004E8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	Reserved.
23	CEC. Read-write,Read,Write-1-to-clear, Volatile. Reset: 0. CEC
22	PLC. Read-write,Read,Write-1-to-clear, Volatile. Reset: 0. PLC
21	PRC. Read-write,Read,Write-1-to-clear, Volatile. Reset: 0. PRC
20:18	Reserved.
17	CSC. Read-write,Read,Write-1-to-clear, Volatile. Reset: 0. CSC
16:14	Reserved.
13:10	PORTSPEED. Read-only. Reset: 0h. PORTSPEED
9	Reserved.
8:5	PLS. Read-only. Reset: 5h. PLS
4	PR. Read-only. Reset: 0. PR
3:2	Reserved.
1	PED. Read-write, Volatile. Reset: 0. PED
0	CCS. Read-only. Reset: 0. CCS

USB[0...1]x0004EC (FCHUSBDCUSB31CNTR0::RSVD1)

Read-only. Reset: 0000_0000h.

Reserved-1

_instUSB0_aliasHOST; USB0x0004EC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004EC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	Reserved.

USB[0...1]x0004F0 (FCHUSBDCUSB31CNTR0::DCCP_LO)

Read-write.

DCCP_LO

_instUSB0_aliasHOST; USB0x0004F0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004F0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:4	DCCPR. Read-write. Reset: 000_0000h. DCCPR
3:0	Reserved.

USB[0...1]x0004F4 (FCHUSBDCUSB31CNTR0::DCCP_HI)

Read-write. Reset: 0000_0000h.

Register DCCP_HI

_instUSB0_aliasHOST; USB0x0004F4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0004F4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	DCCPR. Read-write. Reset: 0000_0000h. DCCPR

USB[0...1]x0004F8 (FCHUSBWDCUSB31CNTR0::DCDDI1)

Read-write.

Register DCDDI1

_instUSB0_aliasHOST; USB0x0004F8; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0004F8; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	VENDORID . Read-write. Reset: 0000h. VENDORID
15:8	Reserved.
7:0	DBCPROTOCOL . Read-write. Reset: 00h. DBCPROTOCOL

USB[0...1]x0004FC (FCHUSBWDCUSB31CNTR0::DCDDI2)

Read-write. Reset: 0000_0000h.

Register DCDDI2

_instUSB0_aliasHOST; USB0x0004FC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0004FC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DEVICEREV . Read-write. Reset: 0000h. DEVICEREV
15:0	PRODUCTID . Read-write. Reset: 0000h. PRODUCTID

USB[0...1]x001000 (FCHUSBWDCUSB31CNTR0::MFINDEX)

Read-only.

Microframe Index Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001000; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001000; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:14	Reserved.
13:0	MICROFRAME_INDEX . Read-only. Reset: 0000h. Description: MICROFRAME_INDEX For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001004 (FCHUSBWDCUSB31CNTR0::RsvdZ)

Read-only.

RsvdZ

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001004; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001004; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:0	Reserved.

USB[0...1]x001020 (FCHUSBWCUSB31CNTR0::IMAN_REGS0_IMAN)

Reset: 0000_0000h.

Interrupter Management Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001020; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001020; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	Reserved.
1	IE. Read-write. Reset: 0. Description: IE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	IP. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: IP Interrupt Pending For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001024 (FCHUSBWCUSB31CNTR0::IMAN_REGS0_IMOD)

Reset: 0000_0FA0h.

Interrupter Moderation Register

The software may use this register to pace (or even out) the delivery of interrupts to the host CPU.

This register provides an assured inter-interrupt delay between interrupts asserted by the xHC, regardless of USB traffic conditions.

To independently validate configuration settings, software may use the following algorithm to convert the inter-interrupt Interval value to the common 'interrupts/sec' performance metric.

_instUSB0_aliasHOST; USB0x001024; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001024; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	IMODC. Read-write, Volatile. Reset: 0000h. Description: Interrupt Moderation Counter (IMODC) - RW. Default = undefined. Down counter. Loaded with the IMODI value whenever IP is cleared to '0', counts down to '0', and stops. The associated interrupt is signaled whenever this counter is '0', the Event Ring is not empty, the IE and IP flags = '1', and EHB = '0'. This counter may be directly written by software at any time to alter the interrupt rate.
15:0	IMODI. Read-write. Reset: 0FA0h. Description: Interrupt Moderation Interval (IMODI) - RW. Default = '4000' (~1ms). Minimum inter-interrupt interval. The interval is specified in 250ns increments. A value of '0' disables interrupt throttling logic and interrupts is generated immediately if IP = '0', EHB = '0', and the Event Ring is not empty.

USB[0...1]x001028 (FCHUSBWCUSB31CNTR0::IMAN_REGS0_ERSTSZ)

Reset: 0000_0000h.

Event Ring Segment Table Size Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001028; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001028; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	Reserved.
15:0	ERS_TABLE_SIZE. Read-write. Reset: 0000h. Description: ERS_TABLE_SIZE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00102C (FCHUSBDCUSB31CNTR0::IMAN_REGS0_RsvdP)

Read-only. Reset: 0000_0000h.

RsvdP

_instUSB0_aliasHOST; USB0x00102C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00102C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:0 Reserved.

USB[0...1]x001030 (FCHUSBDCUSB31CNTR0::IMAN_REGS0_ERSTBA_LO)

Reset: 0000_0000h.

ERSTBA_LO

_instUSB0_aliasHOST; USB0x001030; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001030; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:6 ERS_TABLE_BAR. Read-write. Reset: 000_0000h. ERS_TABLE_BAR

5:0 Reserved.

USB[0...1]x001034 (FCHUSBDCUSB31CNTR0::IMAN_REGS0_ERSTBA_HI)

Read-write. Reset: 0000_0000h.

ERSTBA_HI

_instUSB0_aliasHOST; USB0x001034; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001034; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:0 ERS_TABLE_BAR. Read-write. Reset: 0000_0000h.

Description: ERS_TABLE_BAR

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001038 (FCHUSBWCUSB31CNTR0::IMAN_REGS0_ERDP_LO)

Reset: 0000_0000h.

ERDP_LO

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001038; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001038; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:4	ERD_PNTR. Read-write. Reset: 000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3	EHB. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: HC OS Owned SemaphoreERS_TABLE_SIZEHC BIOS Owned Semaphore USB SMI Enable SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count - HC OS Owned Semaphore - HC BIOS Owned Semaphore - USB SMI Enable - SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count
2:0	DESI. Read-write. Reset: 0h. Description: DESI - For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00103C (FCHUSBDCUSB31CNTR0::IMAN_REGS0_ERDP_HI)

Read-write. Reset: 0000_0000h.

ERDP_HI

_instUSB0_aliasHOST; USB0x00103C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00103C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERD_PNTR. Read-write. Reset: 0000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001040 (FCHUSBDCUSB31CNTR0::IMAN_REGS1_IMAN)

Reset: 0000_0000h.

Interrupter Management Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001040; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001040; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	Reserved.
1	IE. Read-write. Reset: 0. Description: IE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	IP. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: IP Interrupt Pending For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001044 (FCHUSBDCUSB31CNTR0::IMAN_REGS1_IMOD)

Reset: 0000_0FA0h.

Interrupter Moderation Register

The software may use this register to pace (or even out) the delivery of interrupts to the host CPU.

This register provides an assured inter-interrupt delay between interrupts asserted by the xHC, regardless of USB traffic conditions.

To independently validate configuration settings, software may use the following algorithm to convert the inter-interrupt Interval value to the common 'interrupts/sec' performance metric.

_instUSB0_aliasHOST; USB0x001044; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001044; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	IMODC. Read-write, Volatile. Reset: 0000h. Description: Interrupt Moderation Counter (IMODC) - RW. Default = undefined. Down counter. Loaded with the IMODI value whenever IP is cleared to '0', counts down to '0', and stops. The associated interrupt is signaled whenever this counter is '0', the Event Ring is not empty, the IE and IP flags = '1', and EHB = '0'. This counter may be directly written by software at any time to alter the interrupt rate.
15:0	IMODI. Read-write. Reset: 0FA0h. Description: Interrupt Moderation Interval (IMODI) - RW. Default = '4000' (~1ms). Minimum inter-interrupt interval. The interval is specified in 250ns increments. A value of '0' disables interrupt throttling logic and interrupts is generated immediately if IP = '0', EHB = '0', and the Event Ring is not empty.

USB[0...1]x001048 (FCHUSBDCUSB31CNTR0::IMAN_REGS1_ERSTSZ)

Reset: 0000_0000h.

Event Ring Segment Table Size Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001048; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001048; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	Reserved.
15:0	ERS_TABLE_SIZE. Read-write. Reset: 0000h. Description: ERS_TABLE_SIZE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00104C (FCHUSBDWCUSB31CNTR0::IMAN_REGS1_RsvdP)

Read-only. Reset: 0000_0000h.

RsvdP

_instUSB0_aliasHOST; USB0x00104C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00104C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	Reserved.

USB[0...1]x001050 (FCHUSBDWCUSB31CNTR0::IMAN_REGS1_ERSTBA_LO)

Reset: 0000_0000h.

ERSTBA_LO

_instUSB0_aliasHOST; USB0x001050; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001050; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:6	ERS_TABLE_BAR. Read-write. Reset: 000_0000h. ERS_TABLE_BAR
5:0	Reserved.

USB[0...1]x001054 (FCHUSBDWCUSB31CNTR0::IMAN_REGS1_ERSTBA_HI)

Read-write. Reset: 0000_0000h.

ERSTBA_HI

_instUSB0_aliasHOST; USB0x001054; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001054; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERS_TABLE_BAR. Read-write. Reset: 0000_0000h.
	Description: ERS_TABLE_BAR
	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001058 (FCHUSBWCUSB31CNTR0::IMAN_REGS1_ERDP_LO)

Reset: 0000_0000h.

ERDP_LO

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001058; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001058; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:4	ERD_PNTR. Read-write. Reset: 000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3	EHB. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: HC OS Owned SemaphoreERS_TABLE_SIZEHC BIOS Owned Semaphore USB SMI Enable SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count - HC OS Owned Semaphore - HC BIOS Owned Semaphore - USB SMI Enable - SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count
2:0	DESI. Read-write. Reset: 0h. Description: DESI - For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00105C (FCHUSBDCUSB31CNTR0::IMAN_REGS1_ERDP_HI)

Read-write. Reset: 0000_0000h.

ERDP_HI

_instUSB0_aliasHOST; USB0x00105C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00105C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERD_PNTR. Read-write. Reset: 0000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001060 (FCHUSBDCUSB31CNTR0::IMAN_REGS2_IMAN)

Reset: 0000_0000h.

Interrupter Management Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001060; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001060; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	Reserved.
1	IE. Read-write. Reset: 0. Description: IE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	IP. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: IP Interrupt Pending For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001064 (FCHUSBWCUSB31CNTR0::IMAN_REGS2_IMOD)

Reset: 0000_0FA0h.

Interrupter Moderation Register

The software may use this register to pace (or even out) the delivery of interrupts to the host CPU.

This register provides an assured inter-interrupt delay between interrupts asserted by the xHC, regardless of USB traffic conditions.

To independently validate configuration settings, software may use the following algorithm to convert the inter-interrupt Interval value to the common 'interrupts/sec' performance metric.

_instUSB0_aliasHOST; USB0x001064; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001064; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	IMODC. Read-write, Volatile. Reset: 0000h. Description: Interrupt Moderation Counter (IMODC) - RW. Default = undefined. Down counter. Loaded with the IMODI value whenever IP is cleared to '0', counts down to '0', and stops. The associated interrupt is signaled whenever this counter is '0', the Event Ring is not empty, the IE and IP flags = '1', and EHB = '0'. This counter may be directly written by software at any time to alter the interrupt rate.
15:0	IMODI. Read-write. Reset: 0FA0h. Description: Interrupt Moderation Interval (IMODI) - RW. Default = '4000' (~1ms). Minimum inter-interrupt interval. The interval is specified in 250ns increments. A value of '0' disables interrupt throttling logic and interrupts is generated immediately if IP = '0', EHB = '0', and the Event Ring is not empty.

USB[0...1]x001068 (FCHUSBWCUSB31CNTR0::IMAN_REGS2_ERSTSZ)

Reset: 0000_0000h.

Event Ring Segment Table Size Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001068; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001068; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	Reserved.
15:0	ERS_TABLE_SIZE. Read-write. Reset: 0000h. Description: ERS_TABLE_SIZE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00106C (FCHUSBDCUSB31CNTR0::IMAN_REGS2_RsvdP)

Read-only. Reset: 0000_0000h.

RsvdP

_instUSB0_aliasHOST; USB0x00106C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00106C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:0 Reserved.

USB[0...1]x001070 (FCHUSBDCUSB31CNTR0::IMAN_REGS2_ERSTBA_LO)

Reset: 0000_0000h.

ERSTBA_LO

_instUSB0_aliasHOST; USB0x001070; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001070; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:6 ERS_TABLE_BAR. Read-write. Reset: 000_0000h. ERS_TABLE_BAR

5:0 Reserved.

USB[0...1]x001074 (FCHUSBDCUSB31CNTR0::IMAN_REGS2_ERSTBA_HI)

Read-write. Reset: 0000_0000h.

ERSTBA_HI

_instUSB0_aliasHOST; USB0x001074; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001074; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:0 ERS_TABLE_BAR. Read-write. Reset: 0000_0000h.

Description: ERS_TABLE_BAR

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001078 (FCHUSBWCUSB31CNTR0::IMAN_REGS2_ERDP_LO)

Reset: 0000_0000h.

ERDP_LO

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001078; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001078; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:4	ERD_PNTR. Read-write. Reset: 000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3	EHB. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: HC OS Owned SemaphoreERS_TABLE_SIZEHC BIOS Owned Semaphore USB SMI Enable SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count - HC OS Owned Semaphore - HC BIOS Owned Semaphore - USB SMI Enable - SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count
2:0	DESI. Read-write. Reset: 0h. Description: DESI - For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00107C (FCHUSBWCUSB31CNTR0::IMAN_REGS2_ERDP_HI)

Read-write. Reset: 0000_0000h.

ERDP_HI

_instUSB0_aliasHOST; USB0x00107C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00107C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERD_PNTR. Read-write. Reset: 0000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001080 (FCHUSBWCUSB31CNTR0::IMAN_REGS3_IMAN)

Reset: 0000_0000h.

Interrupter Management Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001080; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001080; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	Reserved.
1	IE. Read-write. Reset: 0. Description: IE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	IP. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: IP Interrupt Pending For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001084 (FCHUSBWCUSB31CNTR0::IMAN_REGS3_IMOD)

Reset: 0000_0FA0h.

Interrupter Moderation Register

The software may use this register to pace (or even out) the delivery of interrupts to the host CPU.

This register provides an assured inter-interrupt delay between interrupts asserted by the xHC, regardless of USB traffic conditions.

To independently validate configuration settings, software may use the following algorithm to convert the inter-interrupt Interval value to the common 'interrupts/sec' performance metric.

_instUSB0_aliasHOST; USB0x001084; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001084; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	IMODC. Read-write, Volatile. Reset: 0000h. Description: Interrupt Moderation Counter (IMODC) - RW. Default = undefined. Down counter. Loaded with the IMODI value whenever IP is cleared to '0', counts down to '0', and stops. The associated interrupt is signaled whenever this counter is '0', the Event Ring is not empty, the IE and IP flags = '1', and EHB = '0'. This counter may be directly written by software at any time to alter the interrupt rate.
15:0	IMODI. Read-write. Reset: 0FA0h. Description: Interrupt Moderation Interval (IMODI) - RW. Default = '4000' (~1ms). Minimum inter-interrupt interval. The interval is specified in 250ns increments. A value of '0' disables interrupt throttling logic and interrupts is generated immediately if IP = '0', EHB = '0', and the Event Ring is not empty.

USB[0...1]x001088 (FCHUSBWCUSB31CNTR0::IMAN_REGS3_ERSTSZ)

Reset: 0000_0000h.

Event Ring Segment Table Size Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001088; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001088; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	Reserved.
15:0	ERS_TABLE_SIZE. Read-write. Reset: 0000h. Description: ERS_TABLE_SIZE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00108C (FCHUSBDCUSB31CNTR0::IMAN_REGS3_RsvdP)

Read-only. Reset: 0000_0000h.

RsvdP

_instUSB0_aliasHOST; USB0x00108C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00108C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:0 Reserved.

USB[0...1]x001090 (FCHUSBDCUSB31CNTR0::IMAN_REGS3_ERSTBA_LO)

Reset: 0000_0000h.

ERSTBA_LO

_instUSB0_aliasHOST; USB0x001090; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001090; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:6 ERS_TABLE_BAR. Read-write. Reset: 000_0000h. ERS_TABLE_BAR

5:0 Reserved.

USB[0...1]x001094 (FCHUSBDCUSB31CNTR0::IMAN_REGS3_ERSTBA_HI)

Read-write. Reset: 0000_0000h.

ERSTBA_HI

_instUSB0_aliasHOST; USB0x001094; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001094; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:0 ERS_TABLE_BAR. Read-write. Reset: 0000_0000h.

Description: ERS_TABLE_BAR

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001098 (FCHUSBWCUSB31CNTR0::IMAN_REGS3_ERDP_LO)

Reset: 0000_0000h.

ERDP_LO

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001098; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001098; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:4	ERD_PNTR. Read-write. Reset: 000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3	EHB. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: HC OS Owned SemaphoreERS_TABLE_SIZEHC BIOS Owned Semaphore USB SMI Enable SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count - HC OS Owned Semaphore - HC BIOS Owned Semaphore - USB SMI Enable - SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count
2:0	DESI. Read-write. Reset: 0h. Description: DESI - For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00109C (FCHUSBWCUSB31CNTR0::IMAN_REGS3_ERDP_HI)

Read-write. Reset: 0000_0000h.

ERDP_HI

_instUSB0_aliasHOST; USB0x00109C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00109C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERD_PNTR. Read-write. Reset: 0000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010A0 (FCHUSBWCUSB31CNTR0::IMAN_REGS4_IMAN)

Reset: 0000_0000h.

Interrupter Management Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x0010A0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010A0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	Reserved.
1	IE. Read-write. Reset: 0. Description: IE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	IP. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: IP Interrupt Pending For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010A4 (FCHUSBWCUSB31CNTR0::IMAN_REGS4_IMOD)

Reset: 0000_0FA0h.

Interrupter Moderation Register

The software may use this register to pace (or even out) the delivery of interrupts to the host CPU.

This register provides an assured inter-interrupt delay between interrupts asserted by the xHC, regardless of USB traffic conditions.

To independently validate configuration settings, software may use the following algorithm to convert the inter-interrupt Interval value to the common 'interrupts/sec' performance metric.

_instUSB0_aliasHOST; USB0x0010A4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010A4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	IMODC. Read-write, Volatile. Reset: 0000h. Description: Interrupt Moderation Counter (IMODC) - RW. Default = undefined. Down counter. Loaded with the IMODI value whenever IP is cleared to '0', counts down to '0', and stops. The associated interrupt is signaled whenever this counter is '0', the Event Ring is not empty, the IE and IP flags = '1', and EHB = '0'. This counter may be directly written by software at any time to alter the interrupt rate.
15:0	IMODI. Read-write. Reset: 0FA0h. Description: Interrupt Moderation Interval (IMODI) - RW. Default = '4000' (~1ms). Minimum inter-interrupt interval. The interval is specified in 250ns increments. A value of '0' disables interrupt throttling logic and interrupts is generated immediately if IP = '0', EHB = '0', and the Event Ring is not empty.

USB[0...1]x0010A8 (FCHUSBWCUSB31CNTR0::IMAN_REGS4_ERSTSZ)

Reset: 0000_0000h.

Event Ring Segment Table Size Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x0010A8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010A8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	Reserved.
15:0	ERS_TABLE_SIZE. Read-write. Reset: 0000h. Description: ERS_TABLE_SIZE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010AC (FCHUSBDCUSB31CNTR0::IMAN_REGS4_RsvdP)

Read-only. Reset: 0000_0000h.

RsvdP

_instUSB0_aliasHOST; USB0x0010AC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010AC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
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31:0	Reserved.
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USB[0...1]x0010B0 (FCHUSBDCUSB31CNTR0::IMAN_REGS4_ERSTBA_LO)

Reset: 0000_0000h.

ERSTBA_LO

_instUSB0_aliasHOST; USB0x0010B0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010B0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
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31:6	ERS_TABLE_BAR. Read-write. Reset: 000_0000h. ERS_TABLE_BAR
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5:0	Reserved.
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USB[0...1]x0010B4 (FCHUSBDCUSB31CNTR0::IMAN_REGS4_ERSTBA_HI)

Read-write. Reset: 0000_0000h.

ERSTBA_HI

_instUSB0_aliasHOST; USB0x0010B4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010B4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
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31:0	ERS_TABLE_BAR. Read-write. Reset: 0000_0000h.
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	Description: ERS_TABLE_BAR
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	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
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USB[0...1]x0010B8 (FCHUSBWCUSB31CNTR0::IMAN_REGS4_ERDP_LO)

Reset: 0000_0000h.

ERDP_LO

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x0010B8; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0010B8; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:4	ERD_PNTR. Read-write. Reset: 000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3	EHB. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: HC OS Owned SemaphoreERS_TABLE_SIZEHC BIOS Owned Semaphore USB SMI Enable SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count - HC OS Owned Semaphore - HC BIOS Owned Semaphore - USB SMI Enable - SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count
2:0	DESI. Read-write. Reset: 0h. Description: DESI - For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010BC (FCHUSBDCUSB31CNTR0::IMAN_REGS4_ERDP_HI)

Read-write. Reset: 0000_0000h.

ERDP_HI

_instUSB0_aliasHOST; USB0x0010BC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010BC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERD_PNTR. Read-write. Reset: 0000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010C0 (FCHUSBDCUSB31CNTR0::IMAN_REGS5_IMAN)

Reset: 0000_0000h.

Interrupter Management Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x0010C0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010C0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	Reserved.
1	IE. Read-write. Reset: 0. Description: IE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	IP. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: IP Interrupt Pending For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010C4 (FCHUSBWCUSB31CNTR0::IMAN_REGS5_IMOD)

Reset: 0000_0FA0h.

Interrupter Moderation Register

The software may use this register to pace (or even out) the delivery of interrupts to the host CPU.

This register provides an assured inter-interrupt delay between interrupts asserted by the xHC, regardless of USB traffic conditions.

To independently validate configuration settings, software may use the following algorithm to convert the inter-interrupt Interval value to the common 'interrupts/sec' performance metric.

_instUSB0_aliasHOST; USB0x0010C4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010C4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	IMODC. Read-write, Volatile. Reset: 0000h. Description: Interrupt Moderation Counter (IMODC) - RW. Default = undefined. Down counter. Loaded with the IMODI value whenever IP is cleared to '0', counts down to '0', and stops. The associated interrupt is signaled whenever this counter is '0', the Event Ring is not empty, the IE and IP flags = '1', and EHB = '0'. This counter may be directly written by software at any time to alter the interrupt rate.
15:0	IMODI. Read-write. Reset: 0FA0h. Description: Interrupt Moderation Interval (IMODI) - RW. Default = '4000' (~1ms). Minimum inter-interrupt interval. The interval is specified in 250ns increments. A value of '0' disables interrupt throttling logic and interrupts is generated immediately if IP = '0', EHB = '0', and the Event Ring is not empty.

USB[0...1]x0010C8 (FCHUSBWCUSB31CNTR0::IMAN_REGS5_ERSTSZ)

Reset: 0000_0000h.

Event Ring Segment Table Size Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x0010C8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010C8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	Reserved.
15:0	ERS_TABLE_SIZE. Read-write. Reset: 0000h. Description: ERS_TABLE_SIZE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010CC (FCHUSBDCUSB31CNTR0::IMAN_REGS5_RsvdP)

Read-only. Reset: 0000_0000h.

RsvdP

_instUSB0_aliasHOST; USB0x0010CC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010CC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
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31:0	Reserved.
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USB[0...1]x0010D0 (FCHUSBDCUSB31CNTR0::IMAN_REGS5_ERSTBA_LO)

Reset: 0000_0000h.

ERSTBA_LO

_instUSB0_aliasHOST; USB0x0010D0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010D0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
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31:6	ERS_TABLE_BAR. Read-write. Reset: 000_0000h. ERS_TABLE_BAR
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5:0	Reserved.
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USB[0...1]x0010D4 (FCHUSBDCUSB31CNTR0::IMAN_REGS5_ERSTBA_HI)

Read-write. Reset: 0000_0000h.

ERSTBA_HI

_instUSB0_aliasHOST; USB0x0010D4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010D4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
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31:0	ERS_TABLE_BAR. Read-write. Reset: 0000_0000h.
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	Description: ERS_TABLE_BAR
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	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
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USB[0...1]x0010D8 (FCHUSBWCUSB31CNTR0::IMAN_REGS5_ERDP_LO)

Reset: 0000_0000h.

ERDP_LO

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x0010D8; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0010D8; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:4	ERD_PNTR. Read-write. Reset: 000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3	EHB. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: HC OS Owned SemaphoreERS_TABLE_SIZEHC BIOS Owned Semaphore USB SMI Enable SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count - HC OS Owned Semaphore - HC BIOS Owned Semaphore - USB SMI Enable - SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count
2:0	DESI. Read-write. Reset: 0h. Description: DESI - For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010DC (FCHUSBDCUSB31CNTR0::IMAN_REGS5_ERDP_HI)

Read-write. Reset: 0000_0000h.

ERDP_HI

_instUSB0_aliasHOST; USB0x0010DC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010DC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERD_PNTR. Read-write. Reset: 0000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010E0 (FCHUSBDCUSB31CNTR0::IMAN_REGS6_IMAN)

Reset: 0000_0000h.

Interrupter Management Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x0010E0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010E0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	Reserved.
1	IE. Read-write. Reset: 0. Description: IE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	IP. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: IP Interrupt Pending For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010E4 (FCHUSBWCUSB31CNTR0::IMAN_REGS6_IMOD)

Reset: 0000_0FA0h.

Interrupter Moderation Register

The software may use this register to pace (or even out) the delivery of interrupts to the host CPU.

This register provides an assured inter-interrupt delay between interrupts asserted by the xHC, regardless of USB traffic conditions.

To independently validate configuration settings, software may use the following algorithm to convert the inter-interrupt Interval value to the common 'interrupts/sec' performance metric.

_instUSB0_aliasHOST; USB0x0010E4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010E4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	IMODC. Read-write, Volatile. Reset: 0000h. Description: Interrupt Moderation Counter (IMODC) - RW. Default = undefined. Down counter. Loaded with the IMODI value whenever IP is cleared to '0', counts down to '0', and stops. The associated interrupt is signaled whenever this counter is '0', the Event Ring is not empty, the IE and IP flags = '1', and EHB = '0'. This counter may be directly written by software at any time to alter the interrupt rate.
15:0	IMODI. Read-write. Reset: 0FA0h. Description: Interrupt Moderation Interval (IMODI) - RW. Default = '4000' (~1ms). Minimum inter-interrupt interval. The interval is specified in 250ns increments. A value of '0' disables interrupt throttling logic and interrupts is generated immediately if IP = '0', EHB = '0', and the Event Ring is not empty.

USB[0...1]x0010E8 (FCHUSBWCUSB31CNTR0::IMAN_REGS6_ERSTSZ)

Reset: 0000_0000h.

Event Ring Segment Table Size Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x0010E8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010E8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	Reserved.
15:0	ERS_TABLE_SIZE. Read-write. Reset: 0000h. Description: ERS_TABLE_SIZE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010EC (FCHUSBWDCUSB31CNTR0::IMAN_REGS6_RsvdP)

Read-only. Reset: 0000_0000h.

RsvdP

_instUSB0_aliasHOST; USB0x0010EC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010EC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	Reserved.

USB[0...1]x0010F0 (FCHUSBWDCUSB31CNTR0::IMAN_REGS6_ERSTBA_LO)

Reset: 0000_0000h.

ERSTBA_LO

_instUSB0_aliasHOST; USB0x0010F0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010F0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:6	ERS_TABLE_BAR. Read-write. Reset: 000_0000h. ERS_TABLE_BAR
5:0	Reserved.

USB[0...1]x0010F4 (FCHUSBWDCUSB31CNTR0::IMAN_REGS6_ERSTBA_HI)

Read-write. Reset: 0000_0000h.

ERSTBA_HI

_instUSB0_aliasHOST; USB0x0010F4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010F4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERS_TABLE_BAR. Read-write. Reset: 0000_0000h.
	Description: ERS_TABLE_BAR
	For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010F8 (FCHUSBWCUSB31CNTR0::IMAN_REGS6_ERDP_LO)

Reset: 0000_0000h.

ERDP_LO

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x0010F8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010F8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:4	ERD_PNTR. Read-write. Reset: 000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3	EHB. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: HC OS Owned SemaphoreERS_TABLE_SIZEHC BIOS Owned Semaphore USB SMI Enable SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count - HC OS Owned Semaphore - HC BIOS Owned Semaphore - USB SMI Enable - SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count
2:0	DESI. Read-write. Reset: 0h. Description: DESI - For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0010FC (FCHUSBDCUSB31CNTR0::IMAN_REGS6_ERDP_HI)

Read-write. Reset: 0000_0000h.

ERDP_HI

_instUSB0_aliasHOST; USB0x0010FC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0010FC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERD_PNTR. Read-write. Reset: 0000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001100 (FCHUSBDCUSB31CNTR0::IMAN_REGS7_IMAN)

Reset: 0000_0000h.

Interrupter Management Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001100; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001100; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	Reserved.
1	IE. Read-write. Reset: 0. Description: IE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
0	IP. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: IP Interrupt Pending For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001104 (FCHUSBDCUSB31CNTR0::IMAN_REGS7_IMOD)

Reset: 0000_0FA0h.

Interrupter Moderation Register

The software may use this register to pace (or even out) the delivery of interrupts to the host CPU.

This register provides an assured inter-interrupt delay between interrupts asserted by the xHC, regardless of USB traffic conditions.

To independently validate configuration settings, software may use the following algorithm to convert the inter-interrupt Interval value to the common 'interrupts/sec' performance metric.

_instUSB0_aliasHOST; USB0x001104; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001104; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	IMODC. Read-write, Volatile. Reset: 0000h. Description: Interrupt Moderation Counter (IMODC) - RW. Default = undefined. Down counter. Loaded with the IMODI value whenever IP is cleared to '0', counts down to '0', and stops. The associated interrupt is signaled whenever this counter is '0', the Event Ring is not empty, the IE and IP flags = '1', and EHB = '0'. This counter may be directly written by software at any time to alter the interrupt rate.
15:0	IMODI. Read-write. Reset: 0FA0h. Description: Interrupt Moderation Interval (IMODI) - RW. Default = '4000' (~1ms). Minimum inter-interrupt interval. The interval is specified in 250ns increments. A value of '0' disables interrupt throttling logic and interrupts is generated immediately if IP = '0', EHB = '0', and the Event Ring is not empty.

USB[0...1]x001108 (FCHUSBDCUSB31CNTR0::IMAN_REGS7_ERSTSZ)

Reset: 0000_0000h.

Event Ring Segment Table Size Register Bit Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001108; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001108; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	Reserved.
15:0	ERS_TABLE_SIZE. Read-write. Reset: 0000h. Description: ERS_TABLE_SIZE For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00110C (FCHUSBDCUSB31CNTR0::IMAN_REGS7_RsvdP)

Read-only. Reset: 0000_0000h.

RsvdP

_instUSB0_aliasHOST; USB0x00110C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00110C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:0 Reserved.

USB[0...1]x001110 (FCHUSBDCUSB31CNTR0::IMAN_REGS7_ERSTBA_LO)

Reset: 0000_0000h.

ERSTBA_LO

_instUSB0_aliasHOST; USB0x001110; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001110; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description31:6 **ERS_TABLE_BAR**. Read-write. Reset: 000_0000h. ERS_TABLE_BAR

5:0 Reserved.

USB[0...1]x001114 (FCHUSBDCUSB31CNTR0::IMAN_REGS7_ERSTBA_HI)

Read-write. Reset: 0000_0000h.

ERSTBA_HI

_instUSB0_aliasHOST; USB0x001114; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x001114; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description31:0 **ERS_TABLE_BAR**. Read-write. Reset: 0000_0000h.**Description:** ERS_TABLE_BAR

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x001118 (FCHUSBDCUSB31CNTR0::IMAN_REGS7_ERDP_LO)

Reset: 0000_0000h.

ERDP_LO

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

_instUSB0_aliasHOST; USB0x001118; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x001118; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:4	ERD_PNTR. Read-write. Reset: 000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
3	EHB. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: HC OS Owned SemaphoreERS_TABLE_SIZEHC BIOS Owned Semaphore USB SMI Enable SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count - HC OS Owned Semaphore - HC BIOS Owned Semaphore - USB SMI Enable - SMI on Host System Error Enable - SMI on OS Ownership Enable - SMI on PCI Command Enable - SMI on BAR Enable - SMI on Event Interrupt - SMI on Host System Error - SMI on OS Ownership Change - SMI on PCI Command - SMI on BAR - Compatible Port Count
2:0	DESI. Read-write. Reset: 0h. Description: DESI - For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00111C (FCHUSBWCUSB31CNTR0::IMAN_REGS7_ERDP_HI)

Read-write. Reset: 0000_0000h.

ERDP_HI

_instUSB0_aliasHOST; USB0x00111C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00111C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ERD_PNTR. Read-write. Reset: 0000_0000h. Description: ERD_PNTR For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002000 (FCHUSBWCUSB31CNTR0::DB_REGS0_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002000; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x002000; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002004 (FCHUSBWCUSB31CNTR0::DB_REGS1_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002004; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002004; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002008 (FCHUSBWCUSB31CNTR0::DB_REGS2_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002008; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002008; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00200C (FCHUSBWCUSB31CNTR0::DB_REGS3_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x00200C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00200C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002010 (FCHUSBWCUSB31CNTR0::DB_REGS4_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002010; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002010; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002014 (FCHUSBWCUSB31CNTR0::DB_REGS5_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002014; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002014; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002018 (FCHUSBWCUSB31CNTR0::DB_REGS6_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002018; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002018; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00201C (FCHUSBWCUSB31CNTR0::DB_REGS7_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x00201C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00201C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002020 (FCHUSBWCUSB31CNTR0::DB_REGS8_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002020; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002020; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002024 (FCHUSBDCUSB31CNTR0::DB_REGS9_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002024; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002024; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002028 (FCHUSBDCUSB31CNTR0::DB_REGS10_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002028; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002028; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00202C (FCHUSBWCUSB31CNTR0::DB_REGS11_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x00202C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00202C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002030 (FCHUSBWCUSB31CNTR0::DB_REGS12_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002030; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x002030; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002034 (FCHUSBDCUSB31CNTR0::DB_REGS13_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002034; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002034; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002038 (FCHUSBDCUSB31CNTR0::DB_REGS14_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002038; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002038; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00203C (FCHUSBWCUSB31CNTR0::DB_REGS15_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x00203C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00203C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002040 (FCHUSBWCUSB31CNTR0::DB_REGS16_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002040; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002040; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002044 (FCHUSBWCUSB31CNTR0::DB_REGS17_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002044; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002044; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002048 (FCHUSBWCUSB31CNTR0::DB_REGS18_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002048; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002048; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00204C (FCHUSBWCUSB31CNTR0::DB_REGS19_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x00204C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00204C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002050 (FCHUSBWCUSB31CNTR0::DB_REGS20_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002050; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002050; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002054 (FCHUSBWCUSB31CNTR0::DB_REGS21_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002054; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002054; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002058 (FCHUSBWCUSB31CNTR0::DB_REGS22_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002058; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002058; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00205C (FCHUSBWCUSB31CNTR0::DB_REGS23_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x00205C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00205C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002060 (FCHUSBWCUSB31CNTR0::DB_REGS24_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002060; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002060; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002064 (FCHUSBWCUSB31CNTR0::DB_REGS25_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002064; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x002064; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002068 (FCHUSBWCUSB31CNTR0::DB_REGS26_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002068; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x002068; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00206C (FCHUSBWCUSB31CNTR0::DB_REGS27_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x00206C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00206C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002070 (FCHUSBWCUSB31CNTR0::DB_REGS28_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002070; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x002070; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002074 (FCHUSBWCUSB31CNTR0::DB_REGS29_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002074; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x002074; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002078 (FCHUSBWCUSB31CNTR0::DB_REGS30_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002078; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x002078; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00207C (FCHUSBWCUSB31CNTR0::DB_REGS31_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x00207C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00207C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002080 (FCHUSBWCUSB31CNTR0::DB_REGS32_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002080; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x002080; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002084 (FCHUSBWCUSB31CNTR0::DB_REGS33_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002084; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002084; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002088 (FCHUSBWCUSB31CNTR0::DB_REGS34_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002088; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002088; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00208C (FCHUSBWCUSB31CNTR0::DB_REGS35_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x00208C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00208C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002090 (FCHUSBWCUSB31CNTR0::DB_REGS36_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002090; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x002090; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002094 (FCHUSBWCUSB31CNTR0::DB_REGS37_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002094; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002094; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002098 (FCHUSBWCUSB31CNTR0::DB_REGS38_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002098; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002098; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00209C (FCHUSBDWCUSB31CNTR0::DB_REGS39_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x00209C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00209C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020A0 (FCHUSBDWCUSB31CNTR0::DB_REGS40_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020A0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020A0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020A4 (FCHUSBDWCUSB31CNTR0::DB_REGS41_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020A4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0020A4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020A8 (FCHUSBDWCUSB31CNTR0::DB_REGS42_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020A8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0020A8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020AC (FCHUSBDCUSB31CNTR0::DB_REGS43_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020AC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020AC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020B0 (FCHUSBDCUSB31CNTR0::DB_REGS44_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020B0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020B0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020B4 (FCHUSBWCUSB31CNTR0::DB_REGS45_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020B4; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020B4; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020B8 (FCHUSBWCUSB31CNTR0::DB_REGS46_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020B8; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020B8; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020BC (FCHUSBWCUSB31CNTR0::DB_REGS47_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020BC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020BC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020C0 (FCHUSBWCUSB31CNTR0::DB_REGS48_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020C0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020C0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020C4 (FCHUSBWCUSB31CNTR0::DB_REGS49_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020C4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0020C4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020C8 (FCHUSBWCUSB31CNTR0::DB_REGS50_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020C8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0020C8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020CC (FCHUSBWCUSB31CNTR0::DB_REGS51_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020CC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020CC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020D0 (FCHUSBWCUSB31CNTR0::DB_REGS52_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020D0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020D0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020D4 (FCHUSBDWCUSB31CNTR0::DB_REGS53_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020D4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0020D4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020D8 (FCHUSBDWCUSB31CNTR0::DB_REGS54_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020D8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0020D8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020DC (FCHUSBDCUSB31CNTR0::DB_REGS55_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020DC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020DC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020E0 (FCHUSBDCUSB31CNTR0::DB_REGS56_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020E0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020E0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020E4 (FCHUSBWCUSB31CNTR0::DB_REGS57_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020E4; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020E4; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020E8 (FCHUSBWCUSB31CNTR0::DB_REGS58_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020E8; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020E8; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020EC (FCHUSBWCUSB31CNTR0::DB_REGS59_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020EC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020EC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020F0 (FCHUSBWCUSB31CNTR0::DB_REGS60_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020F0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020F0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020F4 (FCHUSBWCUSB31CNTR0::DB_REGS61_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020F4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0020F4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020F8 (FCHUSBWCUSB31CNTR0::DB_REGS62_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020F8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x0020F8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	DB_STREAM_ID. Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET. Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x0020FC (FCHUSBWCUSB31CNTR0::DB_REGS63_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x0020FC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x0020FC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x002100 (FCHUSBWCUSB31CNTR0::DB_REGS64_DB)

Reset: 0000_0000h.

Doorbell Register Bit Field Definitions

For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

Programming this field with random data will cause side effect i.e. Register Access will fail (Timeout) if the pipe clock is not running or reset is asserted . Bit Bash register testing is not recommended.

_instUSB0_aliasHOST; USB0x002100; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x002100; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	DB_STREAM_ID . Read-write,Reserved. Reset: 0000h. Description: DB_STREAM_ID For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.
15:8	Reserved.
7:0	DB_TARGET . Read-write,Reserved. Reset: 00h. Description: DB_TARGET For a description of this standard USB register field, see the eXtensible Host Controller Interface for Universal Serial Bus (USB) Specification 3.1.

USB[0...1]x00C100 (FCHUSBWCUSB31CNTR0::GSBUSCFG0)

Read-write.

Global SoC Bus Configuration Register 0

This register configures system bus DMA options for the master bus, which may be configured as AHB, AXI, or Native. Options include burst length and cache type (bufferable/posted, cacheable/snoop, and so on). The application can program this register upon power-on, or a change in mode of operation after the DMA engine is halted.

_instUSB0_aliasHOST; USB0x00C100; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C100; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:28	DATRDREQINFO. Read-write. Reset: 0h. Description: DATRDREQINFO AHB-prot/AXI-cache/OCP-ReqInfo for Data Read (DatRdReqInfo) Input to BUS-GM.
27:24	DESRDREQINFO. Read-write. Reset: 0h. Description: DESRDREQINFO AHB-prot/AXI-cache/OCP-ReqInfo for Descriptor Read (DesRdReqInfo). Input to BUS-GM.
23:20	DATWRREQINFO. Read-write. Reset: 0h. Description: DATWRREQINFO AHB-prot/AXI-cache/OCP-ReqInfo for Data Write (DatWrReqInfo). Note: GSBUSCFG0[23:20] needs to keep 0 if GBMUCTL[2] (axi_storder_en)=1 Input to BUS-GM.
19:16	DESWRREQINFO. Read-write. Reset: 0h. Description: DESWRREQINFO AHB-prot/AXI-cache/OCP-ReqInfo for Descriptor Write (DesWrReqInfo) Input to BUS-GM.
15:12	Reserved.
11	DATBIGEND. Read-write. Reset: 0. Description: Data Access is Big Endian This bit controls the endian mode for data accesses. - Little-endian - Big-endian In big-endian mode, DMA access (both read and write) for packet data a Byte Invariant Big-Endian mode Note: Since AXI requires byte invariant endianness, setting DescBigend and DatBigEnd to one causes an address invariant transform to be applied, which is not appropriate. See section 9.3 and 9.4 of the AMBA AXI Specification. Hence for an AXI master (DWC_USB31_MBUS_TYPE=1), this bit must be set to zero.
10	DESBIGEND. Read-write. Reset: 0.

	Description: Descriptor Access is Big Endian This bit controls the endian mode for descriptor accesses. - Little-endian - Big-endian In big-endian mode, DMA access (both read and write) for descriptors uses a Byte Invariant Big-Endian mode Data is considered as 'embedded data' in the descriptors if the Immediate Data (IDT) bit in a Transfer TRB is set to 1. In host mode, if the system uses different endian modes for data and descriptors, the controller treats 'embedded data' as descriptor (not as data) in terms of endian mode handling. If this is not the expectation of the system, the software must manipulate the 'embedded data' accordingly. Note: Since AXI requires byte invariant endianness, setting DescBigend and DatBigEnd to one causes an address invariant transform to be applied, which is not appropriate. See section 9.3 and 9.4 of the AMBA AXI Specification. Hence for an AXI master (DWC_USB31_MBUS_TYPE=1), this bit must be set to zero.
9:8	Reserved.
7	INCR256BRSTENA. Read-write. Reset: 0. Description: INCR256 Burst Type Enable Input to BUS-GM. For the AXI configuration, if software set this bit to 1, the AXI master uses INCR to do the 256-beat burst.
6	INCR128BRSTENA. Read-write. Reset: 0. Description: INCR128 Burst Type Enable Input to BUS-GM; For the AXI configuration, if software set this bit to 1, the AXI master uses INCR to do the 128-beat burst.
5	INCR64BRSTENA. Read-write. Reset: 1. Description: INCR64 Burst Type Enable - Input to BUS-GM; For the AXI configuration, if software set this bit to 1, the AXI master uses INCR to do the 64-beat burst.
4	INCR32BRSTENA. Read-write. Reset: 1. Description: INCR32 Burst Type Enable Input to BUS-GM; For the AXI configuration, if software set this bit to 1, the AXI master uses INCR to do the 32-beat burst.
3	INCR16BRSTENA. Read-write. Reset: 1. Description: INCR16 Burst Type Enable Input to BUS-GM. For the AHB/AXI configuration, if software set this bit to '1', the AHB/AXI master uses INCR to do the 16-beat burst.
2	INCR8BRSTENA. Read-write. Reset: 1. Description: INCR8 Burst Type Enable Input to BUS-GM; For the AHB/AXI configuration, if software set this bit to "1", the AHB/AXI master uses INCR to do the 8-beat burst.
1	INCR4BRSTENA. Read-write. Reset: 1.

	Description: INCR4 Burst Type Enable Input to BUS-GM; For the AXI configuration, when this bit is enabled the controller is allowed to do bursts of beat length 1, 2 and 4. It is highly recommended that this bit is enabled to prevent descriptor reads and writes from being broken up into separate transfers.
0	INCRBRSTENA. Read-write. Reset: 1. Description: Undefined Length INCR Burst Type Enable (INCRBrstEna) Input to BUS-GM; This bit determines the set of burst lengths the master interface uses. It works in conjunction with the GSBUSCFG0[7:1] enables (INCR256/128/64/32/16/8/4). 0: INCRX burst mode HBURST (for AHB configurations) and ARLEN/AWLEN (for AXI configurations) do not use INCR, except in case of non-aligned burst transfers. In the case of address-aligned transfers, they use only the following burst lengths: - 1 - 2, 4 (if GSBUSCFG0.INCR4BrstEna = 1) - 8 (if GSBUSCFG0.INCR8BrstEna = 1) - 16 (if GSBUSCFG0.INCR16BrstEna = 1) - 32 (if GSBUSCFG0.INCR32BrstEna = 1) - 64 (if GSBUSCFG0.INCR64BrstEna = 1) - 128 (if GSBUSCFG0.INCR128BrstEna = 1) - 256 (if GSBUSCFG0.INCR256BrstEna = 1) Note: In case of non-address-aligned transfers, INCR may get generated at the beginning and end of the transfers to align the address boundaries, even though INCR is disabled. 1: INCR (undefined length) burst mode - AHB configurations: HBURST uses SINGLE or INCR of any length less than or equal to the largest-enabled burst length of INCR4/8/16. - AXI configurations: ARLEN/AWLEN uses any length less than or equal to the largest-enabled burst length of INCR32/64/128/256. For cache line-aligned applications, this bit is typically set to 0 to ensure that the master interface uses only power-of-2 burst lengths (as enabled via GSBUSCFG0[7:0]).

USB[0...1]x00C104 (FCHUSBWDCUSB31CNTR0::GSBUSCFG1)**Global SoC Bus Configuration Register 1**

_instUSB0_aliasHOST; USB0x00C104; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C104; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:17	Reserved.
16:13	ExtdPipeTransLimit. Read-write. Reset: 3h. Description: AXI Pipelined Transfers Extended Burst Request Limit The {ExtdPipeTransLimit, PipeTransLimit} fields controls the number of outstanding extended pipelined transfer requests the AXI master pushes to the AXI slave. When the AXI master reaches this limit, it does not make any more requests on the AXI ARADDR and AWADDR buses until the associated data phases complete. This field is encoded as follows: - 'h0: 1 request - 'h1: 2 requests - 'h2: 3 requests - 'h3: 4 requests - and, so on Note: This field must be programmed to a value lesser than DWC_USB31_MAXI_REQUESTS.
12	EN1KPAGE. Read-write. Reset: 0. Description: 1k Page Boundary Enable - When this bit is disabled, the AXI master (DMA data) breaks transfers at the 4k page boundary. - When this bit is enabled, the AXI master (DMA data) breaks transfers at the 1k page boundary.
11:8	PipeTransLimit. Read-write. Reset: Fh. Description: AXI Pipelined Transfers Burst Request Limit The {ExtdPipeTransLimit, PipeTransLimit} fields controls the number of outstanding pipelined transfer requests the AXI master pushes to the AXI slave. When the AXI master reaches this limit, it does not make any more requests on the AXI ARADDR and AWADDR buses until the associated data phases complete. This field is encoded as follows: - 'h0: 1 request - 'h1: 2 requests - 'h2: 3 requests - 'h3: 4 requests - and, so on Note: This field must be programmed to a value lesser than DWC_USB31_MAXI_REQUESTS.
7:0	Reserved.

USB[0...1]x00C108 (FCHUSBDCUSB31CNTR0::GTXTHRCFG)

Read-write.

Global Tx Threshold Control Register

_instUSB0_aliasHOST; USB0x00C108; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C108; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:27 Reserved.

26 **UsbTxPktCntSel.** Read-write. Reset: 1.**Description:** USB Async ESS Transmit Packet Threshold Enable - Host/Device Modes

This field enables/disables the USB transmission multi-packet thresholding for Async endpoints:

- 0: USB transmission multi-packet thresholding is disabled; the controller starts transmission on the USB as soon as one packet has been fetched into the corresponding TXFIFO.
- 1: USB transmission multi-packet thresholding is enabled. The controller can only start transmission on the USB after USB Transmit Packet Count amount of packets for the USB transaction (burst) are already in the corresponding TXFIFO. This mode is only used for Enhanced SuperSpeed.

In device mode, the controller can send ERDY on the USB after USB Transmit Packet Count amount of packets for the USB transaction (burst) are already in the corresponding TXFIFO of a Bulk IN endpoint

25:21 **UsbTxPktCnt.** Read-write. Reset: 04h.**Description:** USB Async ESS Transmit Packet Threshold Count - Host/Device Modes

This field specifies the number of packets that must be in the TXFIFO before the controller can start transmission for the corresponding USB transaction (burst) for Async endpoints. This field is only valid when the USB Async ESS Transmit Packet Threshold Enable field is set to one.

For device mode, this field specifies the number of packets that must be in the TXFIFO before the controller can start transmission of ERDY for the corresponding USB transaction (burst) of Bulk IN endpoint.

Valid values for this field is 1 to 16.

Note:

- In host mode, this field must be less than or equal to the USB Maximum TX Burst Size field.
- In device mode, this field must be less than or equal to the USB endpoint maximum Burst Size amongst all endpoints.

20:16 **UsbMaxTxBurstSize.** Read-write. Reset: 10h.**Description:** USB Async ESS Maximum TX Burst Size - Host Mode Only

When USBTxPktCntSel is 1, this field specifies the Maximum ESS Bulk OUT burst that the controller can execute. When the system bus is slower than the USB, TXFIFO can underrun during a long burst.

You can program a smaller value to this field to limit the TX burst size that the controller can execute.

It only applies to ESS Bulk OUT endpoints in the host mode. Valid values are from 1 to 16.

Note: This field can only be set to 2, 4, 8 or 16.

15 **UsbTxThrNumPktSel_HS_Prd.** Read-write. Reset: 1.

	Description: USB HS High Bandwidth Periodic Transmit Packet Threshold Enable - Host Mode Only This field enables/disables the USB HS High Bandwidth Periodic transmission multi-packet thresholding> - 0: USB HS High Bandwidth Periodic transmission multi-packet thresholding is disabled; the controller can start transmission on the USB as soon as one packet has been fetched into the corresponding TXFIFO. - 1: USB HS High Bandwidth Periodic transmission multi-packet thresholding is enabled. The controller can start transmission on the USB only after USB HS High Bandwidth Periodic Transmit Packet Threshold Count number of packets for the Periodic High Bandwidth USB transaction is already in the corresponding TXFIFO. This mode is valid in host mode for high speed high bandwidth periodic endpoints. Note: The intent of using this bit is only when the system bus is slow and not able to cater to the data rate of High-Bandwidth endpoint (3076 bytes/ufame). If the system bus is slow, and if all the transactions of a High-Bandwidth endpoint could not be executed successfully, it would likely cause a compliance test failure (section 5.4.x of xHCI Compliance Test Specification). This issue is seen only with older PCI based USB-IF compliance devices.
14:13	UsbTxThrNumPkt_HS_Prd. Read-write. Reset: 3h. Description: USB HS High Bandwidth Periodic Transmit Packet Threshold Count - Host Mode Only This field specifies the number of packets that must be available in the TXFIFO before the controller can start transmission for the Periodic High Bandwidth USB transaction. This field is only valid when the USB HS High Bandwidth Periodic Transmit Packet Threshold Enable field is set to one. Valid values are from 1 to 3.
12:11	Reserved.
10	UsbTxThrNumPktSel_Prd. Read-write. Reset: 1. Description: USB Periodic ESS Transmit Packet Threshold Enable - Host Mode Only This field enables/disables the USB Periodic transmission multi-packet thresholding: - 0: USB transmission multi-packet thresholding is disabled; the controller starts transmission on the USB as soon as one packet has been fetched into the corresponding TXFIFO. - 1: USB transmission multi-packet thresholding is enabled. The controller can start transmission on the USB only after USB Periodic ESS Transmit Packet Threshold Count amount of packets for the USB transaction (burst) has been fetched into the corresponding TXFIFO. It is used only for ESS.
9:5	UsbTxThrNumPkt_Prd. Read-write. Reset: 08h. Description: USB Periodic ESS Transmit Packet Threshold Count - Host Mode Only This field specifies the number of packets that must be in the TXFIFO before the controller can start transmission for the corresponding USB transaction (burst). This field is valid only when the USB Periodic ESS Transmit Packet Threshold Enable field (UsbTxThrNumPktSel_Prd) is set to one. Valid values are from 1 to 16. Note: This field must be less than or equal to the USB Maximum Periodic TX Burst Size field.
4:0	UsbMaxTxBurstSize_Prd. Read-write. Reset: 10h. Description: USB Maximum Periodic ESS TX Burst Size - Host Mode Only When UsbTxThrNumPktSel_Prd is 1, this field specifies the Maximum Periodic OUT burst the controller can execute. When the system bus is slower than the USB, TXFIFO can underrun during a long burst. Note: This field can only be set to 2, 4, 8 or 16.

USB[0...1]x00C10C (FCHUSBWDCUSB31CNTR0::GRXTHRCFG)

Read-write.

Global Rx Threshold Control Register

There is an issue when ACK TP with NumP=0 followed by ACK TP with NumP=1 without ERDY TP sent by the device controller during a Burst BULK OUT transfer. This may cause third-party 3.0 host controllers to keep waiting for the ERDY TP.

The USB 3.0 Specification states that "When an endpoint is not in a flow control condition, it shall not send an ERDY TP unless the endpoint is a Bulk endpoint that supports streams." In this case, after the device sent the ACK TP (numP=1), the endpoint was not in the flow control, so it did not send an ERDY.

The device would have sent ERDY if the next OUT packet was not received. When the next OUT packet was received, at that time there was enough buffer space to accept it, so the device accepted the packet by informing host that it is no longer in the flow control. The Host should wait for the responses for all the OUT packets to return and then decide if the endpoint is still in flow control or not.

The USB 3.1 Specification supersedes all the USB 3.0 specification. The errata states that "If the host continues, or resumes, transactions to an endpoint, the endpoint shall re-evaluate its flow control state and respond appropriately." However, there are no ECNs on the USB 3.0 for this issue.

To work around this issue, the Global Rx Threshold mode must be disable by setting GRXTHRCFG.UsbRxPktCntSel = '0'. Instead, software can program the DCFG.NUMP mode (where fixed NUMP is transmitted always) instead of the RX threshold based numP mode to prevent the device from sending ACK TP with NumP=0. The NUMP in the ACK TP is the minimum value of (DCFG.NUMP, bMaxBurstSize) for each endpoint.

IN Transfer Concurrency: When threshold mode is enabled, the Host controller disables IN transfer concurrency. The Host controller will execute IN transfers to one endpoint at a time in order to adhere to the maximum throughput limit specified by the threshold settings.

_instUSB0_aliasHOST; USB0x00C10C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C10C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:27	Reserved.
26	UsbRxPktCntSel. Read-write. Reset: 0. Description: USB Async ESS Receive Packet Threshold Enable - Host/Device Modes This field enables/disables the USB reception multi-packet thresholding: - 0: The controller can only start reception on the USB when the RXFIFO has space for at least one packet. - 1: The controller can only start reception on the USB when the RXFIFO has space for at least UsbRxPktCnt amount of packets. This mode is only used for Enhanced SuperSpeed.
25:21	UsbRxPktCnt. Read-write. Reset: 04h. Description: USB Async ESS Receive Packet Threshold Count - Host/Device Modes In host mode, this field specifies the space (in terms of the number of packets) that must be available in the RXFIFO before the controller can start the corresponding USB RX transaction (burst). This field is valid only when the USB Async ESS Receive Packet Threshold Enable field is set to 1. The valid values for this field are from 1 to 16. Note: This field must be less than or equal to the USB Maximum Receive Burst Size field.
20:16	UsbMaxRxBurstSize. Read-write. Reset: 10h.

	Description: USB Async ESS Maximum Receive Burst Size - Host/Device Modes In host mode, this field specifies the Maximum Bulk IN burst the DWC_usb31 controller can perform. When the system bus is slower than the USB, RXFIFO can overrun during a long burst. You can program a smaller value to this field to limit the RX burst size that the controller can perform. It only applies to ESS Bulk endpoints in the host mode. This field is valid only when USBRxPktCntSel is 1. The valid values for this field are from 1 to 16.
15	UsbRxThrNumPktSel_HS_Prd. Read-write. Reset: 1. Description: USB HS High Bandwidth Periodic Receive Packet Threshold Enable - Host Mode Only This field enables/disables the USB reception multi-packet thresholding: - 0: USB HS High Bandwidth Periodic Receive multi-packet thresholding is disabled; The controller can start reception on the USB as soon as the RXFIFO has space for one packet. - 1: USB HS High Bandwidth Periodic Receive multi-packet thresholding is enabled; the controller can only start reception on the USB when the RXFIFO has space for at least UsbRxThrNumPkt_HS_Prd amount of packets. This mode is valid in host mode for high speed high bandwidth periodic endpoints. Note: The intent of using this bit is only when the system bus is slow and not able to cater to the data rate of High-Bandwidth endpoint (3076 bytes/ufame). If the system bus is slow, and if all the transactions of a High-Bandwidth endpoint could not be executed successfully, it would likely cause a compliance test failure (section 5.4.x of xHCI Compliance Test Specification). This issue is seen only with older PCI based USB-IF compliance devices.
14:13	UsbRxThrNumPkt_HS_Prd. Read-write. Reset: 1h. Description: USB HS High Bandwidth Periodic Receive Packet Threshold Count - Host Mode Only This field specifies the maximum number of packet space needed in the RXFIFO before the controller can start a HS Periodic High Bandwidth USB transaction. This field is valid only when USBRxPktCntSel_HS_Prd is 1. The valid values for this field are from 1 to 3.
12:11	Reserved.
10	UsbRxThrNumPktSel_Prd. Read-write. Reset: 1. Description: USB Periodic ESS Receive Packet Threshold Enable - Host Mode Only. This field should be programmed to 0 in Device mode. This field enables/disables the Periodic ESS USB reception multi-packet thresholding: - 0: USB Periodic ESS Receive multi-packet thresholding is disabled; the controller can start reception on the USB as soon as the RXFIFO has space for one packet. - 1: USB Periodic ESS Receive multi-packet thresholding is enabled; the controller can only start reception on the USB when the RXFIFO has space for at least UsbRxThrNumPkt_Prd amount of packets. This mode is only used for ESS.
9:5	UsbRxThrNumPkt_Prd. Read-write. Reset: 04h. Description: USB Periodic ESS Receive Packet Threshold Count - Host Mode Only. This field should be programmed to 0 in Device mode. This field specifies the minimum number of packet space needed in the RXFIFO before the controller can start a ESS periodic high-bandwidth USB transaction. This field is valid only when the USB Periodic ESS Receive Packet Threshold Enable field is set to 1. Valid values are from 1 to 16. Note: This field must be less than or equal to the USB Maximum Periodic RX Burst Size field.
4:0	UsbMaxRxBurstSize_Prd. Read-write. Reset: 10h.

	Description: USB Maximum Periodic ESS RX Burst Size - Host Mode Only. This field should be programmed to 0 in Device mode.
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	When UsbRxThrNumPktSel_Prd is 1, this field specifies the Maximum ESS Periodic IN burst the controller can execute. When the system bus is slower than the USB, RXFIFO can overrun during a long burst.
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USB[0...1]x00C110 (FCHUSBWDCUSB31CNTR0::GCTL)**Global Core Control Register**

_instUSB0_aliasHOST; USB0x00C110; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C110; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR] , NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:19	PWRDNSCALE. Read-write. Reset: 000Ch. Description: Power Down Scale The USB31 core_suspend_clk input replaces core_pipe_pclk as a clock source to a small part of the DWC_usb31 controller that operates when the ESS PHY is in its lowest power (P3) state, and therefore does not provide a clock. The Power Down Scale field specifies how many core_suspend_clk periods fit into a 16 kHz clock period. When performing the division, round up the remainder. For example, when using an 8-bit/16-bit/32-bit PHY and 25-MHz Suspend clock, Power Down Scale = 25000 kHz/16 kHz = 13'd1563 (rounder up) The LTSSM uses Suspend clock for 12-ms and 100-ms timers during suspend mode. According to the USB 3.1 specification, the accuracy on these timers is 0% to +50%. - 12 ms + 0~+50% accuracy = 18 ms (Range is 12 ms - 18 ms) - 100 ms + 0~+50% accuracy = 150 ms (Range is 100 ms - 150 ms). The suspend clock accuracy requirement is: - (12,000/62.5) * (GCTL[31:19]) * actual Suspend clock period must be between 12,000 and 18,000 - (100,000/62.5) * (GCTL[31:19]) * actual Suspend clock period must be between 100,000 and 150,000 For example, if core_suspend_clk frequency varies from 7.5 MHz to 10.5MHz, then the value needs to programmed is: Power Down Scale = 10500/16 = 657 (rounded up; and fastest frequency used). Same requirement is applicable to suspend_clk Notes: When Hibernation is enabled, - This input is a minimum 32-kHz, maximum 200-kHz clock that is provided to the PMU and the controller. In the PMU, it is used to filter out glitches from the PHY, to time PIPE3 receiver detection, and to set PORTSC bits while in hibernation. In the controller, it is used to allow the sticky state to be restored before the bus_clk, USB 2.0 PHY clock, and USB 3.1 PHY clock are available. - The suspend clock must be always running . - When PMU filters are enabled (DWC_USB31_EN_PMU_FILTER=1), the maximum suspend_clk frequency is only 200 kHz because the suspend_clk is used to filter glitches from the PHY interfaces. A pulse is considered to be a glitch if it is less than 2.5us long (derived from USB 2.0). Therefore, sampling a signal on two consecutive rising edges of a 200-kHz clock ensures that the signal is stable for more than 2.5us. - When PMU filters are disabled (DWC_USB31_EN_PMU_FILTER=0), the suspend_clk frequency may be faster (up to 30 MHz).
18	MASTERFILTBYPASS. Read-write. Reset: 0.

	Description: Master Filter Bypass When this bit is set to 1'b1, irrespective of the parameter `DWC_USB31_EN_BUS_FILTERS chosen, all the filters in the DWC_usb31_filter module (utmiotg_iddig, utmiotg_sessend, utmisrp_bvalid, utmiotg_vbusvalid & pipe_PowerPresent) are bypassed. The double synchronizers to mac_clk preceding the filters are also bypassed. For enabling the filters, this bit must be 1'b0.
17	BYPSETADDR. Read-write. Reset: 0. Description: Bypass SetAddress in Device Mode This bit for simulation purposes only. In the actual hardware, this bit must be set to 1'b0. When BYPSETADDR bit is set, the device controller uses the value in the DCFG[DevAddr] bits directly for comparing the device address in the tokens. For simulation, you can use this feature to avoid sending an actual SET ADDRESS control transfer on the USB, and make the device controller respond to a new address. When the xHCI Debug capability is enabled and this bit is set, the Debug Target immediately enters the configured state without requiring the Debug Host to send a SetAddress or SetConfig request.
16	U2RSTECN. Read-write. Reset: 0. Description: U2RSTECN If the enhanced super speed connection fails during POLL or LMP exchange, the device connects at non-ESS mode. On USB 2.0 reset device attempts three more times to connect at ESS, even if it previously failed to operate in ESS mode. This bit controls whether to attempt rx.detect eight times VS one time. Device controller on USB 2.0 reset checks for receiver termination eight times if this bit is set to zero, or only once if the bit is set to one Note: This bit is applicable only in device mode.
15:14	FRMSCLDWN. Read-write. Reset: 0h. Description: FRMSCLDWN This field scales down device view of a SOF/USOF/ITP duration. For ESS/HS mode: - Value of 2'h3 implements interval to be 15.625 us - Value of 2'h2 implements interval to be 31.25 us - Value of 2'h1 implements interval to be 62.5 us - Value of 2'h0 implements interval to be 125us For FS mode, the scale-down value is multiplied by 8. When xHCI Debug Capability is enabled, this field also scales down the MaxPacketSize of the IN and OUT bulk endpoint to allow more traffic during simulation. It can only be changed from a non-zero value during simulation. - 2'h0: 1024 bytes - 2'h1: 512 bytes - 2'h2: 256 bytes - 2'h3: 128 bytes
13:12	PRTCAPDIR. Read-write, Volatile. Reset: 1h.

	Description: PRTCAPDIR: Port Capability Direction - 2'b01: for Host configurations Programming this field with random data causes the controller to keep toggling between the host mode and the device mode. If the configured value is set to host mode, VBUS is driven immediately after power-on reset. If the end user plugs in the core into a PC and the application configures the controller as a device later, the VBUS will be driven from both sides of the cable. This will potentially damage the PC host.
11	CORESOFTRSET. Read-write. Reset: 0. Description: Core Soft Reset - 1'b0 - No soft reset - 1'b1 - Soft reset to core Clears the interrupts and all the CSRs except the following registers: - GCTL - GUCTL - GSTS - USB31_IP_NAME - GGPIIO - GUID - GUSB2PHYCFGn registers - GUSB3PIPECTLn registers - DCFG - DCTL - DEVTEN - DSTS When you reset PHYs (using GUSB3PHYCFG or GUSB3PIPECTL registers), you must keep the controller in reset state until PHY clocks are stable. This controls the bus, RAM, and MAC domain resets. Note: Programming this field with random data will reset the internal logic of the host controller.
10	Reserved.
9	U1U2TimerScale. Read-write. Reset: 0. Description: Disable U1/U2 timer Scaledown If set to '1' along with GCTL[5:4] (ScaleDown) = 2'b11, disables the scale down of U1/U2 inactive timer values. This is for simulation mode only.
8	DEBUGATTACH. Read-write. Reset: 0. Description: Debug Attach When this bit is set, - ESS Link proceeds directly to the Polling link state (after RUN/STOP in the DCTL register is asserted) without checking remote termination; - Link LFPS polling timeout is infinite; - Polling timeout during TS1 is infinite (in case link is waiting for TXEQ to finish). This register is applicable only in device mode.
7:6	RAMCLKSEL. Read-write, Volatile. Reset: 0h.

	Description: RAM Clock Select The ram_clk_out must be connected back to ram_clk_in for this feature to work. This register field setting can be changed only in device mode. - 2'b00: bus clock - 2'b01: pipe clock - 2'b10: In device mode, pipe/2 clock. - 2'b11: In device mode, selects mac2_clk as ram_clk when 8-bit UTMI or ULPI is used. (Not supported in 16-bit UTMI mode) In device mode, upon a USB reset and USB disconnect, the hardware clears these bits to 2'b00. Note: - In host mode, this register field setting should not be modified. A value of 2 can be chosen only if the pipe data width is 8 or 16 bits. In this case the when the ram_clk is switched to pipe_clk, it uses pipe_clk/2 instead of pipe_clk. If a value of 3 is chosen for RAMClkSel, then when ram_clk is switched to pipe_clk, then pipe_clk is used without any divider.
5:4	SCALEDOWN. Read-write. Reset: 0h. Description: Scale-Down Mode When Scale-Down mode is enabled for simulation, the controller uses scaled-down timing values, resulting in faster simulations. When Scale-Down mode is disabled, actual timing values are used. This is required for hardware operation. HS/FS/LS Modes - 2'b00: Disables all scale-downs. Actual timing values are used. - 2'b01: Reserved - 2'b10: Reserved - 2'b11: Enables scale-down of all timing values of USB events. These include Speed enumeration, HNP/SRP, and suspend and resume. ESS Mode - 2'b00: Disables all scale-downs. Actual timing values are used. - 2'b01: Enables scaled down SS timing and repeat values including: -- (1) Number of TxEq training sequences reduce to 8; -- (2) LFPS polling burst time reduce to 256 nS; -- (3) LFPS warm reset receive reduce to 30 uS. - 2'b10: No TxEq training sequences are sent. Overrides Bit 4. - 2'b11: Enables bit 0 and bit 1 scale-down timing values.
3	DISSCRAMBLE. Read-write. Reset: 0. Description: Disable Scrambling Request to disable scrambling to Link Partner on next transition to Recovery or Polling.
2	U2EXIT_LFPS. Read-write. Reset: 0.

	Description: U2EXIT_LFPS If this bit is, - 0: the link treats 248ns LFPS as a valid U2 exit. - 1: the link waits for 8us of LFPS before it detects a valid U2 exit. This bit is added to improve interoperability with a third party host controller. This host controller in U2 state while performing receiver detection generates an LFPS glitch of about 4us duration. This causes the device to exit from U2 state because the LFPS filter value is 248ns. With the new functionality enabled, the device can stay in U2 while ignoring this glitch from the host controller.
1	GblHibernationEn. Read-write. Reset: 1. Description: GblHibernationEn This bit enables hibernation. If hibernation is not enabled through this bit, the PMU immediately accepts the D0->D3 and D3->D0 power state change requests, but does not save or restore any controller state. In addition, the PMUs never drive the PHY interfaces and let the controller continue to drive the PHY interfaces. - Note: This field is only read-write when hibernation is enabled i.e. [DWC_USB31_EN_PWROPT==2] else its read-only
0	DSBLCLKGTNG. Read-write. Reset: 0. Description: Disable Clock Gating If this bit is set to 1, internal clock gating is disabled

USB[0...1]x00C114 (FCHUSBDCUSB31CNTR0::GPMSTS)**Global Power Management Status Register**

This debug register gives information on which event caused the hibernation exit. It provides internal status and state machine information.

_instUSB0_aliasHOST; USB0x00C114; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C114; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:28	PortSel. Write-only. Reset: 0h. Description: Global Power Management Status Register, PortSel This field selects the port number.
27:17	Reserved.
16:12	U3Wakeup. Read-only. Reset: 00h. Description: U3Wakeup This field gives the following USB 3.1 port wakeup conditions: - Bit [12]: Overcurrent Detected - Bit [13]: Resume Detected - Bit [14]: Connect Detected - Bit [15]: Disconnect Detected - Bit [16]: Last Connection State
11:10	Reserved.
9:0	U2Wakeup. Read-only. Reset: 000h. Description: U2Wakeup This field indicates the following USB 2.0 port wakeup conditions: - Bit [0]: Overcurrent Detected - Bit [1]: Resume Detected - Bit [2]: Connect Detected - Bit [3]: Disconnect Detected - Bit [4]: Last Connection State - Bit [5]: ID Change Detected - Bit [6]: SRP Request Detected - Bit [7]: ULPI Interrupt Detected - Bit [8]: USB Reset Detected - Bit [9]: Resume Detected Changed

USB[0...1]x00C118 (FCHUSBWCUSB31CNTR0::GSTS)**Global Status Register**

_instUSB0_aliasHOST; USB0x00C118; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C118; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)] , NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:20	CBELT. Read-only. Reset: 7E8h. Description: Current BELT Value In Host mode, this field indicates the minimum value of all received device BELT values and the BELT value that is set by the Set Latency Tolerance Value command.
19:8	Reserved.
7	Host_IP. Read-only. Reset: 0. Description: Host Interrupt Pending: This field indicates that there is a pending interrupt pertaining to xHC in the Host event queue.
6	Device_IP. Read-only. Reset: 0. Description: Device Interrupt Pending This field is not valid in host mode.
5	CSRTIMEOUT. Read-write,Read,Write-1-to-clear,Volatile. Description: CSR Timeout When this bit is 1'b1, it indicates that the software performed a write or read to a controller register that could not be completed within the CSR access time. For more details, refer to the <link>07_Registers_additional_info.fm:csr_timeout_mech,"CSR Timeout Mechanism"</link> section.
4	BUSERRADDRVLD. Read-write,Read,Write-1-to-clear,Volatile. Reset: 0. Description: Bus Error Address Valid (BusErrAddrVld) Indicates that the GBUSERRADDR register is valid and reports the first bus address that encounters a bus error. Note: Only supported in AHB and AXI configurations.
3:2	Reserved.
1:0	CURMOD. Read-only. Reset: 1h. Description: Current Mode of Operation (CurMod) Indicates the current mode of operation: - 2'b01: Host mode

USB[0...1]x00C11C (FCHUSBDCUSB31CNTR0::GUCTL1)**Global User Control Register 1**

_instUSB0_aliasHOST; USB0x00C11C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C11C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	DEV_DECOUPLE_L1L2_EVT. Read-write. Reset: 0. Description: DEV_DECOUPLE_L1L2_EVT This bit is not applicable in host mode.
30	DS_RXDET_MAX_TOUT_CTRL. Read-write. Reset: 1. Description: DS_RXDET_MAX_TOUT_CTRL This bit is used to control the tRxDetectTimeoutDFP timer for the Enhanced SuperSpeed link. - 0: 12ms is used as tRxDetectTimeoutDFP - 1: 120ms is used as the tRxDetectTimeoutDFP This bit is used only in host mode. For more details, refer to the USB 3.1 Specification.
29	FILTER_SE0_FSLS_EOP. Read-write. Reset: 1. Description: FILTER_SE0_FSLS_EOP - 0: Single sampling (utmi/ulpi clock) of linestate is checked for SE0 detection. - 1: Feature enabled; Two samplings of linestate are checked for SE0 detection This bit is applicable for FS/LS operation. If this feature is enabled, SE0 on the linestate is validated for two consecutive utmi/ulpi clock edges for EOP detection. This feature is applicable only in FS in device mode and FS/LS mode of operation in host mode. - Device mode (FS): If GUCTL1.FILTER_SE0_FSLS_EOP is set, then for device LPM handshake, the controller ignores single SE0 glitch on the linestate during transmit. Only two or more SE0 is considered as a valid EOP on FS port. - Host mode (FS/LS): If GUCTL1.FILTER_SE0_FSLS_EOP is set, then the controller ignores single SE0 glitch on the linestate during transmit. Only two or more SE0 is considered as a valid EOP on FS/LS port. Enable this feature if linestate has SE0 glitches during transmission. This bit is quasi-static, that is, it must not be changed during operation.
28	TX_IPGAP_LINECHECK_DIS. Read-write. Reset: 0. Description: TX_IPGAP_LINECHECK_DIS - 0: The linestate transitioning from J to idle for HS mode treated as end of current packet. - 1: Feature enabled; For detecting the HS end of packet, a fixed delay is used instead of linestate transition. This bit is applicable for HS operation of U2MAC. If this feature is enabled, then the 2.0 MAC operating in HS ignores the UTMI/ULPI linestate during transmission of a token (during token-to-token and token-to-data IPGAP). When enabled, the controller implements a fixed 40-bit TxEndDelay after the packet is given on UTMI and ignores the linestate during this time. This feature is applicable only in HS mode of operation. - Device mode: If GUCTL1.TX_IPGAP_LINECHECK_DIS is set, then for device LPM handshake, the controller ignores the linestate after TX and waits for a fixed number of clocks (40 bit times equivalent) after transmitting ACK on utmi. - Host mode: If GUCTL1.TX_IPGAP_LINECHECK_DIS is set, then the ipgap (between token to token/data) is added by 40 bit times of TXENDDELAY, and linestate is ignored during this 40 bit times delay. Enable this bit if linestate does not reflect the expected line state (J) during transmission. This bit is quasi-static, that is, it must not be changed during operation.
27	DEV_TRB_OUT_SPR_IND. Read-write. Reset: 0.

	Description: DEV_TRB_OUT_SPR_IND - 0: Feature disabled; OUT TRB status does not set the Short Packet received bit - 1: Feature enabled; OUT TRB status indicates Short Packet This bit is applicable for device mode only (and ignored in host mode). This feature can be enabled if the device application (software/hardware) wants to know whether a short packet is received for an OUT in the TRB status itself, so that a bit is set in the TRB writeback in the buf_size dword. Bit[26] - SPR of the {trbstatus, RSVD, SPR, PCM1, bufsize} dword will be set during an OUT transfer TRB write back if this is the last TRB used for that transfer descriptor. This bit is quasi-static, that is, it must not be changed during device operation.
26:25	Reserved.
24	DEV_L1_EXIT_BY_HW. Read-write. Reset: 0. Description: DEV_L1_EXIT_BY_HW - 0: Disables device L1 hardware exit logic - 1: Feature enabled This bit is applicable for device mode (2.0) only. This field enables device controller sending remote wakeup for L1 if the device becomes ready for sending/accepting data when in L1 state. If the host expects the device to send remote wakeup signaling to resume after going into L1 in flow controlled state, then this bit can be set to send the remote wake signal automatically when the device controller becomes ready. This hardware remote wake feature is applicable only to bulk and interrupt transfers, and not for Isoch/Control - When control transfers are in progress, the LPM will be rejected (NYET response). Only after control transfers are completed (either with ACK/STALL), LPM will be accepted - For Isoch transfers, the host needs to do the wake-up and start the transfer. Device controller will not do remote-wakeup when Isoch endpoints get ready. The device SW needs to keep the GUSB2PHYCFG[EnbLSlpM] reset in order to keep the PHY clock to be running for keeping track of SOF intervals. - When L1 hibernation is enabled, the controller will not do automatic exit for hibernation requests through L1. This bit is quasi-static, it must not be changed during device operation.
23:21	IP_GAP_ADD_ON. Read-write. Reset: 0h. Description: This register field is used to add on to the default inter packet gap setting in the USB 2.0 MAC. It should be programmed to a non-zero value only in case where you need to increase the default inter packet delay calculations in the USB 2.0 MAC module The inter packet delay is increased by number of utmi/ulpi clock cycles of this field value.
20:15	Reserved.
14	HW_LPM_CAP_DISABLE. Read-write. Reset: 0. Description: Disable hardware LPM capability in the xHCI capability register. When set, it disables hardware LPM capability in xHCI capability register.
13	HW_LPM_HLE_DISABLE. Read-write. Reset: 0. Description: Disable hardware LPM function in all USB 2.0 ports. When set, it disables hardware LPM function in all USB 2.0 ports.
12	DisUSB2RefClkGtng. Read-only. Reset: 1. Description: Disable ref_clk gating for USB 2.0 PHY (DisUSB2RefClkGtng) If ref_clk gating is disabled, then the ref_clk input cannot be turned off to the USB 2.0 PHY and controller. This is independent of the GCTL[DisClkGtng] setting. - 1'b0: ref_clk gating enabled for USB 2.0 PHY - 1'b1: ref_clk gating disabled for USB 2.0 PHY
11	DisRefClkGtng. Read-only. Reset: 1.

	Description: Disable ref_clk gating (DisRefClkGtng) If the ref_clk gating is disabled then input ref_clk cannot be turned off to SSPHY and controller. This is independent of GCTL[DisClkGtng] setting. - 1'b0: ref_clk gating Enabled for SSPHY - 1'b1: ref_clk gating Disabled for SSPHY
10	RESUME_OPMODE_HS_HOST. Read-write. Reset: 0. Description: RESUME_OPMODE_HS_HOST This bit is used only in host mode, and is for 2.0 opmode behaviour in HS Resume. - When this bit is set to '1', the utmi/ulpi opmode will be changed to "normal" along with HS terminations after EOR. This is to support certain legacy UTMI/ULPI PHYs. - When this bit is set to '0', the utmi/ulpi opmode will be changed to "normal" 2us after HS terminations change after EOR.
9	Reserved.
8	L1_SUSP_THRLD_EN_FOR_HOST. Read-write. Reset: 0. Description: L1_SUSP_THRLD_EN_FOR_HOST This bit is used only in host mode. The host controller asserts the utmi_l1_suspend_n and utmi_sleep_n output signals (see Table 5-18 on page 363) as follows: The controller asserts the utmi_l1_suspend_n signal to put the PHY into deep low-power mode in L1 when both of the following are true: - The device accepted BESL/BESLD value is greater than or equal to the value in L1_SUSP_THRLD_FOR_HOST field. - The L1_SUSP_THRLD_EN_FOR_HOST bit is set to 1'b1. The controller asserts utmi_sleep_n on L1 when one of the following is true: - The device accepted BESL/BESLD value is less than the value in L1_SUSP_THRLD_FOR_HOST field. - The L1_SUSP_THRLD_EN_FOR_HOST bit is set to 1'b0.
7:4	L1_SUSP_THRLD_FOR_HOST. Read-write. Reset: 0h. Description: L1_SUSP_THRLD_FOR_HOST This field is effective only when the L1_SUSP_THRLD_EN_FOR_HOST bit is set to 1. For more details, refer to the description of the L1_SUSP_THRLD_EN_FOR_HOST bit. Note: Program this register field based on the UTMI/ULPI wakeup time in L1 suspend/sleep. In addition, for PCIe®-based implementation, consider the DBESLD register value also.
3	HC_ERRATA_ENABLE. Read-write. Reset: 1. Description: Host ELD Enable (HELDn) When this bit is set to 1, it enables the Exit Latency Delta (ELD) support defined in the xHCI 1.1. This bit is used only in the host mode. This bit has to be set to 1 in Host mode.
2	HC_PARCHK_DISABLE. Read-write. Reset: 0.

	Description: Host Parameter Check Disable (HParChkDisable) When this bit is set to '0', the xHC checks that the input slot/EP context fields comply to the xHCI Specification. Upon detection of a parameter error during command execution, the xHC generates an event TRB with completion code indicating 'PARAMETER ERROR'. When the bit is set to '1', the xHC does not perform parameter checks and does not generate 'PARAMETER ERROR' completion code.
1	Reserved.
0	LOA_FILTER_EN. Read-write. Reset: 0. Description: LOA_FILTER_EN If this bit is set, the USB 2.0 port babble is checked at least three consecutive times before the port is disabled. This prevents false triggering of the babble condition when using low quality cables. Note: This bit is valid only in host mode.

USB[0...1]x00C124 (FCHUSBWCUSB31CNTR0::GGPIO)

Reset: 0000_0000h.

Global General Purpose Input/Output Register

The application can use this register for general purpose input and output ports or for debugging.

_instUSB0_aliasHOST; USB0x00C124; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C124; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	GPO. Read-write. Reset: 0000h. Description: General Purpose Output This field's value is driven out on the gp_out[15:0] controller output port.
15:0	GPI. Read-only. Reset: 0000h. Description: General Purpose Input This field's read value reflects the gp_in[15:0] controller input value.

USB[0...1]x00C128 (FCHUSBWCUSB31CNTR0::GUID)

Read-write. Reset: 0000_1022h.

Global User ID Register

This is a read/write register containing the User ID. The power-on value for this register is specified as the User Identification Register. This register can be used in the following ways:

- To store the version or revision of your system;
- To store hardware configurations that are outside the controller;
- As a scratch register.

_instUSB0_aliasHOST; USB0x00C128; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C128; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:0	USERID. Read-write. Reset: 0000_1022h. Description: USERID Application-programmable ID field.

USB[0...1]x00C12C (FCHUSBDCUSB31CNTR0::GUCTL)

Read-write.

Global User Control Register

This register provides a few options for the software to control the controller behavior in Host mode.

_instUSB0_aliasHOST; USB0x00C12C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C12C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description31:22 **REFCLKPER.** Read-write. Reset: 029h.**Description:** REFCLKPER

This field indicates the period of ref_clk, in terms of nano seconds.

The valid programmable values of this field are as follows:

- 'h19: 25ns (integer corresponding to the supported ref_clk frequencies of 39.7 MHz)
- 'h29: 41ns (integer corresponding to the supported ref_clk frequency of 24 MHz)
- 'h32: 50ns (integer corresponding to the supported ref_clk frequency of 20 MHz)
- 'h34: 52ns (integer corresponding to the supported ref_clk frequency of 19.2MHz)
- 'h3A: 58ns (integer corresponding to the supported ref_clk frequency of 17 MHz)
- 'h3E: 62ns (integer corresponding to the supported ref_clk frequency of 16 MHz)

This field needs to be updated during power-on initialization.

21 **NoExtrDI.** Read-write. Reset: 0.**Description:** No Extra Delay Between SOF and the First Packet(NoExtrDI)

Some HS devices misbehave when the host sends a packet immediately after a SOF. However, adding an extra delay between a SOF and the first packet can reduce the USB data rate and performance.

This bit is used to control whether the host must wait for 2 microseconds before it sends the first packet after a SOF, or not. User can set this bit to one to improve the performance if those problematic devices are not a concern in the user's host environment.

- 1'b0: Host waits for 2 microseconds after a SOF before it sends the first USB packet.
- 1'b1: Host doesn't wait after a SOF before it sends the first USB packet.

20 **DMAIgnoreHCE.** Read-write. Reset: 0. DMA Ignore HCE19 **IgnoreHCETimeout.** Read-write. Reset: 0. IgnoreHCETimeout18 **EN_EXTD_TBC_CAP.** Read-write. Reset: 1. When set, the Extended TBC Capability is reported in HCCPARAMS2 if the DWC_USB31_EXTD_TBC_CAP_EN parameter is enabled.17 **SprsCtrlTransEn.** Read-write. Reset: 1.**Description:** Sparse Control Transaction Enable

Some devices are slow in responding to Control transfers. Scheduling multiple transactions in one microframe/frame can cause these devices to misbehave.

If this bit is set to 1'b1, the host controller schedules each phase of a Control transfer in different microframes/frames.

16 **ResBwHSEPS.** Read-write. Reset: 1.

	Description: Reserving 85% Bandwidth for HS Periodic EPs (ResBwHSEPS) - 1'b0: HC reserves 80% of the bandwidth for periodic EPs. - 1'b1: HC relaxes the bandwidth to 85% to accommodate two high-speed high-bandwidth ISOC EPs. USB 2.0 required 80% bandwidth allocated for ISOC traffic. If two high-bandwidth ISOC devices (HD Webcams) are connected and each device requires 1024-bytes X 3 packets per micro-frame, then the bandwidth required is around 82%. If this bit is set, then it is possible to connect two Webcams of 1024bytes X 3 payload per micro-frame each. Otherwise, you may have to reduce the resolution of the Webcams.
15	Reserved.
14	USBHstInImmRetryEn. Read-write. Reset: 1. Description: Host IN Immediate Retry (USBHstInImmRetryEn) When set, this field enables the Immediate Retry feature. For IN transfers (non-isochronous) that encounter data packets with CRC errors or internal overrun scenarios, the immediate retry feature causes the Host controller to reply to the device with a non-terminating retry ACK (that is, an ACK transaction packet with Retry = 1 and NumP != 0). If the Immediate Retry feature is disabled, the controller will respond with a terminating retry ACK (that is, an ACK transaction packet with Retry = 1 and NumP = 0). - 1'b0: Immediate Retry Disabled - 1'b1: Immediate Retry Enabled
13	Reserved.
12	ExtCapSupptEN. Read-write. Reset: 0. Description: External Extended Capability Support Enable (ExtCapSuptEN) When set, this field enables extended capabilities to be implemented outside the controller. When the ExtCapSuptEN is set and the Debug Capability is enabled, the Next Capability pointer in "Debug Capability" returns 16. A read to the first DWORD of the last internal extended capability (the "xHCI Supported Protocol Capability for USB 3.1" when the Debug Capability is not enabled) returns a value of 4 in the Next Capability Pointer field. This indicates to software that there is another capability four DWORDs after this capability (for example, at address N+16 where N is the address of this DWORD). If enabled, an external address decoder that snoops the xHC slave interface must be implemented. If it sees an access to N+16 or greater, the slave access is re-routed to a piece of hardware which returns the external capability pointer register of the new capability and also handles reads/writes to this new capability and the side effects. If disabled, a read to the first DWORD of the last internal extended capability returns 0 in the 'Next Capability Pointer field. This indicates there are no more capabilities.
11	InsrtExtrFSBODI. Read-write. Reset: 1. Description: Insert Extra Delay Between FS Bulk OUT Transactions (InsrtExtrFSBODI). Some FS devices are slow to receive Bulk OUT data and can get stuck when there are consecutive Bulk OUT transactions with short inter-transaction delays. This bit is used to control whether the host inserts extra delay between consecutive Bulk OUT transactions to a FS Endpoint. - 1'b0: Host doesn't insert extra delay between consecutive Bulk OUT transactions to a FS Endpoint. - 1'b1: Host inserts about 12us extra delay between consecutive Bulk OUT transactions to a FS Endpoint to work around the device issue. Note: Setting this bit to one will reduce the Bulk OUT transfer performance for FS devices.
10:0	DTOUT. Read-write. Reset: 040h.

	Description: Device Timeout (DTOUT) This field is Host mode parameter which determines how long the host waits for response from Enhanced SuperSpeed Device before considering the transaction to be timeout. Each count indicates duration in terms of 125us. For example a value of 1 indicates timeout at the minimum of 125us and maximum of 250us. The maximum value that can be programmed is 200 which sets 25ms as timeout time (minimum is 25ms and maximum is 25.125ms).
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USB[0...1]x00C130 (FCHUSBWCUSB31CNTR0::GBUSERRADDRLO)

Read-only. Reset: 0000_0000h.

Global SoC Bus Error Address Register - Low

When the AHB or AXI Master Bus returns an "Error" response, the "SoC Bus Error" is generated. In the Host mode, the host_system_err port indicates this condition. In addition, it is also indicated in the USBSTS.HSE field.

Note for AXI configuration:

Due to the nature of AXI, it is possible that multiple AXI transactions are active at a time. The DWC_usb31 controller does not keep track of the start address of all outstanding transactions. Instead, it keeps track of the start address of the DMA transfer associated with all active transactions. It is this address that is reported in the GBUSERRADDR when a bus error occurs.

For example, if the DWC_usb31 controller initiates a DMA transfer to write 1k of packet data starting at buffer address 0xABCD0000, and this DMA is broken up into multiple 256B bursts on the AXI, then if a bus error occurs on any of these associated AXI transfers, the GBUSERRADDR reflects the DMA start address of 0xABCD0000 regardless of which AXI transaction received the error.

_instUSB0_aliasHOST; USB0x00C130; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C130; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	BUSERRADDR. Read-only. Reset: 0000_0000h. Description: Bus Address - Low (BusAddrLo) This register contains the lower 32 bits of the first bus address that encountered a SoC bus error. It is valid when the GSTS.BusErrAddrVld field is 1. It can only be cleared by resetting the controller. Note: Only supported in AHB and AXI configurations.

USB[0...1]x00C134 (FCHUSBWCUSB31CNTR0::GBUSERRADDRHI)

Read-only. Reset: 0000_0000h.

Global SoC Bus Error Address Register - High

When the AHB or AXI Master Bus returns an "Error" response, the "SoC Bus Error" is generated. In the Host mode, the host_system_err port indicates this condition. In addition, it is also indicated in the USBSTS.HSE field.

Note for AXI configuration:

Due to the nature of AXI, it is possible that multiple AXI transactions are active at a time. The DWC_usb31 controller does not keep track of the start address of all outstanding transactions. Instead, it keeps track of the start address of the DMA transfer associated with all active transactions. It is this address that is reported in the GBUSERRADDR when a bus error occurs.

For example, if the DWC_usb31 controller initiates a DMA transfer to write 1k of packet data starting at buffer address 0xABCD0000, and this DMA is broken up into multiple 256B bursts on the AXI, then if a bus error occurs on any of these associated AXI transfers, the GBUSERRADDR reflects the DMA start address of 0xABCD0000 regardless of which AXI transaction received the error.

_instUSB0_aliasHOST; USB0x00C134; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C134; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	BUSERRADDR. Read-only. Reset: 0000_0000h. Description: Bus Address - High (BusAddrHi) This register contains the higher 32 bits of the first bus address that encountered a SoC bus error. It is valid when the GSTS.BusErrAddrVld field is 1. It can only be cleared by resetting the controller. Note: Only supported in AHB and AXI configurations.

USB[0...1]x00C138 (FCHUSBWCUSB31CNTR0::GPRTBIMAPLO)

Reset: 0000_0010h.

Global ESS Port to Bus Instance Mapping Register - Low

This register specifies the Enhanced SuperSpeed USB instance number to which each USB 3.1 port is connected. Software can program this register to specify how USB 3.1 ports are connected to ESS USB instances.

For a configuration with the number of USB 3.1 ports same as the number of ESS Bus Instances, the remapping should not be performed during a debug session. If remapping is performed for some reason, then the debug host must be connected to a port which has a dedicated ESS Bus Instance.

For example, if `DWC_USB31_HOST_NUM_U3_ROOT_PORTS=3` and `DWC_USB31_NUM_ESS_USB_INSTANCES=3`, and software maps the first ESS port to the first ESS BI and the second/third port to the second BI, then the debug host can be connected to the first port only.

The reset value for each field in the GPRTBIMAP register, which is {GPRTBIMAPHI,GPRTBIMAPLO} register is calculated as shown below:

for ($i=0$; $i < \text{DWC_USB31_HOST_NUM_U3_ROOT_PORTS}$; $i=i+1$)

$\text{GPRTBIMAP}[4*i+3:4*i] = i \% \text{DWC_USB31_NUM_ESS_USB_INSTANCES}$;

Note:

- Register fields are read-write with respect to number of port instantiated.
- For ex. If `DWC_USB31_HOST_NUM_U3_ROOT_PORTS==3`, then GPRTBIMAPLO[11:0] i.e. fields (BINUM1/BINUM2/BINUM3) are read-write and all remaining fields are read-only.
- GPRTBIMAP register is not applicable for USB 2.0-only mode.

`_instUSB0_aliasHOST; USB0x00C138; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}`

`_instUSB1_aliasHOST; USB1x00C138; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}`

Bits	Description
31:28	BINUM8. Read-only. Reset: 0h. Description: BINUM8: ESS USB Instance Number for Port 8. Application-programmable ID field.
27:24	BINUM7. Read-only. Reset: 0h. Description: BINUM7: ESS USB Instance Number for Port 7. Application-programmable ID field.
23:20	BINUM6. Read-only. Reset: 0h. Description: BINUM6: ESS USB Instance Number for Port 6. Application-programmable ID field.
19:16	BINUM5. Read-only. Reset: 0h. Description: BINUM5: ESS USB Instance Number for Port 5. Application-programmable ID field.
15:12	BINUM4. Read-only. Reset: 0h. Description: BINUM4: ESS USB Instance Number for Port 4. Application-programmable ID field.

11:8	BINUM3. Read-only. Reset: 0h.
	Description: BINUM3: ESS USB Instance Number for Port 3.
	Application-programmable ID field.
7:4	BINUM2. Read-write. Reset: 1h.
	Description: BINUM2: ESS USB Instance Number for Port 2.
	Application-programmable ID field.
3:0	BINUM1. Read-write. Reset: 0h.
	Description: BINUM1: ESS USB Instance Number for Port 1.
	Application-programmable ID field.

USB[0...1]x00C13C (FCHUSBDCUSB31CNTR0::GPRTBIMAPHI)

Read-only.

Global ESS Port to Bus Instance Mapping Register - High

This register specifies the Enhanced SuperSpeed USB instance number to which each USB 3.1 port is connected. Software can program this register to specify how USB 3.1 ports are connected to ESS USB instances.

For a configuration with the number of USB 3.1 ports same as the number of ESS Bus Instances, the remapping should not be performed during a debug session. If remapping is performed for some reason, then the debug host must be connected to a port which has a dedicated ESS Bus Instance.

For example, if `DWC_USB31_HOST_NUM_U3_ROOT_PORTS=3` and `DWC_USB31_NUM_ESS_USB_INSTANCES=3`, and software maps the first ESS port to the first ESS BI and the second/third port to the second BI, then the debug host can be connected to the first port only.

The reset value for each field in the GPRTBIMAP register, which is {GPRTBIMAPHI,GPRTBIMAPLO} register is calculated as shown below:

for ($i=0$; $i < \text{DWC_USB31_HOST_NUM_U3_ROOT_PORTS}$; $i=i+1$)

$\text{GPRTBIMAP}[4*i+3:4*i] = i \% \text{DWC_USB31_NUM_ESS_USB_INSTANCES}$;

Note:

- Register fields are read-write with respect to number of port instantiated.
- For ex. If `DWC_USB31_HOST_NUM_U3_ROOT_PORTS=11`, then `GPRTBIMAPHI[11:0]` i.e. fields (BINUM9/BINUM10/BINUM11) are read-write and all remaining fields are read-only.
- GPRTBIMAP register is not applicable for USB 2.0-only mode.

`_instUSB0_aliasHOST; USB0x00C13C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}`

`_instUSB1_aliasHOST; USB1x00C13C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}`

Bits	Description
31:28	Reserved.
27:24	BINUM15. Read-only. Reset: 0h. Description: BINUM15: ESS USB Instance Number for Port 15. Application-programmable ID field.
23:20	BINUM14. Read-only. Reset: 0h. Description: BINUM14: ESS USB Instance Number for Port 14. Application-programmable ID field.
19:16	BINUM13. Read-only. Reset: 0h. Description: BINUM13: ESS USB Instance Number for Port 13. Application-programmable ID field.
15:12	BINUM12. Read-only. Reset: 0h. Description: BINUM12: ESS USB Instance Number for Port 12. Application-programmable ID field.
11:8	BINUM11. Read-only. Reset: 0h.

	Description: BINUM11: ESS USB Instance Number for Port 11. Application-programmable ID field.
7:4	BINUM10. Read-only. Reset: 0h. Description: BINUM10: ESS USB Instance Number for Port 10. Application-programmable ID field.
3:0	BINUM9. Read-only. Reset: 0h. Description: BINUM9: ESS USB Instance Number for Port 9. Application-programmable ID field.

USB[0...1]x00C140 (FCHUSBDCUSB31CNTR0::GHWPARAMS0)

Read-only. Reset: 4020_8049h.

Global Hardware Parameters Register 0

This register contains the hardware configuration options.

_instUSB0_aliasHOST; USB0x00C140; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C140; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:24	ghwparams0_31_24. Read-only. Reset: 40h. `DWC_USB31_AWIDTH
23:16	ghwparams0_23_16. Read-only. Reset: 20h. `DWC_USB31_SDWIDTH
15:8	ghwparams0_15_8. Read-only. Reset: 80h. `DWC_USB31_MDWIDTH
7:6	ghwparams0_7_6. Read-only. Reset: 1h. `DWC_USB31_SBUS_TYPE
5:3	ghwparams0_5_3. Read-only. Reset: 1h. `DWC_USB31_MBUS_TYPE
2:0	ghwparams0_2_0. Read-only. Reset: 1h. `DWC_USB31_MODE

USB[0...1]x00C144 (FCHUSBDCUSB31CNTR0::GHWPARAMS1)

Read-only. Reset: 8203_2486h.

Global Hardware Parameters Register 1

This register contains the hardware configuration options.

_instUSB0_aliasHOST; USB0x00C144; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00C144; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31	ghwparams1_31 . Read-only. Reset: 1. `DWC_USB31_EN_DBC
30	ghwparams1_30 . Read-only. Reset: 0. `DWC_USB31_RM_OPT_FEATURES
29	ghwparams1_29 . Read-only. Reset: 0. Reserved
28	ghwparams1_28 . Read-only. Reset: 0. `DWC_USB31_RAM_BUS_CLKS_SYNC
27	ghwparams1_27 . Read-only. Reset: 0. `DWC_USB31_MAC_RAM_CLKS_SYNC
26	ghwparams1_26 . Read-only. Reset: 0. `DWC_USB31_MAC_PHY_CLKS_SYNC
25:24	ghwparams1_25_24 . Read-only. Reset: 2h. `DWC_USB31_EN_PWROPT
23	ghwparams1_23 . Read-only. Reset: 0. `DWC_USB31_SPRAM_TYP
22:17	ghwparams1_22_17 . Read-only. Reset: 01h. `DWC_USB31_DEVICE_NUM_INT
16:14	ghwparams1_16_14 . Read-only. Reset: 4h. `DWC_USB31_ASPACEWIDTH
13:11	ghwparams1_13_11 . Read-only. Reset: 4h. `DWC_USB31_REQINFOWIDTH
10:8	ghwparams1_10_8 . Read-only. Reset: 4h. `DWC_USB31_DATAINFOWIDTH
7:4	ghwparams1_7_4 . Read-only. Reset: 8h. `DWC_USB31_BURSTWIDTH
3:0	ghwparams1_3_0 . Read-only. Reset: 6h. `DWC_USB31_IDWIDTH

USB[0...1]x00C148 (FCHUSBDCUSB31CNTR0::GHWPARAMS2)

Read-only. Reset: 0000_1022h.

Global Hardware Parameters Register 2

This register contains the hardware configuration options.

_instUSB0_aliasHOST; USB0x00C148; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00C148; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:0	ghwparams2_31_0 . Read-only. Reset: 0000_1022h. `DWC_USB31_USERID

USB[0...1]x00C14C (FCHUSBDCUSB31CNTR0::GHWPARAMS3)

Read-only. Reset: 0410_8086h.

Global Hardware Parameters Register 3

This register contains the hardware configuration options.

_instUSB0_aliasHOST; USB0x00C14C; USB0={NBIFEPFNCFG::BASE_ADDR_2 dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1 dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C14C; USB1={NBIFEPFNCFG::BASE_ADDR_2 dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1 dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	ghwparams3_31. Read-only. Reset: 0. Reserved
30:23	ghwparams3_30_23. Read-only. Reset: 08h. `DWC_USB31_CACHE_TOTAL_XFER_RESOURCES
22:18	ghwparams3_22_18. Read-only. Reset: 04h. `DWC_USB31_NUM_IN_EPS
17:12	ghwparams3_17_12. Read-only. Reset: 08h. `DWC_USB31_NUM_EPS
11	ghwparams3_11. Read-only. Reset: 0. Reserved
10	ghwparams3_10. Read-only. Reset: 0. DWC_USB31_VENDOR_CTL_INTERFACE
9:8	ghwparams3_9_8. Read-only. Reset: 0h. Reserved
7:6	ghwparams3_7_6. Read-only. Reset: 2h. `DWC_USB31_HSPHY_DWIDTH
5:4	ghwparams3_5_4. Read-only. Reset: 0h. Reserved
3:2	ghwparams3_3_2. Read-only. Reset: 1h. `DWC_USB31_HSPHY_INTERFACE
1:0	ghwparams3_1_0. Read-only. Reset: 2h. `DWC_USB31_SSPHY_INTERFACE

USB[0...1]x00C150 (FCHUSBDCUSB31CNTR0::GHWPARAMS4)

Read-only. Reset: 4844_2010h.

Global Hardware Parameters Register 4

This register contains the hardware configuration options.

_instUSB0_aliasHOST; USB0x00C150; USB0={NBIFEPFNCFG::BASE_ADDR_2 dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1 dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C150; USB1={NBIFEPFNCFG::BASE_ADDR_2 dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1 dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:28	ghwparams4_31_28. Read-only. Reset: 4h. `DWC_USB31_BMU_LSP_DEPTH
27:23	ghwparams4_27_23. Read-only. Reset: 10h. `DWC_USB31_BMU_PTL_DEPTH
22	ghwparams4_22. Read-only. Reset: 1. `DWC_USB31_EN_ISOC_SUPT
21	ghwparams4_21. Read-only. Reset: 0. `DWC_USB31_EXT_BUFF_CONTROL
20:17	ghwparams4_20_17. Read-only. Reset: 2h. `DWC_USB31_NUM_ESS_USB_INSTANCES
16:13	ghwparams4_16_13. Read-only. Reset: 1h. Description: `DWC_USB31_HIBER_SCRATCHBUFS This field is the number of external scratchpad buffers the controller requires to save its internal state in device mode. Each buffer is assumed to be 4KB. The scratchpad buffer array must have this many buffer pointers.
12:6	ghwparams4_12_6. Read-only. Reset: 00h. Reserved
5:0	ghwparams4_5_0. Read-only. Reset: 10h. `DWC_USB31_CACHE_TRBS_PER_TRANSFER

USB[0...1]x00C154 (FCHUSBDCUSB31CNTR0::GHWPARAMS5)

Read-only. Reset: 643C_8210h.

Global Hardware Parameters Register 5

This register contains the hardware configuration options.

_instUSB0_aliasHOST; USB0x00C154; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00C154; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:29	ghwparams5_31_29 . Read-only. Reset: 3h. `DWC_USB31_NUM_RAMs
28:23	ghwparams5_28_23 . Read-only. Reset: 08h. `DWC_USB31_DFQ_FIFO_DEPTH
22:17	ghwparams5_22_17 . Read-only. Reset: 1Eh. `DWC_USB31_DWQ_FIFO_DEPTH
16:11	ghwparams5_16_11 . Read-only. Reset: 10h. `DWC_USB31_TXQ_FIFO_DEPTH
10:5	ghwparams5_10_5 . Read-only. Reset: 10h. `DWC_USB31_RXQ_FIFO_DEPTH
4:0	ghwparams5_4_0 . Read-only. Reset: 10h. `DWC_USB31_BMU_BUSGM_DEPTH

USB[0...1]x00C158 (FCHUSBDCUSB31CNTR0::GHWPARAMS6)

Read-only. Reset: 1C00_003Fh.

Global Hardware Parameters Register 6

This register contains the hardware configuration options.

_instUSB0_aliasHOST; USB0x00C158; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00C158; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	ghwparams6_31_16 . Read-only. Reset: 1C00h. `DWC_USB31_RAM0_DEPTH
15	BusFltrsSupport . Read-only. Reset: 0. `DWC_USB31_EN_BUS_FILTERS
14:10	ghwparams6_14_10 . Read-only. Reset: 00h. Reserved
9:8	ghwparams6_9_8 . Read-only. Reset: 0h. Reserved
7	ghwparams6_7 . Read-only. Reset: 0. `DWC_USB31_EN_FPGA
6	ghwparams6_6 . Read-only. Reset: 0. `DWC_USB31_EN_DBG_PORTS
5:0	ghwparams6_5_0 . Read-only. Reset: 3Fh. `DWC_USB31_PSQ_FIFO_DEPTH

USB[0...1]x00C15C (FCHUSBDCUSB31CNTR0::GHWPARAMS7)

Read-only. Reset: 1E00_1E00h.

Global Hardware Parameters Register 7

This register contains the hardware configuration options.

_instUSB0_aliasHOST; USB0x00C15C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00C15C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	ghwparams7_31_16 . Read-only. Reset: 1E00h. `DWC_USB31_RAM2_DEPTH
15:0	ghwparams7_15_0 . Read-only. Reset: 1E00h. `DWC_USB31_RAM1_DEPTH

USB[0...1]x00C160 (FCHUSBWDCUSB31CNTR0::GDBGFIFOSPACE)

Global Debug Queue/FIFO Space Available Register

_instUSB0_aliasHOST; USB0x00C160; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C160; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	SPACE_AVAILABLE. Read-only. Reset: 00A1h. SPACE_AVAILABLE
15:9	Reserved.
8:0	FIFO_QUEUE_SELECT. Read-write. Reset: 000h. Description: FIFO/Queue Select - FIFO/Queue Select[8:5] indicates the FIFO/Queue Type - FIFO/Queue Select[4:0] indicates the FIFO/Queue Number For example, 9'b0_0010_0001 refers to RxFIFO_1 and 9'b0_0101_1110 refers to TxReqQ_30. - 9'b0_0001_1111 to 9'b0_0000_0000: TxFIFO_31 to TxFIFO_0 - 9'b0_0011_1111 to 9'b0_0010_0000: RxFIFO_31 to RxFIFO_0 - 9'b0_0101_1111 to 9'b0_0100_0000: TxReqQ_31 to TxReqQ_0 - 9'b0_0111_1111 to 9'b0_0110_0000: RxReqQ_31 to RxReqQ_0 - 9'b0_1001_1111 to 9'b0_1000_0000: RxInfoQ_31 to RxInfoQ_0 - 9'b0_1010_0000: DescFetchQ_0 (for backwards compatibility) - 9'b0_1010_0001: EventQ_0 (for backwards compatibility) - 9'b0_1010_0010: ProtocolStatusQ_0 - 9'b0_1101_1111 to 9'b0_1110_0000: DescFetchQ_31 to DescFetchQ_0 - 9'b0_1111_1111 to 9'b0_1110_0000: WriteBack/EventQ_31 to WriteBack/EventQ_0 - 9'b1_0000_0111 to 9'b1_0000_0000: AuxEventQ_7 to AuxEventQ_0 (if EN_SEPARATE_DESC_QUEUES=1)

USB[0...1]x00C164 (FCHUSBWDCUSB31CNTR0::GBMUCTL)

Global BMU Control Register

_instUSB0_aliasHOST; USB0x00C164; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C164; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:6	Reserved.
5	separate_psq_en. Read-write. Reset: 1. Description: When DWC_USB31_EN_SEPARATE_PSQ_PER_DIR is enabled Separate internal process queue and state machine per direction is enabled when the bit is set to '1', BMU will process the plr_msg_type[PSQ_DIR] bit and pushes the messages to separate PSQ based on direction including special handling of control endpoint messages. when the bit is set to '0', BMU will ignore the plr_msg_type[PSQ_DIR] bit for all message types and pushes the messages to single PSQ to retain the legacy mode of operation.
4:2	Reserved.
1	active_id_en. Read-write. Reset: 0. Active Id enabled
0	Reserved.

USB[0...1]x00C16C (FCHUSBWCUSB31CNTR0::GDBGBMU)

Read-only.

Global Debug BMU Register

_instUSB0_aliasHOST; USB0x00C16C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C16C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:8	BMU_BCU. Read-only. BMU_BCU Debug information
7:4	BMU_DCU. Read-only. BMU_DCU Debug information
3:0	BMU_CCU. Read-only. BMU_CCU Debug information

USB[0...1]x00C170 (FCHUSBWCUSB31CNTR0::GDBGLSPMUX)

Global Debug LSP MUX Register in host mode

This register is for internal use only.

_instUSB0_aliasHOST; USB0x00C170; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C170; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	Reserved.
23:16	logic_analyzer_trace. Read-write. Description: logic_analyzer_trace Port MUX Select For details on how the mux controls the debug traces, refer to the "assign logic_analyzer_trace =" code section in the DWC_usb31.v file. A value of 6'h3F drives 0s on the logic_analyzer_trace signal. If you plan to OR (instead using a mux) this signal with other trace signals in your system to generate a common trace signal, you can use this feature.
15	Reserved.
14:0	LSPSELECT. Read-write. Description: LSP Select In host mode: - [14:0] - Selects the LSP debug information presented in the GDBGGLSP register. In device mode: - [3:0] - Device Endpoint Select (EPSELECT) Selects the Endpoint debug information presented in the GDBGEPINFO registers in device mode - [7:4] - Device LSP Select (DEVSELECT) Selects the LSP debug information presented in the GDBGGLSP register - [14:8] - Host LSP Select Selects the LSP debug information presented in the GDBGGLSP register

USB[0...1]x00C174 (FCHUSBDWCUSB31CNTR0::GDBG LSP)

Read-only.

Global Debug LSP Register

This register is for internal use only.

_instUSB0_aliasHOST; USB0x00C174; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C174; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
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31:0	LSPDEBUG. Read-only. LSP Debug Information
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USB[0...1]x00C178 (FCHUSBDWCUSB31CNTR0::GDBGEPINFO0)

Read-only.

Global Debug Endpoint Information Register 0

This register is for internal use only.

_instUSB0_aliasHOST; USB0x00C178; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C178; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
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31:0	EPDEBUG. Read-only. Endpoint Debug Information
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USB[0...1]x00C17C (FCHUSBDWCUSB31CNTR0::GDBGEPINFO1)

Read-only.

Global Debug Endpoint Information Register 1

This register is for internal use only.

_instUSB0_aliasHOST; USB0x00C17C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C17C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
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31:0	EPDEBUG. Read-only. Endpoint Debug Information, bits[63:32]
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USB[0...1]x00C180 (FCHUSBWCUSB31CNTR0::GPRTBIMAP_HSLO)

Reset: 0000_0000h.

Global High-Speed Port to Bus Instance Mapping Register - Low

This register specifies the High Speed USB instance number to which each USB 3.1 port is connected. The software can program this register to specify how USB 3.1 ports are connected to HS USB instances.

The Reset Value for each field in the GPRTBIMAP_HS register, which is {GPRTBIMAP_HSHI,GPRTBIMAP_HSLO} register is calculated as shown below:

for (i=0; i<`DWC_USB31_HOST_NUM_U2_ROOT_PORTS; i=i+1)

GPRTBIMAP_HS[4*i+3:4*i] = i%`DWC_USB31_NUM_HS_USB_INSTANCES;

Note:

- Register fields are read-write with respect to number of port instantiated.
- For ex. If DWC_USB31_HOST_NUM_U2_ROOT_PORTS==3, then GPRTBIMAP_HSLO[11:0] i.e. fields (BINUM1/BINUM2/BINUM3) are read-write and all remaining fields are read-only.

_instUSB0_aliasHOST; USB0x00C180; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C180; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:28	BINUM8. Read-only. Reset: 0h. Description: BINUM8: HS USB Instance Number for Port 8. Application-programmable ID field.
27:24	BINUM7. Read-only. Reset: 0h. Description: BINUM7: HS USB Instance Number for Port 7. Application-programmable ID field.
23:20	BINUM6. Read-only. Reset: 0h. Description: BINUM6 USB Instance Number for Port 6. Application-programmable ID field.
19:16	BINUM5. Read-only. Reset: 0h. Description: BINUM5: HS USB Instance Number for Port 5. Application-programmable ID field.
15:12	BINUM4. Read-only. Reset: 0h. Description: BINUM4: HS USB Instance Number for Port 4. Application-programmable ID field.
11:8	BINUM3. Read-only. Reset: 0h. Description: BINUM3: HS USB Instance Number for Port 3. Application-programmable ID field.
7:4	BINUM2. Read-write. Reset: 0h. Description: BINUM2: HS USB Instance Number for Port 2. Application-programmable ID field.
3:0	BINUM1. Read-write. Reset: 0h. Description: BINUM1: HS USB Instance Number for Port 1. Application-programmable ID field.

USB[0...1]x00C184 (FCHUSBWCUSB31CNTR0::GPRTBIMAP_HSHI)

Read-only.

Global High-Speed Port to Bus Instance Mapping Register - High

This register specifies the High Speed USB instance number to which each USB 3.1 port is connected. The software can program this register to specify how USB 3.1 ports are connected to HS USB instances.

The Reset Value for each field in the GPRTBIMAP_HS register is calculated as shown below:

for (i=0; i<`DWC_USB31_HOST_NUM_U2_ROOT_PORTS; i=i+1)

GPRTBIMAP_HS[4*i+3:4*i] = i%`DWC_USB31_NUM_HS_USB_INSTANCES;

Note:

- Register fields are read-write with respect to number of port instantiated.
- For ex. If DWC_USB31_HOST_NUM_U2_ROOT_PORTS==3, then GPRTBIMAP_HSHI[11:0] i.e. fields (BINUM9/BINUM10/BINUM11) are read-write and all remaining fields are read-only.

_instUSB0_aliasHOST; USB0x00C184; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C184; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:28	Reserved.
27:24	BINUM15. Read-only. Reset: 0h. Description: BINUM15: HS USB Instance Number for Port 15. Application-programmable ID field.
23:20	BINUM14. Read-only. Reset: 0h. Description: BINUM14: HS USB Instance Number for Port 14. Application-programmable ID field.
19:16	BINUM13. Read-only. Reset: 0h. Description: BINUM13: HS USB Instance Number for Port 13. Application-programmable ID field.
15:12	BINUM12. Read-only. Reset: 0h. Description: BINUM12: HS USB Instance Number for Port 12. Application-programmable ID field.
11:8	BINUM11. Read-only. Reset: 0h. Description: BINUM11: HS USB Instance Number for 11. Application-programmable ID field.
7:4	BINUM10. Read-only. Reset: 0h. Description: BINUM10: HS USB Instance Number for Port 10. Application-programmable ID field.
3:0	BINUM9. Read-only. Reset: 0h. Description: BINUM9: HS USB Instance Number for Port 9. Application-programmable ID field.

USB[0...1]x00C188 (FCHUSBWCUSB31CNTR0::GPRTBIMAP_FSLO)

Reset: 0000_0000h.

Global Full/Low-Speed Port to Bus Instance Mapping Register - Low

This register specifies the Full/Low Speed USB instance number to which each USB 3.1 port is connected. Software can program this register to specify how USB 3.1 ports are connected to FS/LS USB instances

The Reset Value for each field in the GPRTBIMAP_FS register, which is {GPRTBIMAP_FSHI,GPRTBIMAP_FSLO} register is calculated as shown below:

for (i=0; i<`DWC_USB31_HOST_NUM_U2_ROOT_PORTS; i=i+1)

GPRTBIMAP_FS[4*i+3:4*i] = i%`DWC_USB31_NUM_FSLS_USB_INSTANCES;

Note:

- Register fields are read-write with respect to number of port instantiated.

- For ex. If DWC_USB31_HOST_NUM_U2_ROOT_PORTS==3, then GPRTBIMAP_FSLO[11:0] i.e. fields (BINUM1/BINUM2/BINUM3) are read-write and all remaining fields are read-only.

_instUSB0_aliasHOST; USB0x00C188; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C188; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:28	BINUM8. Read-only. Reset: 0h. Description: BINUM8: FS/LS USB Instance Number for Port 8. Application-programmable ID field.
27:24	BINUM7. Read-only. Reset: 0h. Description: BINUM7: FS/LS USB Instance Number for Port 7. Application-programmable ID field.
23:20	BINUM6. Read-only. Reset: 0h. Description: BINUM6: FS/LS USB Instance Number for Port 6. Application-programmable ID field.
19:16	BINUM5. Read-only. Reset: 0h. Description: BINUM5: FS/LS USB Instance Number for Port 5. Application-programmable ID field.
15:12	BINUM4. Read-only. Reset: 0h. Description: BINUM4: FS/LS USB Instance Number for Port 4. Application-programmable ID field.
11:8	BINUM3. Read-only. Reset: 0h. Description: BINUM3: FS/LS USB Instance Number for Port 3. Application-programmable ID field.
7:4	BINUM2. Read-write. Reset: 0h. Description: BINUM2: FS/LS USB Instance Number for Port 2. Application-programmable ID field.
3:0	BINUM1. Read-write. Reset: 0h.

	Description: BINUM1: FS/LS USB Instance Number for Port 1.
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	Application-programmable ID field.
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USB[0...1]x00C18C (FCHUSBDCUSB31CNTR0::GPRTBIMAP_FSHI)

Read-only.

Global Full/Low-Speed Port to Bus Instance Mapping Register - High

This register specifies the Full/Low Speed USB instance number to which each USB 3.1 port is connected. Software can program this register to specify how USB 3.1 ports are connected to FS/LS USB instances

The Reset Value for each field in the GPRTBIMAP_FS register, which is {GPRTBIMAP_FSHI, GPRTBIMAP_FSLO} register is calculated as shown below:

for (i=0; i<`DWC_USB31_HOST_NUM_U2_ROOT_PORTS; i=i+1)

GPRTBIMAP_FS[4*i+3:4*i] = i%`DWC_USB31_NUM_FSLS_USB_INSTANCES;

Note:

- Register fields are read-write with respect to number of port instantiated.
- For ex. If DWC_USB31_HOST_NUM_U2_ROOT_PORTS==3, then GPRTBIMAP_FSHI[11:0] i.e. fields (BINUM9/BINUM10/BINUM11) are read-write and all remaining fields are read-only.

_instUSB0_aliasHOST; USB0x00C18C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C18C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:28	Reserved.
27:24	BINUM15. Read-only. Reset: 0h. Description: BINUM15: FS/LS USB Instance Number for Port 15. Application-programmable ID field
23:20	BINUM14. Read-only. Reset: 0h. Description: BINUM14: FS/LS USB Instance Number for Port 14. Application-programmable ID field
19:16	BINUM13. Read-only. Reset: 0h. Description: BINUM13: FS/LS USB Instance Number for Port 13. Application-programmable ID field
15:12	BINUM12. Read-only. Reset: 0h. Description: BINUM12: FS/LS USB Instance Number for Port 12. Application-programmable ID field
11:8	BINUM11. Read-only. Reset: 0h. Description: BINUM11: FS/LS USB Instance Number for Port 11. Application-programmable ID field
7:4	BINUM10. Read-only. Reset: 0h. Description: BINUM10: FS/LS USB Instance Number for Port 10. Application-programmable ID field
3:0	BINUM9. Read-only. Reset: 0h. Description: BINUM9: FS/LS USB Instance Number for Port 9. Application-programmable ID field.

USB[0...1]x00C190 (FCHUSBDWCUSB31CNTR0::GHMSOCBWOR)

Read-write. Reset: 0000_0000h.

Global Host Mode SoC Bandwidth Override Register

_instUSB0_aliasHOST; USB0x00C190; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C190; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	ovrd_port_soc_bw. Read-write. Reset: 0. Description: Overrides the SoC bandwidth port values When this field is set 1, the ovrd_soc_common_rd_wr_bus, ovrd_soc_rd_uF_kB_bandwidth, and ovrd_soc_wr_uF_kB_bandwidth register values are used instead common_soc_rd_wr, soc_rd_uF_kB_bandwidth, and soc_wr_uF_kB_bandwidth input ports.
30	ovrd_common_soc_rd_wr. Read-write. Reset: 0. Description: Override value for the common_soc_rd_wr port - 1'b0: Separate SoC Bus for Read and Write - 1'b1: Common SoC Bus for Read and Write
29:15	ovrd_soc_wr_uF_kB_bandwidth. Read-write. Reset: 0000h. Description: SoC Bus Write Bandwidth in KiloBytes/Micro-Frame Override value for the soc_wr_uF_kB_bandwidth port
14:0	ovrd_soc_rd_uF_kB_bandwidth. Read-write. Reset: 0000h. Description: SoC Bus Read Bandwidth in KiloBytes/Micro-Frame Override value for the soc_rd_uF_kB_bandwidth port

USB[0...1]x00C1A0 (FCHUSBDWCUSB31CNTR0::USB31_VER_NUMBER)

Read-only. Reset: 3139_302Ah.

USB31 IP VERSION

- This register reflects the current corekit release number in ASCII format

_instUSB0_aliasHOST; USB0x00C1A0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C1A0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	VERSION_NUM. Read-only. Reset: 3139_302Ah. Description: VERSION_NUM Indicates the corekit release version number in ASCII format - For example Release number '1.90a' is represented as 0x3139302A The USB31_VER_NUMBER register in conjunction with the USB31_VER_TYPE register identifies the current release. Software uses this register to configure the release-specific features in the driver.

USB[0...1]x00C1A4 (FCHUSBDCUSB31CNTR0::USB31_VER_TYPE)

Read-only. Reset: 6761_2A2Ah.

USB31 IP VERSION TYPE

- This register reflects the current release type of the IP

_instUSB0_aliasHOST; USB0x00C1A4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C1A4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description**31:0** **VERSIONTYPE**. Read-only. Reset: 6761_2A2Ah.**Description:** VERSIONTYPE

Indicated the corekit release version type number in ASCII format

- For example:

Release number '1.80a-lp01' is represented as 0x6C703031. For generally available release, lower 2-bytes of the register is always represented with ASCII value of '*' which is '2A'

The USB31_VER_NUMBER register is in conjunction with the USB31_VER_NUMBER register identifies the current release.

Software uses this register to configure the release-specific features in the driver.

USB[0...1]x00C1B0 (FCHUSBDCUSB31CNTR0::GSYSBLKWINCTRL)

Read-write. Reset: 0000_0000h.

Global System Bus Blocking Window Control

_instUSB0_aliasHOST; USB0x00C1B0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C1B0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	sys_blocking_ok_all_the_time. Read-write. Reset: 0. Always assert ok_to_block_sysbus=1
30	disable_no_blocking_window. Read-write. Reset: 0. Setting this bit does not check uf_start_no_blocking_time and uf_end_no_blocking_time
29	erst_prefetching_en. Read-write. Reset: 0. Description: Enable ERST Entry Prefetching When enabled, the host controller will prefetch one ERST entry into the cache based on the erst_prefetching_watermark. When not enabled, the host controller will prefetch one ERST entry into the cache when the current segment has only 1 TRB left. When LPDDR4 is used, enabling this feature can avoid the corner case in which the LSP is running out of event TRB and fetching of a new ERST entry is blocked by re-training. This corner case may cause QOS issue on ISOC EPs. Enabling this feature may cause the compliant test when the event ring size is changed by the driver.
28:24	erst_prefetching_watermark. Read-write. Reset: 00h. Description: ERST Prefetching Watermark Controls when the host controller will prefetch the next ERST entry into the cache before the current one runs out of TRBs. Only have effect when the erst_prefetching_en set to 1. The maximum valid value is 31 When it is less than 2 (0 or 1), the event ring handler will start the prefetching of the next ERST when the current segment has only 1 TRB left otherwise , the ERST prefetching starts when current the current segment has the number of TRBs less than this register value When LPDDR4 is used, Setting erst_prefetching_en to 1 and setting this register a larger value an avoid the corner case in which the LSP is running out of event TRB but fetching of a new ERST entry is blocked by re-training. This corner case may cause QOS issue on ISOC EPs. Setting it to larger than 15 may cause the compliant test failure on the event ring size change test .
23:16	Reserved.
15:8	end_no_blocking_time. Read-write. Reset: 00h. Description: System bus no blocking time window (in usec) at end of uframe. Valid value: 0 to 63
7:0	beginining_no_blocking_time. Read-write. Reset: 00h. Description: System bus no blocking time window (in usec) at start of uframe. Valid value: 0 to 63

USB[0...1]x00C1B4 (FCHUSBDCUSB31CNTR0::GPCIEL1EXTLAT)Global [PCIe](#) L1 exit Latency Register

_instUSB0_aliasHOST; USB0x00C1B4; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C1B4; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:30	pcie_l1_exit_mode_ctrl . Read-write. Reset: 0h. pcie_l1_exit_mode_ctrl
29:12	Reserved.
11:0	pcie_l1_exit_latency . Read-write. Reset: 000h. Description: pcie_l1_exit_latency This is the time (in usec) the PCIe bus needed to be waked up from L1 and resume normal operation Valid value: 0-125

USB[0...1]x00C200 (FCHUSBDCUSB31CNTR0::GUSB2PHYCFG_REGS0_GUSB2PHYCFG)

Reset: 0010_2540h.

Global USB2 PHY Configuration Register

The application must program this register before starting any transactions on either the SoC bus or the USB.

In Device-only configurations, only one register is needed.

In Host mode, per-port registers are implemented.

_instUSB0_aliasHOST; USB0x00C200; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C200; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	PHYSOFTTRST. Read-write. Reset: 0. Description: UTMI PHY Soft Reset (PHYSoftRst) Causes the usb2phy_reset signal to be asserted to reset a UTMI PHY. Not applicable to ULPI because ULPI PHYs are reset via their FunctionControl.Reset register, and the controller automatically writes to this register when the controller is reset (vcc_reset_n, USBCMD.HCRST, DCTL.SoftReset, or GCTL.SoftReset)
30	Reserved.
29	ULPI_LPM_WITH_OPMODE_CHK. Read-write. Reset: 0. Description: ULPI_LPM_WITH_OPMODE_CHK Support the LPM over ULPI without NOPID token to the ULPI PHY. If this bit is set, the ULPI PHY is expected to qualify the EXT PID with OPMODE=2'b00 for LPM and not treat it as a NOPID. Check with your PHY vendor about your PHY behavior. This bit is valid only when the DWC_USB31_HSPHY_INTERFACE parameter is 2 or 3. - 1'b0: A NOPID is sent before sending an EXTPID for LPM; - 1'b1: An EXTPID is sent without previously sending a NOPID; Note: This bit is valid only in host mode.
28:27	HSIC_CON_WIDTH_ADJ. Read-only. Reset: 0h. Description: HSIC_CON_WIDTH_ADJ This bit is used in the HSIC device mode of operation. It can be used to control the connect duration. The encoding is as follows: - 0: 3 times the strobe period - 1: 4 times the strobe period - 2: 5 times the strobe period - 3: 6 times the strobe period
26	INV_SEL_HSIC. Read-only. Reset: 0.

	Description: INV_SEL_HSIC The application driver uses this bit to control the HSIC enable/disable function. When set to '1', this bit overrides and functionally inverts the "if_select_hsic" input signal. If {INV_SEL_HSIC, if_select_hsic} is: - 00: HSIC Capability is disabled. - 01: HSIC Capability is enabled. - 10: HSIC Capability is enabled. - 11: HSIC Capability is disabled. If the controller operates as non-HSIC-capable, it can only connect to non-HSIC-capable PHYs. If it operates as HSIC-capable, it can connect to HSIC-capable PHYs. This bit is reserved if the DWC_USB31_ENABLE_HSIC parameter is set to '0'. When selecting the HSIC feature, set the host side to HSIC mode first, then set the device mode side. If the device side is set to HSIC mode first and if the host does not see a connection in HSIC mode, then you must de-select the device HSIC mode and select it again using the if_select_hsic setting or register bit GUSB2PHYCFGn[26] to ensure that the device can connect to the host.
25	OVRD_FSLS_DISC_TIME. Read-write. Reset: 0. Description: Overriding the FS/LS disconnect time to 32us. If this value is 0, the FS/LS disconnect time is set to 2.5us as per the USB specification. If this value is non-0, then the disconnect detection time is set to 32us. Normally this value is set to 0. But if the 2.0 PHYs introduce noise on UTMI linestate and cause SE0 glitches longer than 2.5us, then a false disconnect condition may get triggered. To avoid interoperability issues with these PHYs, this bit can be set to 1
24:22	LSTRD. Read-write. Reset: 0h. Description: LS Turnaround Time (LSTRDTIM) This field indicates the value of the Rx-to-Tx packet gap for LS devices. The encoding is as follows: - 0: 2 bit times - 1: 2.5 bit times - 2: 3 bit times - 3: 3.5 bit times - 4: 4 bit times - 5: 4.5 bit times - 6: 5 bit times - 7: 5.5 bit times Note: - For normal operation (to work with most LS devices), it is recommended to set the value of this field to 3'h0 (2 bit times). - The programmable LS device inter-packet gap and turnaround delays are provided to support some legacy LS devices that might require different delays than the default/fixed ones. For instance, the AOpen LS mouse requires 3 bit times of inter-packet gap to work correctly. - Include your PHY delays when programming the LSIPD/LSTRDTIM values. For example, if your PHY TxEndDelay in LS mode is 30 UTMI/ULPI CLKS, then subtract this delay (~1 LS bit time) from the delay requirement of the device.
21:19	LSIPD. Read-write. Reset: 2h.

	Description: LS Inter-Packet Time (LSIPD) This field indicates the value of Tx-to-Tx packet gap for LS devices. The encoding is as follows: - 0: 2 bit times - 1: 2.5 bit times - 2: 3 bit times - 3: 3.5 bit times - 4: 4 bit times - 5: 4.5 bit times - 6: 5 bit times - 7: 5.5 bit times Note: - For normal operation (to work with most LS devices), it is recommended to set this field to 3'h2 (3 bit times). - The programmable LS device inter-packet gap and turnaround delays are provided to support some legacy LS devices that might require different delays than the default/fixed ones. For instance, the AOpen LS mouse requires 3 bit times of inter-packet gap to work correctly. - Include your PHY delays when programming the LSIPD/LSTRDTIM values. For example, if your PHY TxEndDelay in LS mode is 30 UTM/ULPI CLKs, then subtract this delay (~1 LS bit time) from the delay requirement of the device.
18	ULPIEXTVBUSINDIACTOR. Read-write. Reset: 0. Description: ULPI External VBUS Indicator (ULPIExtVbusIndicator) Indicates the ULPI PHY VBUS indicator. - 1'b0: PHY uses an internal VBUS valid comparator. - 1'b1: PHY uses an external VBUS valid comparator. Valid only when parameter DWC_USB31_HSPHY_INTERFACE = 2 or 3
17	ULPIEXTVBUSDRV. Read-write. Reset: 0. Description: ULPI External VBUS Drive (ULPIExtVbusDrv) Selects supply source to drive 5V on VBUS, in the ULPI PHY. - 1'b0: PHY drives VBUS with internal charge pump - 1'b1: PHY drives VBUS with an external supply (Only when parameter DWC_USB31_HSPHY_INTERFACE = 2 or 3)
16	Reserved.
15	ULPIAUIORES. Read-write. Reset: 0. Description: ULPI Auto Resume (ULPIAutoRes) This field is valid for host mode only. Sets the AutoResume bit in Interface Control register on the ULPI PHY. - 1'b0: PHY does not use the AutoResume feature. - 1'b1: PHY uses the AutoResume feature. Valid only when parameter DWC_USB31_HSPHY_INTERFACE = 2 or 3
14	eUSB2OPMODE. Read-write. Reset: 0. Description: eUSB 2.0 UTMI interface enhancement (eUSB2OPMODE) This field is valid in host mode of operation. If this bit is set opmode and utmi_txvalid change on same clock when opmode switches to 2 for test mode operation. - 1'b0: eUSB UTMI interface enhancement disabled. - 1'b1: eUSB UTMI interface enhancement enabled.

13:10	USBTRDTIM. Read-write. Reset: 9h. Description: USB 2.0 Turnaround Time (USBTrdTim) Note: This field is valid only in device mode.
9	XCVRDLY. Read-write. Reset: 0. Description: Transceiver Delay: - This bit is valid only in device mode.
8	ENBLSLPM. Read-write. Reset: 1. Description: Enable utmi_sleep_n and utmi_l1_suspend_n The application uses this bit to control utmi_sleep_n and utmi_l1_suspend_n assertion to the PHY in the L1 state. - 1'b0: utmi_sleep_n and utmi_l1_suspend_n assertion from the controller is not transferred to the external PHY. - 1'b1: utmi_sleep_n and utmi_l1_suspend_n assertion from the controller is transferred to the external PHY. Note: In Device mode - Before issuing any device endpoint command when operating in 2.0 speeds, disable this bit and enable it after the command completes. Without disabling this bit, if a command is issued when the device is in L1 state and if mac2_clk (utmi_clk/ulpi_clk) is gated off, the command will not get completed.
7	PHYSEL. Write-only. Reset: 0. Description: USB 2.0 High-Speed PHY or USB 1.1 Full-Speed Serial Transceiver Select The application uses this bit to select a high-speed PHY or a full-speed transceiver. - 1'b0: USB 2.0 high-speed UTMI+ or ULPI PHY. This bit is always 0, with Write Only access. - 1'b1: USB 1.1 full-speed serial transceiver. This bit is always 1, with Write Only access. If both interface types are selected (that is, parameters' values are not zero), the application uses this bit to select the active interface is active, with Read-Write bit access.
6	SUSPENDUSB20. Read-write. Reset: 1. Description: Suspend USB2.0 HS/FS/LS PHY When set, USB 2.0 PHY enters Suspend mode if Suspend conditions are valid. Note: In Device mode, before issuing any device endpoint command when operating in 2.0 speeds, disable this bit and enable it after the command completes. Without disabling this bit, if a command is issued when the device is in L2 state and if mac2_clk (utmi_clk/ulpi_clk) is gated off, the command will not get completed.
5	FSINTF. Read-only. Reset: 0. Description: Full-Speed Serial Interface Select (FSIntf) The application uses this bit to select a unidirectional or bidirectional USB 1.1 full-speed serial transceiver interface. - 1'b0: 6-pin unidirectional full-speed serial interface. This bit is set to 0 with Read Only access. - 1'b1: 3-pin bidirectional full-speed serial interface. This bit is set to 0 with Read Only access. Note: USB 1.1 full-speed serial interface is not supported. This bit always reads as 1'b0.
4	ULPI_UTMI_Sel. Read-only. Reset: 0. Description: ULPI or UTMI+ Select (ULPI_UTMI_Sel) The application uses this bit to select a UTMI+ or ULPI Interface. - 1'b0: UTMI+ Interface - 1'b1: ULPI Interface This bit is writable only if UTMI+ and ULPI is specified for High-Speed PHY Interface(s) (DWC_USB31_HSPHY_INTERFACE = 3). Otherwise, this bit is read-only and the value depends on the interface selected through DWC_USB31_HSPHY_INTERFACE.
3	PHYIF. Read-write. Reset: 0.

	Description: PHY Interface (PHYIf) If UTMI+ is selected, the application uses this bit to configure the controller to support a UTMI+ PHY with an 8- or 16-bit interface. - 1'b0: 8 bits - 1'b1: 16 bits ULPI Mode: 1'b0 Note: - All the enabled 2.0 ports must have the same clock frequency as Port0 clock frequency (utmi_clk[0]). - The UTMI 8-bit and 16-bit modes cannot be used together for different ports at the same time (that is, all the ports must be in 8-bit mode, or all of them must be in 16-bit mode, at a time). - If any of the USB 2.0 ports is selected as ULPI port for operation, then all the USB 2.0 ports must be operating at 60 MHz.
2:0	TOutCal. Read-write. Reset: 0h. Description: HS/FS Timeout Calibration (TOutCal) The number of PHY clocks, as indicated by the application in this field, is multiplied by a bit-time factor; this factor is added to the high-speed/full-speed interpacket timeout duration in the controller to account for additional delays introduced by the PHY. This may be required, since the delay introduced by the PHY in generating the linestate condition may vary among PHYs. The USB standard timeout value for high-speed operation is 736 to 816 (inclusive) bit times. The USB standard timeout value for full-speed operation is 16 to 18 (inclusive) bit times. The application must program this field based on the speed of connection. The number of bit times added per PHY clock are: High-speed operation: - One 30-MHz PHY clock = 16 bit times - One 60-MHz PHY clock = 8 bit times Full-speed operation: - One 30-MHz PHY clock = 0.4 bit times - One 60-MHz PHY clock = 0.2 bit times - One 48-MHz PHY clock = 0.25 bit times

USB[0...1]x00C204 (FCHUSBWCUSB31CNTR0::GUSB2PHYCFG_REGS1_GUSB2PHYCFG)

Reset: 0010_2540h.

Global USB2 PHY Configuration Register

The application must program this register before starting any transactions on either the SoC bus or the USB.

In Device-only configurations, only one register is needed.

In Host mode, per-port registers are implemented.

_instUSB0_aliasHOST; USB0x00C204; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C204; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	PHYSOFTTRST. Read-write. Reset: 0. Description: UTMI PHY Soft Reset (PHYSoftRst) Causes the usb2phy_reset signal to be asserted to reset a UTMI PHY. Not applicable to ULPI because ULPI PHYs are reset via their FunctionControl.Reset register, and the controller automatically writes to this register when the controller is reset (vcc_reset_n, USBCMD.HCRST, DCTL.SoftReset, or GCTL.SoftReset)
30	Reserved.
29	ULPI_LPM_WITH_OPMODE_CHK. Read-write. Reset: 0. Description: ULPI_LPM_WITH_OPMODE_CHK Support the LPM over ULPI without NOPID token to the ULPI PHY. If this bit is set, the ULPI PHY is expected to qualify the EXT PID with OPMODE=2'b00 for LPM and not treat it as a NOPID. Check with your PHY vendor about your PHY behavior. This bit is valid only when the DWC_USB31_HSPHY_INTERFACE parameter is 2 or 3. - 1'b0: A NOPID is sent before sending an EXTPID for LPM; - 1'b1: An EXTPID is sent without previously sending a NOPID; Note: This bit is valid only in host mode.
28:27	HSIC_CON_WIDTH_ADJ. Read-only. Reset: 0h. Description: HSIC_CON_WIDTH_ADJ This bit is used in the HSIC device mode of operation. It can be used to control the connect duration. The encoding is as follows: - 0: 3 times the strobe period - 1: 4 times the strobe period - 2: 5 times the strobe period - 3: 6 times the strobe period
26	INV_SEL_HSIC. Read-only. Reset: 0.

	Description: INV_SEL_HSIC The application driver uses this bit to control the HSIC enable/disable function. When set to '1', this bit overrides and functionally inverts the "if_select_hsic" input signal. If {INV_SEL_HSIC, if_select_hsic} is: - 00: HSIC Capability is disabled. - 01: HSIC Capability is enabled. - 10: HSIC Capability is enabled. - 11: HSIC Capability is disabled. If the controller operates as non-HSIC-capable, it can only connect to non-HSIC-capable PHYs. If it operates as HSIC-capable, it can connect to HSIC-capable PHYs. This bit is reserved if the DWC_USB31_ENABLE_HSIC parameter is set to '0'. When selecting the HSIC feature, set the host side to HSIC mode first, then set the device mode side. If the device side is set to HSIC mode first and if the host does not see a connection in HSIC mode, then you must de-select the device HSIC mode and select it again using the if_select_hsic setting or register bit GUSB2PHYCFGn[26] to ensure that the device can connect to the host.
25	OVRD_FSLS_DISC_TIME. Read-write. Reset: 0. Description: Overriding the FS/LS disconnect time to 32us. If this value is 0, the FS/LS disconnect time is set to 2.5us as per the USB specification. If this value is non-0, then the disconnect detection time is set to 32us. Normally this value is set to 0. But if the 2.0 PHYs introduce noise on UTMI linestate and cause SE0 glitches longer than 2.5us, then a false disconnect condition may get triggered. To avoid interoperability issues with these PHYs, this bit can be set to 1
24:22	LSTRD. Read-write. Reset: 0h. Description: LS Turnaround Time (LSTRDTIM) This field indicates the value of the Rx-to-Tx packet gap for LS devices. The encoding is as follows: - 0: 2 bit times - 1: 2.5 bit times - 2: 3 bit times - 3: 3.5 bit times - 4: 4 bit times - 5: 4.5 bit times - 6: 5 bit times - 7: 5.5 bit times Note: - For normal operation (to work with most LS devices), it is recommended to set the value of this field to 3'h0 (2 bit times). - The programmable LS device inter-packet gap and turnaround delays are provided to support some legacy LS devices that might require different delays than the default/fixed ones. For instance, the AOpen LS mouse requires 3 bit times of inter-packet gap to work correctly. - Include your PHY delays when programming the LSIPD/LSTRDTIM values. For example, if your PHY TxEndDelay in LS mode is 30 UTMI/ULPI CLKS, then subtract this delay (~1 LS bit time) from the delay requirement of the device.
21:19	LSIPD. Read-write. Reset: 2h.

	Description: LS Inter-Packet Time (LSIPD) This field indicates the value of Tx-to-Tx packet gap for LS devices. The encoding is as follows: - 0: 2 bit times - 1: 2.5 bit times - 2: 3 bit times - 3: 3.5 bit times - 4: 4 bit times - 5: 4.5 bit times - 6: 5 bit times - 7: 5.5 bit times Note: - For normal operation (to work with most LS devices), it is recommended to set this field to 3'h2 (3 bit times). - The programmable LS device inter-packet gap and turnaround delays are provided to support some legacy LS devices that might require different delays than the default/fixed ones. For instance, the AOpen LS mouse requires 3 bit times of inter-packet gap to work correctly. - Include your PHY delays when programming the LSIPD/LSTRDTIM values. For example, if your PHY TxEndDelay in LS mode is 30 UTM/ULPI CLKs, then subtract this delay (~1 LS bit time) from the delay requirement of the device.
18	ULPIEXTVBUSINDIACTOR. Read-write. Reset: 0. Description: ULPI External VBUS Indicator (ULPIExtVbusIndicator) Indicates the ULPI PHY VBUS indicator. - 1'b0: PHY uses an internal VBUS valid comparator. - 1'b1: PHY uses an external VBUS valid comparator. Valid only when parameter DWC_USB31_HSPHY_INTERFACE = 2 or 3
17	ULPIEXTVBUSDRV. Read-write. Reset: 0. Description: ULPI External VBUS Drive (ULPIExtVbusDrv) Selects supply source to drive 5V on VBUS, in the ULPI PHY. - 1'b0: PHY drives VBUS with internal charge pump - 1'b1: PHY drives VBUS with an external supply (Only when parameter DWC_USB31_HSPHY_INTERFACE = 2 or 3)
16	Reserved.
15	ULPIAUTORES. Read-write. Reset: 0. Description: ULPI Auto Resume (ULPIAutoRes) This field is valid for host mode only. Sets the AutoResume bit in Interface Control register on the ULPI PHY. - 1'b0: PHY does not use the AutoResume feature. - 1'b1: PHY uses the AutoResume feature. Valid only when parameter DWC_USB31_HSPHY_INTERFACE = 2 or 3
14	eUSB2OPMODE. Read-write. Reset: 0. Description: eUSB 2.0 UTMI interface enhancement (eUSB2OPMODE) This field is valid in host mode of operation. If this bit is set opmode and utmi_txvalid change on same clock when opmode switches to 2 for test mode operation. - 1'b0: eUSB UTMI interface enhancement disabled. - 1'b1: eUSB UTMI interface enhancement enabled.

13:10	USBTRDTIM. Read-write. Reset: 9h. Description: USB 2.0 Turnaround Time (USBTrdTim) Note: This field is valid only in device mode.
9	XCVRDLY. Read-write. Reset: 0. Description: Transceiver Delay: - This bit is valid only in device mode.
8	ENBLSLPM. Read-write. Reset: 1. Description: Enable utmi_sleep_n and utmi_l1_suspend_n The application uses this bit to control utmi_sleep_n and utmi_l1_suspend_n assertion to the PHY in the L1 state. - 1'b0: utmi_sleep_n and utmi_l1_suspend_n assertion from the controller is not transferred to the external PHY. - 1'b1: utmi_sleep_n and utmi_l1_suspend_n assertion from the controller is transferred to the external PHY. Note: In Device mode - Before issuing any device endpoint command when operating in 2.0 speeds, disable this bit and enable it after the command completes. Without disabling this bit, if a command is issued when the device is in L1 state and if mac2_clk (utmi_clk/ulpi_clk) is gated off, the command will not get completed.
7	PHYSEL. Write-only. Reset: 0. Description: USB 2.0 High-Speed PHY or USB 1.1 Full-Speed Serial Transceiver Select The application uses this bit to select a high-speed PHY or a full-speed transceiver. - 1'b0: USB 2.0 high-speed UTMI+ or ULPI PHY. This bit is always 0, with Write Only access. - 1'b1: USB 1.1 full-speed serial transceiver. This bit is always 1, with Write Only access. If both interface types are selected (that is, parameters' values are not zero), the application uses this bit to select the active interface is active, with Read-Write bit access.
6	SUSPENDUSB20. Read-write. Reset: 1. Description: Suspend USB2.0 HS/FS/LS PHY When set, USB 2.0 PHY enters Suspend mode if Suspend conditions are valid. Note: In Device mode, before issuing any device endpoint command when operating in 2.0 speeds, disable this bit and enable it after the command completes. Without disabling this bit, if a command is issued when the device is in L2 state and if mac2_clk (utmi_clk/ulpi_clk) is gated off, the command will not get completed.
5	FSINTF. Read-only. Reset: 0. Description: Full-Speed Serial Interface Select (FSIntf) The application uses this bit to select a unidirectional or bidirectional USB 1.1 full-speed serial transceiver interface. - 1'b0: 6-pin unidirectional full-speed serial interface. This bit is set to 0 with Read Only access. - 1'b1: 3-pin bidirectional full-speed serial interface. This bit is set to 0 with Read Only access. Note: USB 1.1 full-speed serial interface is not supported. This bit always reads as 1'b0.
4	ULPI_UTMI_Sel. Read-only. Reset: 0. Description: ULPI or UTMI+ Select (ULPI_UTMI_Sel) The application uses this bit to select a UTMI+ or ULPI Interface. - 1'b0: UTMI+ Interface - 1'b1: ULPI Interface This bit is writable only if UTMI+ and ULPI is specified for High-Speed PHY Interface(s) (DWC_USB31_HSPHY_INTERFACE = 3). Otherwise, this bit is read-only and the value depends on the interface selected through DWC_USB31_HSPHY_INTERFACE.
3	PHYIF. Read-write. Reset: 0.

	Description: PHY Interface (PHYIf) If UTMI+ is selected, the application uses this bit to configure the controller to support a UTMI+ PHY with an 8- or 16-bit interface. - 1'b0: 8 bits - 1'b1: 16 bits ULPI Mode: 1'b0 Note: - All the enabled 2.0 ports must have the same clock frequency as Port0 clock frequency (utmi_clk[0]). - The UTMI 8-bit and 16-bit modes cannot be used together for different ports at the same time (that is, all the ports must be in 8-bit mode, or all of them must be in 16-bit mode, at a time). - If any of the USB 2.0 ports is selected as ULPI port for operation, then all the USB 2.0 ports must be operating at 60 MHz.
2:0	TOutCal. Read-write. Reset: 0h. Description: HS/FS Timeout Calibration (TOutCal) The number of PHY clocks, as indicated by the application in this field, is multiplied by a bit-time factor; this factor is added to the high-speed/full-speed interpacket timeout duration in the controller to account for additional delays introduced by the PHY. This may be required, since the delay introduced by the PHY in generating the linestate condition may vary among PHYs. The USB standard timeout value for high-speed operation is 736 to 816 (inclusive) bit times. The USB standard timeout value for full-speed operation is 16 to 18 (inclusive) bit times. The application must program this field based on the speed of connection. The number of bit times added per PHY clock are: High-speed operation: - One 30-MHz PHY clock = 16 bit times - One 60-MHz PHY clock = 8 bit times Full-speed operation: - One 30-MHz PHY clock = 0.4 bit times - One 60-MHz PHY clock = 0.2 bit times - One 48-MHz PHY clock = 0.25 bit times

USB[0...1]x00C2C0 (FCHUSBWDCUSB31CNTR0::GUSB3PIPECTL_REGS0_GUSB3PIPECTL)

Reset: 0102_0102h.

Global USB 3.1 PIPE Control Register

The application uses this register to configure the USB 3.1 PHY and PIPE interface.

Device-only configuration requires only one register. In Host mode, registers are implemented for each port.

Note:

- GUSB3PIPECTLn registers are not applicable for USB 2.0-only mode.

_instUSB0_aliasHOST; USB0x00C2C0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C2C0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31	PHYSoftRst. Read-write. Reset: 0. Description: USB 3.1 PHY Soft Reset PHY soft-reset; to issue PHY reset, software should set this bit and reset this bit after meeting PHY reset timing.
30	HstPrtCmpl. Read-write. Reset: 0. Description: HstPrtCmpl This feature tests the PIPE PHY compliance patterns without having to have a test fixture on the USB 3.1 cable. This bit enables placing the SS port link into a compliance state. In compliance lab testing, the SS port link enters compliance after failing the first polling sequence after power on. Set this bit to 0, when you run compliance tests. The sequence for using this functionality is as follows: - 1. Disconnect any plugged in devices. - 2. Perform USBCMD.HCRST or power-on-chip reset. - 3. Set PORTSC.PP=0. - 4. Set GUSB3PIPECTL.HstPrtCmpl=1. This places the link into compliance state. To advance the compliance pattern, follow this sequence (toggle the set GUSB3PIPECTL.HstPrtCmpl): - 1. Set GUSB3PIPECTL.HstPrtCmpl=0. - 2. Set GUSB3PIPECTL.HstPrtCmpl=1. This advances the link to the next compliance pattern. To exit from the compliance state perform USBCMD.HCRST or power-on-chip reset.
29	U2P3ok. Read-write. Reset: 0. Description: Enable P3 entry during U2/SSInactive (U2P3ok). This is recommended only if your PHY's P3 exit time is close to P2 exit time. Putting the ESS PHY in P3 during U2 would affect ESS Asynchronous endpoint performance and also this could prevent U2 entry if there are ESS periodic endpoints and their BInterval is in the sub milli-second range. rammer's Guide for more details. - 0: During link state U2/ESS.Inactive, put PHY in P2 (Default) - 1: During link state U2/ESS.Inactive, put PHY in P3 (Not recommended). Note: For a port, if GUSB3PIPECTL[7]=1 and GUSB3PIPECTL[29]=1, set GUSB3PIPECTL[11] to 1.
28	DisRxDetP3. Read-write. Reset: 0.

	Description: Disabled receiver detection in P3 (DisRxDetP3) - 0: If PHY is in P3 and controller needs to perform receiver detection, The controller performs receiver detection in P3. - 1: If PHY is in P3 and controller needs to perform receiver detection, The controller changes the PHY power state to P2 and then performs receiver detection. After receiver detection, the cores changes PHY power state to P3.
27	Ux_exit_in_Px. Read-write. Reset: 0. Description: Ux Exit in Px (Ux_exit_in_Px) - 0: The controller does U1/U2/U3 exit in PHY power state P0. - 1: The controller does U1/U2/U3 exit in PHY power state P1/P2/P3 respectively. This bit is added for SS PHY workaround where SS PHY injects a glitch on pipe_RxElecIdle while receiving Ux exit LFPS, and pipe_PowerDown change is in progress. Note: This bit is used by third-party SS PHY.
26	ping_enhancement_en. Read-write. Reset: 0. Description: Ping Enhancement Enable (ping_enhancement_en) When set, the Downstream port U1 ping receive timeout becomes 500 ms instead of 300 ms. Minimum Ping.LFPS receive duration is 8 ns (one mac31_clk). This field is valid for the downstream port only. Note: This bit is used by third-party SS PHY.
25	u1u2exitfail_to_recov. Read-write. Reset: 0. Description: U1U2exitfail to Recovery (u1u2exitfail_to_recov) When set, and U1/U2 LFPS handshake fails, the LTSSM transitions from U1/U2 to Recovery instead of SS Inactive. If Recovery fails, then the LTSSM can enter SS.Inactive. This is an enhancement only. It prevents interoperability issue if the remote link does not do proper handshake.
24	request_p1p2p3. Read-write. Reset: 1. Description: Always Request P1/P2/P3 for U1/U2/U3 (request_p1p2p3) When set, the controller always requests PHY power change from P0 to P1/P2/P3 during U0 to U1/U2/U3 transition. If this bit is 0, and immediate Ux exit (remotely initiated, or locally initiated) happens, the controller does not request P1/P2/P3 power state change. Note: For third-party SS PHY, check with your PHY vendor.
23	StartRxDetU3RxDet. Write-only, Volatile. Reset: 0. Description: Start Receiver Detection in U3/Rx.Detect (StartRxdetU3RxDet) If DWC_USB31_GUSB3PIPECTL_INIT[22] is set, and the link is in either U3 or Rx.Detect state, the controller starts receiver detection on the rising edge of this bit. This can only be used for Downstream ports. This bit must be set to '0' for Upstream ports. This feature must not be enabled for normal operation.
22	DisRxDetU3RxDet. Read-write. Reset: 0. Description: Disable Receiver Detection in U3/Rx.Det When set, the controller does not handle receiver detection in either U3 or Rx.Detect states. DWC_USB31_GUSB3PIPECTL_INIT[23] must be used to start receiver detection manually. This bit can only be used for the downstream port. This bit must be set to "0" for Upstream ports. This feature must not be enabled for normal operation.
21:19	DelayP1P2P3. Read-write. Reset: 0h.

	Description: Delay P1P2P3 Delay P0 to P1/P2/P3 request when entering U1/U2/U3 until (DWC_USB31_GUSB3PIPECTL_INIT[21:19]*8) 8B10B error occurs, or Pipe3_RxValid drops to 0. DWC_USB31_GUSB3PIPECTL_INIT[18] must be 1 to enable this functionality.
18	DELAYP1TRANS. Read-write. Reset: 0. Description: Delay PHY power change from P0 to P1/P2/P3 when link state changing from U0 to U1/U2/U3 respectively. - 1'b1: When entering U1/U2/U3, delay the transition to P1/P2/P3 until the pipe signals, pipe_RxElecIdle is 1 and pipe_RxValid is 0 - 1'b0: When entering U1/U2/U3, transition to P1/P2/P3 without checking for pipe_RxElecIdle and pipe_RxValid.
17	SUSPENDENABLE. Read-write. Reset: 1. Description: Suspend USB 3.1 ESS PHY (Suspend_en) When set, and if Suspend conditions are valid, the USB 3.1 PHY enters Suspend mode. For DRD configurations, it is recommended that this bit is set to '0'. If it is set to '1', then the application must clear this bit after power-on reset. Application needs to set it to '1' after the controller initialization is completed. If software is not going to change the controller from device to host mode or vice versa after power-on reset, then the software does not need to clear this bit. This ensures that the PHY clock is available when the software changes the GCTL.PRTCAPDIR register field. For all other configurations, this bit can be set to '1' during controller configuration. Must be 1 if need special support of P3, P3.CPM and P4 in several LTSSM states.
16:15	DATWIDTH. Read-only, Volatile. Reset: 0h. Description: PIPE Data Width (DatWidth) - 2'b00: 32 bits - 2'b01: 16 bits - 2'b10: 8 bits One clock after reset, these bits receive the value seen on the pipe_DataBusWidth.
14	AbortRxDetInU2. Read-write. Reset: 0. Description: Abort Rx Detect in U2 (AbortRxDetInU2) When set, and the link state is U2, then the controller will abort receiver detection if it receives U2 exit LFPS from the remote link partner. This bit is for the downstream port only. Note: This bit is used by third-party SS PHY.
13	SkipRxDet. Read-write. Reset: 0. Description: Skip Rx Detect: When set, the controller skips Rx Detection if pipe_RxElecIdle is low. Skip is defined as waiting for the appropriate timeout, then repeating the operation.
12	LFPS0Algn. Read-write. Reset: 0. Description: LFPS P0 Align When this bit is set, LFPS during U1/U2/U3 exit is extended until data is ready at the PIPE to ensure the 20-ns gap between LFPS and SS/SSP.
11	P3P2TranOK. Read-write. Reset: 0.

	Description: P3 P2 Transitions OK (P3P2TranOK) When set, the controller transitions directly from Phy power state P2 to P3 or from state P3 to P2. When not set, P0 is always entered as an intermediate state during transitions between P2 and P3, as defined in the PIPE4 Specification. According to the PIPE4 Specification, any direct transition between P3 and P2 is illegal. Note: This bit is used by third-party SS PHY.
10	P3ExSigP2. Read-write. Reset: 0. Description: P3 Exit Signal in P2 (P3ExSigP2) When this bit is set, the controller always changes the PHY power state to P2, before attempting a U3 exit handshake. Note: This bit is used by third-party SS PHY.
9	LFPSFILTER. Read-write. Reset: 0. Description: LFPS Filter (LFPSFilt) When set, filter LFPS reception with pipe_RxValid in PHY power state P0, that is, ignore LFPS reception from the PHY unless both pipe_Rxelecidle and pipe_RxValid are deasserted.
8	RX_DETECT_to_Polling_LFPS_Control. Read-write. Reset: 1. Description: RX_DETECT to Polling.LFPS Control - 1'b0: Enables a 400us delay to start Polling LFPS after RX_DETECT. This allows VCM offset to settle to a proper level. - 1'b1: Disables the 400us delay to start Polling LFPS after RX_DETECT. During controller certification with third party PHY it is observed that the PHY is not able to meet the Tx AC common mode voltage active (VTX-CM-ACPP_ACTIVE <100mv) if the link starts polling within 80us from the time rx.detect is performed. To meet this VTX-CM-ACPP_ACTIVE specification, the polling must be delayed further. If the PHY does not have issue then they can set this bit to 1 which allows polling to start within 80us.
7	Reserved.
6	TX_SWING. Read-write. Reset: 0. Description: Tx Swing (TxSwing) Refer to the PIPE Specification.
5:3	TX_MARGIN. Read-write. Reset: 0h. Description: Tx Margin[2:0] (TxMargin) Refer to Table 5-3 of the PIPE4 Specification.
2:1	SS_TX_DE_EMPHASIS. Read-write. Reset: 1h. Description: Tx Deemphasis (TxDeemphasis) The value driven to the PHY is controlled by the LTSSM during USB 3.1 Compliance mode. (Refer to Table 5-3 of the PIPE Specification.)
0	ELASTIC_BUFFER_MODE. Read-write. Reset: 0. Description: Elastic Buffer Mode (ElasticBufferMode) (Refer to Table 5-3 of the PIPE Specification.)

USB[0...1]x00C2C4 (FCHUSBWDCUSB31CNTR0::GUSB3PIPECTL_REGS1_GUSB3PIPECTL)

Reset: 0102_0102h.

Global USB 3.1 PIPE Control Register

The application uses this register to configure the USB 3.1 PHY and PIPE interface.

Device-only configuration requires only one register. In Host mode, registers are implemented for each port.

Note:

- GUSB3PIPECTLn registers are not applicable for USB 2.0-only mode.

_instUSB0_aliasHOST; USB0x00C2C4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C2C4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	PHYSoftRst. Read-write. Reset: 0. Description: USB 3.1 PHY Soft Reset PHY soft-reset; to issue PHY reset, software should set this bit and reset this bit after meeting PHY reset timing.
30	HstPrtCmpl. Read-write. Reset: 0. Description: HstPrtCmpl This feature tests the PIPE PHY compliance patterns without having to have a test fixture on the USB 3.1 cable. This bit enables placing the SS port link into a compliance state. In compliance lab testing, the SS port link enters compliance after failing the first polling sequence after power on. Set this bit to 0, when you run compliance tests. The sequence for using this functionality is as follows: - 1. Disconnect any plugged in devices. - 2. Perform USBCMD.HCRST or power-on-chip reset. - 3. Set PORTSC.PP=0. - 4. Set GUSB3PIPECTL.HstPrtCmpl=1. This places the link into compliance state. To advance the compliance pattern, follow this sequence (toggle the set GUSB3PIPECTL.HstPrtCmpl): - 1. Set GUSB3PIPECTL.HstPrtCmpl=0. - 2. Set GUSB3PIPECTL.HstPrtCmpl=1. This advances the link to the next compliance pattern. To exit from the compliance state perform USBCMD.HCRST or power-on-chip reset.
29	U2P3ok. Read-write. Reset: 0. Description: Enable P3 entry during U2/SSInactive (U2P3ok). This is recommended only if your PHYs P3 exit time is close to P2 exit time. Putting the ESS PHY in P3 during U2 would affect ESS Asynchronous endpoint performance and also this could prevent U2 entry if there are ESS periodic endpoints and their BInterval is in the sub milli-second range. rammer's Guide for more details. - 0: During link state U2/ESS.Inactive, put PHY in P2 (Default) - 1: During link state U2/ESS.Inactive, put PHY in P3 (Not recommended). Note: For a port, if GUSB3PIPECTL[7]=1 and GUSB3PIPECTL[29]=1, set GUSB3PIPECTL[11] to 1.
28	DisRxDetP3. Read-write. Reset: 0.

	Description: Disabled receiver detection in P3 (DisRxDetP3) - 0: If PHY is in P3 and controller needs to perform receiver detection, The controller performs receiver detection in P3. - 1: If PHY is in P3 and controller needs to perform receiver detection, The controller changes the PHY power state to P2 and then performs receiver detection. After receiver detection, the cores changes PHY power state to P3.
27	Ux_exit_in_Px. Read-write. Reset: 0. Description: Ux Exit in Px (Ux_exit_in_Px) - 0: The controller does U1/U2/U3 exit in PHY power state P0. - 1: The controller does U1/U2/U3 exit in PHY power state P1/P2/P3 respectively. This bit is added for SS PHY workaround where SS PHY injects a glitch on pipe_RxElecIdle while receiving Ux exit LFPS, and pipe_PowerDown change is in progress. Note: This bit is used by third-party SS PHY.
26	ping_enhancement_en. Read-write. Reset: 0. Description: Ping Enhancement Enable (ping_enhancement_en) When set, the Downstream port U1 ping receive timeout becomes 500 ms instead of 300 ms. Minimum Ping.LFPS receive duration is 8 ns (one mac31_clk). This field is valid for the downstream port only. Note: This bit is used by third-party SS PHY.
25	u1u2exitfail_to_recov. Read-write. Reset: 0. Description: U1U2exitfail to Recovery (u1u2exitfail_to_recov) When set, and U1/U2 LFPS handshake fails, the LTSSM transitions from U1/U2 to Recovery instead of SS Inactive. If Recovery fails, then the LTSSM can enter SS.Inactive. This is an enhancement only. It prevents interoperability issue if the remote link does not do proper handshake.
24	request_p1p2p3. Read-write. Reset: 1. Description: Always Request P1/P2/P3 for U1/U2/U3 (request_p1p2p3) When set, the controller always requests PHY power change from P0 to P1/P2/P3 during U0 to U1/U2/U3 transition. If this bit is 0, and immediate Ux exit (remotely initiated, or locally initiated) happens, the controller does not request P1/P2/P3 power state change. Note: For third-party SS PHY, check with your PHY vendor.
23	StartRxDetU3RxDet. Write-only, Volatile. Reset: 0. Description: Start Receiver Detection in U3/Rx.Detect (StartRxdetU3RxDet) If DWC_USB31_GUSB3PIPECTL_INIT[22] is set, and the link is in either U3 or Rx.Detect state, the controller starts receiver detection on the rising edge of this bit. This can only be used for Downstream ports. This bit must be set to '0' for Upstream ports. This feature must not be enabled for normal operation.
22	DisRxDetU3RxDet. Read-write. Reset: 0. Description: Disable Receiver Detection in U3/Rx.Det When set, the controller does not handle receiver detection in either U3 or Rx.Detect states. DWC_USB31_GUSB3PIPECTL_INIT[23] must be used to start receiver detection manually. This bit can only be used for the downstream port. This bit must be set to "0" for Upstream ports. This feature must not be enabled for normal operation.
21:19	DelayP1P2P3. Read-write. Reset: 0h.

	Description: Delay P1P2P3 Delay P0 to P1/P2/P3 request when entering U1/U2/U3 until (DWC_USB31_GUSB3PIPECTL_INIT[21:19]*8) 8B10B error occurs, or Pipe3_RxValid drops to 0. DWC_USB31_GUSB3PIPECTL_INIT[18] must be 1 to enable this functionality.
18	DELAYP1TRANS. Read-write. Reset: 0. Description: Delay PHY power change from P0 to P1/P2/P3 when link state changing from U0 to U1/U2/U3 respectively. - 1'b1: When entering U1/U2/U3, delay the transition to P1/P2/P3 until the pipe signals, pipe_RxElecIdle is 1 and pipe_RxValid is 0 - 1'b0: When entering U1/U2/U3, transition to P1/P2/P3 without checking for pipe_RxElecIdle and pipe_RxValid.
17	SUSPENDENABLE. Read-write. Reset: 1. Description: Suspend USB 3.1 ESS PHY (Suspend_en) When set, and if Suspend conditions are valid, the USB 3.1 PHY enters Suspend mode. For DRD configurations, it is recommended that this bit is set to '0'. If it is set to '1', then the application must clear this bit after power-on reset. Application needs to set it to '1' after the controller initialization is completed. If software is not going to change the controller from device to host mode or vice versa after power-on reset, then the software does not need to clear this bit. This ensures that the PHY clock is available when the software changes the GCTL.PRTCAPDIR register field. For all other configurations, this bit can be set to '1' during controller configuration. Must be 1 if need special support of P3, P3.CPM and P4 in several LTSSM states.
16:15	DATWIDTH. Read-only, Volatile. Reset: 0h. Description: PIPE Data Width (DatWidth) - 2'b00: 32 bits - 2'b01: 16 bits - 2'b10: 8 bits One clock after reset, these bits receive the value seen on the pipe_DataBusWidth.
14	AbortRxDetInU2. Read-write. Reset: 0. Description: Abort Rx Detect in U2 (AbortRxDetInU2) When set, and the link state is U2, then the controller will abort receiver detection if it receives U2 exit LFPS from the remote link partner. This bit is for the downstream port only. Note: This bit is used by third-party SS PHY.
13	SkipRxDet. Read-write. Reset: 0. Description: Skip Rx Detect: When set, the controller skips Rx Detection if pipe_RxElecIdle is low. Skip is defined as waiting for the appropriate timeout, then repeating the operation.
12	LFPS0Algn. Read-write. Reset: 0. Description: LFPS P0 Align When this bit is set, LFPS during U1/U2/U3 exit is extended until data is ready at the PIPE to ensure the 20-ns gap between LFPS and SS/SSP.
11	P3P2TranOK. Read-write. Reset: 0.

	Description: P3 P2 Transitions OK (P3P2TranOK) When set, the controller transitions directly from Phy power state P2 to P3 or from state P3 to P2. When not set, P0 is always entered as an intermediate state during transitions between P2 and P3, as defined in the PIPE4 Specification. According to the PIPE4 Specification, any direct transition between P3 and P2 is illegal. Note: This bit is used by third-party SS PHY.
10	P3ExSigP2. Read-write. Reset: 0. Description: P3 Exit Signal in P2 (P3ExSigP2) When this bit is set, the controller always changes the PHY power state to P2, before attempting a U3 exit handshake. Note: This bit is used by third-party SS PHY.
9	LFPSFILTER. Read-write. Reset: 0. Description: LFPS Filter (LFPSFilt) When set, filter LFPS reception with pipe_RxValid in PHY power state P0, that is, ignore LFPS reception from the PHY unless both pipe_Rxelecidle and pipe_RxValid are deasserted.
8	RX_DETECT_to_Polling_LFPS_Control. Read-write. Reset: 1. Description: RX_DETECT to Polling.LFPS Control - 1'b0: Enables a 400us delay to start Polling LFPS after RX_DETECT. This allows VCM offset to settle to a proper level. - 1'b1: Disables the 400us delay to start Polling LFPS after RX_DETECT. During controller certification with third party PHY it is observed that the PHY is not able to meet the Tx AC common mode voltage active (VTX-CM-ACPP_ACTIVE <100mv) if the link starts polling within 80us from the time rx.detect is performed. To meet this VTX-CM-ACPP_ACTIVE specification, the polling must be delayed further. If the PHY does not have issue then they can set this bit to 1 which allows polling to start within 80us.
7	Reserved.
6	TX_SWING. Read-write. Reset: 0. Description: Tx Swing (TxSwing) Refer to the PIPE Specification.
5:3	TX_MARGIN. Read-write. Reset: 0h. Description: Tx Margin[2:0] (TxMargin) Refer to Table 5-3 of the PIPE4 Specification.
2:1	SS_TX_DE_EMPHASIS. Read-write. Reset: 1h. Description: Tx Deemphasis (TxDeemphasis) The value driven to the PHY is controlled by the LTSSM during USB 3.1 Compliance mode. (Refer to Table 5-3 of the PIPE Specification.)
0	ELASTIC_BUFFER_MODE. Read-write. Reset: 0. Description: Elastic Buffer Mode (ElasticBufferMode) (Refer to Table 5-3 of the PIPE Specification.)

USB[0...1]x00C300 (FCHUSBWCUSB31CNTR0::GTXFIFOSIZ_REGS0_GTXFIFOSIZ0)

Reset: 0000_00A1h.

Global Transmit FIFO Size Register

This register specifies the RAM start address and depth (both in MDWIDTH-bit words) for each implemented TxFIFO. The number of TxFIFOs depends on the configuration parameters including the number of Device IN Endpoints, number of Host Bus Instances, and presence of Debug Capability.

The register default values for each mode are assigned based on the maximum packet size, number of packets to be buffered, speed of host bus instance, bus latency, and mode of operation (host, device, or, DBC). Upon reset and mode transitions, hardware automatically programs these registers to the default values. Consequently, there is typically no need for the software to modify the pre-defined default values.

For the debug capability mode, the currently mapped EP0 IN and EP1 IN TxFIFO numbers can be read from the GFIFOPRIDBC register.

_instUSB0_aliasHOST; USB0x00C300; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C300; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	TXFSTADDR_N. Read-write. Reset: 0000h. Description: Transmit FIFO RAM Start Address This field contains the memory start address for TxFIFO in MDWIDTH-bit words.
15	Reserved.
14:0	TXFDEP_N. Read-write. Reset: 00A1h. Description: TxFIFO Depth This field contains the depth of TxFIFO in MDWIDTH-bit words. - Minimum value: 32 - Maximum value: 32,767

USB[0...1]x00C304 (FCHUSBWCUSB31CNTR0::GTXFIFOSIZ_REGS0_GTXFIFOSIZ1)

Reset: 00A1_8083h.

Register GTXFIFOSIZ 1

_instUSB0_aliasHOST; USB0x00C304; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C304; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	TXFSTADDR_N. Read-write. Reset: 00A1h. Description: Transmit FIFO RAM Start Address This field contains the memory start address for TxFIFO in MDWIDTH-bit words.
15	Reserved.
14:0	TXFDEP_N. Read-write. Reset: 0083h. TXFDEP_N

USB[0...1]x00C308 (FCHUSBWDCUSB31CNTR0::GTXFIFOSIZ_REGS0_GTXFIFOSIZ2)

Reset: 0124_0064h.

Transmit FIFOn RAM Start Address

This field contains the memory start address for TxFIFOn in MDWIDTH-bit words.

_instUSB0_aliasHOST; USB0x00C308; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C308; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description31:16 **TXFSTADDR_N**. Read-write. Reset: 0124h.**Description:** Transmit FIFOn RAM Start Address

This field contains the memory start address for TxFIFOn in MDWIDTH-bit words.

15 Reserved.

14:0 **TXFDEP_N**. Read-write. Reset: 0064h.**Description:** TxFIFO Depth

This field contains the depth of TxFIFOn in MDWIDTH-bit words.

- Minimum value: 32

- Maximum value: 32,767

USB[0...1]x00C30C (FCHUSBWDCUSB31CNTR0::GTXFIFOSIZ_REGS0_GTXFIFOSIZ3)

Reset: 0188_80C4h.

Register GTXFIFOSIZ 3

_instUSB0_aliasHOST; USB0x00C30C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C30C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description31:16 **TXFSTADDR_N**. Read-write. Reset: 0188h.**Description:** Transmit FIFOn RAM Start Address

This field contains the memory start address for TxFIFOn in MDWIDTH-bit words.

15 Reserved.

14:0 **TXFDEP_N**. Read-write. Reset: 00C4h.**Description:** TXFDEP_N: TxFIFO Depth (TxFDep_n)

This field contains the depth of TxFIFOn in MDWIDTH-bit words.

- Minimum value: 32

- Maximum value: 32,767

USB[0...1]x00C310 (FCHUSBDWCUSB31CNTR0::GTXFIFOSIZ_REGS0_GTXFIFOSIZ4)

Reset: 024C_0631h.

Register GTXFIFOSIZ 4

_instUSB0_aliasHOST; USB0x00C310; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C310; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	TXFSTADDR_N. Read-write. Reset: 024Ch. TXFSTADDR_N
15	Reserved.
14:0	TXFDEP_N. Read-write. Reset: 0631h. Description: TXFDEP_N: TxFIFO Depth (TxFDep_n) This field contains the depth of TxFIFO in MDWIDTH-bit words. - Minimum value: 32 - Maximum value: 32,767

USB[0...1]x00C314 (FCHUSBDWCUSB31CNTR0::GTXFIFOSIZ_REGS0_GTXFIFOSIZ5)

Reset: 087D_8631h.

Register GTXFIFOSIZ 5

_instUSB0_aliasHOST; USB0x00C314; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C314; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	TXFSTADDR_N. Read-write. Reset: 087Dh. Transmit FIFO RAM Start Address
15	Reserved.
14:0	TXFDEP_N. Read-write. Reset: 0631h. Description: TxFIFO Depth This field contains the depth of TxFIFO in MDWIDTH-bit words. - Minimum value: 32 - Maximum value: 32,767

USB[0...1]x00C318 (FCHUSBDWCUSB31CNTR0::GTXFIFOSIZ_REGS0_GTXFIFOSIZ6)

Reset: 0EAE_0631h.

Register GTXFIFOSIZ 6

_instUSB0_aliasHOST; USB0x00C318; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C318; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	TXFSTADDR_N. Read-write. Reset: 0EAEh. Description: Transmit FIFO RAM Start Address This field contains the memory start address for TxFIFO in MDWIDTH-bit words.
15	Reserved.
14:0	TXFDEP_N. Read-write. Reset: 0631h. Description: TxFIFO Depth This field contains the depth of TxFIFO in MDWIDTH-bit words. - Minimum value: 32 - Maximum value: 32,767

USB[0...1]x00C31C (FCHUSBDCUSB31CNTR0::GTXFIFOSIZ_REGS0_GTXFIFOSIZ7)

Reset: 14DF_8631h.

Register GTXFIFOSIZ 7

_instUSB0_aliasHOST; USB0x00C31C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C31C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	TXFSTADDR_N. Read-write. Reset: 14DFh. Description: Transmit FIFOn RAM Start Address This field contains the memory start address for TxFIFOn in MDWIDTH-bit words.
15	Reserved.
14:0	TXFDEP_N. Read-write. Reset: 0631h. Description: TXFDEP_N This field contains the depth of TxFIFOn in MDWIDTH-bit words. - Minimum value: 32 - Maximum value: 32,767

USB[0...1]x00C320 (FCHUSBDCUSB31CNTR0::GTXFIFOSIZ_REGS0_GTXFIFOSIZ8)

Reset: 1B10_0013h.

Register GTXFIFOSIZ 8

_instUSB0_aliasHOST; USB0x00C320; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C320; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	TXFSTADDR_N. Read-write. Reset: 1B10h. Description: Transmit FIFOn RAM Start Address This field contains the memory start address for TxFIFOn in MDWIDTH-bit words.
15	Reserved.
14:0	TXFDEP_N. Read-write. Reset: 0013h. Description: TxFIFO Depth This field contains the depth of TxFIFOn in MDWIDTH-bit words. - Minimum value: 32 - Maximum value: 32,767

USB[0...1]x00C380 (FCHUSBDWCUSB31CNTR0::GRXFIFOSIZ_REGS0_GRXFIFOSIZ0)

Reset: 0000_0084h.

Global Receive FIFO Size Register

This register specifies the RAM start address and depth (both in MDWIDTH-bit words) for each implemented RxFIFO. The number of RxFIFOs depends on the configuration parameters including the number of Host Bus Instances and presence of Debug Capability; device mode requires only one RxFIFO.

The register default values for each mode are assigned based on the maximum packet size, number of packets to be buffered, speed of the host bus instance, bus latency, and mode of operation (host, device, or DBC). Upon reset and mode transitions, hardware automatically programs these registers to the default values. Consequently, there is typically no need for the software to modify the pre-defined default values.

For the debug capability mode, the currently mapped RxFIFO number can be read from the GFIFOPRIDBC register.

_instUSB0_aliasHOST; USB0x00C380; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C380; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	RXFSTADDR_N. Read-write. Reset: 0000h. Description: RxFIFO RAM Start Address (RxFStAddr_n) This field contains the memory start address for RxFIFO in MDWIDTH-bit words.
15	Reserved.
14:0	RXFDEP_N. Read-write. Reset: 0084h. Description: RxFIFO Depth (RxFDep_n) This field contains the depth of RxFIFO in MDWIDTH-bit words. - Minimum value: 32 - Maximum value: 16,384

USB[0...1]x00C384 (FCHUSBDWCUSB31CNTR0::GRXFIFOSIZ_REGS0_GRXFIFOSIZ1)

Reset: 0084_8084h.

Register

_instUSB0_aliasHOST; USB0x00C384; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00C384; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	RXFSTADDR_N. Read-write. Reset: 0084h. RXFSTADDR_N
15	Reserved.
14:0	RXFDEP_N. Read-write. Reset: 0084h. Description: RxFIFO Depth This field contains the depth of RxFIFO in MDWIDTH-bit words. - Minimum value: 32 - Maximum value: 16,384

USB[0...1]x00C388 (FCHUSBDCUSB31CNTR0::GRXFIFOSIZ_REGS0_GRXFIFOSIZ2)

Reset: 0108_0064h.

Register

_instUSB0_aliasHOST; USB0x00C388; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C388; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description**31:16** **RXFSTADDR_N.** Read-write. Reset: 0108h.**Description:** RAM Start Address

This field contains the memory start address for RxFIFO in MDWIDTH-bit words.

15 Reserved.**14:0** **RXFDEP_N.** Read-write. Reset: 0064h.**Description:** RxFIFO Depth

This field contains the depth of RxFIFO in MDWIDTH-bit words.

- Minimum value: 32

- Maximum value: 16,384

USB[0...1]x00C38C (FCHUSBDCUSB31CNTR0::GRXFIFOSIZ_REGS0_GRXFIFOSIZ3)

Reset: 016C_80C4h.

Register

_instUSB0_aliasHOST; USB0x00C38C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C38C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description**31:16** **RXFSTADDR_N.** Read-write. Reset: 016Ch. RXFSTADDR_N**15** Reserved.**14:0** **RXFDEP_N.** Read-write. Reset: 00C4h.**Description:** RxFIFO Depth

This field contains the depth of RxFIFO in MDWIDTH-bit words.

- Minimum value

- Maximum value: 16,384

USB[0...1]x00C390 (FCHUSBDCUSB31CNTR0::GRXFIFOSIZ_REGS0_GRXFIFOSIZ4)

Reset: 0230_0608h.

Register

_instUSB0_aliasHOST; USB0x00C390; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C390; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description31:16 **RXFSTADDR_N**. Read-write. Reset: 0230h.**Description:** Transmit FIFO RAM Start Address

This field contains the memory start address for TxFIFO in MDWIDTH-bit words.

15 Reserved.

14:0 **RXFDEP_N**. Read-write. Reset: 0608h.**Description:** RXFDEP_N: RxFIFO Depth (RxFDep_n)

This field contains the depth of RxFIFO in MDWIDTH-bit words.

- Minimum value: 32

- Maximum value: 16,384

USB[0...1]x00C394 (FCHUSBDCUSB31CNTR0::GRXFIFOSIZ_REGS0_GRXFIFOSIZ5)

Reset: 0838_8608h.

Register

_instUSB0_aliasHOST; USB0x00C394; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C394; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description31:16 **RXFSTADDR_N**. Read-write. Reset: 0838h.**Description:** Transmit FIFO RAM Start Address

This field contains the memory start address for TxFIFO in MDWIDTH-bit words.

15 Reserved.

14:0 **RXFDEP_N**. Read-write. Reset: 0608h.**Description:** RxFIFO Depth

This field contains the depth of RxFIFO in MDWIDTH-bit words.

- Minimum value: 32

- Maximum value: 16,384

USB[0...1]x00C398 (FCHUSBDCUSB31CNTR0::GRXFIFOSIZ_REGS0_GRXFIFOSIZ6)

Reset: 0E40_0608h.

Register

_instUSB0_aliasHOST; USB0x00C398; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C398; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	RXFSTADDR_N. Read-write. Reset: 0E40h. Description: RxFIFO RAM Start Address This field contains the memory start address for RxFIFO in MDWIDTH-bit words.
15	Reserved.
14:0	RXFDEP_N. Read-write. Reset: 0608h. Description: RxFIFO Depth This field contains the depth of RxFIFO in MDWIDTH-bit words. - Minimum value: 32 - Maximum value: 16,384

USB[0...1]x00C39C (FCHUSBDCUSB31CNTR0::GRXFIFOSIZ_REGS0_GRXFIFOSIZ7)

Reset: 1448_8608h.

Register

_instUSB0_aliasHOST; USB0x00C39C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C39C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	RXFSTADDR_N. Read-write. Reset: 1448h. Description: RxFIFO RAM Start Address This field contains the memory start address for RxFIFO in MDWIDTH-bit words.
15	Reserved.
14:0	RXFDEP_N. Read-write. Reset: 0608h. Description: RxFIFO Depth This field contains the depth of RxFIFO in MDWIDTH-bit words. - Minimum value: 32 - Maximum value: 16,384

USB[0...1]x00C600 (FCHUSBDCUSB31CNTR0::GHWPARAMS8)

Read-only. Reset: 0000_1AD3h.

Global Hardware Parameters Register 8

This register contains the hardware configuration options.

_instUSB0_aliasHOST; USB0x00C600; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C600; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	ghwparams8_32_0. Read-only. Reset: 0000_1AD3h. `DWC_USB31_DCACHE_DEPTH_INFO

USB[0...1]x00C604 (FCHUSBWCUSB31CNTR0::GSMACCTL)

Read-write. Reset: 0000_0000h.

Global SMAC CONTROL REGISTER

_instUSB0_aliasHOST; USB0x00C604; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C604; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	dbg_tis_index . Read-write. Reset: 0h. SMAC bus instance index
29:26	dbg_ptl_addr . Read-write. Reset: 0h. PTL debug address
25:18	Reserved.
17	hostin_signle_en . Read-write. Reset: 0. Host: Limit a maximum of one active IN transfer on USB at a time
16	hostout_single_en . Read-write. Reset: 0. Host: Limit a maximum of one active OUT transfer on USB at a time
15:4	Reserved.
3	single_psqdir_en . Read-write. Reset: 0. Description: Enable PSQ message type [PSQDIR] sorting If this bit is set, enables PSQ message sorting with PSQDIR: 0: PSQDIR=0 for all messages 1: PSQDIR=0 for OUT endpoints, CTRL endpoints, non-transaction headers (ie Vendor LMP) PSQDIR=1 for IN endpoints
2	host_mask_nump0 . Read-write. Reset: 0. Description: Mask TPACK nump0 (stream) with nump1 LSP workaround - for host stream transfers, if TP.ACK.nump=0 is received, then convert to nump=1 to avoid flow control If this bit is set, the controller converts TP.ACK.nump=0 to TP.ACK.nump=1. - 0: Normal operation - 1: Convert TP.ACK nump=0 to nump=1
1	ignore_babble . Read-write. Reset: 0. Description: Ignore Receive babble conditions When the controller receives data packets, a payload that exceeds the allowed Max Packet Size is considered a babble condition. If this bit is set, the controller ignores babble conditions. - 0: detect receive payload babble conditions - 1: ignore receive payload babble conditions
0	single_TBT_req . Read-write. Reset: 0. Description: Disable pipelined Transmit Data The transmit data path utilizes an internal pipeline to reduce interpacket delay during burst transmit. Setting this bit disables the pipeline mechanism. Interpacket delay is increased if pipelining is disabled. - 0: enable transmit data pipelining - 1: disable transmit data pipelining

USB[0...1]x00C608 (FCHUSBDCUSB31CNTR0::GUCTL2)

Read-write. Reset: 1809_4246h.

Global User Control Register 2

_instUSB0_aliasHOST; USB0x00C608; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C608; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:25	PERIODIC_UF_THR. Read-write. Reset: 0Ch. Description: Periodic microseconds threshold Indicates the number of microseconds before the end of the microframe that the LSP stops issuing periodic requests to the PTL. A value of n corresponds to n microseconds, which is (125-n)/125% approximately. For example, if you need a threshold of approximately 10% of one uframe, then program this field as $0.1 * 125\mu s \approx 12\mu s$.
24:18	ASYNC_UF_THR. Read-write. Reset: 02h. Description: Asynchronous microseconds threshold Indicates the number of microseconds before the end of the microframe that the LSP stops issuing asynchronous requests to the PTL. A value of n corresponds to n microseconds, which is (125-n)/125% approximately. For example, if you need a threshold of approximately 2% of one uframe, then program this field as $0.02 * 125\mu s \approx 2\mu s$.
17:11	PERIODIC_TXDMA_UF_THR. Read-write. Reset: 28h. Description: Indicates the number of microseconds before the end of the microframe that the LSP needs to start issuing TX DMA for periodic OUT endpoints scheduled in the next microframe. A value of n corresponds to n microseconds, which is (125-n)/125% approximately. For example, if you need a threshold of approximately 32% of one uframe, then program this field as $0.32 * 125\mu s \approx 40\mu s$.
10:9	SVC_OPP_PER_SSP. Read-write. Reset: 1h. Description: Service opportunities to Enhanced SuperSpeedPlus bulk endpoints Indicates the number of Service Opportunities (SOs) that will be allocated to Enhanced SuperSpeedPlus bulk endpoints. The encoding used is as follows: - 0: maxburstsize service opportunities - 1: 2* maxburstsize service opportunities - 2: 4*maxburstsize service opportunities - 3: maxburstsize/2 service opportunities where, Maxburstsize is the maximum number of packets the endpoint can send/receive as part of the burst.
8:7	SVC_OPP_PER_SS. Read-write. Reset: 0h.

	Description: Service opportunities to SuperSpeed bulk endpoints Indicates the number of Service Opportunities (SOs) that will be allocated to SuperSpeed bulk endpoints. The encoding used is as follows: - 0: maxburstsize service opportunities - 1: 2* maxburstsize service opportunities - 2: 4*maxburstsize service opportunities - 3: maxburstsize/2 service opportunities where, Maxburstsize is the maximum number of packets the endpoint can send/receive as part of the burst.
6:5	SVC_OPP_PER_HS. Read-write. Reset: 2h. Description: Service opportunities to HS bulk endpoints Indicates the number of Service Opportunities (SOs) that will be allocated to HS bulk endpoints. The encoding used is as follows: - 0: 1 service opportunity - 1: 1 service opportunities - 2: 3 service opportunities - 3: 4 service opportunities
4:3	SVC_OPP_PER_FSLS. Read-write. Reset: 0h. Description: Service opportunities to FS/LS bulk endpoints Indicates the number of Service Opportunities (SOs) that will be allocated to FS/LS bulk endpoints. The encoding used is as follows: - 0: 1 service opportunity - 1: 1 service opportunities - 2: 3 service opportunities - 3: 4 service opportunities
2:1	MAX_REISU_CNT. Read-write. Reset: 3h. Description: Indicates the number of re-issue opportunities that will be allocated to OUT transactions that meet the following criteria: - Bulk EP types - Receives a link abort due to: -- Lack of credits -- Link in U1, U2 or Recovery -- Not able to fit all the packets into the remainder of the microframe before the ITP Valid values are 0, 1, 2 or 3.
0	SCHEDDULE_LT_THR. Read-write. Reset: 0. Indicates that the Host LSP schedulers should schedule transactions before all the TRBs have been fetched. This may increase performance by allowing transactions to start earlier, but may also have a negative effect because more TRQ space will be used.

USB[0...1]x00C60C (FCHUSBWDCUSB31CNTR0::GUCTL3)

Read-write.

Global User Control Register 3

_instUSB0_aliasHOST; USB0x00C60C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C60C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:25	Reserved.
24:22	CFGEP_CMD_SWITCHING_TIME_CTRL. Read-write. Reset: 0h. Description: Configure EP command switching time control Bit 24 : Enable the fast EP configure mode - 1: fast EP configure enabled . Only consider the ISOC EP service interval in determining the switching time - 0: fast EP configure disabled. Consider the ISOC and INTR EP service interval in determining the switching time Bit [23:22] Maximum switching frame boundary - 2'b00 : Maximum Switching Frame boundary is 32ms . Fast EP configure enabled/disabled based on bit 24 guctl3[24:22]=b000 is legacy switching mode - 2'b01 : Maximum Switching Frame boundary is 1ms . Fast EP configure enabled/disabled based on bit 24 - 2'b10 : Maximum Switching Frame boundary is 4ms . Fast EP configure enabled/disabled based on bit 24 - 2'b11 : Maximum Switching Frame boundary is 8ms . Fast EP configure enabled/disabled based on bit 24
21	BLOCK_CONCURRENT_IN_CTRL_XFERS. Read-write. Reset: 1. Description: Block concurrent IN transactions during Control transfers This bit controls the way in which concurrent IN is handled for control transfers - 0: Concurrent IN transfers are blocked during control transfers - 1: Concurrent IN transfers are not blocked during control transfers
20	SSBI_SINGLE_EP_MODE_DISABLE. Read-write. Reset: 0. Description: SS BI Single EP mode Disable When set, this field disables single EP mode in the SS BI. - 1'b0: Single EP mode in SS BI enabled - 1'b1: Single EP mode in SS BI disabled
19	DISEXTBUSCLKGT_U1U2L1. Read-write. Reset: 0. Description: Disable External Bus Clock Gating in U1/U2/L1 state This bit is only used when the DWC_USB31_EXT_BUS_CLK_OFF parameter is set to 1. This bit is used to generate the pmgt_ext_bus_clk_off signal when the following conditions occur: - GUCTL3[DISEXTBUSCLKGT] is cleared - GUCTL3[DISEXTBUSCLKGT_U1U2L1] is cleared - Link is in RXDET, DISABLE, INACTIVE, or U3 state and the controller is idle - All USB 2.0 ports are in L2 state, that is, UTMI suspend is asserted
18	DISEXTBUSCLKGT. Read-write. Reset: 0. Description: Disable External Bus Clock Gating This bit is only used when the DWC_USB31_EXT_BUS_CLK_OFF parameter is set to 1. This bit is used to generate the pmgt_ext_bus_clk_off signal when the following conditions occur: - GUCTL3[DISEXTBUSCLKGT] is cleared - Link is in RXDET, DISABLE, INACTIVE, U1, U2, or U3 state and the controller is idle - All USB 2.0 ports are in LPM-L1 or L2 state, that is, UTMI suspend, l1_suspend and sleep is asserted
17	RSVD_PERI_BANDWIDTH_FS. Read-write. Reset: 0.

	Description: 85percent bandwidth reservation for periodic transfer for FS bus behind HS hub When not set, the bandwidth reserved for periodic transfers on a FS bus behind a HS hub is 85% . Setting it to 1 will increase reserved bandwidth to 90% When SSsplit in uframe 7 is disabled , the bandwidth available to periodic transfer is less . Setting reserved bandwidth to be 90% may cause over scheduling - 1'b0: 85% of FS bus bandwidth reserved for periodic transfer when the FS bus is behind HS hub - 1'b1: 90% of FS bus bandwidth reserved for periodic transfer when the FS bus is behind HS hub
16	USB20_RETRY_DISABLE. Read-write. Description: USB2.0 Internal Retry Disable When set, this field disables the internal retry feature on the USB 2.0 bus. - 1'b0: Internal retry feature enabled - 1'b1: Internal retry feature disabled
15:13	Reserved.
12:9	SVC_OPP_PER_HS_SEP. Read-write. Reset: 3h. Description: Service opportunities for HS bulk endpoints in single endpoint mode Indicates the number of Service Opportunities (SOs) that will be allocated to the HS bulk endpoint when it is the only one asynchronous endpoint active for that bus instance. A value of n corresponds to n Service Opportunities, and the valid values of n are 1 to 13.
8	BARB_BURST_ENABLE. Read-write. Reset: 1. Description: LSP BARB Burst Support This bit is valid only in DRD or host mode when using LSP EHST (((DWC_USB31_MODE == 2) (DWC_USB31_MODE == 1)) && (DWC_USB31_CURRENT_GEN_HST_LSP == 2)). - 1'b1: The lbc interface between LSP and BMU supports burst transactions. - 1'b0: The lbc interface between LSP and BMU does not support burst transactions.
7	INTR_EP_PING_ENABLE. Read-write. Reset: 0. Description: Interrupt EP PING Support - 1'b1: Host sends PING for SS/SSP interrupt EPs. - 1'b0: Host does not send PING for SS/SSP interrupt EPs.
6:0	PERIODIC_UF_THR_FSLs. Read-write. Reset: 0Ch. Description: Periodic microseconds threshold Indicates the number of microseconds before the end of the microframe that the LSP cannot issue periodic requests to the PTL. A value of n corresponds to n microseconds, which is (125-n)/125% approximately. For example, if you need a threshold of approximately 10% of one uframe, then program this field as $0.1 * 125\mu s \approx 12\mu s$.

USB[0...1]x00C618 (FCHUSBWCUSB31CNTR0::GTXFIFOPRIHST)**Global Host TXFIFO DMA Priority Register (GTXFIFOPRIHST)**

This register specifies the relative DMA priority level among the Host TXFIFOs (one per USB bus instance) within the associated speed group (SS or HS/FSLs). Each register bit[n] controls the priority (1: high, 0: low) of TXFIFO[n] within a speed group. When multiple TXFIFOs compete for DMA service at a given time (i.e., multiple TXQs contain TX DMA requests and their corresponding TXFIFOs have space available), the TX DMA arbiter grants access on a packet-basis in the following manner:

- 1. Among the FIFOs in the same speed group (SS or HS/FSLs):
 - a. High-priority TXFIFOs are granted access using round-robin arbitration
 - b. Low-priority TXFIFOs are granted access using round-robin arbitration only after the high-priority TXFIFOs have no further processing to do (that is, either the TXQs are empty or the corresponding TXFIFOs are full).
- 2. The TX DMA arbiter prioritizes the SS speed group or HS/FSLs speed group according to the ratio programmed in the GDMABLRATIO register.

For scatter-gather packets, the arbiter grants successive DMA requests to the same FIFO until the entire packet is completed.

This register is present only when the controller is configured to operate in the host mode (includes DRD). The register size corresponds to the number of configured USB bus instances; for example, in the default configuration, there are 3 USB bus instances (1 SS, 1 HS, and 1 FSLs).

_instUSB0_aliasHOST; USB0x00C618; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C618; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:9	Reserved.
8:0	gtxfifoprihst . Read-write. Reset: 000h. Host TxFIFO priority

USB[0...1]x00C61C (FCHUSBWDCUSB31CNTR0::GRXFIFOPRIHST)

Global Host RXFIFO DMA Priority Register

This register specifies the relative DMA priority level among the Host RXFIFOs (one per USB bus instance) within the associated speed group (SS or HS/FSLs). Each register bit[n] controls the priority (1: high, 0: low) of RXFIFO[n] within a speed group. When multiple RXFIFOs compete for DMA service at a given time (i.e., multiple RXQs contain RX DMA requests and their corresponding RXFIFOs have data available), the RX DMA arbiter grants access on a packet-basis in the following manner:

- 1. Among the FIFOs in the same speed group (SS or HS/FSLs):
 - a. High-priority RXFIFOs are granted access using round-robin arbitration
 - b. Low-priority RXFIFOs are granted access using round-robin arbitration only after high-priority RXFIFOs have no further processing to do (that is, either the RXQs are empty or the corresponding RXFIFOs do not have the required data).
- 2. The RX DMA arbiter prioritizes the SS speed group or HS/FSLs speed group according to the ratio programmed in the GDMAHLRATIO register.

For scatter-gather packets, the arbiter grants successive DMA requests to the same FIFO until the entire packet is completed.

This register is present only when the controller is configured to operate in the host mode (includes DRD). The register size corresponds to the number of configured USB bus instances; for example, in the default configuration, there are 3 USB bus instances (1 SS, 1 HS, and 1 FSLs).

_instUSB0_aliasHOST; USB0x00C61C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C61C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:8	Reserved.
7:0	grxfifoprihst . Read-write. Reset: 00h. Host RxFIFO priority

USB[0...1]x00C620 (FCHUSBWDCUSB31CNTR0::GFIFOPRIDBC)

Global Host Debug Capability DMA Priority Register

This register specifies the relative priority of the RXFIFOs and TXFIFOs associated with the DbC mode. It overrides the priority assigned in the corresponding indexes of the Host RXFIFO and TXFIFO DMA priority registers, when the DbC mode is enabled.

Priority settings are specified in relation to the low-priority SS speed group:

- 1. Normal priority indicates that the DbC FIFOs are considered identical to the Host SS low-priority FIFOs.
- 2. Low priority indicates that the DbC FIFOs are considered to have lower priority than all Host SS FIFOs.
- 3. High priority indicates that the DbC FIFOs are considered higher priority than the Host SS low-priority FIFOs but lower priority than the Host SS high-priority FIFOs.

This register is present only when the controller is configured to operate in Host Debug Capability (DbC) mode.

_instUSB0_aliasHOST; USB0x00C620; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C620; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	Reserved.
1:0	gfifopridbc . Read-write. Reset: 1h. Host DbC DMA priority

USB[0...1]x00C624 (FCHUSBWCUSB31CNTR0::GDMAHLRATIO)**Global Host FIFO DMA High-Low Priority Ratio Register**

This register specifies the relative priority of the SS FIFOs with respect to the HS/FSLs FIFOs. The DMA arbiter prioritizes the HS/FSLs round-robin arbiter group every DMA High-Low Priority Ratio grants as indicated in the register separately for TX and RX.

To illustrate, consider that all FIFOs are requesting access simultaneously, and the ratio is 4. SS gets priority for 4 packets, HS/FSLs gets priority for 1 packet, SS gets priority for 4 packets, HS/FSLs gets priority for 1 packet, and so on.

If FIFOs from both speed groups are not requesting access simultaneously then,

- if SS got grants 4 out of the last 4 times, then HS/FSLs get the priority on any future request.
- if HS/FSLs got the grant last time, SS gets the priority on the next request.
- if there is a valid request on either SS or HS/FSLs, a grant is always awarded; there is no idle.

This register is present if the controller is configured to operate in host mode (includes DRD).

_instUSB0_aliasHOST; USB0x00C624; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C624; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:29	Reserved.
28:24	hstrxfifo_mac . Read-write. Reset: 0Ah.
	Description: RX MAC ESS Priority Count Specifies the Global Host FIFO MAC access Ratio Register's (GDMAHLRATIO) RX SS:HSFSLs Ratio power-on initialization value (bit[28:24]). This register specifies MAC write access count relation between ESS FIFOs vs HS+FSLs FIFOs. Specifically, FIFO access arbiter prioritizes the HS+FSLs FIFOs for one clock after the specified number of write access to the SS FIFOs.
23:21	Reserved.
20:16	hsttxfifo_mac . Read-write. Reset: 0Ah.
	Description: TX MAC ESS Priority Count Specifies the Global Host FIFO MAC access Ratio Register's (GDMAHLRATIO) TX SS:HSFSLs Ratio power-on initialization value (bit[20:16]). This register specifies MAC read access count relation between ESS FIFOs vs HS+FSLs FIFOs. Specifically, FIFO read access arbiter prioritizes the HS+FSLs FIFOs for one clock after the specified number of read access to the SS FIFOs.
15:13	Reserved.
12:8	hstrxfifo_dma . Read-write. Reset: 01h.

	Description: RX DMA ESS Priority Count Specifies the Global Host FIFO DMA access Ratio Register's (GDMAHLRATIO) RX SS:HSFSLs Ratio power-on initialization value (bit[12:8]). This register specifies the relative priority of the SS FIFOs vs. the HS+FSLs FIFOs. Specifically, the DMA arbiter prioritizes the HS/FSLs round-robin arbiter group for one packet after the specified number of packet grants to the ESS round-robin arbiter group. When a standard driver is used, such as the xHCI driver from Microsoft®, this register must be initialized to meet system requirements before synthesizing the controller. If you are developing your own xHCI host driver, then this register can be configured by your driver. It is recommended to keep this value as 1 so that HS/FSLs gets same priority as ESS. The HS/FSLs bandwidth requirement is negligible compared to ESS, so keeping this value as 1 ensures USB 2.0 operation is not affected by high bandwidth ESS. - Enabled : (DWC_USB31_MODE == 1 DWC_USB31_MODE == 2) && DWC_USB31_MBUS_TYPE!=4 && (DWC_USB31_EN_USB2_ONLY==0)
7:5	Reserved.
4:0	hsttxfifo_dma. Read-write. Reset: 01h. Description: TX DMA ESS Priority Count Specifies the Global Host FIFO DMA access Ratio Register's (GDMAHLRATIO) TX SS:HSFSLs Ratio power-on initialization value (bit[4:0]). This register specifies the access weight of the SS FIFOs vs. the HS+FSLs FIFOs. Specifically, the DMA arbiter prioritizes the HS/FSLs round-robin arbiter group for one packet after the specified number of packet grants to the ESS round-robin arbiter group. For more information, see the "GDMAHLRATIO" section. When a standard driver is used, such as the xHCI driver from Microsoft, this register must be initialized to meet system requirements before synthesizing the controller. If you are developing your own xHCI host driver, then this register can be configured by your driver. It is recommended to keep this value as 1 so that HS/FSLs gets same priority as ESS. The HS/FSLs bandwidth requirement is negligible compared to ESS, so keeping this value as 1 ensures USB 2.0 operation is not affected by high bandwidth ESS. -Enabled : (DWC_USB31_MODE == 1 DWC_USB31_MODE == 2) && DWC_USB31_MBUS_TYPE!=4 && (DWC_USB31_EN_USB2_ONLY==0)

USB[0...1]x00C628 (FCHUSBWCUSB31CNTR0::GOSTDDMA_ASYNC)

	Read-write. Reset: 1212_0D12h.
	Global Number of Async Outstanding DMA Register
	_instUSB0_aliasHOST; USB0x00C628; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}
	_instUSB1_aliasHOST; USB1x00C628; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}
Bits	Description
31:24	OSTDRX_ASYNC. Read-write. Reset: 12h. Number of outstanding async RX DMA
23:16	OSTDTX_ASYNC_ACTIVE. Read-write. Reset: 12h. Number of outstanding async TX DMA for active FIFO/BI on USB
15:8	OSTDTX_ASYNC_INACTIVE. Read-write. Reset: 0Dh. Number of outstanding async TX DMA for inactive FIFO/BI on USB
7:0	OSTDTX_ASYNC. Read-write. Reset: 12h. Number of outstanding async TX DMA

USB[0...1]x00C62C (FCHUSBDCUSB31CNTR0::GOSTDDMA_PRD)

Read-write. Reset: 1818_1218h.

Global Number of Periodic Outstanding DMA Register

_instUSB0_aliasHOST; USB0x00C62C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C62C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	OSTDRX_PRD . Read-write. Reset: 18h. Number of outstanding periodic RX DMA
23:16	OSTDTX_PRD_ACTIVE . Read-write. Reset: 18h. Number of outstanding periodic TX DMA for active FIFO/BI on USB
15:8	OSTDTX_PRD_INACTIVE . Read-write. Reset: 12h. Number of outstanding periodic TX DMA for inactive FIFO/BI on USB
7:0	OSTDTX_PRD . Read-write. Reset: 18h. Number of outstanding periodic TX DMA

USB[0...1]x00C630 (FCHUSBWCUSB31CNTR0::GFLADJ)

Read-write.

Global Frame Length Adjustment Register

This register provides options for the software to control the controller behavior with respect to SOF (Start of Frame) and ITP (Isochronous Timestamp Packet) timers and frame timer functionality. It provides an option to override the fladj_30mhz_reg sideband signal. This also facilitates hardware LPM in host mode with the SOF or ITP counters being run from the ref_clk signal.

_instUSB0_aliasHOST; USB0x00C630; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C630; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	GFLADJ_REFCLK_240MHZDECR_PLS1. Read-write. Reset: 0. Description: GFLADJ_REFCLK_240MHZDECR_PLS1 This field indicates that the decrement value that the controller applies for each ref_clk must be GFLADJ_REFCLK_240MHZ_DECR and GFLADJ_REFCLK_240MHZ_DECR +1 alternatively on each ref_clk. Set this bit to a '1' only if the fractional component of 240/ref_frequency is greater than or equal to 0.5. Examples: If the ref_clk is 19.2 MHz then - GUCTL.REF_CLK_PERIOD = 52 - GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = (240/19.2) = 12.5 - GFLADJ.GFLADJ_REFCLK_240MHZDECR_PLS1 = 1 If the ref_clk is 24 MHz then - GUCTL.REF_CLK_PERIOD = 41 - GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = (240/24) = 10 - GFLADJ.GFLADJ_REFCLK_240MHZDECR_PLS1 = 0
30:24	GFLADJ_REFCLK_240MHZ_DECR. Read-write. Reset: 0Ah. Description: This field indicates the decrement value that the controller applies for each ref_clk in order to derive a frame timer in terms of a 240-MHz clock. The value is derived as follows: $\text{GFLADJ_REFCLK_240MHZ_DECR} = 240/\text{ref_clk_frequency}$ Examples: If the ref_clk is 24 MHz then - GUCTL.REF_CLK_PERIOD = 41 - GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = 240/24 = 10 If the ref_clk is 17 MHz then - GUCTL.REF_CLK_PERIOD = 58 - GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = 240/17 = 14
23	GFLADJ_REFCLK_LPM_SEL. Read-write. Reset: 1. Description: The ref_clk frequencies supported are 16/17/19.2/20/24/39.7 MHz. - Host mode: This bit internally enables the functionality of running SOF/ITP counters on the ref_clk. - Device mode with DWC_USB31_LPM_SUSP_OFF = 1: This bit internally enables the functionality of running SOF counters on the ref_clk in 2.0 speeds.
22	Reserved.
21:8	GFLADJ_REFCLK_FLADJ. Read-write. Reset: 07F0h.

	Description: This field indicates the frame length adjustment to be applied to the SOF/ITP counter that is running on the ref_clk. This register value is used to adjust the SOF and ITP interval. The value is derived as follows: $\text{FLADJ_REF_CLK_FLADJ} = ((125000/\text{ref_clk_period_integer}) - (125000/\text{ref_clk_period})) * \text{ref_clk_period}$ where - the ref_clk_period_integer is the integer value of the ref_clk period got by truncating the decimal (fractional) value that is programmed in the GUCTL.REF_CLK_PERIOD field. - the ref_clk_period is the ref_clk period including the fractional value. Example: If the ref_clk is 24 MHz then - GUCTL.REF_CLK_PERIOD = 41 - GFLADJ.GLADJ_REFCLK_FLADJ = $((125000/41) - (125000/41.6666)) * 41.6666 = 2032$ (ignoring the fractional value)
7	GFLADJ_30MHZ_SDBND_SEL. Read-write. Reset: 0. Description: GFLADJ_30MHZ_SDBND_SEL This field selects whether to use the input signal fladj_30mhz_reg or the GFLADJ.GFLADJ_30MHZ to adjust the frame length for the SOF/ITP. When this bit is set to, - 1, the controller uses the register field GFLADJ.GFLADJ_30MHZ value - 0, the controller uses the input signal fladj_30mhz_reg value
6	Reserved.
5:0	GFLADJ_30MHZ. Read-write. Reset: 20h. Description: GFLADJ_30MHZ This field indicates the value that is used for frame length adjustment instead of considering from the sideband input signal fladj_30mhz_reg. This enables post-silicon frame length adjustment in case the input signal fladj_30mhz_reg is connected to a wrong value or is not valid. For details on how to set this value, refer to section 5.2.4, "Frame Length Adjustment Register (FLADJ)," of the xHCI Specification.

USB[0...1]x00C634 (FCHUSBWDCUSB31CNTR0::GUCTL4)

Reset: 0001_FFFFh.

Global User Control Register 4

_instUSB0_aliasHOST; USB0x00C634; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C634; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	Reserved.
30:26	SSP_BWD_OVHD_ADJ. Read-write,Reserved. Reset: 00h. Description: SS/SSP bandwidth adjust control GUCTL4[30]=1 indicates GUCTL4[29:26] will be subtracted from the overhead for each uframe(125us) GUCTL4[30]=0 indicates GUCTL4[29:26] will be added the overhead for each uframe(125us) GUCTL4[29:26] indicates duration in terms of 1us.
25:21	Reserved.
20:19	LOA_EOP_CHECK_CLKS_WORD. Read-write. Reset: 0h. Description: Number of clock cycles to check for EOP SE0 for LOA logic Note: - This field only affects USB2.0 FS operation in UTMI 16bit mode. The U2 root hub logic checks for the programmed number of clocks of SE0 to validate an EOP for a packet. - The actual number of clocks checked will be the value programmed plus 1. So a value 1 will results is checking for 2 clocks of SE0. Value of 3 will do 4 clocks of SE0 check and so on. - The default value is 1, which results in 2 clocks of SE0 check.
18:17	LOA_EOP_CHECK_CLKS_BYTE. Read-write. Reset: 0h. Description: Number of clock cycles to check for EOP SE0 for LOA logic Note: - This field only affects USB2.0 FS operation in UTMI 8bit mode. The U2 root hub logic checks for the programmed number of clocks of SE0 to validate an EOP for a packet. - The actual number of clocks checked will be the value programmed plus 1. So a value 1 will results is checking for 2 clocks of SE0. Value of 3 will do 4 clocks of SE0 check and so on. - The default value is 1, which results in 2 clocks of SE0 check.
16:0	CSR_TIMEOUT_VL. Read-write. Reset: 1_FFFFh. Description: CSR Timeout Value This register field determines the CSR access timeout value. Note: - It is not recommended to change this register field value except for simulation and debug purpose. - Bit-bash testing is not recommended for this register field due to side-effects of early timeout access with ram_clk/link_clk domain registers.

USB[0...1]x00C638 (FCHUSBWDCUSB31CNTR0::GUCTL5)

Read-write. Reset: 0000_0000h.

Global User Control Register 5

_instUSB0_aliasHOST; USB0x00C638; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C638; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:0	Reserved.

USB[0...1]x00C640 (FCHUSBWCUSB31CNTR0::GUSB2RHBCTL_REGS0_GUSB2RHBCTL)

Reset: 0000_0000h.

Global USB2 Root Hub Control Register

In Device-only configurations, only one register is needed.

In Host mode, per-port registers are implemented.

_instUSB0_aliasHOST; USB0x00C640; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C640; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:20	Reserved.
19:12	OVRD_FS_INT_PKT_DEL. Read-write. Reset: 00h. Description: Overriding the internal FS interpacket delay. This field indicates the value of Tx-to-Tx packet gap for FS devices. The encoding is as follows: - 0: The FS Tx-to-Tx interpacket delay is calculated internally. This value is not used. - Non-0: The FS Tx-to-Tx interpacket delay is OVRD_FS_INT_PKT_DEL number of UTMI clocks. Note: - For normal operation (to work with most FS devices), no change is required to this field. - The programmable FS device Tx-to-Tx interpacket delay is provided to support any legacy FS devices that might require different delays than the default/fixed ones. - Include your PHY delays when programming this value. For example, if your PHY TxEndDelay in FS mode is five UTMI/ULPI clocks, then subtract this delay from the delay requirement of the device.
11:4	OVRD_HS_INT_PKT_DEL. Read-write. Reset: 00h. Description: Overriding the internal HS interpacket delay. This field indicates the value of Tx-to-Tx packet gap for HS devices. The encoding is as follows: - 0: The HS Tx-to-Tx interpacket delay is calculated internally. This value is not used. - Non-0: The HS Tx-to-Tx interpacket delay is OVRD_HS_INT_PKT_DEL number of UTMI clocks. This will also override the TX_IPGAP_LINECHECK_DIS setting. Note: - For normal operation (to work with most HS devices), no change is required to this field. - When this is set to a non-0 value, it is required to set the TX_IPGAP_LINECHECK_DIS to 0. - The programmable HS device Tx-to-Tx interpacket delay is provided to support any legacy HS devices that might require different delays than the default/fixed ones. - Include your PHY delays when programming this value. For example, if your PHY TxEndDelay in HS mode is five UTMI/ULPI clocks, then subtract this delay from the delay requirement of the device.
3:0	OVRD_L1TIMEOUT. Read-write. Reset: 0h. Description: Overriding the driver programmed L1TIMEOUT value. If this value is 0, the L1 Timeout value is taken from the xHCI PORTHLPMC register. If this value is non-0, then this will override the L1 Timeout value programmed in the xHCI PORTHLPMC register. In that case the actual L1 Timeout would be $2^{\text{OVRD_L1TIMEOUT-1}} * 8\mu\text{s}$. (1=8us, 2=16us, 3=32us etc)

USB[0...1]x00C644 (FCHUSBWCUSB31CNTR0::GUSB2RHBCTL_REGS1_GUSB2RHBCTL)

Reset: 0000_0000h.

Global USB2 Root Hub Control Register

In Device-only configurations, only one register is needed.

In Host mode, per-port registers are implemented.

_instUSB0_aliasHOST; USB0x00C644; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00C644; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:20	Reserved.
19:12	OVRD_FS_INT_PKT_DEL. Read-write. Reset: 00h. Description: Overriding the internal FS interpacket delay. This field indicates the value of Tx-to-Tx packet gap for FS devices. The encoding is as follows: - 0: The FS Tx-to-Tx interpacket delay is calculated internally. This value is not used. - Non-0: The FS Tx-to-Tx interpacket delay is OVRD_FS_INT_PKT_DEL number of UTMI clocks. Note: - For normal operation (to work with most FS devices), no change is required to this field. - The programmable FS device Tx-to-Tx interpacket delay is provided to support any legacy FS devices that might require different delays than the default/fixed ones. - Include your PHY delays when programming this value. For example, if your PHY TxEndDelay in FS mode is five UTMI/ULPI clocks, then subtract this delay from the delay requirement of the device.
11:4	OVRD_HS_INT_PKT_DEL. Read-write. Reset: 00h. Description: Overriding the internal HS interpacket delay. This field indicates the value of Tx-to-Tx packet gap for HS devices. The encoding is as follows: - 0: The HS Tx-to-Tx interpacket delay is calculated internally. This value is not used. - Non-0: The HS Tx-to-Tx interpacket delay is OVRD_HS_INT_PKT_DEL number of UTMI clocks. This will also override the TX_IPGAP_LINECHECK_DIS setting. Note: - For normal operation (to work with most HS devices), no change is required to this field. - When this is set to a non-0 value, it is required to set the TX_IPGAP_LINECHECK_DIS to 0. - The programmable HS device Tx-to-Tx interpacket delay is provided to support any legacy HS devices that might require different delays than the default/fixed ones. - Include your PHY delays when programming this value. For example, if your PHY TxEndDelay in HS mode is five UTMI/ULPI clocks, then subtract this delay from the delay requirement of the device.
3:0	OVRD_L1TIMEOUT. Read-write. Reset: 0h. Description: Overriding the driver programmed L1TIMEOUT value. If this value is 0, the L1 Timeout value is taken from the xHCI PORTHLPMC register. If this value is non-0, then this will override the L1 Timeout value programmed in the xHCI PORTHLPMC register. In that case the actual L1 Timeout would be $2^{<OVRD_L1TIMEOUT-1>} * 8\mu s$. (1=8us, 2=16us, 3=32us etc)

USB[0...1]x00CC00 (FCHUSBDCUSB31CNTR0::RSVD0_REG)

Read-only. Reset: 0000_0000h.

Reserved Register

_instUSB0_aliasHOST; USB0x00CC00; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00CC00; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description31:0 **rsvd_field.** Read-only. Reset: 0000_0000h. Reserved Register**USB[0...1]x00D000 (FCHUSBDCUSB31CNTR0::LINK_REGS0_LU1LFPSRXTIM)**

Read-write. Reset: 3030_2126h.

U1_LFPS_RX_TIMER_REG

_instUSB0_aliasHOST; USB0x00D000; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D000; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description31:24 **gen2_u1_lfps_exit_rx_clk.** Read-write. Reset: 30h.**Description:** gen2 U1_LFPS_EXIT_RX_CLKS

- Time to recognize Ux exit request from the remote partner.

- This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS.

-- 1: 6.4ns

-- 2: 12.8ns

-- 3: 19.2ns, and so on

23:16 **gen2_u1_exit_rsp_rx_clk.** Read-write. Reset: 30h.**Description:** gen2 U1_EXIT_RESP_RX_CLKS

- Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful.

- This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS.

-- 1: 6.4ns

-- 2: 12.8ns

-- 3: 19.2ns, and so on

15:8 **gen1_u1_lfps_exit_rx_clk.** Read-write. Reset: 21h.**Description:** gen1 U1_LFPS_EXIT_RX_CLKS

- Time to recognize Ux exit request from remote partner.

- This field is encoded as the Gen1 link_clk (8ns) count for the LFPS.

-- 1: 8ns

-- 2: 16ns

-- 3: 24ns, and so on

7:0 **gen1_u1_exit_rsp_rx_clk.** Read-write. Reset: 26h.**Description:** gen1 U1_EXIT_RESP_RX_CLKS

- Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful.

- This field is encoded as the Gen1 link_clk (8ns) count for the LFPS.

-- 1: 8ns

-- 2: 16ns

-- 3: 24ns, and so on

USB[0...1]x00D004 (FCHUSBWCUSB31CNTR0::LINK_REGS0_LU1LFPSTXTIM)

Reset: 0094_0076h.

U1 LFPS TX TIMER REGISTER

_instUSB0_aliasHOST; USB0x00D004; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D004; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	Reserved.
29:16	gen2_u1_exit_resp_tx_clk. Read-write. Reset: 0094h. Description: gen2 U1_EXIT_RESP_TX_CLKS - Remotely initiated Ux exit: Local link will transmit response this long before initiating transition to Recovery (t13 - t11) - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on
15:14	Reserved.
13:0	gen1_u1_exit_rsp_tx_clk. Read-write. Reset: 0076h. Description: gen1 U1_EXIT_RESP_TX_CLKS - Remotely initiated Ux exit: Local link will transmit response this long before initiating transition to Recovery (t13 - t11) - This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D008 (FCHUSBDCUSB31CNTR0::LINK_REGS0_LU2LFPSRXTIM)

Read-write. Reset: 3030_2626h.

U2 LFPS RX TIMER REGISTER

_instUSB0_aliasHOST; USB0x00D008; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D008; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	gen2_u2_lfps_exit_rx_clk. Read-write. Reset: 30h. Description: gen2 U2_LFPS_EXIT_RX_CLKS - Time to recognize Ux exit request from remote partner. - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on
23:16	gen2_u2_exit_rsp_rx_clk. Read-write. Reset: 30h. Description: gen2 U2_EXIT_RESP_RX_CLKS - Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful. - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on
15:8	gen1_u2_lfps_exit_rx_clk. Read-write. Reset: 26h. Description: gen1 U2_LFPS_EXIT_RX_CLKS - Time to recognize Ux exit request from remote partner. - This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
7:0	gen1_u2_exit_rsp_rx_clk. Read-write. Reset: 26h. Description: gen1 U2_EXIT_RESP_RX_CLKS - Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful. - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on

USB[0...1]x00D00C (FCHUSBDWCUSB31CNTR0::LINK_REGS0_LU2LFPSTXTIM)

Reset: 0000_0052h.

U2 LFPS TX TIMER REG REGISTER

_instUSB0_aliasHOST; USB0x00D00C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR],
NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D00C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR],
NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:14	Reserved.
13:0	u2_exit_rsp_tx_us. Read-write. Reset: 0052h. Description: U2_EXIT_RESP_TX_US - Remotely initiated Ux exit: Local link will transmit response this long before initiating transition to Recovery. - When P3 in U2 is enabled, software needs to program this according to suspend_clk - This field is encoded as the number of us count for the LFPS. -- 1 : 1us -- 2 : 2us -- 3 : 3us, and so on

USB[0...1]x00D010 (FCHUSBDCUSB31CNTR0::LINK_REGS0_LU3LFPSRXTIM)

Read-write. Reset: 9C06_7D05h.

U3 LFPS RX TIMER REGS REGISTER

_instUSB0_aliasHOST; USB0x00D010; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D010; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	gen2_u3_lfps_exit_rx_clk. Read-write. Reset: 9Ch. Description: gen2 U3_LFPS_EXIT_RX_CLKS - Time to recognize Ux exit request from remote partner. - When PHY is not in P3, the value in this field is multiplied by 8 to get the U3_LFPS_EXIT_RX_CLKS - This field is encoded as follows: -- 1: $1 \times 8 \times 6.4\text{ns} = 51.2\text{ns}$ -- 2: $2 \times 8 \times 6.4\text{ns} = 102.4\text{ns}$ -- 3: $3 \times 8 \times 6.4\text{ns} = 153.6\text{ns}$, and so on
23:16	gen2_u3_exit_rsp_rx_clk. Read-write. Reset: 06h. Description: gen2 U3_EXIT_RESP_RX_CLKS - Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful. - When PHY is not in P3, the value in this field is multiplied by 8 to get the U3_EXIT_RESP_RX_CLKS - This field is encoded as follows: -- 1: $1 \times 8 \times 6.4\text{ns} = 51.2\text{ns}$ -- 2: $2 \times 8 \times 6.4\text{ns} = 102.4\text{ns}$ -- 3: $3 \times 8 \times 6.4\text{ns} = 153.6\text{ns}$, and so on
15:8	gen1_u3_lfps_exit_rx_clk. Read-write. Reset: 7Dh. Description: gen1 U3_LFPS_EXIT_RX_CLKS - Time to recognize Ux exit request from remote partner. - When PHY is not in P3, the value in this field is multiplied by 8 to get the U3_LFPS_EXIT_RX_CLKS. - This field is encoded as follows: -- 1: $1 \times 8 \times 8.0\text{ns} = 64\text{ns}$ -- 2: $2 \times 8 \times 8.0\text{ns} = 128\text{ns}$ -- 3: $3 \times 8 \times 8.0\text{ns} = 196\text{ns}$, and so on
7:0	gen1_u3_exit_rsp_rx_clk. Read-write. Reset: 05h. Description: gen1 U3_EXIT_RESP_RX_CLKS - Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful. - When PHY is not in P3, the value in this field is multiplied by 8 to get the U3_EXIT_RESP_RX_CLKS. - This field is encoded as follows: -- 1: $1 \times 8 \times 8.0\text{ns} = 64\text{ns}$ -- 2: $2 \times 8 \times 8.0\text{ns} = 128\text{ns}$ -- 3: $3 \times 8 \times 8.0\text{ns} = 196\text{ns}$, and so on

USB[0...1]x00D014 (FCHUSBWCUSB31CNTR0::LINK_REGS0_LU3LFPSTXTIM)

Read-write. Reset: 30D4_2710h.

U3 LFPS TX TIMER REGS REGISTER

_instUSB0_aliasHOST; USB0x00D014; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D014; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	Reserved.
29:16	gen2_u3_exit_tx_clk. Read-write. Reset: 30D4h. Description: gen2 U3_EXIT_RESP_TX_CLKS - Remotely initiated U3 exit: Local link will transmit response this long before initiating transition to Recovery. - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on
15:14	Reserved.
13:0	gen1_u3_exit_rsp_tx_clk. Read-write. Reset: 2710h. Description: gen1 U3_EXIT_RESP_TX_CLKS - Remotely initiated U3 exit: Local link will transmit response this long before initiating transition to Recovery. - This field is encoded as the Gen1 link_clk (8.0ns) count for the LFPS. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D018 (FCHUSBDCUSB31CNTR0::LINK_REGS0_LPINGLFPSTIM)

Read-write. Reset: 4284_3603h.

PING LFPS TIMER REGISTER

_instUSB0_aliasHOST; USB0x00D018; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D018; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:26	gen2_lfps_ping_burst_clk. Read-write. Reset: 10h. Description: gen2 LFPS_PING_BURST_CLKS: TX ping.LFPS tBurst This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. - 1: 6.4ns - 2: 12.8ns - 3: 19.2ns, and so on
25:20	gen2_lfps_ping_max_clk. Read-write. Reset: 28h. Description: gen2 LFPS_PING_MAX_CLKS: Max RX ping.LFPS tBurst This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. - 1: 6.4ns - 2: 12.8ns - 3: 19.2ns, and so on
19:16	gen2_lfps_ping_min_clk. Read-write. Reset: 4h. Description: gen2 LFPS_PING_MIN_CLKS: Min RX ping.LFPS tBurst This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. - 1: 6.4ns - 2: 12.8ns - 3: 19.2ns, and so on
15:10	gen1_lfps_ping_burst_clk. Read-write. Reset: 0Dh. Description: gen1 LFPS_PING_BURST_CLKS: TX ping.LFPS tBurst This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. - 1: 8ns - 2: 16ns - 3: 24ns, and so on
9:4	gen1_lfps_ping_max_clk. Read-write. Reset: 20h. Description: gen1 LFPS_PING_MAX_CLKS: Max RX ping.LFPS tBurst This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. - 1: 8ns - 2: 16ns - 3: 24ns, and so on
3:0	gen1_lfps_ping_min_clk. Read-write. Reset: 3h. Description: gen1 LFPS_PING_MIN_CLKS: Minx RX ping.LFPS tBurst This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. - 1: 8ns - 2: 16ns - 3: 24ns, and so on

USB[0...1]x00D01C (FCHUSBDCUSB31CNTR0::LINK_REGS0_LPOLLFPSTXTIM)

Reset: 0023_287Dh.

POLL LFPS TX TIMER REGISTER

_instUSB0_aliasHOST; USB0x00D01C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D01C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:22	Reserved.
21:11	lfps_poll_space_clk. Read-write. Reset: 465h. Description: LFPS_POLL_SPACE_CLKS: space between Polling.LFPS This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. - 1: 8ns - 2: 16ns - 3: 24ns, and so on
10:0	lfps_poll_burst_clk. Read-write. Reset: 07Dh. Description: LFPS_POLL_BURST_CLKS: TX Polling.LFPS tBurst This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. - 1: 8ns - 2: 16ns - 3: 24ns, and so on

USB[0...1]x00D020 (FCHUSBWCUSB31CNTR0::LINK_REGS0_LSKIPFREQ)

Reset: 4E92_8162h.

SKIP FREQUENCY REGISTER

_instUSB0_aliasHOST; USB0x00D020; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D020; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	Reserved.
30:28	u1_resid_timer_us. Read-write. Reset: 4h. Description: Programmable U1 MIN RESIDENCY TIMER This field specifies U1 MIN RESIDENCY TIMER value in us. Set to 0 to disable the timer.
27	en_pm_timer_us. Read-write. Reset: 1. Description: Enable us PM timers This field enables the use of PM_LC_TIMER and PM_ENTRY_TIMER in us.
26:24	pm_lc_timer_us. Read-write. Reset: 6h. Description: Programmable PM_LC_TIMER This field specifies PM_LC_TIMER value in us.
23:20	pm_entry_timer_us. Read-write. Reset: 9h. Description: Programmable PM_ENTRY_TIMER This field specifies PM_ENTRY_TIMER value in us.
19:12	gen2_skp_freq. Read-write. Reset: 28h. Description: gen2 SKP frequency This field specifies the average number of blocks the link sends before inserting a SKP OS.
11:0	gen1_skp_freq. Read-write. Reset: 162h. Description: gen1 SKP frequency This field specifies the average number of symbols the link sends before inserting a SKP OS.

USB[0...1]x00D024 (FCHUSBWCUSB31CNTR0::LINK_REGSG0_LLUCTL)

Read-write. Reset: 19AB_E0E0h.

TX TS1 COUNT REGISTER

- This register provides an option to increase the number of TS1 sent in Polling.Active or Recovery.Active.
- The total number of TS1 during Polling.Active or Recovery.Active is based on the following formula:

$$\text{Total TS1} = 20 \times 2^{(\text{tx_ts1_cnt})}$$

- This give more opportunity for the remote PHY to achive symbol lock.

_instUSB0_aliasHOST; USB0x00D024; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D024; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	Reserved.
30	inverse_sync_header. Read-write. Reset: 0. Description: For Gen2 polarity detection, link uses data block (0011b) sync header for SYNC OS instead of control block (1100b). Set this bit if the third-party PHY does not correct the sync header of the SYNC OS in case of inverse polarity.
29	support_p4_pg. Read-write. Reset: 0. Description: PHY P4 Power gate mode (PG) is enabled. Set this bit if PHY support PG mode in P4.
28	support_p4. Read-write. Reset: 1. Description: Support PHY P3.CPM and P4. When this bit is set, controller puts PHY into P3.CPM or P4 in certain states.
27:24	delay_tx_gen1_dp. Read-write. Reset: 9h. Description: delay_tx_gen1_dp. When this field is set to a non-zero value, the link delays the transmission of a large Gen1 DP for 32 pipe_pclk cycles if it detects a header being received at the PIU.
23	DisRxDet_LTSSM_Timer_Ovrrd. Read-write. Reset: 1. Description: DisRxDet_LTSSM_Timer_Ovrrd. When DisRxDetU3RxDet is asserted in Polling or U1, the timeout expires immediately
22	force_dpp_truncate. Read-write. Reset: 0. Description: force_dpp_truncate. Truncate Gen1 DPP if there is a pending LGOOD/LBAD and DPP is large. When this bit is set, the logic for detecting recovery due to expiring pending_HP_timer is not used.
21	en_dpp_truncate. Read-write. Reset: 1. Description: en_dpp_truncate. This bit enables the logic for detecting recovery due to expiring pending_HP_timer and truncates DPP when necessary to avoid recovery.
20:16	pending_hp_timer_us. Read-write. Reset: 0Bh. Description: pending_hp_timer_us. Programmable PENDING_HP_TIMER in us. This field is used when LUCTL[15] is set.
15	en_us_hp_timer. Read-write. Reset: 1.

	Description: en_us_hp_timer. This bit enables programmable PENDING_HP_TIMER in us
14:13	ring_buf_d_delay. Read-write. Reset: 3h. Description: ring_buf_d_delay. This programs the counter for data valid signal in source ring buffer
12	U2P3CPMok. Read-write. Reset: 0. Description: P3CPM OK for U2/SSInactive (U2P3CPMok) - 0: During link state U2/ESS.Inactive, put PHY in P2 (Default) - 1: During link state U2/ESS.Inactive, put PHY in P3CPM. Note: For a port, if both GUSB3PIPECTL[29]=1 and LLUCTL[12]=1, LLUCTL[12]=1 takes priority.
11	en_reset_pipe_after_phy_mux. Read-write. Reset: 0. Description: en_reset_pipe_after_phy_mux. controller issues 3.1 PHY reset after DisRxDetU3RxDet is de-asserted
10	force_gen1. Read-write. Reset: 0. Force Gen1
9	gen2_loopback_entry_mode. Read-write. Reset: 0. Description: Gen2 loopback entry mode. If this bit is set, the controller will enter loopback in Gen2 after successfully detecting remote termination.
8	gen1_loopback_entry_mode. Read-write. Reset: 0. Description: Gen1 loopback entry mode. If this bit is set, the controller will enter loopback in Gen1 after successfully detecting remote termination.
7	mask_pipe_reset. Read-write. Reset: 1. Description: Mask pipe reset. If this bit is set, controller will block pipe_reset_n from going to PHY when DisRxDetU3RxDet=1
6	delay_ux_after_lpma. Read-write. Reset: 1. Description: delay_ux_after_lpma. - If this bit is set to 0, after sending LPMA, Link LTSSM delays 10 link_clk cycles before entering Ux. - If this bit is set to 1, after sending LPMA, Link LTSSM delays 20 link_clk cycles before entering Ux.
5	no_ux_exit_p0_trans. Read-write. Reset: 1. Description: no_ux_exit_p0_trans. - If this bit is 0, Link LTSSM detects Ux_exit LFPS during P0 transition. - If this bit is set to 1, the Link LTSSM does not detect Ux_exit LFPS when P0 transition is happening.
4:0	tx_ts1_cnt. Read-write. Reset: 00h. Additional TX_TS1_count

USB[0...1]x00D028 (FCHUSBWDCUSB31CNTR0::LINK_REGS0_LPTMDPDELAY)

Read-write. Reset: 30C0_09C4h.

PTM DATAPATH DELAY REGISTER

This register provides an option to override the default timing parameters used in PTM calculation.

_instUSB0_aliasHOST; USB0x00D028; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D028; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:27	rx_path_delay_gen2. Read-write. Reset: 06h. Description: rx_path_delay_gen2 RX path delay in unit of 8ns
26:22	tx_path_delay_gen2. Read-write. Reset: 03h. Description: tx_path_delay_gen2 TX path delay in unit of 8ns
21:10	p3cpmp4_residency. Read-write. Reset: 002h. Description: Minimum number of suspend_clk periods that the PHY needs to stay in P3.CPM or P4 before exiting P3.CPM or P4.
9:5	rx_path_delay. Read-write. Reset: 0Eh. Description: rx_path_delay Gen1 RX path delay (including PHY and PCS) in terms of 8ns.
4:0	tx_path_delay. Read-write. Reset: 04h. Description: tx_path_delay Gen1 TX path delay (including PHY and PCS) in terms of 8ns.

USB[0...1]x00D02C (FCHUSBWDCUSB31CNTR0::LINK_REGS0_LSCDTIM1)

Reset: 0047_E1F4h.

SCD TIMER 1 REGISTER

_instUSB0_aliasHOST; USB0x00D02C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D02C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	Reserved.
23:12	scd_bit0_rpt_max_clk. Read-write. Reset: 47Eh. Description: SCD_BIT0_REPEAT_MAX_CLKS - max duration of SCD bit 0 tRepeat in term of CLKs, used in SCD detection FSM - This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
11:0	cd_bit0_rpt_min_clk. Read-write. Reset: 1F4h. Description: CD_BIT0_REPEAT_MIN_CLKS - min duration of SCD bit 0 tRepeat in term of CLKs, used in SCD detection FSM - This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. -- 1 : 8ns -- 2 : 16ns -- 3 : 24ns, and so on

USB[0...1]x00D030 (FCHUSBWDCUSB31CNTR0::LINK_REGS0_LSCDTIM2)

Reset: 0075_3546h.

SCD TIMER2 REGISTER

_instUSB0_aliasHOST; USB0x00D030; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D030; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	Reserved.
23:12	scd_bit1_rpt_max_clk. Read-write. Reset: 753h. Description: SCD_BIT1_REPEAT_MAX_CLKS - max duration of SCD bit 1 tRepeat in term of CLKs, used in SCD detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
11:0	cd_bit1_rpt_min_clk. Read-write. Reset: 546h. Description: CD_BIT1_REPEAT_MIN_CLKS - min duration of SCD bit 1 tRepeat in term of CLKs, used in SCD detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D034 (FCHUSBWDCUSB31CNTR0::LINK_REGS0_LSCDTIM3)

Read-write. Reset: 5DC2_F07Dh.

SCD TIMER3 REGISTER

_instUSB0_aliasHOST; USB0x00D034; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D034; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:20	lfps_scd_space1_clk. Read-write. Reset: 5DCh. Description: LFPS_SCD_SPACE1_CLKS - Duration of SCD space 1 used in SCD TX FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
19:8	lfps_scd_space0_clk. Read-write. Reset: 2F0h. Description: LFPS_SCD_SPACE0_CLKS - Duration of SCD space 0 used in SCD TX FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
7:0	lfps_scd_burst_clk. Read-write. Reset: 7Dh. Description: LFPS_SCD_BURST_CLKS - Duration of tBurst used in SCD TX FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D038 (FCHUSBWDCUSB31CNTR0::LINK_REGS0_LSCDTIM4)

Reset: 0000_0EA6h.

SCD TIMER4 REGISTER

_instUSB0_aliasHOST; USB0x00D038; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D038; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:12	Reserved.
11:0	lfps_scd_last_space. Read-write. Reset: EA6h. Description: LFPS_SCD_LAST_SPACE_CLKS - Duration of the last space for each SCD - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D03C (FCHUSBWDCUSB31CNTR0::LINK_REGS0_LLPBMTIM1)

Read-write. Reset: F096_6B32h.

LPBM TIMER1 REGISTER

_instUSB0_aliasHOST; USB0x00D03C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D03C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	lfps_lbps1_burst_max_clk. Read-write. Reset: F0h. Description: LFPS_LBPS1_BURST_MAX_CLKS - Max duration of LPBM bit 1, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
23:16	lfps_lbps1_burst_min_clk. Read-write. Reset: 96h. Description: LFPS_LBPS1_BURST_MIN_CLKS - Min duration of LPBM bit 1, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
15:8	lfps_lbps0_burst_max_clk. Read-write. Reset: 6Bh. Description: LFPS_LBPS0_BURST_MAX_CLKS - Max duration of LPBM bit 0, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
7:0	lfps_lbps0_burst_min_clk. Read-write. Reset: 32h. Description: LFPS_LBPS0_BURST_MIN_CLKS - Min duration of LPBM bit 0, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D040 (FCHUSBDCUSB31CNTR0::LINK_REGS0_LLPBMTIM2)

Reset: 0002_6CF5h.

LPBM TIMER2 REGISTER

_instUSB0_aliasHOST; USB0x00D040; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D040; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:18	Reserved.
17:9	lfps_lbpm_tpwm_max_clk. Read-write. Reset: 136h. Description: LFPS_LBPM_TPWM_MAX_CLKS - Max value of TPWM, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
8:0	lfps_lbpm_tpwm_min_clk. Read-write. Reset: 0F5h. Description: LFPS_LBPM_TPWM_MIN_CLKS - Min value of TPWM, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D044 (FCHUSBDCUSB31CNTR0::LINK_REGS0_LLPBMTXTIM)

Reset: 1130_C850h.

LPBM TX TIMER REGISTER

_instUSB0_aliasHOST; USB0x00D044; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D044; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	Reserved.
29:20	lfps_lbpm_tpwm_clk. Read-write. Reset: 113h. Description: LFPS LBPS TPWM CLKS - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
19:18	Reserved.
17:8	lfps_lbpm_burst1_clk. Read-write. Reset: 0C8h. Description: LFPS LBPS BURST1 CLKS - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
7:0	lfps_lbpm_burst0_clk. Read-write. Reset: 50h. Description: LFPS LBPS BURST0 CLKS - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D048 (FCHUSBWCUSB31CNTR0::LINK_REGS0_LLINKERRINJ)

Read-write. Reset: 0000_0000h.

LINK ERROR TYPE INJECT REGISTER

_instUSB0_aliasHOST; USB0x00D048; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D048; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	pipe_txdata. Read-write. Reset: 0. Description: PIPE TxData - Inject single bit error to TxData. - This includes additional logic to ensure IDLE symbols at the end of recovery aren't corrupted so that link won't enter recovery loop.
30	TX_insert_defer. Read-write. Reset: 0. TX Insert Deferred Bit
29	TX_insert_delay. Read-write. Reset: 0. TX Insert Delayed Bit
28	TX_TS2. Read-write. Reset: 0. TX_TS2 : Corrupt entire TS2
27	TX_TS1. Read-write. Reset: 0. TX_TS1 : Corrupt entire TS1
26	TX_TSEQ. Read-write. Reset: 0. TX_TSEQ : Corrupt entire TSEQ
25	TX_LFR. Read-write. Reset: 0. Description: TX_LFR - Transmitted DPH will have invalid LFR
24	TX_delay_credit_rl. Read-write. Reset: 0. Description: TX Delay credit release - when this bit is set, link will wait for the amount of microsecond specified in the Error Enable Counter register [15:0]
23	TX_one_srt_frame_sybl. Read-write. Reset: 0. Description: TX DPP all start framing symbol - all framing symbols of DPP (SDP and EPF) will be corrupted
22	TX_all_end_frame_sybl. Read-write. Reset: 0. Description: TX DPP all end framing symbol - all framing symbols of DPP (END and EPF) will be corrupted
21	TX_one_end_frame_sybl. Read-write. Reset: 0. Description: TX DPP one end framing symbol - one framing symbol of DPP (END) will be corrupted
20	TX_all_frame_sybl. Read-write. Reset: 0. Description: TX TPH/DPH all framing symbol - all framing symbols of TPH/DPH (SHP, DPHP, EPF) will be corrupted
19	TX_one_frame_sybl. Read-write. Reset: 0. Description: TX TPH/DPH one framing symbol - One framing symbol of TPH/DPH (SHP or DPHP) will be corrupted
18	TX_CRC32. Read-write. Reset: 0. Description: TX_CRC32 - Transmitted DPP will have invalid CRC32
17	TX_CRC16. Read-write. Reset: 0. Description: TX_CRC16 - Transmitted header will have invalid CRC16
16	TX_CRC5. Read-write. Reset: 0. Description: TX_CRC5 - Transmitted header will have CRC error in Link Control Word
15	Rx_pipe_rxdata. Read-write. Reset: 0.

	Description: Rx_pipe_rxdata - Inject single bit error to RxData. - This includes additional logic to ensure IDLE symbols at the end of recovery aren't corrupted so that link won't enter recovery loop.
14	RX_insert_defer. Read-write. Reset: 0. RX_insert_defer
13	RX_insert_delay. Read-write. Reset: 0. RX_insert_delay
12	RX_TS2. Read-write. Reset: 0. Description: RX_TS2 - Corrupt entire TS2
11	RX_TS1. Read-write. Reset: 0. Description: RX_TS1 - Corrupt entire TS1
10	RX_TSEQ. Read-write. Reset: 0. Description: RX_TSEQ - Corrupt entire TSEQ
9	RX_LFR. Read-write. Reset: 0. Description: RX_LFR - Received DPH will have invalid LFR
8	RX_all_start_frame_sybl. Read-write. Reset: 0. Description: DPP all start framing symbol - Before 64-bit received data is forwarded to parser, all framing symbols (SDP or EPF) will be corrupted. - Note: because the entire framing symbol may not be aligned and presented on the 64-bit received data, 1 - 4 framing symbols will be corrupted
7	RX_one_srt_frame_sybl. Read-write. Reset: 0. Description: DPP one start framing symbol - Before received data is forwarded to parser in RPKT, only one framing symbol (SDP or EPF) will be corrupted
6	RX_all_end_frame_sybl. Read-write. Reset: 0. Description: DPP all end framing symbol - Before 64-bit received data is forwarded to parser, all framing symbols (END or EPF) will be corrupted. - Note: Because the entire framing symbol may not be aligned and presented on the 64-bit received data, 1 - 4 framing symbols will be corrupted
5	RX_one_end_frame_sybl. Read-write. Reset: 0. Description: RX DPP one end framing symbol - Before received data is forwarded to parser in RPKT, only one framing symbol (END or EPF) will be corrupted
4	RX_all_frame_sybl. Read-write. Reset: 0. Description: RX TPH/DPH all framing symbol - Before 64-bit received data is forwarded to parser, all framing symbols (SHP or EPF or DPHP) will be corrupted. - Note: Because the entire framing symbol may not be aligned and presented on the 64-bit received data, 1 - 4 framing symbols will be corrupted
3	RX_one_frame_sybl. Read-write. Reset: 0. Description: RX TPH/DPH one framing symbol - Before received data is forwarded to parser in RPKT, only one framing symbol (SHP or EPF or DPHP) will be corrupted
2	RX_CRC32. Read-write. Reset: 0. Description: RX_CRC32 - Received DPP will have invalid CRC32 value
1	RX_CRC16. Read-write. Reset: 0. Description: RX_CRC16 - Received header will have invalid CRC16 value

0	RX_CRC5. Read-write. Reset: 0.
	Description: RX_CRC5 - Received header will have invalid CRC5 value in Link Control Word

USB[0...1]x00D04C (FCHUSBDCUSB31CNTR0::LINK_REGS0_LLINKERRINJEN)

Read-write. Reset: 0000_0000h.

LINK ERROR INJECT ENABLE REGISTER

- This register is for injecting N number of errors for every M number of packets

_instUSB0_aliasHOST; USB0x00D04C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D04C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	disable_inj_err_cnt. Read-write. Reset: 0000h. Description: Disable injected error count - This error specifies the interval of error free between 2 error injection periods. - In the case of PIPE data error injection, this field specify the number of microsecond between every bit error injection
15:0	B2B_err_cnt. Read-write. Reset: 0000h. Description: B2B error count - This counter indicates the number of error to be injected consecutively. - For instance, the RX CRC5 error type is selected and this counter value is 3, then the next 3 received headers will have invalid CRC5

USB[0...1]x00D050 (FCHUSBWCUSB31CNTR0::LINK_REGSO_GDBGLTSSM)

Reset: 4101_0440h.

Global Debug LTSSM Register

Note: GDBGLTSSM register is not applicable for USB 2.0-only mode.

- Bit Bash test should not be done on this debug register.

_instUSB0_aliasHOST; USB0x00D050; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D050; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	Reserved.
30	RxElecIdle . Read-only, Volatile. Reset: 1. Description: RxElecIdle For description of RxElecIdle, see table 5-4, "Status Interface Signals" of the PIPE4 Specification.
29:27	Reserved.
26	LTDBTIMEOUT . Read-only. Reset: 0. LTDB Timeout (LTDBTimeout)
25:22	LTDBLINKSTATE . Read-only. Reset: 4h. LTDB Link State (LTDBLinkState)
21:18	LTDBSUBSTATE . Read-only, Volatile. Reset: 0h. LTDB Sub-State (LTDBSubState)
17	ELASTICBUFFERMODE . Read-only. Reset: 0. Description: Elastic Buffer Mode (ElasticBufferMode) For field definition, refer to Table 5-3 of the PIPE specification.
16	TXELECLDLE . Read-only. Reset: 1. Description: Tx Elec Idle (TxElecIdle) For field definition, refer to Table 5-3 of the PIPE specification.
15	RXPOLARITY . Read-only. Reset: 0. Description: Rx Polarity (RxPolarity) For field definition, refer to Table 5-3 of the PIPE specification.
14	TxDetRxLoopback . Read-only. Reset: 0. Description: Tx Detect Rx/Loopback (TxDetRxLoopback) For field definition, refer to Table 5-3 of the PIPE specification.
13:11	LTDBPhyCmdState . Read-only, Volatile. Reset: 0h. Description: LTSSM PHY command State (LTDBPhyCmdState) - 000: PHY_IDLE (PHY command state is in IDLE. No PHY request pending) - 001: PHY_DET (Request to start Receiver detection) - 010: PHY_DET_3 (Wait for Phy_Status (Receiver detection)) - 011: PHY_PWR_DLY (Delay Pipe3_PowerDown P0 -> P1/P2/P3 request) - 100: PHY_PWR_A (Delay for internal logic) - 101: PHY_PWR_B (Wait for Phy_Status(Power state change request))
10:9	POWERDOWN . Read-only, Volatile. Reset: 2h. Description: POWERDOWN (PowerDown) For field definition, refer to Table 5-3 of the PIPE Specification.
8	RXEQTRAIN . Read-only. Reset: 0. Description: RxEq Train For field definition, refer to Table 5-3 of the PIPE Specification.

7:6	TXDEEMPHASIS. Read-only. Reset: 1h. Description: TXDEEMPHASIS (TxDeemphasis) For field definition, refer to Table 5-3 of the PIPE Specification.
5:3	LTDBClkState. Read-only. Reset: 0h. Description: LTSSM Clock State (LTDBClkState) In multi-port host configuration, the port number is defined by Port-Select[3:0] field in the GDBGFIFOSPACE register. Note: GDBGLTSSM register is not applicable for USB 2.0-only mode. - 000: CLK_NORM (PHY is in non-P3 state and PCLK is running) - 001: CLK_TO_P3 (P3 entry request to PHY); - 010: CLK_WAIT1 (Wait for Phy_Status (P3 request)); - 011: CLK_P3 (PHY is in P3 and PCLK is not running); - 100: CLK_TO_P0 (P3 exit request to PHY); - 101: CLK_WAIT2 (Wait for Phy_Status (P3 exit request))
2	TXSWING. Read-only. Reset: 0. Description: Tx Swing (TxSwing) For field definition, refer to Table 5-3 of the PIPE Specification.
1	RXTERMINATION. Read-only. Reset: 0. Rx Termination (RxTermination)
0	TXONESZEROS. Read-only. Reset: 0. Description: Tx Ones/Zeros (TxOnesZeros) For field definition, refer to Table 5-3 of the PIPE Specification.

USB[0...1]x00D054 (FCHUSBWCUSB31CNTR0::LINK_REGS0_GDBG_LNMCC)

Read-only. Reset: 0000_0000h.

Global Debug LNMCC Register

Bit Bash test should not be done on this debug register.

_instUSB0_aliasHOST; USB0x00D054; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D054; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	LNMCCLSERC. Read-only. Reset: 0000h. Description: This field indicates the value of link soft error counter for the port selected in the GDBGFIFOSPACE.PortSelect field. This field is for debug purposes only.
15:9	Reserved.
8:0	LNMCCLBERC. Read-only. Reset: 000h. Description: This field indicates the bit error rate information for the port selected in the GDBGFIFOSPACE.PortSelect field. This field is for debug purposes only.

USB[0...1]x00D058 (FCHUSBWCUSB31CNTR0::LINK_REGS0_LLINKDBGCTRL)

Reset: 8000_0000h.

LINK DEBUG CONTROL REGISTER

- This register is used for link debug purpose.

_instUSB0_aliasHOST; USB0x00D058; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D058; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31	retry_DP. Read-write. Reset: 1. Link retries full DP instead of DPH in Gen2. When link needs to resend header due to LBAD or recovery replay, enabling this bit makes link to send full DPH + DPP. This workaround is for a non-compliance 3.1 device
30	trigger_end_sts. Read-only. Reset: 0. Trigger End Status
29	trigger_start_sts. Read-only. Reset: 0. Trigger Start Status
28	link_state_trigger_descrambled. Read-write. Reset: 0. Description: Link State Trigger Descrambled Debug Control When set, the controller writes the descrambled received data into the RAM. The received data is the output of the descrambler. The link starts writing debug data based on the following conditions: - Link_state and sub_state match the link_state_trigger and sub_state_trigger field, respectively. After this condition occurs, the link internal debug_byte_count starts incrementing for every link clock, based on the number of bytes to be written to the RAM. For example, if MDWIDTH=128, the debug_byte_count increments by 16 every link_clk. - Link internal debug_byte_count reaches the start_count_trigger field. The link stops writing debug data when the link internal debug_byte_count reaches the stop_count_trigger field. If stop_count is zero, the link writes out data continuously.
27	link_state_trigger_scrambled. Read-write. Reset: 0. Description: Link State Trigger Scrambled Debug Control When set, controller writes the PIPE received data into the RAM. The received data is scrambled. The link starts writing debug data based on the following conditions: - Link_state and sub_state match the link_state_trigger and sub_state_trigger field, respectively. After this condition occurs, the link internal debug_byte_count starts incrementing for every link clock, based on the number of bytes to be written to RAM. For example, if MDWIDTH=64, the debug_byte_count increments by 8 every link_clk. - Link internal debug_byte_count reaches the start_count_trigger field The link stops writing debug data when the link internal debug_byte_count reaches the stop_count_trigger field. If stop_count is zero, the link writes out data continuously.
26	pipe_txdetectrxlb_trigger. Read-write. Reset: 0. Description: PIPE TXDETECTRXLB Trigger Debug Control The link starts writing debug data based on the following conditions: - Link_state and sub_state match the link_state_trigger and sub_state_trigger field, respectively - Assertion of pipe_TxDetectRxLoopbk
25	pipe_rxlecidle_trigger. Read-write. Reset: 0.

	Description: PIPE RXELECIdle Trigger Debug Control The link start writing debug data based on the following conditions: - Link_state and sub_state match the link_state_trigger and sub_state_trigger field, respectively - Change state of pipe_RxElecIdle
24	pipe_phystatus_trigger. Read-write. Reset: 0. Description: PIPE PHYSTATUS Trigger Debug Control The link start writing debug data based on the following conditions: - Link_state and sub_state match the link_state_trigger and sub_state_trigger field, respectively - Assertion of pipe_PhyStatus
23:20	sub_state_trigger. Read-write. Reset: 0h. Sub-state Trigger
19:16	link_state_trigger. Read-write. Reset: 0h. Link State Trigger
15:12	peri_rsc_rxfifo_number. Read-write. Reset: 0h. Periodic Resource RXFIFO Number
11:8	peri_rsc_txfifo_number. Read-write. Reset: 0h. Periodic Resource TXFIFO Number
7:4	rxfifo_number. Read-write. Reset: 0h. RXFIFO Number
3:0	txfifo_number. Read-write. Reset: 0h. TXFIFO Number

USB[0...1]x00D05C (FCHUSBWCUSB31CNTR0::LINK_REGS0_LLINKDBGCNTTRIG)

Read-write. Reset: 0000_0000h.

LINK DEBUG [COUNT](#) TRIGGER REGISTER

This register is used in conjunction with the LINK_DEBUG_CTRL_REG (LINKDGBCTRL). For more information, refer to the LINK_DEBUG_CTRL_REG description.

_instUSB0_aliasHOST; USB0x00D05C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D05C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	stp_cnt. Read-write. Reset: 0000h. Description: Stop Count This field is used for triggering purpose based on the selection in the LINK_DEBUG_CTRL_REG. If this field is zero, link writes out debug data continuously. SW needs to take into account of the RX RAM size to program this field. (stop_count start_count) is less than total RX RAM size.
15:0	srt_cnt. Read-write. Reset: 0000h. Description: Start Count This field is used for triggering purpose based on the selection in the LINK_DEBUG_CTRL_REG

USB[0...1]x00D060 (FCHUSBDWCUSB31CNTR0::LINK_REGS0_LCSR_TX_DEEMPH)

Reset: 0000_8C45h.

LCSR_TX_DEEMPH REGISTER

This register is for setting TX deemphasis used in normal operation in gen2.

_instUSB0_aliasHOST; USB0x00D060; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D060; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}**Bits Description**

31:18 Reserved.

17:0 **csr_tx_deemph_field_1**. Read-write. Reset: 0_8C45h. 18 bits of TX deemphasis, defined in PIPE4 spec, in normal operation (non-compliance mode)**USB[0...1]x00D064 (FCHUSBDWCUSB31CNTR0::LINK_REGS0_LCSR_TX_DEEMPH_1)**

Reset: 0000_0E45h.

LCSR_TX_DEEMPH_1 REGISTER

This register is for setting TX deemphasis used in compliance mode, pattern 13

_instUSB0_aliasHOST; USB0x00D064; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D064; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}**Bits Description**

31:18 Reserved.

17:0 **csr_tx_deemph_1_field_1**. Read-write. Reset: 0_0E45h. 18 bits of TX deemphasis used by controller in gen2 compliance pattern 13**USB[0...1]x00D068 (FCHUSBDWCUSB31CNTR0::LINK_REGS0_LCSR_TX_DEEMPH_2)**

Reset: 0000_8D80h.

LCSR_TX_DEEMPH_2 REGISTER

This register is for setting TX deemphasis used in compliance mode, pattern 14

_instUSB0_aliasHOST; USB0x00D068; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D068; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}**Bits Description**

31:18 Reserved.

17:0 **csr_tx_deemph_2_field_1**. Read-write. Reset: 0_8D80h. 18 bits of TX deemphasis used by controller in gen2 compliance pattern 14**USB[0...1]x00D06C (FCHUSBDWCUSB31CNTR0::LINK_REGS0_LCSR_TX_DEEMPH_3)**

Reset: 0000_0F80h.

LCSR_TX_DEEMPH_3 REGISTER

This register is for setting TX deemphasis used in compliance mode, pattern 16

_instUSB0_aliasHOST; USB0x00D06C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D06C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}**Bits Description**

31:18 Reserved.

17:0 **csr_tx_deemph_3_field_1**. Read-write. Reset: 0_0F80h. 18 bits of TX deemphasis used by controller in gen2 compliance pattern 16

USB[0...1]x00D070 (FCHUSBWCUSB31CNTR0::LINK_REGSG0_LCSRPTMDEBUG1)

Read-only. Reset: 0000_0000h.

LCSRPTMDEBUG1 REGISTER: This register stores LDM t1 and LDM t32 values

In Host mode this register read value is 32'h0.

_instUSB0_aliasHOST; USB0x00D070; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D070; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	Reserved.
29:17	LDM_T32. Read-only. Reset: 0000h. Description: LDM_T32 This is Response Delay value captured from the LMP TS response field. The unit of this field is tIsochTimestampGranularity
16:0	LDM_T1. Read-only. Reset: 0_0000h. Description: LDM_T1 This is the 't1' timestamp, which is the time the link transmits the LMP TS request. The unit of this field is nanosecond (ns). Note: - The internal timer counts down from 125000 to 0, and restarts at 125000 (therefore, t1 > t4, except in case of a roll over). - This value does not take into account the internal TX pipeline delay. uses the following formula for adjusting 't1' based on the pipeline delay: $TX_PATH_DELAY(tIsochTimestampGranularity) = (gen1 \ ? \ 2 : 2) + LPTMDPDELAY[3:0]$

USB[0...1]x00D074 (FCHUSBDWCUSB31CNTR0::LINK_REGS0_LCSRPTMDEBUG2)

Read-only. Reset: 0000_0000h.

LCSRPTMDELAY2 REGISTER : This register stores LSM t4 value

In Host mode this register read value is 32'h0.

_instUSB0_aliasHOST; USB0x00D074; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D074; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:17 Reserved.

16:0 **LDM_T4**. Read-only. Reset: 0_0000h.**Description:** LDM_T4

This is the 't4' timestamp, which is the time the link receives the LMP TS response.

The unit of this field is nanosecond (ns).

Note:

- The internal timer counts down 125000 to 0, and restarts at 125000 (therefore, $t_1 > t_4$, except the case of roll over)

- This value does not take into account the internal RX pipeline delay. uses this formula for adjusting 't4' based on the pipeline delay:

RX_PATH_DELAY(tIsochTimestampGranularity) = (gen1 ? 4 : 5) + LPTMDPDELAY[7:4]

USB[0...1]x00D078 (FCHUSBDWCUSB31CNTR0::LINK_REGS0_LPTMDPDELAY2)

Reset: 0005_1565h.

PTM DATAPATH DELAY REGISTER2

- This register provide an option to override the default timing parameters used in PTM calculation.

_instUSB0_aliasHOST; USB0x00D078; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D078; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:20 Reserved.

19:15 **link_rx_delay_gen2**. Read-write. Reset: 0Ah.**Description:** link_rx_delay_gen2

RX path delay in unit of 8ns

14:10 **link_tx_delay_gen2**. Read-write. Reset: 05h.**Description:** link_tx_delay_gen2

RX path delay in unit of 8ns

9:5 **link_rx_delay_gen1**. Read-write. Reset: 0Bh.**Description:** link_rx_delay_gen1

RX path delay in unit of 8ns

4:0 **link_tx_delay_gen1**. Read-write. Reset: 05h.**Description:** link_tx_delay_gen1

TX path delay in unit of 8ns

USB[0...1]x00D080 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LU1LFPSRXTIM)

Read-write. Reset: 3030_2126h.

U1_LFPS_RX_TIMER_REG

_instUSB0_aliasHOST; USB0x00D080; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D080; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	gen2_u1_lfps_exit_rx_clk. Read-write. Reset: 30h. Description: gen2 U1_LFPS_EXIT_RX_CLKS - Time to recognize Ux exit request from the remote partner. - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on
23:16	gen2_u1_exit_rsp_rx_clk. Read-write. Reset: 30h. Description: gen2 U1_EXIT_RESP_RX_CLKS - Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful. - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on
15:8	gen1_u1_lfps_exit_rx_clk. Read-write. Reset: 21h. Description: gen1 U1_LFPS_EXIT_RX_CLKS - Time to recognize Ux exit request from remote partner. - This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
7:0	gen1_u1_exit_rsp_rx_clk. Read-write. Reset: 26h. Description: gen1 U1_EXIT_RESP_RX_CLKS - Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful. - This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D084 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LU1LFPSTXTIM)

Reset: 0094_0076h.

U1 LFPS TX TIMER REGISTER

_instUSB0_aliasHOST; USB0x00D084; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D084; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	Reserved.
29:16	gen2_u1_exit_resp_tx_clk. Read-write. Reset: 0094h. Description: gen2 U1_EXIT_RESP_TX_CLKS - Remotely initiated Ux exit: Local link will transmit response this long before initiating transition to Recovery (t13 - t11) - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on
15:14	Reserved.
13:0	gen1_u1_exit_rsp_tx_clk. Read-write. Reset: 0076h. Description: gen1 U1_EXIT_RESP_TX_CLKS - Remotely initiated Ux exit: Local link will transmit response this long before initiating transition to Recovery (t13 - t11) - This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D088 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LU2LFPSRXTIM)

Read-write. Reset: 3030_2626h.

U2 LFPS RX TIMER REGISTER

_instUSB0_aliasHOST; USB0x00D088; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D088; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	gen2_u2_lfps_exit_rx_clk. Read-write. Reset: 30h. Description: gen2 U2_LFPS_EXIT_RX_CLKS - Time to recognize Ux exit request from remote partner. - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on
23:16	gen2_u2_exit_rsp_rx_clk. Read-write. Reset: 30h. Description: gen2 U2_EXIT_RESP_RX_CLKS - Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful. - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on
15:8	gen1_u2_lfps_exit_rx_clk. Read-write. Reset: 26h. Description: gen1 U2_LFPS_EXIT_RX_CLKS - Time to recognize Ux exit request from remote partner. - This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
7:0	gen1_u2_exit_rsp_rx_clk. Read-write. Reset: 26h. Description: gen1 U2_EXIT_RESP_RX_CLKS - Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful. - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on

USB[0...1]x00D08C (FCHUSBDWCUSB31CNTR0::LINK_REGS1_LU2LFPSTXTIM)

Reset: 0000_0052h.

U2 LFPS TX TIMER REG REGISTER

_instUSB0_aliasHOST; USB0x00D08C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR],
NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D08C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR],
NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:14	Reserved.
13:0	u2_exit_rsp_tx_us. Read-write. Reset: 0052h. Description: U2_EXIT_RESP_TX_US - Remotely initiated Ux exit: Local link will transmit response this long before initiating transition to Recovery. - When P3 in U2 is enabled, software needs to program this according to suspend_clk - This field is encoded as the number of us count for the LFPS. -- 1 : 1us -- 2 : 2us -- 3 : 3us, and so on

USB[0...1]x00D090 (FCHUSBWCUSB31CNTR0::LINK_REGS1_LU3LFPSRXTIM)

Read-write. Reset: 9C06_7D05h.

U3 LFPS RX TIMER REGS REGISTER

_instUSB0_aliasHOST; USB0x00D090; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D090; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	gen2_u3_lfps_exit_rx_clk. Read-write. Reset: 9Ch. Description: gen2 U3_LFPS_EXIT_RX_CLKS - Time to recognize Ux exit request from remote partner. - When PHY is not in P3, the value in this field is multiplied by 8 to get the U3_LFPS_EXIT_RX_CLKS - This field is encoded as follows: -- 1: $1 \times 8 \times 6.4\text{ns} = 51.2\text{ns}$ -- 2: $2 \times 8 \times 6.4\text{ns} = 102.4\text{ns}$ -- 3: $3 \times 8 \times 6.4\text{ns} = 153.6\text{ns}$, and so on
23:16	gen2_u3_exit_rsp_rx_clk. Read-write. Reset: 06h. Description: gen2 U3_EXIT_RESP_RX_CLKS - Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful. - When PHY is not in P3, the value in this field is multiplied by 8 to get the U3_EXIT_RESP_RX_CLKS - This field is encoded as follows: -- 1: $1 \times 8 \times 6.4\text{ns} = 51.2\text{ns}$ -- 2: $2 \times 8 \times 6.4\text{ns} = 102.4\text{ns}$ -- 3: $3 \times 8 \times 6.4\text{ns} = 153.6\text{ns}$, and so on
15:8	gen1_u3_lfps_exit_rx_clk. Read-write. Reset: 7Dh. Description: gen1 U3_LFPS_EXIT_RX_CLKS - Time to recognize Ux exit request from remote partner. - When PHY is not in P3, the value in this field is multiplied by 8 to get the U3_LFPS_EXIT_RX_CLKS. - This field is encoded as follows: -- 1: $1 \times 8 \times 8.0\text{ns} = 64\text{ns}$ -- 2: $2 \times 8 \times 8.0\text{ns} = 128\text{ns}$ -- 3: $3 \times 8 \times 8.0\text{ns} = 196\text{ns}$, and so on
7:0	gen1_u3_exit_rsp_rx_clk. Read-write. Reset: 05h. Description: gen1 U3_EXIT_RESP_RX_CLKS - Locally initiated Ux exit: Minimum LFPS reception from remote to consider Ux exit handshake successful. - When PHY is not in P3, the value in this field is multiplied by 8 to get the U3_EXIT_RESP_RX_CLKS. - This field is encoded as follows: -- 1: $1 \times 8 \times 8.0\text{ns} = 64\text{ns}$ -- 2: $2 \times 8 \times 8.0\text{ns} = 128\text{ns}$ -- 3: $3 \times 8 \times 8.0\text{ns} = 196\text{ns}$, and so on

USB[0...1]x00D094 (FCHUSBWCUSB31CNTR0::LINK_REGS1_LU3LFPSTXTIM)

Read-write. Reset: 30D4_2710h.

U3 LFPS TX TIMER REGS REGISTER

_instUSB0_aliasHOST; USB0x00D094; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D094; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	Reserved.
29:16	gen2_u3_exit_tx_clk. Read-write. Reset: 30D4h. Description: gen2 U3_EXIT_RESP_TX_CLKS - Remotely initiated U3 exit: Local link will transmit response this long before initiating transition to Recovery. - This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. -- 1: 6.4ns -- 2: 12.8ns -- 3: 19.2ns, and so on
15:14	Reserved.
13:0	gen1_u3_exit_rsp_tx_clk. Read-write. Reset: 2710h. Description: gen1 U3_EXIT_RESP_TX_CLKS - Remotely initiated U3 exit: Local link will transmit response this long before initiating transition to Recovery. - This field is encoded as the Gen1 link_clk (8.0ns) count for the LFPS. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D098 (FCHUSBDCUSB31CNTR0::LINK_REGS1_LPINGLFPSTIM)

Read-write. Reset: 4284_3603h.

PING LFPS TIMER REGISTER

_instUSB0_aliasHOST; USB0x00D098; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D098; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:26	gen2_lfps_ping_burst_clk. Read-write. Reset: 10h. Description: gen2 LFPS_PING_BURST_CLKS: TX ping.LFPS tBurst This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. - 1: 6.4ns - 2: 12.8ns - 3: 19.2ns, and so on
25:20	gen2_lfps_ping_max_clk. Read-write. Reset: 28h. Description: gen2 LFPS_PING_MAX_CLKS: Max RX ping.LFPS tBurst This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. - 1: 6.4ns - 2: 12.8ns - 3: 19.2ns, and so on
19:16	gen2_lfps_ping_min_clk. Read-write. Reset: 4h. Description: gen2 LFPS_PING_MIN_CLKS: Min RX ping.LFPS tBurst This field is encoded as the Gen2 link_clk (6.4ns) count for the LFPS. - 1: 6.4ns - 2: 12.8ns - 3: 19.2ns, and so on
15:10	gen1_lfps_ping_burst_clk. Read-write. Reset: 0Dh. Description: gen1 LFPS_PING_BURST_CLKS: TX ping.LFPS tBurst This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. - 1: 8ns - 2: 16ns - 3: 24ns, and so on
9:4	gen1_lfps_ping_max_clk. Read-write. Reset: 20h. Description: gen1 LFPS_PING_MAX_CLKS: Max RX ping.LFPS tBurst This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. - 1: 8ns - 2: 16ns - 3: 24ns, and so on
3:0	gen1_lfps_ping_min_clk. Read-write. Reset: 3h. Description: gen1 LFPS_PING_MIN_CLKS: Minx RX ping.LFPS tBurst This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. - 1: 8ns - 2: 16ns - 3: 24ns, and so on

USB[0...1]x00D09C (FCHUSBDCUSB31CNTR0::LINK_REGS1_LPOLLFPSTXTIM)

Reset: 0023_287Dh.

POLL LFPS TX TIMER REGISTER

_instUSB0_aliasHOST; USB0x00D09C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR],
NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D09C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR],
NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:22	Reserved.
21:11	lfps_poll_space_clk. Read-write. Reset: 465h. Description: LFPS_POLL_SPACE_CLKS: space between Polling.LFPS This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. - 1: 8ns - 2: 16ns - 3: 24ns, and so on
10:0	lfps_poll_burst_clk. Read-write. Reset: 07Dh. Description: LFPS_POLL_BURST_CLKS: TX Polling.LFPS tBurst This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. - 1: 8ns - 2: 16ns - 3: 24ns, and so on

USB[0...1]x00D0A0 (FCHUSBWCUSB31CNTR0::LINK_REGS1_LSKIPFREQ)

Reset: 4E92_8162h.

SKIP FREQUENCY REGISTER_instUSB0_aliasHOST; USB0x00D0A0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D0A0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31	Reserved.
30:28	u1_resid_timer_us. Read-write. Reset: 4h. Description: Programmable U1 MIN RESIDENCY TIMER This field specifies U1 MIN RESIDENCY TIMER value in us. Set to 0 to disable the timer.
27	en_pm_timer_us. Read-write. Reset: 1. Description: Enable us PM timers This field enables the use of PM_LC_TIMER and PM_ENTRY_TIMER in us.
26:24	pm_lc_timer_us. Read-write. Reset: 6h. Description: Programmable PM_LC_TIMER This field specifies PM_LC_TIMER value in us.
23:20	pm_entry_timer_us. Read-write. Reset: 9h. Description: Programmable PM_ENTRY_TIMER This field specifies PM_ENTRY_TIMER value in us.
19:12	gen2_skp_freq. Read-write. Reset: 28h. Description: gen2 SKP frequency This field specifies the average number of blocks the link sends before inserting a SKP OS.
11:0	gen1_skp_freq. Read-write. Reset: 162h. Description: gen1 SKP frequency This field specifies the average number of symbols the link sends before inserting a SKP OS.

USB[0...1]x00D0A4 (FCHUSBDCUSB31CNTR0::LINK_REGS1_LLUCTL)

Read-write. Reset: 19AB_E0E0h.

TX TS1 COUNT REGISTER

- This register provides an option to increase the number of TS1 sent in Polling.Active or Recovery.Active.

- The total number of TS1 during Polling.Active or Recovery.Active is based on the following formula:

Total TS1 = 20 x 2^(tx_ts1_cnt)

- This give more opportunity for the remote PHY to achive symbol lock.

_instUSB0_aliasHOST; USB0x00D0A4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0A4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	Reserved.
30	inverse_sync_header. Read-write. Reset: 0. Description: For Gen2 polarity detection, link uses data block (0011b) sync header for SYNC OS instead of control block (1100b). Set this bit if the third-party PHY does not correct the sync header of the SYNC OS in case of inverse polarity.
29	support_p4_pg. Read-write. Reset: 0. Description: PHY P4 Power gate mode (PG) is enabled. Set this bit if PHY support PG mode in P4.
28	support_p4. Read-write. Reset: 1. Description: Support PHY P3.CPM and P4. When this bit is set, controller puts PHY into P3.CPM or P4 in certain states.
27:24	delay_tx_gen1_dp. Read-write. Reset: 9h. Description: delay_tx_gen1_dp. When this field is set to a non-zero value, the link delays the transmission of a large Gen1 DP for 32 pipe_pclk cycles if it detects a header being received at the PIU.
23	DisRxDet_LTSSM_Timer_Ovrrd. Read-write. Reset: 1. Description: DisRxDet_LTSSM_Timer_Ovrrd. When DisRxDetU3RxDet is asserted in Polling or U1, the timeout expires immediately
22	force_dpp_truncate. Read-write. Reset: 0. Description: force_dpp_truncate. Truncate Gen1 DPP if there is a pending LGOOD/LBAD and DPP is large. When this bit is set, the logic for detecting recovery due to expiring pending_HP_timer is not used.
21	en_dpp_truncate. Read-write. Reset: 1. Description: en_dpp_truncate. This bit enables the logic for detecting recovery due to expiring pending_HP_timer and truncates DPP when necessary to avoid recovery.
20:16	pending_hp_timer_us. Read-write. Reset: 0Bh. Description: pending_hp_timer_us. Programmable PENDING_HP_TIMER in us. This field is used when LUCTL[15] is set.
15	en_us_hp_timer. Read-write. Reset: 1.

	Description: en_us_hp_timer. This bit enables programmable PENDING_HP_TIMER in us
14:13	ring_buf_d_delay. Read-write. Reset: 3h. Description: ring_buf_d_delay. This programs the counter for data valid signal in source ring buffer
12	U2P3CPMok. Read-write. Reset: 0. Description: P3CPM OK for U2/SSInactive (U2P3CPMok) - 0: During link state U2/ESS.Inactive, put PHY in P2 (Default) - 1: During link state U2/ESS.Inactive, put PHY in P3CPM. Note: For a port, if both GUSB3PIPECTL[29]=1 and LLUCTL[12]=1, LLUCTL[12]=1 takes priority.
11	en_reset_pipe_after_phy_mux. Read-write. Reset: 0. Description: en_reset_pipe_after_phy_mux. controller issues 3.1 PHY reset after DisRxDetU3RxDet is de-asserted
10	force_gen1. Read-write. Reset: 0. Force Gen1
9	gen2_loopback_entry_mode. Read-write. Reset: 0. Description: Gen2 loopback entry mode. If this bit is set, the controller will enter loopback in Gen2 after successfully detecting remote termination.
8	gen1_loopback_entry_mode. Read-write. Reset: 0. Description: Gen1 loopback entry mode. If this bit is set, the controller will enter loopback in Gen1 after successfully detecting remote termination.
7	mask_pipe_reset. Read-write. Reset: 1. Description: Mask pipe reset. If this bit is set, controller will block pipe_reset_n from going to PHY when DisRxDetU3RxDet=1
6	delay_ux_after_lpma. Read-write. Reset: 1. Description: delay_ux_after_lpma. - If this bit is set to 0, after sending LPMA, Link LTSSM delays 10 link_clk cycles before entering Ux. - If this bit is set to 1, after sending LPMA, Link LTSSM delays 20 link_clk cycles before entering Ux.
5	no_ux_exit_p0_trans. Read-write. Reset: 1. Description: no_ux_exit_p0_trans. - If this bit is 0, Link LTSSM detects Ux_exit LFPS during P0 transition. - If this bit is set to 1, the Link LTSSM does not detect Ux_exit LFPS when P0 transition is happening.
4:0	tx_ts1_cnt. Read-write. Reset: 00h. Additional TX_TS1_count

USB[0...1]x00D0A8 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LPTMDPDELAY)

Read-write. Reset: 30C0_09C4h.

PTM DATAPATH DELAY REGISTER

This register provides an option to override the default timing parameters used in PTM calculation.

_instUSB0_aliasHOST; USB0x00D0A8; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D0A8; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:27	rx_path_delay_gen2. Read-write. Reset: 06h. Description: rx_path_delay_gen2 RX path delay in unit of 8ns
26:22	tx_path_delay_gen2. Read-write. Reset: 03h. Description: tx_path_delay_gen2 TX path delay in unit of 8ns
21:10	p3cpmp4_residency. Read-write. Reset: 002h. Description: Minimum number of suspend_clk periods that the PHY needs to stay in P3.CPM or P4 before exiting P3.CPM or P4.
9:5	rx_path_delay. Read-write. Reset: 0Eh. Description: rx_path_delay Gen1 RX path delay (including PHY and PCS) in terms of 8ns.
4:0	tx_path_delay. Read-write. Reset: 04h. Description: tx_path_delay Gen1 TX path delay (including PHY and PCS) in terms of 8ns.

USB[0...1]x00D0AC (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LSCDTIM1)

Reset: 0047_E1F4h.

SCD TIMER 1 REGISTER_instUSB0_aliasHOST; USB0x00D0AC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D0AC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:24	Reserved.
23:12	scd_bit0_rpt_max_clk. Read-write. Reset: 47Eh. Description: SCD_BIT0_REPEAT_MAX_CLKS - max duration of SCD bit 0 tRepeat in term of CLKs, used in SCD detection FSM - This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
11:0	cd_bit0_rpt_min_clk. Read-write. Reset: 1F4h. Description: CD_BIT0_REPEAT_MIN_CLKS - min duration of SCD bit 0 tRepeat in term of CLKs, used in SCD detection FSM - This field is encoded as the Gen1 link_clk (8ns) count for the LFPS. -- 1 : 8ns -- 2 : 16ns -- 3 : 24ns, and so on

USB[0...1]x00D0B0 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LSCDTIM2)

Reset: 0075_3546h.

SCD TIMER2 REGISTER_instUSB0_aliasHOST; USB0x00D0B0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D0B0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:24	Reserved.
23:12	scd_bit1_rpt_max_clk. Read-write. Reset: 753h. Description: SCD_BIT1_REPEAT_MAX_CLKS - max duration of SCD bit 1 tRepeat in term of CLKs, used in SCD detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
11:0	cd_bit1_rpt_min_clk. Read-write. Reset: 546h. Description: CD_BIT1_REPEAT_MIN_CLKS - min duration of SCD bit 1 tRepeat in term of CLKs, used in SCD detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D0B4 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LSCDTIM3)

Read-write. Reset: 5DC2_F07Dh.

SCD TIMER3 REGISTER_instUSB0_aliasHOST; USB0x00D0B4; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D0B4; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:20	lfps_scd_space1_clk. Read-write. Reset: 5DCh. Description: LFPS_SCD_SPACE1_CLKS - Duration of SCD space 1 used in SCD TX FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
19:8	lfps_scd_space0_clk. Read-write. Reset: 2F0h. Description: LFPS_SCD_SPACE0_CLKS - Duration of SCD space 0 used in SCD TX FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
7:0	lfps_scd_burst_clk. Read-write. Reset: 7Dh. Description: LFPS_SCD_BURST_CLKS - Duration of tBurst used in SCD TX FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D0B8 (FCHUSBDCUSB31CNTR0::LINK_REGS1_LSCDTIM4)

Reset: 0000_0EA6h.

SCD TIMER4 REGISTER

_instUSB0_aliasHOST; USB0x00D0B8; USB0={NBIFEPFNCFG::BASE_ADDR_2 dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1 dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0B8; USB1={NBIFEPFNCFG::BASE_ADDR_2 dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1 dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:12	Reserved.
11:0	lfps_scd_last_space. Read-write. Reset: EA6h. Description: LFPS_SCD_LAST_SPACE_CLKS - Duration of the last space for each SCD - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D0BC (FCHUSBDCUSB31CNTR0::LINK_REGS1_LLPBMTIM1)

Read-write. Reset: F096_6B32h.

LPBM TIMER1 REGISTER

_instUSB0_aliasHOST; USB0x00D0BC; USB0={NBIFEPFNCFG::BASE_ADDR_2 dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1 dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0BC; USB1={NBIFEPFNCFG::BASE_ADDR_2 dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1 dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	lfps_lbps1_burst_max_clk. Read-write. Reset: F0h. Description: LFPS_LBPS1_BURST_MAX_CLKS - Max duration of LPBM bit 1, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
23:16	lfps_lbps1_burst_min_clk. Read-write. Reset: 96h. Description: LFPS_LBPS1_BURST_MIN_CLKS - Min duration of LPBM bit 1, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
15:8	lfps_lbps0_burst_max_clk. Read-write. Reset: 6Bh. Description: LFPS_LBPS0_BURST_MAX_CLKS - Max duration of LPBM bit 0, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
7:0	lfps_lbps0_burst_min_clk. Read-write. Reset: 32h. Description: LFPS_LBPS0_BURST_MIN_CLKS - Min duration of LPBM bit 0, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D0C0 (FCHUSBDCUSB31CNTR0::LINK_REGS1_LLPBMTIM2)

Reset: 0002_6CF5h.

LPBM TIMER2 REGISTER

_instUSB0_aliasHOST; USB0x00D0C0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0C0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:18	Reserved.
17:9	lfps_lbpm_tpwm_max_clk. Read-write. Reset: 136h. Description: LFPS_LBPM_TPWM_MAX_CLKS - Max value of TPWM, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
8:0	lfps_lbpm_tpwm_min_clk. Read-write. Reset: 0F5h. Description: LFPS_LBPM_TPWM_MIN_CLKS - Min value of TPWM, used in LPBM detection FSM - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D0C4 (FCHUSBDCUSB31CNTR0::LINK_REGS1_LLPBMTXTIM)

Reset: 1130_C850h.

LPBM TX TIMER REGISTER

_instUSB0_aliasHOST; USB0x00D0C4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0C4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	Reserved.
29:20	lfps_lbpm_tpwm_clk. Read-write. Reset: 113h. Description: LFPS LBPM TPWM CLKS - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
19:18	Reserved.
17:8	lfps_lbpm_burst1_clk. Read-write. Reset: 0C8h. Description: LFPS LBPM BURST1 CLKS - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on
7:0	lfps_lbpm_burst0_clk. Read-write. Reset: 50h. Description: LFPS LBPM BURST0 CLKS - This field is encoded as the Gen1 link_clk (8ns) count. -- 1: 8ns -- 2: 16ns -- 3: 24ns, and so on

USB[0...1]x00D0C8 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LLINKERRINJ)

Read-write. Reset: 0000_0000h.

LINK ERROR TYPE INJECT REGISTER

_instUSB0_aliasHOST; USB0x00D0C8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0C8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	pipe_txdata. Read-write. Reset: 0. Description: PIPE TxData - Inject single bit error to TxData. - This includes additional logic to ensure IDLE symbols at the end of recovery aren't corrupted so that link won't enter recovery loop.
30	TX_insert_defer. Read-write. Reset: 0. TX Insert Deferred Bit
29	TX_insert_delay. Read-write. Reset: 0. TX Insert Delayed Bit
28	TX_TS2. Read-write. Reset: 0. TX_TS2 : Corrupt entire TS2
27	TX_TS1. Read-write. Reset: 0. TX_TS1 : Corrupt entire TS1
26	TX_TSEQ. Read-write. Reset: 0. TX_TSEQ : Corrupt entire TSEQ
25	TX_LFR. Read-write. Reset: 0. Description: TX_LFR - Transmitted DPH will have invalid LFR
24	TX_delay_credit_rl. Read-write. Reset: 0. Description: TX Delay credit release - when this bit is set, link will wait for the amount of microsecond specified in the Error Enable Counter register [15:0]
23	TX_one_srt_frame_sybl. Read-write. Reset: 0. Description: TX DPP all start framing symbol - all framing symbols of DPP (SDP and EPF) will be corrupted
22	TX_all_end_frame_sybl. Read-write. Reset: 0. Description: TX DPP all end framing symbol - all framing symbols of DPP (END and EPF) will be corrupted
21	TX_one_end_frame_sybl. Read-write. Reset: 0. Description: TX DPP one end framing symbol - one framing symbol of DPP (END) will be corrupted
20	TX_all_frame_sybl. Read-write. Reset: 0. Description: TX TPH/DPH all framing symbol - all framing symbols of TPH/DPH (SHP, DPHP, EPF) will be corrupted
19	TX_one_frame_sybl. Read-write. Reset: 0. Description: TX TPH/DPH one framing symbol - One framing symbol of TPH/DPH (SHP or DPHP) will be corrupted
18	TX_CRC32. Read-write. Reset: 0. Description: TX_CRC32 - Transmitted DPP will have invalid CRC32
17	TX_CRC16. Read-write. Reset: 0. Description: TX_CRC16 - Transmitted header will have invalid CRC16
16	TX_CRC5. Read-write. Reset: 0. Description: TX_CRC5 - Transmitted header will have CRC error in Link Control Word
15	Rx_pipe_rxdata. Read-write. Reset: 0.

	Description: Rx_pipe_rxdata - Inject single bit error to RxData. - This includes additional logic to ensure IDLE symbols at the end of recovery aren't corrupted so that link won't enter recovery loop.
14	RX_insert_defer. Read-write. Reset: 0. RX_insert_defer
13	RX_insert_delay. Read-write. Reset: 0. RX_insert_delay
12	RX_TS2. Read-write. Reset: 0. Description: RX_TS2 - Corrupt entire TS2
11	RX_TS1. Read-write. Reset: 0. Description: RX_TS1 - Corrupt entire TS1
10	RX_TSEQ. Read-write. Reset: 0. Description: RX_TSEQ - Corrupt entire TSEQ
9	RX_LFR. Read-write. Reset: 0. Description: RX_LFR - Received DPH will have invalid LFR
8	RX_all_start_frame_sybl. Read-write. Reset: 0. Description: DPP all start framing symbol - Before 64-bit received data is forwarded to parser, all framing symbols (SDP or EPF) will be corrupted. - Note: because the entire framing symbol may not be aligned and presented on the 64-bit received data, 1 - 4 framing symbols will be corrupted
7	RX_one_srt_frame_sybl. Read-write. Reset: 0. Description: DPP one start framing symbol - Before received data is forwarded to parser in RPKT, only one framing symbol (SDP or EPF) will be corrupted
6	RX_all_end_frame_sybl. Read-write. Reset: 0. Description: DPP all end framing symbol - Before 64-bit received data is forwarded to parser, all framing symbols (END or EPF) will be corrupted. - Note: Because the entire framing symbol may not be aligned and presented on the 64-bit received data, 1 - 4 framing symbols will be corrupted
5	RX_one_end_frame_sybl. Read-write. Reset: 0. Description: RX DPP one end framing symbol - Before received data is forwarded to parser in RPKT, only one framing symbol (END or EPF) will be corrupted
4	RX_all_frame_sybl. Read-write. Reset: 0. Description: RX TPH/DPH all framing symbol - Before 64-bit received data is forwarded to parser, all framing symbols (SHP or EPF or DPHP) will be corrupted. - Note: Because the entire framing symbol may not be aligned and presented on the 64-bit received data, 1 - 4 framing symbols will be corrupted
3	RX_one_frame_sybl. Read-write. Reset: 0. Description: RX TPH/DPH one framing symbol - Before received data is forwarded to parser in RPKT, only one framing symbol (SHP or EPF or DPHP) will be corrupted
2	RX_CRC32. Read-write. Reset: 0. Description: RX_CRC32 - Received DPP will have invalid CRC32 value
1	RX_CRC16. Read-write. Reset: 0. Description: RX_CRC16 - Received header will have invalid CRC16 value

0	RX_CRC5. Read-write. Reset: 0.
	Description: RX_CRC5 - Received header will have invalid CRC5 value in Link Control Word

USB[0...1]x00D0CC (FCHUSBDCUSB31CNTR0::LINK_REGS1_LLINKERRINJEN)

Read-write. Reset: 0000_0000h.

LINK ERROR INJECT ENABLE REGISTER

- This register is for injecting N number of errors for every M number of packets

_instUSB0_aliasHOST; USB0x00D0CC; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0CC; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:16	disable_inj_err_cnt. Read-write. Reset: 0000h. Description: Disable injected error count - This error specifies the interval of error free between 2 error injection periods. - In the case of PIPE data error injection, this field specify the number of microsecond between every bit error injection
15:0	B2B_err_cnt. Read-write. Reset: 0000h. Description: B2B error count - This counter indicates the number of error to be injected consecutively. - For instance, the RX CRC5 error type is selected and this counter value is 3, then the next 3 received headers will have invalid CRC5

USB[0...1]x00D0D0 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_GDBGLTSSM)

Reset: 4101_0440h.

Global Debug LTSSM Register

Note: GDBGLTSSM register is not applicable for USB 2.0-only mode.

- Bit Bash test should not be done on this debug register.

_instUSB0_aliasHOST; USB0x00D0D0; USB0={NBIFEPFNCFG::BASE_ADDR_2 dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1 dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0D0; USB1={NBIFEPFNCFG::BASE_ADDR_2 dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1 dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	Reserved.
30	RxElecIdle . Read-only, Volatile. Reset: 1. Description: RxElecIdle For description of RxElecIdle, see table 5-4, "Status Interface Signals" of the PIPE4 Specification.
29:27	Reserved.
26	LTDBTIMEOUT . Read-only. Reset: 0. LTDB Timeout (LTDBTimeout)
25:22	LTDBLINKSTATE . Read-only. Reset: 4h. LTDB Link State (LTDBLinkState)
21:18	LTDBSUBSTATE . Read-only, Volatile. Reset: 0h. LTDB Sub-State (LTDBSubState)
17	ELASTICBUFFERMODE . Read-only. Reset: 0. Description: Elastic Buffer Mode (ElasticBufferMode) For field definition, refer to Table 5-3 of the PIPE specification.
16	TXELECLDLE . Read-only. Reset: 1. Description: Tx Elec Idle (TxElecIdle) For field definition, refer to Table 5-3 of the PIPE specification.
15	RXPOLARITY . Read-only. Reset: 0. Description: Rx Polarity (RxPolarity) For field definition, refer to Table 5-3 of the PIPE specification.
14	TxDetRxLoopback . Read-only. Reset: 0. Description: Tx Detect Rx/Loopback (TxDetRxLoopback) For field definition, refer to Table 5-3 of the PIPE specification.
13:11	LTDBPhyCmdState . Read-only, Volatile. Reset: 0h. Description: LTSSM PHY command State (LTDBPhyCmdState) - 000: PHY_IDLE (PHY command state is in IDLE. No PHY request pending) - 001: PHY_DET (Request to start Receiver detection) - 010: PHY_DET_3 (Wait for Phy_Status (Receiver detection)) - 011: PHY_PWR_DLY (Delay Pipe3_PowerDown P0 -> P1/P2/P3 request) - 100: PHY_PWR_A (Delay for internal logic) - 101: PHY_PWR_B (Wait for Phy_Status(Power state change request))
10:9	POWERDOWN . Read-only, Volatile. Reset: 2h. Description: POWERDOWN (PowerDown) For field definition, refer to Table 5-3 of the PIPE Specification.
8	RXEQTRAIN . Read-only. Reset: 0. Description: RxEq Train For field definition, refer to Table 5-3 of the PIPE Specification.

7:6	TXDEEMPHASIS. Read-only. Reset: 1h. Description: TXDEEMPHASIS (TxDeemphasis) For field definition, refer to Table 5-3 of the PIPE Specification.
5:3	LTDBClkState. Read-only. Reset: 0h. Description: LTSSM Clock State (LTDBClkState) In multi-port host configuration, the port number is defined by Port-Select[3:0] field in the GDBGFIFOSPACE register. Note: GDBGLTSSM register is not applicable for USB 2.0-only mode. - 000: CLK_NORM (PHY is in non-P3 state and PCLK is running) - 001: CLK_TO_P3 (P3 entry request to PHY); - 010: CLK_WAIT1 (Wait for Phy_Status (P3 request)); - 011: CLK_P3 (PHY is in P3 and PCLK is not running); - 100: CLK_TO_P0 (P3 exit request to PHY); - 101: CLK_WAIT2 (Wait for Phy_Status (P3 exit request))
2	TXSWING. Read-only. Reset: 0. Description: Tx Swing (TxSwing) For field definition, refer to Table 5-3 of the PIPE Specification.
1	RXTERMINATION. Read-only. Reset: 0. Rx Termination (RxTermination)
0	TXONESZEROS. Read-only. Reset: 0. Description: Tx Ones/Zeros (TxOnesZeros) For field definition, refer to Table 5-3 of the PIPE Specification.

USB[0...1]x00D0D4 (FCHUSBDCUSB31CNTR0::LINK_REGS1_GDBGLNMCC)

Read-only. Reset: 0000_0000h.

Global Debug LNMCC Register

Bit Bash test should not be done on this debug register.

_instUSB0_aliasHOST; USB0x00D0D4; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D0D4; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	LNMCCLSERC. Read-only. Reset: 0000h. Description: This field indicates the value of link soft error counter for the port selected in the GDBGFIFOSPACE.PortSelect field. This field is for debug purposes only.
15:9	Reserved.
8:0	LNMCCLBERC. Read-only. Reset: 000h. Description: This field indicates the bit error rate information for the port selected in the GDBGFIFOSPACE.PortSelect field. This field is for debug purposes only.

USB[0...1]x00D0D8 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LLINKDBGCTRL)

Reset: 8000_0000h.

LINK DEBUG CONTROL REGISTER

- This register is used for link debug purpose.

_instUSB0_aliasHOST; USB0x00D0D8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0D8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31	retry_DP . Read-write. Reset: 1. Link retries full DP instead of DPH in Gen2. When link needs to resend header due to LBAD or recovery replay, enabling this bit makes link to send full DPH + DPP. This workaround is for a non-compliance 3.1 device
30	trigger_end_sts . Read-only. Reset: 0. Trigger End Status
29	trigger_start_sts . Read-only. Reset: 0. Trigger Start Status
28	link_state_trigger_descrambled . Read-write. Reset: 0. Description: Link State Trigger Descrambled Debug Control When set, the controller writes the descrambled received data into the RAM. The received data is the output of the descrambler. The link starts writing debug data based on the following conditions: - Link_state and sub_state match the link_state_trigger and sub_state_trigger field, respectively. After this condition occurs, the link internal debug_byte_count starts incrementing for every link clock, based on the number of bytes to be written to the RAM. For example, if MDWIDTH=128, the debug_byte_count increments by 16 every link_clk. - Link internal debug_byte_count reaches the start_count_trigger field. The link stops writing debug data when the link internal debug_byte_count reaches the stop_count_trigger field. If stop_count is zero, the link writes out data continuously.
27	link_state_trigger_scrambled . Read-write. Reset: 0. Description: Link State Trigger Scrambled Debug Control When set, controller writes the PIPE received data into the RAM. The received data is scrambled. The link starts writing debug data based on the following conditions: - Link_state and sub_state match the link_state_trigger and sub_state_trigger field, respectively. After this condition occurs, the link internal debug_byte_count starts incrementing for every link clock, based on the number of bytes to be written to RAM. For example, if MDWIDTH=64, the debug_byte_count increments by 8 every link_clk. - Link internal debug_byte_count reaches the start_count_trigger field The link stops writing debug data when the link internal debug_byte_count reaches the stop_count_trigger field. If stop_count is zero, the link writes out data continuously.
26	pipe_txdetectrxlb_trigger . Read-write. Reset: 0. Description: PIPE TXDETECTRXLB Trigger Debug Control The link starts writing debug data based on the following conditions: - Link_state and sub_state match the link_state_trigger and sub_state_trigger field, respectively - Assertion of pipe_TxDetectRxLoopbk
25	pipe_rxlecidle_trigger . Read-write. Reset: 0.

	Description: PIPE RXELECIdle Trigger Debug Control The link start writing debug data based on the following conditions: - Link_state and sub_state match the link_state_trigger and sub_state_trigger field, respectively - Change state of pipe_RxElecIdle
24	pipe_phystatus_trigger. Read-write. Reset: 0. Description: PIPE PHYSTATUS Trigger Debug Control The link start writing debug data based on the following conditions: - Link_state and sub_state match the link_state_trigger and sub_state_trigger field, respectively - Assertion of pipe_PhyStatus
23:20	sub_state_trigger. Read-write. Reset: 0h. Sub-state Trigger
19:16	link_state_trigger. Read-write. Reset: 0h. Link State Trigger
15:12	peri_rsc_rxfifo_number. Read-write. Reset: 0h. Periodic Resource RXFIFO Number
11:8	peri_rsc_txfifo_number. Read-write. Reset: 0h. Periodic Resource TXFIFO Number
7:4	rxfifo_number. Read-write. Reset: 0h. RXFIFO Number
3:0	txfifo_number. Read-write. Reset: 0h. TXFIFO Number

USB[0...1]x00D0DC (FCHUSBWCUSB31CNTR0::LINK_REGS1_LLINKDBGCNTTRIG)

Read-write. Reset: 0000_0000h.

LINK DEBUG [COUNT](#) TRIGGER REGISTER

This register is used in conjunction with the LINK_DEBUG_CTRL_REG (LINKDGBCTRL). For more information, refer to the LINK_DEBUG_CTRL_REG description.

_instUSB0_aliasHOST; USB0x00D0DC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D0DC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
31:16	stp_cnt. Read-write. Reset: 0000h. Description: Stop Count This field is used for triggering purpose based on the selection in the LINK_DEBUG_CTRL_REG. If this field is zero, link writes out debug data continuously. SW needs to take into account of the RX RAM size to program this field. (stop_count start_count) is less than total RX RAM size.
15:0	srt_cnt. Read-write. Reset: 0000h. Description: Start Count This field is used for triggering purpose based on the selection in the LINK_DEBUG_CTRL_REG

USB[0...1]x00D0E0 (FCHUSBDCUSB31CNTR0::LINK_REGS1_LCSR_TX_DEEMPH)

Reset: 0000_8C45h.

LCSR_TX_DEEMPH REGISTER

This register is for setting TX deemphasis used in normal operation in gen2.

_instUSB0_aliasHOST; USB0x00D0E0; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D0E0; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}**Bits Description**

31:18 Reserved.

17:0 **csr_tx_deemph_field_1**. Read-write. Reset: 0_8C45h. 18 bits of TX deemphasis, defined in PIPE4 spec, in normal operation (non-compliance mode)**USB[0...1]x00D0E4 (FCHUSBDCUSB31CNTR0::LINK_REGS1_LCSR_TX_DEEMPH_1)**

Reset: 0000_0E45h.

LCSR_TX_DEEMPH_1 REGISTER

This register is for setting TX deemphasis used in compliance mode, pattern 13

_instUSB0_aliasHOST; USB0x00D0E4; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D0E4; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}**Bits Description**

31:18 Reserved.

17:0 **csr_tx_deemph_1_field_1**. Read-write. Reset: 0_0E45h. 18 bits of TX deemphasis used by controller in gen2 compliance pattern 13**USB[0...1]x00D0E8 (FCHUSBDCUSB31CNTR0::LINK_REGS1_LCSR_TX_DEEMPH_2)**

Reset: 0000_8D80h.

LCSR_TX_DEEMPH_2 REGISTER

This register is for setting TX deemphasis used in compliance mode, pattern 14

_instUSB0_aliasHOST; USB0x00D0E8; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D0E8; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}**Bits Description**

31:18 Reserved.

17:0 **csr_tx_deemph_2_field_1**. Read-write. Reset: 0_8D80h. 18 bits of TX deemphasis used by controller in gen2 compliance pattern 14**USB[0...1]x00D0EC (FCHUSBDCUSB31CNTR0::LINK_REGS1_LCSR_TX_DEEMPH_3)**

Reset: 0000_0F80h.

LCSR_TX_DEEMPH_3 REGISTER

This register is for setting TX deemphasis used in compliance mode, pattern 16

_instUSB0_aliasHOST; USB0x00D0EC; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D0EC; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}**Bits Description**

31:18 Reserved.

17:0 **csr_tx_deemph_3_field_1**. Read-write. Reset: 0_0F80h. 18 bits of TX deemphasis used by controller in gen2 compliance pattern 16

USB[0...1]x00D0F0 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LCSRPTMDEBUG1)

Read-only. Reset: 0000_0000h.

LCSRPTMDEBUG1 REGISTER: This register stores LDM t1 and LDM t32 values

In Host mode this register read value is 32'h0.

_instUSB0_aliasHOST; USB0x00D0F0; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0F0; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	Reserved.
29:17	LDM_T32. Read-only. Reset: 0000h. Description: LDM_T32 This is Response Delay value captured from the LMP TS response field. The unit of this field is tIsochTimestampGranularity
16:0	LDM_T1. Read-only. Reset: 0_0000h. Description: LDM_T1 This is the 't1' timestamp, which is the time the link transmits the LMP TS request. The unit of this field is nanosecond (ns). Note: - The internal timer counts down from 125000 to 0, and restarts at 125000 (therefore, t1 > t4, except in case of a roll over). - This value does not take into account the internal TX pipeline delay. uses the following formula for adjusting 't1' based on the pipeline delay: $TX_PATH_DELAY(tIsochTimestampGranularity) = (gen1 \ ? \ 2 : 2) + LPTMDPDELAY[3:0]$

USB[0...1]x00D0F4 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LCSRPTMDEBUG2)

Read-only. Reset: 0000_0000h.

LCSRPTMDELAY2 REGISTER : This register stores LSM t4 value

In Host mode this register read value is 32'h0.

_instUSB0_aliasHOST; USB0x00D0F4; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0F4; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:17 Reserved.

16:0 **LDM_T4**. Read-only. Reset: 0_0000h.**Description:** LDM_T4

This is the 't4' timestamp, which is the time the link receives the LMP TS response.

The unit of this field is nanosecond (ns).

Note:

- The internal timer counts down 125000 to 0, and restarts at 125000 (therefore, t1 > t4, except the case of roll over)

- This value does not take into account the internal RX pipeline delay. uses this formula for adjusting 't4' based on the pipeline delay:

RX_PATH_DELAY(tIsochTimestampGranularity) = (gen1 ? 4 : 5) + LPTMDPDELAY[7:4]

USB[0...1]x00D0F8 (FCHUSBWDCUSB31CNTR0::LINK_REGS1_LPTMDPDELAY2)

Reset: 0005_1565h.

PTM DATAPATH DELAY REGISTER2

- This register provide an option to override the default timing parameters used in PTM calculation.

_instUSB0_aliasHOST; USB0x00D0F8; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D0F8; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description

31:20 Reserved.

19:15 **link_rx_delay_gen2**. Read-write. Reset: 0Ah.**Description:** link_rx_delay_gen2

RX path delay in unit of 8ns

14:10 **link_tx_delay_gen2**. Read-write. Reset: 05h.**Description:** link_tx_delay_gen2

RX path delay in unit of 8ns

9:5 **link_rx_delay_gen1**. Read-write. Reset: 0Bh.**Description:** link_rx_delay_gen1

RX path delay in unit of 8ns

4:0 **link_tx_delay_gen1**. Read-write. Reset: 05h.**Description:** link_tx_delay_gen1

TX path delay in unit of 8ns

USB[0...1]x00D800 (FCHUSBDWCUSB31CNTR0::RHBDBG_REGS0_BU31RHBDBG)

Reset: 0000_0008h.

U31 ROOT HUB DEBUG REG

_instUSB0_aliasHOST; USB0x00D800; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D800; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:4	Reserved.
3	tpcfg_tout_ctrl. Read-write. Reset: 1. Description: tpcfg_tout_ctrl This bit controls the 3.x port configuration timeout duration. - 1: The port configuration timeout counter resets when the link is not in U0. - 0: The port configuration timeout counter does not reset if the link goes to recovery or exits U0.
2:1	pcap. Read-write. Reset: 0h. pcap
0	ovrcur. Read-write. Reset: 0. Description: ovrcur Overcurrent injection.

USB[0...1]x00D804 (FCHUSBDWCUSB31CNTR0::RHBDBG_REGS1_BU31RHBDBG)

Reset: 0000_0008h.

U31 ROOT HUB DEBUG REG

_instUSB0_aliasHOST; USB0x00D804; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D804; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:4	Reserved.
3	tpcfg_tout_ctrl. Read-write. Reset: 1. Description: tpcfg_tout_ctrl This bit controls the 3.x port configuration timeout duration. - 1: The port configuration timeout counter resets when the link is not in U0. - 0: The port configuration timeout counter does not reset if the link goes to recovery or exits U0.
2:1	pcap. Read-write. Reset: 0h. pcap
0	ovrcur. Read-write. Reset: 0. Description: ovrcur Overcurrent injection.

USB[0...1]x00D84C (FCHUSBDWCUSB31CNTR0::BRAMHIADDR)**RAM HIGHER ADDRESS REGISTER**

_instUSB0_aliasHOST; USB0x00D84C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D84C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:21	Reserved.
20:18	ram_select. Read-write. Reset: 0h. Chip select for RAM 0-4
17:12	ramhiaddr. Read-write. Reset: 00h. Partial decoding bits for RAM higher address
11:0	Reserved.

USB[0...1]x00D850 (FCHUSBDCUSB31CNTR0::BRSERRCNT)

RAM Single Error Count. 6 bit count for each of 5 RAMS

_instUSB0_aliasHOST; USB0x00D850; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D850; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	Reserved.
29:24	ram4serrcnt. Read-write,Read,Write-1-to-clear,Volatile. RAM4 Single Error Count
23:18	ram3serrcnt. Read-write,Read,Write-1-to-clear,Volatile. RAM3 Single Error Count
17:12	ram2serrcnt. Read-write,Read,Write-1-to-clear,Volatile. RAM2 Single Error Count
11:6	ram1serrcnt. Read-write,Read,Write-1-to-clear,Volatile. RAM1 Single Error Count
5:0	ram0serrcnt. Read-write,Read,Write-1-to-clear,Volatile. RAM0 Single Error Count

USB[0...1]x00D854 (FCHUSBDCUSB31CNTR0::BRMERRCNT)

RAM Multiple Error Count. 6 bit count for each of 5 RAMS

_instUSB0_aliasHOST; USB0x00D854; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D854; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:30	Reserved.
29:24	ram4merrcnt. Read-write,Read,Write-1-to-clear,Volatile. RAM4 Multiple Error Count
23:18	ram3merrcnt. Read-write,Read,Write-1-to-clear,Volatile. RAM3 Multiple Error Count
17:12	ram2merrcnt. Read-write,Read,Write-1-to-clear,Volatile. RAM2 Multiple Error Count
11:6	ram1merrcnt. Read-write,Read,Write-1-to-clear,Volatile. RAM1 Multiple Error Count
5:0	ram0merrcnt. Read-write,Read,Write-1-to-clear,Volatile. RAM0 Multiple Error Count

USB[0...1]x00D858 (FCHUSBDCUSB31CNTR0::BRAMECCERR)

RAM ECC Error Register

_instUSB0_aliasHOST; USB0x00D858; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D858; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:12	Reserved.
11	ramserr. Read-write,Read,Write-1-to-clear,Volatile. RAM Single Error
10	rammerr. Read-write,Read,Write-1-to-clear,Volatile. RAM Multiple Error
9:5	ramserrvec. Read-write,Read,Write-1-to-clear,Volatile. RAM0 Single Error Vector
4:0	rammerrvec. Read-write,Read,Write-1-to-clear,Volatile. RAM Multiple Error Vector

USB[0...1]x00D85C (FCHUSBDCUSB31CNTR0::BRERRCTL)

RAM ERROR CONTROL REGISTER

_instUSB0_aliasHOST; USB0x00D85C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D85C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:2	Reserved.
1	rserrclr. Read-write. Reset: 0. RAM Single Error Clear
0	rmerrclr. Read-write. Reset: 0. RAM Multiple Error Clear

USB[0...1]x00D860 (FCHUSBDWCUSB31CNTR0::BRAM0ADDRERR)

Read-only.

Address of RAM0 which had uncorrectable error

_instUSB0_aliasHOST; USB0x00D860; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D860; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description

31:16 Reserved.

15:0 **ram0errloc.** Read-only. Reset: 0000h. RAM0 Address Error Location**USB[0...1]x00D864 (FCHUSBDWCUSB31CNTR0::BRAM1ADDRERR)**

Read-only.

RAM1 Address Error Register

_instUSB0_aliasHOST; USB0x00D864; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D864; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description

31:16 Reserved.

15:0 **ram1errloc.** Read-only. Reset: 0000h. RAM1 address error location**USB[0...1]x00D868 (FCHUSBDWCUSB31CNTR0::BRAM2ADDRERR)**

Read-only.

RAM2 Address Error Register

_instUSB0_aliasHOST; USB0x00D868; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D868; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description

31:16 Reserved.

15:0 **ram2errloc.** Read-only. Reset: 0000h. RAM2 address error location**USB[0...1]x00D86C (FCHUSBDWCUSB31CNTR0::BRAM3ADDRERR)**

Read-only.

RAM3 Address Error Register

_instUSB0_aliasHOST; USB0x00D86C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D86C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description

31:16 Reserved.

15:0 **ram3errloc.** Read-only. Reset: 0000h. RAM3 address error location**USB[0...1]x00D870 (FCHUSBDWCUSB31CNTR0::BRAM4ADDRERR)**

Read-only.

RAM4 Address Error Register

_instUSB0_aliasHOST; USB0x00D870; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D870; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)],
NBIFEPFNCFG::[BASE_ADDR_1](#) dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description

31:16 Reserved.

15:0 **ram4errloc.** Read-only. Reset: 0000h. RAM4 address error location

USB[0...1]x00D900 (FCHUSBWCUSB31CNTR0::BLOOPBCKCTRL)

Reset: 0000_0000h.

Loopback Control Register

_instUSB0_aliasHOST; USB0x00D900; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D900; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:8	Reserved.
7:4	loopback_prtnum. Read-write. Reset: 0h. Description: Loopback Port Number Valid for Host Mode Loopback
3:2	loopback_level. Read-write. Reset: 0h. Description: Loopback Level - 2'b00 : SSP PIPE - 2'b01 : SSP PHY - 2'b10 : Reserved - 2'b11 : Reserved
1	loopback_mode. Read-write. Reset: 0. Description: Loopback Mode - 1'b0 :SW loopback Mode - 1'b1 : BIST HW Loopback Mode
0	loopback_mode_en. Read-write. Reset: 0. Enable Loopback Mode

USB[0...1]x00D904 (FCHUSBWCUSB31CNTR0::BLOOPBCKTFERSZ)

Reset: 0000_0000h.

Loopback Transfer Size Register

_instUSB0_aliasHOST; USB0x00D904; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D904; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits	Description
31:24	Reserved.
23:0	loopback_xfer_sz. Read-write. Reset: 00_0000h. Description: Loopback Transfer Size The loopback logic uses this field information for size of data to transfer. In SW mode, the transfer size in TRB and loopback transfer size should match MAX_PACKET_SIZE value. In BIST mode, the loopback transfer size should match MAX_PACKET_SIZE value configured.

USB[0...1]x00D908 (FCHUSBWDCUSB31CNTR0::BBISTDATAPATSEED)

Read-write. Reset: 0000_0000h.

BIST Data Pattern Seed Register

- Used only in BIST MODE

_instUSB0_aliasHOST; USB0x00D908; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D908; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description31:0 **BIST_pattern_seed**. Read-write. Reset: 0000_0000h. Data is repeated or inverted or incremented or used as seed for LSFR**USB[0...1]x00D90C (FCHUSBWDCUSB31CNTR0::BBISTCTRL)**

Reset: 0000_0000h.

BIST Control Register

- Used only in BIST MODE

_instUSB0_aliasHOST; USB0x00D90C; USB0={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio0_aliasHOST[BASE_ADDR]}

_instUSB1_aliasHOST; USB1x00D90C; USB1={NBIFEPFNCFG::BASE_ADDR_2_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR], NBIFEPFNCFG::BASE_ADDR_1_dev0_func4_instNBIF0_nbio1_aliasHOST[BASE_ADDR]}

Bits Description31 **start_BIST_test**. Read-write. Reset: 0.**Description:** Start **BIST** Test

- 1'b0 : Stop Test

- 1'b1 : Start Test

This bit is cleared when the test is completed.

30 **BIST_test_type**. Read-write, Volatile. Reset: 0.**Description:** **BIST** Test Type

- 1'b0 : BIST Transfer loopback test

- 1'b1 : BIST Register/RAM test

29 **fail**. Read-write, Volatile. Reset: 0. fail

28:27 Reserved.

26:24 **BIST_pattern**. Read-write. Reset: 0h.**Description:** **BIST** Pattern

- 3'b000: Use same pattern

- 3'b001: Alternate Invert (If start will AA, then it will perform AA/55 BIST pattern)

- 3'b010: Increment Pattern

- 3'b011: LSFR

- 3'b100: Shift right (Walking 1 and Walking 0 testing)

- others: Reserved

23:0 **BIST_iteration_cnt**. Read-write. Reset: 00_0000h.**Description:** **BIST** Iteration Count

- 0: Continuous till stop/fail

- x(others): Iteration till count x stop/fail

USB[0...1]x00D910 (FCHUSBDWCUSB31CNTR0::BBISTXFERSTS0)

Read-only.

[BIST](#) Transfer Status Register 0

Note : BIST Transfer Status contains 4 DWORD distributed over four 32 bit register

BBISTXFERSTS0,BBISTXFERSTS1,BBISTXFERSTS2,BBISTXFERSTS3

_instUSB0_aliasHOST; USB0x00D910; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D910; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
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31:24	Reserved.
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23:0	bist_pending_trans_size . Read-only. Reset: 00_0000h. BIST Pending Transfer Size
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USB[0...1]x00D914 (FCHUSBDWCUSB31CNTR0::BBISTXFERSTS1)

Read-only.

[BIST](#) Transfer Status Register 1

Note : BIST Transfer Status contains 4 DWORD distributed over four 32 bit register

BBISTXFERSTS0,BBISTXFERSTS1,BBISTXFERSTS2,BBISTXFERSTS3

_instUSB0_aliasHOST; USB0x00D914; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D914; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
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31:24	Reserved.
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23:0	failed_pending_trans_size . Read-only. Reset: 00_0000h. Failed Pending Transfer Size
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USB[0...1]x00D918 (FCHUSBDWCUSB31CNTR0::BBISTXFERSTS2)

Read-only.

[BIST](#) Transfer Status Register 2

Note : BIST Transfer Status contains 4 DWORD distributed over four 32 bit register

BBISTXFERSTS0,BBISTXFERSTS1,BBISTXFERSTS2,BBISTXFERSTS3

_instUSB0_aliasHOST; USB0x00D918; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D918; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
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31:24	Reserved.
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23:0	failed_iteration . Read-only. Reset: 00_0000h. Failed Iteration Count
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USB[0...1]x00D91C (FCHUSBDWCUSB31CNTR0::BBISTXFERSTS3)

Read-only.

[BIST](#) Transfer Status Register 3

Note : BIST Transfer Status contains 4 DWORD distributed over four 32 bit register

BBISTXFERSTS0,BBISTXFERSTS1,BBISTXFERSTS2,BBISTXFERSTS3

_instUSB0_aliasHOST; USB0x00D91C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}_instUSB1_aliasHOST; USB1x00D91C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
------	-------------

31:4	Reserved.
------	-----------

3:0	loopback_statemachine . Read-only. Reset: 0h. Loopback state machine
-----	---

USB[0...1]x00D920 (FCHUSBWCUSB31CNTR0::BBISTEXPDATASTS0)

Read-only. Reset: 0000_0000h.

[BIST](#) Expected Data Status Register 0

Note : BIST Expected Data Status indicates the expected data when BIST fails which is contained in 4 DWORD distributed over four 32 bit register

BBISTEXPDATASTS0,BBISTEXPDATASTS1,BBISTEXPDATASTS2,BBISTEXPDATASTS3

- Only Expected Data [DWC_USB31_MDWIDTH-1:0] is valid, other bits are don't care if any.

_instUSB0_aliasHOST; USB0x00D920; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D920; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description31:0 **exp_data.** Read-only. Reset: 0000_0000h. Expected Data-0**USB[0...1]x00D924 (FCHUSBWCUSB31CNTR0::BBISTEXPDATASTS1)**

Read-only. Reset: 0000_0000h.

[BIST](#) Expected Data Status Register 1

Note : BIST Expected Data Status indicates the expected data when BIST fails which is contained in 4 DWORD distributed over four 32 bit register

BBISTEXPDATASTS0,BBISTEXPDATASTS1,BBISTEXPDATASTS2,BBISTEXPDATASTS3

- Only Expected Data [DWC_USB31_MDWIDTH-1:0] is valid, other bits are don't care if any.

_instUSB0_aliasHOST; USB0x00D924; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D924; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description31:0 **exp_data.** Read-only. Reset: 0000_0000h. Expected Data-1**USB[0...1]x00D928 (FCHUSBWCUSB31CNTR0::BBISTEXPDATASTS2)**

Read-only. Reset: 0000_0000h.

[BIST](#) Expected Data Status Register 2

Note : BIST Expected Data Status indicates the expected data when BIST fails which is contained in 4 DWORD distributed over four 32 bit register

BBISTEXPDATASTS0,BBISTEXPDATASTS1,BBISTEXPDATASTS2,BBISTEXPDATASTS3

- Only Expected Data [DWC_USB31_MDWIDTH-1:0] is valid, other bits are don't care if any.

_instUSB0_aliasHOST; USB0x00D928; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D928; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description31:0 **exp_data.** Read-only. Reset: 0000_0000h. Expected Data-2

USB[0...1]x00D92C (FCHUSBDWCUSB31CNTR0::BBISTEXPDATASTS3)

Read-only. Reset: 0000_0000h.

[BIST](#) Expected Data Status Register 3

Note : BIST Expected Data Status indicates the expected data when BIST fails which is contained in 4 DWORD distributed over four 32 bit register

BBISTEXPDATASTS0,BBISTEXPDATASTS1,BBISTEXPDATASTS2,BBISTEXPDATASTS3

- Only Expected Data [DWC_USB31_MDWIDTH-1:0] is valid, other bits are don't care if any.

_instUSB0_aliasHOST; USB0x00D92C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D92C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description31:0 **exp_data**. Read-only. Reset: 0000_0000h. Expected Data-3**USB[0...1]x00D930 (FCHUSBDWCUSB31CNTR0::BBISTRVCVDDATASTS0)**

Read-only. Reset: 0000_0000h.

[BIST](#) Received Data Status Register 0

Note : BIST Received Data Status indicates the received data when BIST fails which is contained in 4 DWORD distributed over four 32 bit register

BBISTRVCVDDATASTS0,BBISTRVCVDDATASTS1,BBISTRVCVDDATASTS2,BBISTRVCVDDATASTS3

- Only Received Data [DWC_USB31_MDWIDTH-1:0] is valid, other bits are don't care if any.

_instUSB0_aliasHOST; USB0x00D930; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D930; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description31:0 **received_data**. Read-only. Reset: 0000_0000h. Received Data-0**USB[0...1]x00D934 (FCHUSBDWCUSB31CNTR0::BBISTRVCVDDATASTS1)**

Read-only. Reset: 0000_0000h.

[BIST](#) Received Data Status Register 1

Note : BIST Received Data Status indicates the received data when BIST fails which is contained in 4 DWORD distributed over four 32 bit register

BBISTRVCVDDATASTS0,BBISTRVCVDDATASTS1,BBISTRVCVDDATASTS2,BBISTRVCVDDATASTS3

- Only Received Data [DWC_USB31_MDWIDTH-1:0] is valid, other bits are don't care if any.

_instUSB0_aliasHOST; USB0x00D934; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D934; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits Description31:0 **received_data**. Read-only. Reset: 0000_0000h. Received Data-1

USB[0...1]x00D938 (FCHUSBDCUSB31CNTR0::BBISTRVCVDDATASTS2)

Read-only. Reset: 0000_0000h.

[BIST](#) Received Data Status Register 2

Note : BIST Received Data Status indicates the received data when BIST fails which is contained in 4 DWORD distributed over four 32 bit register

BBISTRVCVDDATASTS0,BBISTRVCVDDATASTS1,BBISTRVCVDDATASTS2,BBISTRVCVDDATASTS3

- Only Received Data [DWC_USB31_MDWIDTH-1:0] is valid, other bits are don't care if any.

_instUSB0_aliasHOST; USB0x00D938; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D938; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
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31:0	received_data. Read-only. Reset: 0000_0000h. Received Data-2
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USB[0...1]x00D93C (FCHUSBDCUSB31CNTR0::BBISTRVCVDDATASTS3)

Read-only. Reset: 0000_0000h.

[BIST](#) Received Data Status Register 3

Note : BIST Received Data Status indicates the received data when BIST fails which is contained in 4 DWORD distributed over four 32 bit register

BBISTRVCVDDATASTS0,BBISTRVCVDDATASTS1,BBISTRVCVDDATASTS2,BBISTRVCVDDATASTS3

- Only Received Data [DWC_USB31_MDWIDTH-1:0] is valid, other bits are don't care if any.

_instUSB0_aliasHOST; USB0x00D93C; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00D93C; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
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31:0	received_data. Read-only. Reset: 0000_0000h. Received Data-3
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USB[0...1]x00DA00 (FCHUSBDCUSB31CNTR0::RSVD_REG)

Read-only. Reset: 0000_0000h.

Reserved Register

This Reserved Register block is added to fulfill the AXI Alignment Check for 4KByte boundary, that is, the lower 12-bits of the end address should be set to all ones.

_instUSB0_aliasHOST; USB0x00DA00; USB0={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio0_aliasHOST[[BASE_ADDR](#)]}

_instUSB1_aliasHOST; USB1x00DA00; USB1={NBIFEPFNCFG::[BASE_ADDR_2](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)], NBIFEPFNCFG::[BASE_ADDR_1](#)_dev0_func4_instNBIF0_nbio1_aliasHOST[[BASE_ADDR](#)]}

Bits	Description
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31:0	rsvd_field. Read-only. Reset: 0000_0000h.
------	--

Description: Reserved Register

This Reserved Register block is added to fulfill the AXI Alignment Check for 4KByte boundary, that is, the lower 12-bits of the end address should be set to all ones.

15.3.3.3 USB 2.0 – Lane Parameter Registers

USB20LANECTRL0x00000000...USB20LANECTRL1x00000400 (USB31::USB20LANEPARAMCTLREGCNTR0PORT::USB_20LANEPARACTL0_CNTR0)	
Read-write. Reset: 3223_3C11h.	
USB2.0 PHY per-lane parameter register0	
_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB20LANECTRL0x0000_0[4,0]00; USB20LANECTRL0=16D1_C000h	
_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB20LANECTRL1x0000_0[4,0]00; USB20LANECTRL1=16F1_C000h	
Bits	Description
31:28	TXVREFTUNE0. Read-write. Reset: 3h. HS DC Voltage Level Adjustment
27:26	USB_20LANEPARACTL0_27_26. Read-write. Reset: 0h. Reserved
25:24	TXRISETUNE0. Read-write. Reset: 2h. HS Transmitter Rise/Fall Time Adjustment
23	TXPREEMPPLSETUNE0. Read-write. Reset: 0. HS Transmitter Pre-Emphasis Duration Control
22	USB_20LANEPARACTL0_22. Read-write. Reset: 0. Reserved
21:20	TXPREEMPAMPTUNE0. Read-write. Reset: 2h. HS Transmitter Pre-Emphasis Current Control
19:16	TXFSLSTUNE0. Read-write. Reset: 3h. FS/LS Source Impedance Adjustment
15	USB_20LANEPARACTL0_15. Read-write. Reset: 0. Reserved
14:12	SQRXTUNE0. Read-write. Reset: 3h. Squelch Threshold Adjustment
11:8	PLLPTUNE0. Read-write. Reset: Ch. PLL Proportional Path Tune
7:6	PLLITUNE0. Read-write. Reset: 0h. PLL Integral Path Tune
5	USB_20LANEPARACTL0_5. Read-write. Reset: 0. Reserved
4	PLLBTUNE0. Read-write. Reset: 1. PLL Bandwidth adjustment
3	USB_20LANEPARACTL0_3. Read-write. Reset: 0. Reserved
2:0	COMPDISTUNE0. Read-write. Reset: 1h. Disconnect Threshold Adjustment
USB20LANECTRL0x00000004...USB20LANECTRL1x00000404 (USB31::USB20LANEPARAMCTLREGCNTR0PORT::USB_20LANEPARACTL1_CNTR0)	
Read-write. Reset: 0000_001Bh.	
USB2.0 PHY per-lane parameter register1	
_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB20LANECTRL0x0000_0[4,0]04; USB20LANECTRL0=16D1_C000h	
_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB20LANECTRL1x0000_0[4,0]04; USB20LANECTRL1=16F1_C000h	
Bits	Description
31:8	USB_20LANEPARACTL1_31_8. Read-write. Reset: 00_0000h. Reserved
7:6	VATESTENB. Read-write. Reset: 0h. Description: Enabling ANALOGTEST or ID0 for analog observations. 2'b00: ID0/ANALOGTEST disabled. 2'b01: ID0 enabled. 2'b10: ANALOGTEST enabled. 2'b11: Invalid.
5:4	VDATEFTUNE. Read-write. Reset: 1h. Data Detect Voltage Adjustment.
3:2	TXRESTUNE0. Read-write. Reset: 2h. USB Source Impedance Adjustment
1:0	TXHSXVTUNE0. Read-write. Reset: 3h. Transmitter High-Speed Crossover Adjustment

15.3.3.4 USB 2.0 – PHY Parameter Registers

USB20PHYCTRL0x00000000...USB20PHYCTRL1x00000400

(USB31::USB20PHYPARAMCTRLREGCNTR0::USB_20PHYPARACTL0_CNTR0)

Reset: 0000_0012h.

USB_20PHY Parameter Ctrl0 register

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB20PHYCTRL0x0000_0[4,0]00; USB20PHYCTRL0=16D1_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB20PHYCTRL1x0000_0[4,0]00; USB20PHYCTRL1=16F1_8000h

Bits	Description
31:11	USB_20PHYPARACTL0_31_11. Read-write. Reset: 00_0000h. Reserved
10	HSSQUELCH. Read-only. Reset: 0. Description: Read-only HSSQUELCH from USB20 PHY for testing
9	HSRXDAT. Read-only. Reset: 0. Description: Read-only HSRXDAT from USB20 PHY for testing
8	HS_TEST_ENABLE. Read-write. Reset: 0. Reserved
7:6	USB_20PHYPARACTL0_7_6. Read-write. Reset: 0h. Reserved
5	Reserved.
4	COMMONONN. Read-write. Reset: 1. Description: Common Block Power-Down Control Function: This signal controls the power-down signals in the Bias, and PLL blocks when the USB 2.0 femtoPHY is in Suspend, or Sleep mode. 1: In Suspend mode, the Bias, and PLL blocks are powered down. In Sleep mode, the Bias and PLL blocks are powered down. 0: In Suspend or Sleep modes, the Bias, and PLL blocks remain powered.
3	USB_20PHYPARACTL0_3. Read-write. Reset: 0. Reserved
2:0	FSEL. Read-write. Reset: 2h. Reference Clock Frequency Select

15.3.3.5 USB 3.1 – Lane Parameter Registers

USB31LANECTRL0x00000000...USB31LANECTRL1x00000400 (USB31::USB31LANEPARAMCTLREGCNTR0::USB_31LANEPARACTL0_CNTR0)	
Read-write. Reset: 0222_2034h.	
USB31PHY loopback enable register	
<u>_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL0x0000_0[4,0]00; USB31LANECTRL0=16D6_C000h</u>	
<u>_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL1x0000_0[4,0]00; USB31LANECTRL1=16F6_C000h</u>	
Bits	Description
31:28	USB_31LANEPARACTL0_31_28. Read-write. Reset: 0h. Reserved
27	USB_31LANEPARACTL0_27. Read-write. Reset: 0. Reserved
26:24	rx_sigdet_lf_threshold_g2. Read-write. Reset: 2h. Threshold for g2 signal detection in low speed path
23	USB_31LANEPARACTL0_23. Read-write. Reset: 0. Reserved
22:20	rx_sigdet_lf_threshold_g1. Read-write. Reset: 2h. Threshold for g1 signal detection in low speed path
19	USB_31LANEPARACTL0_19. Read-write. Reset: 0. Reserved
18:16	rx_sigdet_hf_threshold_g2. Read-write. Reset: 2h. Threshold for g2 signal detection in high speed path
15	USB_31LANEPARACTL0_15. Read-write. Reset: 0. Reserved
14:12	rx_sigdet_hf_threshold_g1. Read-write. Reset: 2h. Threshold for g1 signal detection in high speed path
11:7	USB_31LANEPARACTL0_11_7. Read-write. Reset: 00h. Reserved
6	rx0_cmn_refclk_mode. Read-write. Reset: 0. pipe_rx0_cmn_refclk_mode
5	rx_sigdet_lf_filter_en. Read-write. Reset: 1. phy_ext_rx_sigdet_lf_filter_en
4	rx_sigdet_lf_en. Read-write. Reset: 1. phy_ext_rx_sigdet_lf_en
3	rx_sigdet_hf_filt_dis. Read-write. Reset: 0. phy_ext_rx_sigdet_hf_filt_dis
2	rx_sigdet_hf_en. Read-write. Reset: 1. phy_ext_rx_sigdet_hf_en
1	laneX_rx2tx_par_lb_en. Read-write. Reset: 0. Description: Parallel (RX to TX) loopback enable. Function: When this signal is asserted, recovered parallel. Connect to lane0_rx2tx_ser_lb_eb input of PHY (RAW PCS + PMA).
0	laneX_tx2rx_ser_lb_en. Read-write. Reset: 0. Description: Analog serial (TX to RX) loopback enable. Function: When this signal is asserted, data from the TX pre-driver is looped back to the RX analog front end (AFE); loss of signal (LOS) is bypassed and based on the txX_data_en input so that rxX_los = !txX_data_en. Connect to lane0_tx2rx_ser_lb_eb input of PHY (RAW PCS + PMA).

USB31LANECTRL0x00000004...USB31LANECTRL1x00000404**(USB31::USB31LANEPARAMCTLREGCNTR0::USB_31LANEPARACTL1_CNTR0)**

Read-write. Reset: 0828_0C73h.

USB31PHY TX_EQ control0 register

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL0x0000_0[4,0]04; USB31LANECTRL0=16D6_C000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL1x0000_0[4,0]04; USB31LANECTRL1=16F6_C000h

Bits	Description
31:30	USB_31LANEPARACTL1_31_30. Read-write. Reset: 0h. Reserved
29:24	tx_eq_post_g2. Read-write. Reset: 08h. Description: Transmitter Post-Emphasis level adjustment control, TX_EQ_POST group2. Function: Control for setting the transmitter driver output post-emphasis (post coefficient) txX_eq_post[5:2] : Integer value (0 to 8) txX_eq_post[1:0] : Reserved (set to 2'b00) Connect to protocol1_ext_tx_eq_post_g2[3:0]
23:18	tx_eq_post_g1. Read-write. Reset: 0Ah. Description: Transmitter Post-Emphasis level adjustment control, TX_EQ_POST group1. Function: Control for setting the transmitter driver output post-emphasis (post coefficient) txX_eq_post[5:2] : Integer value (0 to 8) txX_eq_post[1:0] : Reserved (set to 2'b00) Connect to protocol1_ext_tx_eq_post_g1[3:0].
17	Reserved.
16	tx_eq_ovrd_g2. Read-write. Reset: 0. Description: Transmitter equalization setting overrides enable, TX_EQ_OVRD group2. Function: Enable external overrides for the per-protocol settings of the PHY's TX equalization inputs. Connect to protocol1_ext_tx_eq_ovrd_g2[0].
15	tx_eq_ovrd_g1. Read-write. Reset: 0. Description: Transmitter equalization setting overrides enable, TX_EQ_OVRD group1. Function: Enable external overrides for the per-protocol settings of the PHY's TX equalization inputs. Connect to protocol1_ext_tx_eq_ovrd_g1[0].
14:12	Reserved.
11:6	tx_eq_main_g2. Read-write. Reset: 31h. Description: Transmitter amplitude adjustment control, TX_EQ_MAIN group2. Function: Control for setting the transmitter driver output amplitude (main coefficient) txX_eq_main[5:0] : Integer value (0 to 24). Note: The maximum number of legs is 24, which corresponds to txX_eq_main[5:0] = 24. Connect to protocol1_ext_tx_eq_main_g2[5:0].
5:0	tx_eq_main_g1. Read-write. Reset: 33h. Description: Transmitter amplitude adjustment control, TX_EQ_MAIN group1. Function: Control for setting the transmitter driver output amplitude (main coefficient) txX_eq_main[5:0] : Integer value (0 to 24). Note: The maximum number of legs is 24, which corresponds to txX_eq_main[5:0] = 24. Connect to protocol1_ext_tx_eq_main_g1[5:0].

USB31LANECTRL0x00000008...USB31LANECTRL1x00000408**(USB31::USB31LANEPARAMCTLREGCNTR0::USB_31LANEPARACTL2_CNTR0)**

Read-write. Reset: 0000_F140h.

USB31PHY TX_EQ control1 register

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL0x0000_0[4,0]08; USB31LANECTRL0=16D6_C000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL1x0000_0[4,0]08; USB31LANECTRL1=16F6_C000h

Bits	Description
31:16	USB_31LANEPARACTL2_31_16. Read-write. Reset: 0000h. Reserved
15:12	tx_iboost_lvl. Read-write. Reset: Fh. Description: Transmitter current boost level, TX_IBOOST_LVL. Function: Controls the TX swing level. The recommended default setting for this input is 4'b1111. The TX swing can be calculated as: $V_{swing} = MAX(v_{ptx}, v_{ptx} + (2 * tx_vboost_vref - v_{ptx}) * (1 + tx_iboost_lvl < 3:0 >) / 16)$ Where tx_vboost_vref is determined by tx_vboost_lvl[2:0]. When tx_vboost_lvl[2:0] = 3'b000, the boosting circuit is disabled and the launch amplitude is determined by the magnitude of the vptxX supply. When tx_vboost_lvl[2:0] = 3'b011, the launch amplitude is 900 mV. Connect to protocol1_ext_tx_iboost_lvl[3:0].
11:6	tx_eq_pre_g2. Read-write. Reset: 05h. Description: Transmitter Pre-Emphasis level adjustment control, TX_EQ_PRE group2. Function: Control for setting the transmitter driver output pre-emphasis (preshoot coefficient) txX_eq_pre[5:2] : Integer value (0 to 6) txX_eq_pre[1:0] : Reserved (set to 2'b00) Connect to protocol1_ext_tx_eq_pre_g2[3:0].
5:0	tx_eq_pre_g1. Read-write. Reset: 00h. Description: Transmitter Pre-Emphasis level adjustment control, TX_EQ_PRE group1. Function: Control for setting the transmitter driver output pre-emphasis (preshoot coefficient) txX_eq_pre[5:2] : Integer value (0 to 6) txX_eq_pre[1:0] : Reserved (set to 2'b00) Connect to protocol1_ext_tx_eq_pre_g1[3:0].

USB31LANECTRL0x0000000C...USB31LANECTRL1x0000040C**(USB31::USB31LANEPARAMCTLREGCNTR0::USB_31LANEPARACTL3_CNTR0)**

Read-write. Reset: 0000_001Bh.

USB31PHY RX adapt and RX eq control register

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL0x0000_0[4,0]0C; USB31LANECTRL0=16D6_C000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL1x0000_0[4,0]0C; USB31LANECTRL1=16F6_C000h

Bits	Description
31:30	USB_31LANEPARACTL3_31_30. Read-write. Reset: 0h. Reserved
29:26	Reserved.
25:22	rx_eq_att_lvl_g2. Read-write. Reset: 0h. Description: Rx equalization attenuation level group2. Function: Reserved; tie off to 1'b0. Connect to protocol1_ext_rx_eq_att_lvl_g2[2:0]. Bit3 is open.
21:18	rx_eq_att_lvl_g1. Read-write. Reset: 0h. Description: Rx equalization attenuation level group1. Function: Reserved; tie off to 1'b0. Connect to protocol1_ext_rx_eq_att_lvl_g1[2:0]. Bit3 is open.
17:14	Reserved.
13:10	rx_eq_afe_gain_g2. Read-write. Reset: 0h. Description: Rx equalization afe gain group2. Function: Controls the gain of the AFE. Connect to protocol1_ext_rx_eq_afe_gain_g2[3:0].
9:6	rx_eq_afe_gain_g1. Read-write. Reset: 0h. Description: Rx equalization afe gain group1. Function: Controls the gain of the AFE. Connect to protocol1_ext_rx_eq_afe_gain_g1[3:0].
5	Reserved.
4	rx_adapt_dfe_en_g2. Read-write. Reset: 1. Description: Rx adaptation dfe enable group2. Function: Enables the RX adaptation and decision feedback equalization (DFE) circuitry, and applies the input setting for DFE Tap1: rxX_eq_dfe_tap1[7:0] When this signal is de-asserted, the change on the input DFE settings can be applied (with a request command using the rxX_req/rxX_ack handshake). When this signal is asserted, the DFE settings are initialized to the inputs and RX DFE adaptation routines are executed to achieve optimal DFE settings. Connect to protocol1_ext_rx_adapt_dfe_en_g2[0].
3	rx_adapt_dfe_en_g1. Read-write. Reset: 1. Description: Rx adaptation dfe enable group1. Function: Enables the RX adaptation and decision feedback equalization (DFE) circuitry, and applies the input setting for DFE Tap1: rxX_eq_dfe_tap1[7:0] When this signal is de-asserted, the change on the input DFE settings can be applied (with a request command using the rxX_req/rxX_ack handshake). When this signal is asserted, the DFE settings are initialized to the inputs and RX DFE adaptation routines are executed to achieve optimal DFE settings. Connect to protocol1_ext_rx_adapt_dfe_en_g1[0].
2	Reserved.
1	rx_adapt_afe_en_g2. Read-write. Reset: 1.

	Description: Rx adaptation afe enable group2. Function: Enables the RX adaptation circuitry and applies the input receiver AFE equalization settings: rxX_eq_att_lv[2:0] rxX_eq_afe_gain[3:0] rxX_eq_ctle_boost[4:0] When this signal is de-asserted, the change on the input AFE settings can be applied (with a request command using the rxX_req/rxX_ack handshake). When this signal is asserted, the AFE equalization settings are initialized to the inputs, but any further change on these inputs are ignored and RX AFE adaptation routines are executed to achieve optimal AFE settings. Connect to protocol1_ext_rx_adapt_afe_en_g2[0]
0	rx_adapt_afe_en_g1. Read-write. Reset: 1. Description: Rx adaptation afe enable group1. Function: Enables the RX adaptation circuitry and applies the input receiver AFE equalization settings: rxX_eq_att_lv[2:0] rxX_eq_afe_gain[3:0] rxX_eq_ctle_boost[4:0] When this signal is de-asserted, the change on the input AFE settings can be applied (with a request command using the rxX_req/rxX_ack handshake). When this signal is asserted, the AFE equalization settings are initialized to the inputs, but any further change on these inputs are ignored and RX AFE adaptation routines are executed to achieve optimal AFE settings. Connect to protocol1_ext_rx_adapt_afe_en_g1[0]

USB31LANECTRL0x00000010...USB31LANECTRL1x00000410

(USB31::USB31LANEPARAMCTLREGCNTR0::USB_31LANEPARACTL4_CNTR0)

Read-write. Reset: 8000_00E7h.

USB31PHY RX eq control1 register

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL0x0000_0[4,0]10; USB31LANECTRL0=16D6_C000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL1x0000_0[4,0]10; USB31LANECTRL1=16F6_C000h

Bits	Description
31	rx_cdr_ssc_en. Read-write. Reset: 1. Description: Connect to rx_cdr_ssc_en of pma. The port name of phy ip(dwc_usb31sspphy_upcs_x4_ns_x4_x4_pipe) to be connected is pipe_rx0_sris_mode_en
30:23	rx_eq_dfe_tap1_g2. Read-write. Reset: 00h. Description: Rx equalization dfe tap1 group2. Function: Controls the value of DFE data Tap1. This is a signed input; two's complement encoded. Connect to protocol1_ext_rx_eq_dfe_tap1_g2[7:0].
22:15	rx_eq_dfe_tap1_g1. Read-write. Reset: 00h. Description: Rx equalization dfe tap1 group1. Function: Controls the value of DFE data Tap1. This is a signed input; two's complement encoded. Connect to protocol1_ext_rx_eq_dfe_tap1_g1[7:0].
14:10	Reserved.
9:5	rx_eq_ctle_boost_g2. Read-write. Reset: 07h. Description: Rx equalization ctile boost group2. Function: Controls the CTLE boost level; binary encoded. Connect to protocol1_ext_rx_eq_ctle_boost_g2[4:0].
4:0	rx_eq_ctle_boost_g1. Read-write. Reset: 07h. Description: Rx equalization ctile boost group1. Function: Controls the CTLE boost level; binary encoded. Connect to protocol1_ext_rx_eq_ctle_boost_g1[4:0].

USB31LANECTRL0x00000014...USB31LANECTRL1x00000414**(USB31::USB31LANEPARAMCTLREGCNTR0::USB_31LANEPARACTL5_CNTR0)**

Read-write. Reset: 0000_0300h.

USB31PHY LANEPARACTL5 control register

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL0x0000_0[4,0]14; USB31LANECTRL0=16D6_C000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL1x0000_0[4,0]14; USB31LANECTRL1=16F6_C000h

Bits	Description
31:11	USB_31LANEPARACTL5_31_11. Read-write. Reset: 00_0000h. Reserved
10:8	rx_los_threshold. Read-write. Reset: 3h. Description: Receiver LOS threshold. Function: Sets the LOS threshold level trip point. It is recommended to set this input to 3'b011 for USB3.1 SS+. rxX_los_threshold LOS threshold 3'b000 Reserved 3'b001 90mVpp 3'b010 120mVpp 3'b011 150mvpp 3'b100 180mVpp 3'b101 210mVpp 3'b110 240mVpp 3'b111 270mVpp Connect to protocol1_ext_rx_los_threshold[2:0].
7:0	Reserved.

USB31LANECTRL0x00000018...USB31LANECTRL1x00000418**(USB31::USB31LANEPARAMCTLREGCNTR0::USB_31LANEPARACTL6_CNTR0)**

Read-write. Reset: 0000_0000h.

USB31PHY LANEPARACTL6 control register

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL0x0000_0[4,0]18; USB31LANECTRL0=16D6_C000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31LANECTRL1x0000_0[4,0]18; USB31LANECTRL1=16F6_C000h

Bits	Description
31:18	USB_31LANEPARACTL6_31_18. Read-write. Reset: 0000h. Reserved
17:12	rx0_idle_los_cnt. Read-write. Reset: 00h. These pins provide a control to disable the CDR in PCIE GEN1 mode when there is electrical idle on rx line without EIOS.
11:8	rx_eq_delta_iq_g2. Read-write. Reset: 0h. Rx equalization delta iq group2
7:4	USB_31LANEPARACTL6_7_4. Read-write. Reset: 0h. Reserved
3:0	rx_eq_delta_iq_g1. Read-write. Reset: 0h. Rx equalization delta iq group1

15.3.3.6 USB 3.1 – PHY Parameter Registers

USB31PHYCTRL0x00000000...USB31PHYCTRL1x00000400 (USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL0_CNTR0)	
Read-write. Reset: 0000_8000h.	
_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]00; USB31PHYCTRL0=16D6_8000h	
_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]00; USB31PHYCTRL1=16F6_8000h	
Bits	Description
31:23	USB_31PHYPARACTL0_31_23. Read-write. Reset: 000h. Reserved -- do not reuse
22:16	USB_31PHYPARACTL0_22_16. Read-write. Reset: 00h. Reserved -- do not reuse
15	mplla_div_clk_en. Read-write. Reset: 1. Description: MPLLA divide clock enable. Function: When asserted, the frequency of the mplla_div_clk output clock is the MPLLA frequency divided by mplla_div_multiplier[6:0]. Fractional division and/or SSC is additionally applied to mplla_div_clk depending on the settings of mplla_ssc_en and mplla_fracn_ctrl[8:0] mplla_ssc_en. Connect to protocol1_ext_mplla_div_clk_en input of UPCS
14	mplla_div8_clk_en. Read-write. Reset: 0. Description: MPLLA divide by 8 enable. Function: When asserted, the frequency of the mplla_word_clk output clock is the MPLLA frequency divided by 8. If mplla_div10_clk_en is also asserted, then divide by 10 takes priority. Connect to protocol1_ext_mplla_div8_clk_en input of UPCS
13	mplla_div16p5_clk_en. Read-write. Reset: 0. Description: MPLLA divide by 16.5 enable. Connect to protocol1_ext_mplla_div16p5_clk_en input of UPCS
12	mplla_div10_clk_en. Read-write. Reset: 0. Description: MPLLA divide by 10 enable. Function: When asserted, the frequency of the mplla_word_clk output clock is the MPLLA frequency divided by 10. This divide by 10 takes priority over mplla_div8_clk_en. Connect to protocol1_ext_mplla_div10_clk_en input of UPCS.
11:1	USB_31PHYPARACTL0_11_1. Read-write. Reset: 000h. Reserved -- do not reuse
0	ext_ctrl_sel. Read-write. Reset: 0. Description: External control select 0: use pcs internal hard-code configuration. 1: use the external configuration from register bits. Connect to the phy_ext_ctrl_sel input of UPCS.

USB31PHYCTRL0x00000004...USB31PHYCTRL1x00000404**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL1_CNTR0)**

Read-write. Reset: 8000_0000h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]04; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]04; USB31PHYCTRL1=16F6_8000h

Bits	Description
31	mplla_ssc_en. Read-write. Reset: 1. Description: MPLLA ssc enable Function: Enables spread-spectrum clock (SSC) generation on the mplla_div_clk output. If the reference clock already has spread spectrum applied, mplla_ssc_en must be de-asserted. Connects to mplla_ssc_en input.
30:23	USB_31PHYPARACTL1_30_23. Read-write. Reset: 00h. Reserved
22:18	ref_range. Read-write. Reset: 00h. Description: Input reference clock frequency range. Function: Specifies the frequency range of the input reference clock (post ref_clk_div2_en division if any). The code mapping of ref_range is as follows: 3'b000 : 20 - 26 MHz 3'b001 : 26.1 - 52 MHz 3'b010 : 52.1 - 78 MHz 3'b011 : 78.1 - 104 MHz 3'b100 : 104.1 - 130 MHz 3'b101 : 130.1 - 156 MHz 3'b110 : 156.1 - 182 MHz 3'b111 : 182.1 - 200 MHz Connects to protocol1_ext_ref_range input of UPCS.
17	ref_clk_mplla_div2_en. Read-write. Reset: 0. Description: MPLLA reference clock divider control. Function: The reference clock used for MPLLA calibration and locking can be divided by 2. This division is applied after ref_clk_div2_en. Hence the total division ratio (from the input reference clock) can be 1, 2 or 4. Connects to protocol1_ext_ref_clk_mplla_div2_en input of UPCS.
16	ref_clk_div2_en. Read-write. Reset: 0. Description: Input reference clock divider control. Function: The reference clock is divided by 2 when asserted. Connects to protocol1_ext_ref_clk_div2_en input of UPCS.
15:14	USB_31PHYPARACTL1_15_14. Read-write. Reset: 0h. Reserved -- do not reuse
13:11	mplla_ssc_range. Read-write. Reset: 0h. Description: Spread-Spectrum clock range. Function: Selects the range of modulation to insert. Applies a fixed offset to the phase accumulator, as follows: mplla_ssc_range PPM Offset 3'b000 -4980 3'b001 -4492 3'b010 -4003 3'b011 -3027 3'b100 4980 3'b101 4492 3'b110 4003 3'b111 3027 Connects to protocol1_ext_mplla_ssc_range input of UPCS.
10:8	mplla_ssc_clk_sel. Read-write. Reset: 0h. Description: MPLLA spread spectrum clock select. Function: MPLLA clock select for generating spread spectrum. Connects to protocol1_ext_mplla_ssc_clk_sel input of UPCS.

7:0	USB_31PHYPARACTL1_7_0. Read-write. Reset: 00h. Reserved -- do not reuse
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USB31PHYCTRL0x00000008...USB31PHYCTRL1x00000408
(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL2_CNTR0)

Read-write. Reset: 0000_0000h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]08; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]08; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:16	USB_31PHYPARACTL2_31_16. Read-write. Reset: 0000h. Reserved
15:13	rx_eq_ctle_pole_g2. Read-write. Reset: 0h. Description: EQ_CTLE_POLE group2. Connects to protocol1_ext_rx_ctle_pole_g2 input of UPCS.
12:10	rx_eq_ctle_pole_g1. Read-write. Reset: 0h. Description: EQ_CTLE_POLE group1. Connects to protocol1_ext_rx_ctle_pole_g1 input of UPCS.
9:0	USB_31PHYPARACTL2_9_0. Read-write. Reset: 00h. Reserved

USB31PHYCTRL0x0000000C...USB31PHYCTRL1x0000040C
(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL3_CNTR0)

Read-write. Reset: 0001_00FEh.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]0C; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]0C; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:28	USB_31PHYPARACTL3_31_28. Read-write. Reset: 0h. Reserved
27:17	rx_los_pwr_up_cnt. Read-write. Reset: 000h. Description: rx LOS power up counter. Connects to protocol1_ext_rx_los_pwr_up_cnt input of UPCS.
16	rx_los_lfps_en. Read-write. Reset: 1. Description: Receiver LOS LFPS enable. Function: Enables the LFPS filter for USB-like LFPS detection. By default, this input should be set to 0. Connects to protocol1_ext_rx_los_lfps_en input of UPCS.
15:12	rx_eq_vga2_gain_g2. Read-write. Reset: 0h. Description: EQ_VGA2_GAIN group2. Connects to protocol1_ext_rx_eq_vga2_gain_g2 input of UPCS
11:8	rx_eq_vga2_gain_g1. Read-write. Reset: 0h. Description: EQ_VGA2_GAIN group1. Connects to protocol1_ext_rx_eq_vga2_gain_g1 input of UPCS.
7:4	rx_eq_vga1_gain_g2. Read-write. Reset: Fh. Description: EQ_VGA1_GAIN group2. Connects to protocol1_ext_rx_eq_vga1_gain_g2 input of UPCS.
3:0	rx_eq_vga1_gain_g1. Read-write. Reset: Eh. Description: EQ_VGA1_GAIN group1. Connects to protocol1_ext_rx_eq_vga1_gain_g1 input of UPCS.

USB31PHYCTRL0x00000010...USB31PHYCTRL1x00000410**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL4_CNTR0)**

Read-write. Reset: 0000_5000h.

`_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]10; USB31PHYCTRL0=16D6_8000h``_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]10; USB31PHYCTRL1=16F6_8000h`

Bits	Description
31:17	USB_31PHYPARACTL4_31_17. Read-write. Reset: 0000h. Reserved
16:12	rx_vref_ctrl. Read-write. Reset: 05h. Description: RX biasing current control, RX_VREF_CTRL. Function: Sets the RX biasing current for RX analog front end. The recommended default setting for this input is 5'b10001. Connects to protocol1_ext_rx_vref_ctrl input of UPCS.
11:0	USB_31PHYPARACTL4_11_0. Read-write. Reset: 000h. Reserved

USB31PHYCTRL0x00000014...USB31PHYCTRL1x00000414**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL5_CNTR0)**

Read-write. Reset: 00B6_45B2h.

`_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]14; USB31PHYCTRL0=16D6_8000h``_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]14; USB31PHYCTRL1=16F6_8000h`

Bits	Description
31:26	USB_31PHYPARACTL5_31_26. Read-write. Reset: 00h. Reserved
25:13	rx_vco_ld_val_g2. Read-write. Reset: 05B2h. Description: RX VCO calibration load value, RX_VCO_LD_VAL group2. Function: This input is used to load internal calibration registers used to perform RX VCO calibration. Connects to protocol1_ext_rx_vco_ld_val_g2 input of UPCS.
12:0	rx_vco_ld_val_g1. Read-write. Reset: 05B2h. Description: RX VCO calibration load value, RX_VCO_LD_VAL group1. Function: This input is used to load internal calibration registers used to perform RX VCO calibration. Connects to protocol1_ext_rx_vco_ld_val_g1 input of UPCS.

USB31PHYCTRL0x00000018...USB31PHYCTRL1x00000418**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL6_CNTR0)**

Read-write. Reset: 16B0_0000h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]18; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]18; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:30	USB_31PHYPARACTL6_31_30. Read-write. Reset: 0h. Reserved
29:25	bs_rx_level. Read-write. Reset: 0Bh. Description: ACJTAG Receiver Sensitivity Level Control, BS_RX_LEVEL. Function: Sets the internal thresholds for the ACJTAG receiver. Connects to protocol1_ext_bs_rx_level input of UPCS.
24	bs_tx_lowswing. Read-write. Reset: 0. Description: TX Boundary Scan Low Swing, BS_TX_LOWSWING. Function: Enables TX JTAG low-swing mode (otherwise, it is in full-swing mode). Connects to protocol1_ext_bs_tx_lowswing input of UPCS.
23	bs_rx_bigswing. Read-write. Reset: 1. Description: RX Boundary Scan Big Swing, BS_RX_BIGSWING. Function: Enables RX ACJTAG big-swing mode. Connects to protocol1_ext_bs_rx_bigswing input of UPCS.
22:20	tx_vboost_lvl. Read-write. Reset: 3h. Description: TX voltage boost level, TX_VBOOST_LVL. Function: Sets the minimum TX differential output level. The recommended default setting for this input is 3'b011. Connects to protocol1_ext_tx_vboost_lvl input of UPCS.
19:0	USB_31PHYPARACTL6_19_0. Read-write. Reset: 0_0000h. Reserved

USB31PHYCTRL0x0000001C...USB31PHYCTRL1x0000041C**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL7_CNTR0)**

Read-write. Reset: 0004_F01Dh.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]1C; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]1C; USB31PHYCTRL1=16F6_8000h

Bits	Description
31	USB_31PHYPARACTL7_31. Read-write. Reset: 0. Reserved
30:28	mplla_tx_clk_div. Read-write. Reset: 0h. connect to phy_ext_mplla_tx_clk_div of PHY
27:24	USB_31PHYPARACTL7_27_24. Read-write. Reset: 0h. Reserved
23:16	mplla_div_multiplier. Read-write. Reset: 04h. Description: MPLLA output frequency multiplier control. Function: Frequency multiplication factor used to generate MPLL clock output from the reference clock input as seen by the MPLL (after ref_clk_div2_en and ref_clk_mplla_div2_en dividers). Connect to protocol1_ext_mplla_div_multiplier
15:0	mplla_bandwidth. Read-write. Reset: F01Dh. Description: MPLLA bandwidth control. Function: Controls the bandwidth for MPLLA. Connects to protocol1_ext_mplla_bandwidth input of UPCS.

USB31PHYCTRL0x00000020...USB31PHYCTRL1x00000420**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL8_CNTR0)**

Read-write. Reset: 0000_0486h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]20; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]20; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:11	USB_31PHYPARACTL8_31_11. Read-write. Reset: 00_0000h. Reserved
10:0	mplla_fracn_ctrl. Read-write. Reset: 486h. Description: MPLLA fractional control. Connects to protocol1_ext_mplla_fracn_ctrl input of UPCS.

USB31PHYCTRL0x00000024...USB31PHYCTRL1x00000424**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL9_CNTR0)**

Read-write. Reset: 0152_003Bh.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]24; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]24; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:25	USB_31PHYPARACTL9_31_25. Read-write. Reset: 00h. Reserved
24	mplla_word_div2_en. Read-write. Reset: 1. connect to phy_ext_mplla_word_div2_en of PHY
23:16	mplla_ssc_freq_cnt_peak. Read-write. Reset: 52h. Mplla ssc freq cnt peak
15:13	USB_31PHYPARACTL9_15_13. Read-write. Reset: 0h. Reserved
12	mplla_ssc_freq_cnt_ovrd_en. Read-write. Reset: 0. Mplla ssc freq cnt ovrd en
11:0	mplla_ssc_freq_cnt_init. Read-write. Reset: 03Bh. Mplla ssc freq cnt init

USB31PHYCTRL0x00000028...USB31PHYCTRL1x00000428**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL10_CNTR0)**

Read-write. Reset: 1004_0202h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]28; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]28; USB31PHYCTRL1=16F6_8000h

Bits	Description
31	USB_31PHYPARACTL10_31. Read-write. Reset: 0. Reserved
30:24	mplla_cp_prop. Read-write. Reset: 10h. connect to phy_ext_mplla_cp_prop of PHY
23	USB_31PHYPARACTL10_23. Read-write. Reset: 0. Reserved
22:16	mplla_cp_int. Read-write. Reset: 04h. connect to phy_ext_mplla_cp_int of PHY
15:11	USB_31PHYPARACTL10_15_11. Read-write. Reset: 00h. Reserved
10:8	tx_term_ctrl. Read-write. Reset: 2h. Tx term control
7:3	USB_31PHYPARACTL10_7_3. Read-write. Reset: 00h. Reserved
2:0	rx_term_ctrl. Read-write. Reset: 2h. Rx term control

USB31PHYCTRL0x0000002C...USB31PHYCTRL1x0000042C**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL11_CNTR0)**

Read-write. Reset: 0387_1801h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]2C; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]2C; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:30	USB_31PHYPARACTL11_31_30. Read-write. Reset: 0h. Reserved
29:23	rx_ref_ld_val_g2. Read-write. Reset: 07h. Description: RX VCO calibration reference load value, RX_REF_LD_VAL group2. Function: This input is used to load internal calibration registers used to perform RX VCO calibration. Connects to phy_ext_rx_ref_ld_val_g2 input of UPCS.
22:16	rx_ref_ld_val_g1. Read-write. Reset: 07h. Description: RX VCO calibration reference load value, RX_REF_LD_VAL group1. Function: This input is used to load internal calibration registers used to perform RX VCO calibration. Connects to phy_ext_rx_ref_ld_val_g1 input of UPCS.
15:4	mplla_multiplier. Read-write. Reset: 180h. Description: MPLLA frequency multiplier control. Function: Multiplies the reference clock to a frequency suitable for intended operating speed. For a list of values and corresponding frequencies. Connects to protocol1_ext_mplla_multiplier input of UPCS.
3:1	ref_ana_mplla_div. Read-write. Reset: 0h. connect to phy_ext_ref_clk_mplla_div input of PHY
0	mplla_div5_clk_en. Read-write. Reset: 1. connect to phy_ext_mplla_div5_clk_en input of PHY

USB31PHYCTRL0x00000030...USB31PHYCTRL1x00000430**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL12_CNTR0)**

Read-write. Reset: 5555_0001h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]30; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]30; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:16	mplla_fracn_quot. Read-write. Reset: 5555h. connect to phy_ext_mplla_fracn_quot of PHY
15:1	USB_31PHYPARACTL12_15_1. Read-write. Reset: 0000h. Reserved
0	mplla_fracn_en. Read-write. Reset: 1. connect to phy_ext_mplla_fracn_en of PHY

USB31PHYCTRL0x00000034...USB31PHYCTRL1x00000434**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL13_CNTR0)**

Read-write. Reset: 0003_0001h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]34; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]34; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:16	mplla_fracn_den. Read-write. Reset: 0003h. connect to phy_ext_mplla_fracn_den of PHY
15:0	mplla_fracn_rem. Read-write. Reset: 0001h. connect to phy_ext_mplla_fracn_rem of PHY

USB31PHYCTRL0x00000038...USB31PHYCTRL1x00000438**(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL14_CNTR0)**

Read-write. Reset: 0002_1556h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]38; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]38; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:21	USB_31PHYPARACTL14_31_21. Read-write. Reset: 000h. Reserved
20:1	mplla_ssc_peak. Read-write. Reset: 1_0AABh. connect to phy_ext_mplla_ssc_peak of PHY
0	mplla_ssc_up_spread. Read-write. Reset: 0. connect to phy_ext_mplla_ssc_up_spread of PHY

USB31PHYCTRL0x0000003C...USB31PHYCTRL1x0000043C
(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL15_CNTR0)

Read-write. Reset: 0041_6666h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]3C; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]3C; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:25	USB_31PHYPARACTL15_31_25. Read-write. Reset: 00h. Reserved
24:23	mplla_freq_vco. Read-write. Reset: 0h. connect to phy_ext_mplla_freq_vco of PHY
22:21	mplla_v2i. Read-write. Reset: 2h. connect to phy_ext_mplla_v2i of PHY
20:0	mplla_ssc_stepsize. Read-write. Reset: 01_6666h. connect to phy_ext_mplla_ssc_stepsize of PHY

USB31PHYCTRL0x00000040...USB31PHYCTRL1x00000440
(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL16_CNTR0)

Reset: 00F8_0011h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]40; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]40; USB31PHYCTRL1=16F6_8000h

Bits	Description
31:26	USB_31PHYPARACTL16_31_26. Read-write. Reset: 00h. Reserved
25	sstx_vregdrv_byp. Read-write. Reset: 0. connect to sstx_vregdrv_byp of PHY
24:22	sup_rx_vco_vref_sel. Read-write. Reset: 3h. connect to sup_rx_vco_vref_sel of PHY
21	sup_pre_hp. Read-write. Reset: 1. connect to sup_pre_hp of PHY
20	pipe_rx_cdr_legacy_en. Read-write. Reset: 1. connect to pipe_rx_cdr_legacy_en of PHY
19	pipe_rx_recal_cont_en. Read-write. Reset: 1. connect to pipe_rx_recal_cont_en of PHY
18	typec_flip_invert. Read-write. Reset: 0. connect to typec_flip_invert of PHY
17	rtune_ack. Read-only. Reset: 0. connect to phy_rtune_ack of PHY
16	rtune_req. Read-write. Reset: 0. connect to phy_rtune_req of PHY
15:13	USB_31PHYPARACTL16_15_13. Read-write. Reset: 0h. Reserved
12:8	cmn_in_generic_bus. Read-write. Reset: 00h. connect to phy_cmn_in_generic_bus of PHY
7:6	dco_range. Read-write. Reset: 0h. connect to phy_ext_dco_range of PHY
5:0	dco_finetune. Read-write. Reset: 11h. connect to phy_ext_dco_finetune of PHY

USB31PHYCTRL0x00000044...USB31PHYCTRL1x00000444
(USB31::USB31PHYPARAMCTRLREGCNTR0::USB_31PHYPARACTL17_CNTR0)

Read-write. Reset: 7F41_0000h.

_inst[USB0CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL0x0000_0[4,0]44; USB31PHYCTRL0=16D6_8000h

_inst[USB1CNTR0PORT[1:0]]_aliasSMN; USB31PHYCTRL1x0000_0[4,0]44; USB31PHYCTRL1=16F6_8000h

Bits	Description
31	USB_31PHYPARACTL17_31. Read-write. Reset: 0. Reserved
30:24	mplla_cp_prop_gs. Read-write. Reset: 7Fh. connect to phy_ext_mplla_cp_prop_gs of PHY
23	USB_31PHYPARACTL17_23. Read-write. Reset: 0. Reserved
22:16	mplla_cp_int_gs. Read-write. Reset: 41h. connect to phy_ext_mplla_cp_int_gs of PHY
15:0	USB_31PHYPARACTL17_15_0. Read-write. Reset: 0000h. Reserved

15.3.3.7 USB 3.1 – USB S0 Container 0 Registers

USBCONTAINER[0...1]x00070000 (USB31::USBCONTAINERS0REGCNTR0::MCAFUNCTRL_CNTR0)	
Read-write. Reset: 0001_0070h.	
MCA function ctrl register	
_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070000; USBCONTAINER[1:0]=16[F,D]8_0000h	
Bits	Description
31:17	MCAFUNCTRL_RESERVED_31_17. Read-write. Reset: 0000h. Reserved
16	MCAINTREQ_EN. Read-write. Reset: 1. Description: MCA IntReq Enable 0: Disable Type3 Error input to MCA IP 1(default): Enable Type3 Error input to MCA IP
15:7	MCAFUNCTRL_RESERVED_15_7. Read-write. Reset: 000h. Reserved
6	MCAERRT3EN. Read-write. Reset: 1. Description: MCA Type3 Error (AXI SLV Err) Input Enable 0: Disable Type3 Error input to MCA IP 1(default): Enable Type3 Error input to MCA IP
5	MCAERRT2EN. Read-write. Reset: 1. Description: MCA Type2 Error (PHY memory parity) Input Enable 0: Disable Type2 Error input to MCA IP 1(default): Enable Type2 Error input to MCA IP
4	MCAERRT1EN. Read-write. Reset: 1. Description: MCA Type1 Error (S0 memory error) Input Enable 0: Disable Type1 Error input to MCA IP 1(default): Enable Type1 Error input to MCA IP
3	MCAFUNCTRL_RESERVED_3. Read-write. Reset: 0. Reserved
2	MCAERRTRIG_MODE. Read-write. Reset: 0. Description: MCA Err Trig Mode 0(default): McaErrTrig drive IpTriggers_IP_DBGU_Triggers[2] directly. 1: Use McaErrTrig to generate a level type error trigger and connect to IpTriggers_IP_DBGU_Triggers[2].
1	MCACLKCOARSEEN_OVRVALUE. Read-write. Reset: 0. Description: MCA clock coarse enable override value 0(default): CoarseEn is 1'b0 when MCACLKCOARSEEN_OVREN is 1'b1. 1: CoarseEn is 1'b1 when MCACLKCOARSEEN_OVREN is 1'b1.
0	MCACLKCOARSEEN_OVREN. Read-write. Reset: 0. Description: MCA clock coarse enable override enable: 0(default): Disable override CoarseEn 1: Enable override CoarseEn The CoarseEn is MCACLKCOARSEEN_OVRDAT

USBCONTAINER[0...1]x00070004**(USB31::USBCONTAINERS0REGCNTR0::MCAERRSEVERITY0_CNTR0)**

Read-write. Reset: 0001_5525h.

[MCA](#) Error Severity Control Register0

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070004; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:20	MCAERRSEVERITY0_RESERVED_31_20. Read-write. Reset: 000h. Reserved
19:16	AXISLVERRSEVERITY. Read-write. Reset: 1h. Description: AXI SLV Error Severity. The bit[3] is reserved. 3'b000=No error, 3'b001=Transparent Error(default) 3'b010=Corrected Error 3'b011=Deferred Error 3'b100=Uncorrected Error 3'b101=Fatal Error
15:12	FEDOUTBERRSEVERITY. Read-write. Reset: 5h. Description: FEDOUTB Error Severity. The bit[3] is reserved. 3'b000=No error, 3'b001=Transparent Error 3'b010=Corrected Error 3'b011=Deferred Error 3'b100=Uncorrected Error 3'b101=Fatal Error(default)
11:8	DEDBERRSEVERITY. Read-write. Reset: 5h. Description: DEDB Error Severity. The bit[3] is reserved. 3'b000=No error, 3'b001=Transparent Error 3'b010=Corrected Error 3'b011=Deferred Error 3'b100=Uncorrected Error 3'b101=Fatal Error(default)
7:4	SECBERRSEVERITY. Read-write. Reset: 2h. Description: SECB Error Severity. The bit[3] is reserved. 3'b000=No error, 3'b001=Transparent Error 3'b010=Corrected Error (default) 3'b011=Deferred Error 3'b100=Uncorrected Error 3'b101=Fatal Error
3:0	PARITYERRSEVERITY. Read-write. Reset: 5h. Description: Parity Error Severity. The bit[3] is reserved. 3'b000=No error, 3'b001=Transparent Error 3'b010=Corrected Error 3'b011=Deferred Error 3'b100=Uncorrected Error 3'b101=Fatal Error (default)

USBCONTAINER[0...1]x00070008**(USB31::USBCONTAINERS0REGCNTR0::MCAERRSEVERITY1_CNTR0)**

Read-write. Reset: 0000_0000h.

[MCA](#) Error Severity Control Register1

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070008; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:0	MCAERRSEVERITY1_RESERVED_31_0. Read-write. Reset: 0000_0000h. Reserved for future Error Severity

USBCONTAINER[0...1]x0007000C**(USB31::USBCONTAINERS0REGCNTR0::MCAERREXCPTTYPE0_CNTR0)**

Read-write. Reset: 0000_03AEh.

[MCA](#) Error Exception Type Control Register0

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x0007000C; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:10	MCAERREXCPTTYPE0_RESERVED_31_10. Read-write. Reset: 00_0000h. Reserved
9:8	AXISLVERREXCPTTYPE. Read-write. Reset: 3h. Description: AXI SLV Error Exception Type 2'b00: PRECISE 2'b01: SYNCHRONOUS 2'b10: ASYNCHRONOUS 2'b11: NONE (default)
7:6	FEDOUTBERREXCPTTYPE. Read-write. Reset: 2h. Description: FEDOUTB Error Exception Type 2'b00: PRECISE 2'b01: SYNCHRONOUS 2'b10: ASYNCHRONOUS (default) 2'b11: NONE
5:4	DEDBERREXCPTTYPE. Read-write. Reset: 2h. Description: DEDB Error Exception Type 2'b00: PRECISE 2'b01: SYNCHRONOUS 2'b10: ASYNCHRONOUS (default) 2'b11: NONE
3:2	SECBERREXCPTTYPE. Read-write. Reset: 3h. Description: SECB Error Exception Type 2'b00: PRECISE 2'b01: SYNCHRONOUS 2'b10: ASYNCHRONOUS 2'b11: NONE (default)
1:0	PARITYERREXCPTTYPE. Read-write. Reset: 2h. Description: Parity Error Exception Type 2'b00: PRECISE 2'b01: SYNCHRONOUS 2'b10: ASYNCHRONOUS (default) 2'b11: NONE

USBCONTAINER[0...1]x00070010**(USB31::USBCONTAINERS0REGCNTR0::MCAERRSTRUCTYPE0_CNTR0)**

Read-write. Reset: 0005_4444h.

[MCA](#) Error Structure Type Control Register0

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070010; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:20	MCAERRSTRUCTYPE0_RESERVED_31_20. Read-write. Reset: 000h. Reserved
19:16	AXISLVERRSTRUCTYPE. Read-write. Reset: 5h. Description: AXI SLV Error Structure Type 4'b0000=DRAM, 4'b0001=Cache, 4'b0010=Tag, 4'b0011=TLB, 4'b0100=OtherArray (incl. flop arrays), 4'b0101=NonArrayError (e.g. Abort, Livelock/Deadlock, etc), (default)
15:12	FEDOUTBERRSTRUCTYPE. Read-write. Reset: 4h. Description: FEDOUTB Error Structure Type 4'b0000=DRAM, 4'b0001=Cache, 4'b0010=Tag, 4'b0011=TLB, 4'b0100=OtherArray (incl. flop arrays), (default) 4'b0101=NonArrayError (e.g. Abort, Livelock/Deadlock, etc)
11:8	DEDBERRSTRUCTYPE. Read-write. Reset: 4h. Description: DEDB Error Structure Type 4'b0000=DRAM, 4'b0001=Cache, 4'b0010=Tag, 4'b0011=TLB, 4'b0100=OtherArray (incl. flop arrays), (default) 4'b0101=NonArrayError (e.g. Abort, Livelock/Deadlock, etc)
7:4	SECBERRSTRUCTYPE. Read-write. Reset: 4h. Description: SECB Error Structure Type 4'b0000=DRAM, 4'b0001=Cache, 4'b0010=Tag, 4'b0011=TLB, 4'b0100=OtherArray (incl. flop arrays), (default) 4'b0101=NonArrayError (e.g. Abort, Livelock/Deadlock, etc)
3:0	PARITYERRSTRUCTYPE. Read-write. Reset: 4h. Description: Parity Error Structure Type 4'b0000=DRAM, 4'b0001=Cache, 4'b0010=Tag, 4'b0011=TLB, 4'b0100=OtherArray (incl. flop arrays), (default) 4'b0101=NonArrayError (e.g. Abort, Livelock/Deadlock, etc)

**USBCONTAINER[0...1]x00070014
(USB31::USBCONTAINERS0REGCNTR0::MCAERRSTRUCTYPE1_CNTR0)**

Read-write. Reset: 0000_0000h.

[MCA](#) Error Structure Type Control Register1[_inst\[USB\[1:0\]\]_aliasSMN](#); USBCONTAINER[1:0]x00070014; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:0	MCAERRSTRUCTYPE1_RESERVED_31_0 . Read-write. Reset: 0000_0000h. Reserved for future Error Structure Type

USBCONTAINER[0...1]x00070018 (USB31::USBCONTAINERS0REGCNTR0::MCAERROPTYPE0_CNTR0)

Read-write. Reset: 0001_0000h.

[MCA](#) Error Operation Type Control Register0

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070018; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:20	MCAERROPTYPE0_RESERVED_31_20. Read-write. Reset: 000h. Reserved
19:16	AXISLVERROPTYPE. Read-write. Reset: 1h. Description: AXI SLV Error Operation Type. The bit[3] is reserved. 3'b000=Read, 3'b001=Write, (default) 3'b010=Prefetch, 3'b011=Evict, 3'b100=Snoop, 3'b111=Generic, all else reserved.
15:12	FEDOUTBERROPTYPE. Read-write. Reset: 0h. Description: FEDOUTB Error Operation Type. The bit[3] is reserved. 3'b000=Read, (default) 3'b001=Write, 3'b010=Prefetch, 3'b011=Evict, 3'b100=Snoop, 3'b111=Generic, all else reserved.
11:8	DEDBERROPTYPE. Read-write. Reset: 0h. Description: DEDB Error Operation Type. The bit[3] is reserved. 3'b000=Read, (default) 3'b001=Write, 3'b010=Prefetch, 3'b011=Evict, 3'b100=Snoop, 3'b111=Generic, all else reserved.
7:4	SECBERROPTYPE. Read-write. Reset: 0h. Description: SECB Error Operation Type. The bit[3] is reserved. 3'b000=Read, (default) 3'b001=Write, 3'b010=Prefetch, 3'b011=Evict, 3'b100=Snoop, 3'b111=Generic, all else reserved.
3:0	PARITYERROPTYPE. Read-write. Reset: 0h. Description: Parity Error Operation Type. The bit[3] is reserved. 3'b000=Read, (default) 3'b001=Write, 3'b010=Prefetch, 3'b011=Evict, 3'b100=Snoop, 3'b111=Generic, all else reserved.

USBCONTAINER[0...1]x0007001C (USB31::USBCONTAINERS0REGCNTR0::MCAERROPTYPE1_CNTR0)

Read-write. Reset: 0000_0000h.

[MCA](#) Error Operation Type Control Register1[_inst\[USB\[1:0\]\]_aliasSMN; USBCONTAINER\[1:0\]x0007001C; USBCONTAINER\[1:0\]=16\[F,D\]8_0000h](#)

Bits	Description
31:0	MCAERROPTYPE1_RESERVED_31_0. Read-write. Reset: 0000_0000h. Reserved for future Error Operation Type

USBCONTAINER[0...1]x00070020**(USB31::USBCONTAINERS0REGCNTR0::MCAERRTRANTYPE0_CNTR0)**

Read-write. Reset: 0000_0255h.

[MCA](#) Error Transaction Type Control Register0[_inst\[USB\[1:0\]\]_aliasSMN; USBCONTAINER\[1:0\]x00070020; USBCONTAINER\[1:0\]=16\[F,D\]8_0000h](#)

Bits	Description
31:10	MCAERRTRANTYPE0_RESERVED_31_10. Read-write. Reset: 00_0000h. Reserved
9:8	AXISLVERRTRANTYPE. Read-write. Reset: 2h. Description: AXI SLV Error Transaction Type 2'b00: Instruction, 2'b01: Data, 2'b10: Generic (default) 2'b11: Reserved.
7:6	FEDOUTBERRTRANTYPE. Read-write. Reset: 1h. Description: FEDOUTB Error Transaction Type 2'b00: Instruction, 2'b01: Data, (default) 2'b10: Generic 2'b11: Reserved.
5:4	DEDBERRTRANTYPE. Read-write. Reset: 1h. Description: DEDB Error Transaction Type 2'b00: Instruction, 2'b01: Data, (default) 2'b10: Generic 2'b11: Reserved.
3:2	SECBERRTRANTYPE. Read-write. Reset: 1h. Description: SECB Error Transaction Type 2'b00: Instruction, 2'b01: Data, (default) 2'b10: Generic 2'b11: Reserved.
1:0	PARITYERRTRANTYPE. Read-write. Reset: 1h. Description: Parity Error Transaction Type 2'b00: Instruction, 2'b01: Data, (default) 2'b10: Generic 2'b11: Reserved.

USBCONTAINER[0...1]x00070024 (USB31::USBCONTAINERS0REGCNTR0::MCAUSBMCMADDR_CNTR0)

Read-write. Reset: 0000_0000h.

[SMN](#) INT MCM ADDRESS defined for USB [MCA](#)[_inst\[USB\[1:0\]\]_aliasSMN; USBCONTAINER\[1:0\]x00070024; USBCONTAINER\[1:0\]=16\[F,D\]8_0000h](#)

Bits	Description
31:8	MCAUSBMCMADDR_RESERVED_31_8. Read-write. Reset: 00_0000h. Reserved
7:0	CFG_SMN_INT_MCM_ADDR. Read-write. Reset: 00h. Description: SMN INT MCM ADDRESS defined for USB MCA [Note2] When implementing, please use a feature for bit width definition for different project.

USBCONTAINER[0...1]x00070028 (USB31::USBCONTAINERS0REGCNTR0::MCAUSBAPERTUREID_CNTR0)

Read-write. Reset: 0000_0000h.

[SMN](#) INT APERTUREID defined for USB [MCA](#)[_inst\[USB\[1:0\]\]_aliasSMN; USBCONTAINER\[1:0\]x00070028; USBCONTAINER\[1:0\]=16\[F,D\]8_0000h](#)

Bits	Description
31:12	MCAUSBAPERTUREID_RESERVED_31_12. Read-write. Reset: 0_0000h. Reserved
11:0	CFG_SMN_INT_APERTUREID. Read-write. Reset: 000h. SMN INT APERTUREID defined for USB MCA

USBCONTAINER[0...1]x0007002C (USB31::USBCONTAINERS0REGCNTR0::MCAUSBINTREQADDR_CNTR0)

Read-write. Reset: 0001_D780h.

[SMN](#) INT REQ ADDRESS defined for USB [MCA](#)[_inst\[USB\[1:0\]\]_aliasSMN; USBCONTAINER\[1:0\]x0007002C; USBCONTAINER\[1:0\]=16\[F,D\]8_0000h](#)

Bits	Description
31:20	MCAUSBINTREQADDR_RESERVED_31_20. Read-write. Reset: 000h. Reserved
19:0	Reg_CFG_SMN_INT_REQ_ADDR. Read-write. Reset: 1_D780h. SMN INT REQ ADDRESS defined for USB MCA

USBCONTAINER[0...1]x00070030**(USB31::USBCONTAINERS0REGCNTR0::MCAUSBMIFOVDCTL_CNTR0)**

Read-write. Reset: 6001_0000h.

[MCA](#) USB RSMU Master IF Override Control Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070030; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31	RSMU_MASTER_IF_OVDEN. Read-write. Reset: 0. Description: RSMU master interface override enable 1'b0: disable RSMU master interface override (default) 1'b1: enable RSMU master interface override
30	RSMU_master_reg_tran_posted_ovd. Read-write. Reset: 1. The override value of RSMU_master_reg_tran_posted
29	RSMU_master_reg_burst_ovd. Read-write. Reset: 1. The override value of RSMU_master_reg_burst
28	RSMU_master_reg_vf_ovd. Read-write. Reset: 0. The override value of RSMU_master_reg_vf
27:26	MCAUSBMIFOVDCTL_RESERVED_27_26. Read-write. Reset: 0h. Reserved
25:20	RSMU_master_reg_vf_id_ovd. Read-write. Reset: 00h. The override value of RSMU_master_reg_vf_id
19:16	RSMU_master_reg_qos_ovd. Read-write. Reset: 1h. The override value of RSMU_master_reg_qos
15	RSMU_master_reg_debug_mask_ovd. Read-write. Reset: 0. The override value of RSMU_master_reg_debug_mask
14:12	RSMU_master_reg_sec_flag_ovd. Read-write. Reset: 0h. The override value of RSMU_master_reg_sec_flag
11:8	RSMU_master_reg_tran_id_ovd. Read-write. Reset: 0h. The override value of RSMU_master_reg_tran_id
7	MCAUSBMIFOVDCTL_RESERVED_7. Read-write. Reset: 0. Reserved
6:0	RSMU_master_reg_unit_id_ovd. Read-write. Reset: 00h. The override value of RSMU_master_reg_unit_id

USBCONTAINER[0...1]x00070034**(USB31::USBCONTAINERS0REGCNTR0::MCAUSBMIFERRSTS_CNTR0)**[MCA](#) USB RSMU Master IF Error Status Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070034; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:2	MCAUSBMIFERRSTS_RESERVED_31_2. Read-write. Reset: 0000_0000h. Reserved
1	MCAUSBMIFERRTYPE. Read,Write-1-to-clear. Description: Write-1-Clear.Used to save RSMU_master_reg_tran_error_type. When RSMU_master_reg_tran_error happens, it is set by RSMU_master_reg_tran_error_type It is cleared by writing 1 to this bit.
0	MCAUSBMIFERR. Read,Write-1-to-clear. Description: Write-1-Clear.Used to save RSMU_master_reg_tran_error. When RSMU_master_reg_tran_error happens, it is set by RSMU_master_reg_tran_error It is cleared by writing 1 to this bit.

USBCONTAINER[0...1]x00070038 (USB31::USBCONTAINERS0REGCNTR0::MCASTATUS_CNTR0)[MCA](#) USB Status Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070038; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:1	MCASTATUS_RESERVED_31_2. Read-write. Reset: 0000_0000h. Reserved
0	MCAERRTRIG. Read,Write-1-to-clear. Description: Write-One-Clear bit. MCAErrTrig Status It is set when there is a pulse output on MCAErrTrig from MCA_IP It is cleared by SW writing 1 to this register bit.

USBCONTAINER[0...1]x00070100 (USB31::USBCONTAINERS0REGCNTR0::S0_LOGGER_CTRL_CNTR0)

Read-write. Reset: 0000_0006h.

S0 Logger Control Register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070100; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:3	S0_LOGGER_CTRL_RESERVED_31_3. Read-write. Reset: 0000_0000h. Reserved
2	ECC_LOGGER_STOP. Read-write. Reset: 1. Description: 0: Run 1: Stop When this bit is set to 1, pause the storage of all the ECC LOGGER registers. Enable the ECC LOGGER record when this bit is 0. This bit is 1 when ECC_LOGGER_EN=0 or ECC_LOGGER_CLR is set.
1	ECC_LOGGER_CLR. Read-write. Reset: 1. Description: 1: Clear When this bit is set to 1, all the ECC logger info registers are cleared to hard-coded default values. This bit will be set if ECC_LOGGER_EN is 0.
0	ECC_LOGGER_EN. Read-write. Reset: 0. Description: Chicken bit for ECC logger function 0: disable 1: enable When 0: ECC_LOGGER_CLR=1, ECC_LOGGER_STOP=1

USBCONTAINER[0...1]x00070104**(USB31::USBCONTAINERS0REGCNTR0::ECC_LOGGER_ERR_CNT_CNTR0)**

Read-only. Reset: 0000_0000h.

ecc logger error counter register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070104; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:16	DEDB_ERR_CNT. Read-only. Reset: 0000h. this field indicates the number of DEDB errors occurs
15:0	SECB_ERR_CNT. Read-only. Reset: 0000h. this field indicates the number of SECB errors occurs

USBCONTAINER[0...1]x00070110 (USB31::USBCONTAINERS0REGCNTR0::ECC_LOGGER_INFO_0)

Read-only. Reset: 0000_0000h.

ecc logger info register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070110; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:28	IDX. Read-only. Reset: 0h. Description: This field indicates the index of the ECC error. 4'h1: 1st ECC Error 4'h2: 2nd ECC Error 4'h3: 3rd ECC Error ... 4'h8: 8th ECC Error Other values: undefined
27	DEDB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with DEDB
26	SECB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with SECB
25:24	RAM_NUM. Read-only. Reset: 0h. This field indicates which RAM this ECC error is related with.
23	WR_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram write access
22	RD_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram read access
21:13	Reserved.
12:0	RAM_ADDR. Read-only. Reset: 0000h. This field indicates the ECC error?s related write or read address.

USBCONTAINER[0...1]x00070114 (USB31::USBCONTAINERS0REGCNTR0::ECC_LOGGER_INFO_1)

Read-only. Reset: 0000_0000h.

ecc logger info register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070114; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:28	IDX. Read-only. Reset: 0h. Description: This field indicates the index of the ECC error. 4'h1: 1st ECC Error 4'h2: 2nd ECC Error 4'h3: 3rd ECC Error ... 4'h8: 8th ECC Error Other values: undefined
27	DEDB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with DEDB
26	SECB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with SECB
25:24	RAM_NUM. Read-only. Reset: 0h. This field indicates which RAM this ECC error is related with.
23	WR_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram write access
22	RD_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram read access
21:13	Reserved.
12:0	RAM_ADDR. Read-only. Reset: 0000h. This field indicates the ECC error?s related write or read address.

USBCONTAINER[0...1]x00070118 (USB31::USBCONTAINERS0REGCNTR0::ECC_LOGGER_INFO_2)

Read-only. Reset: 0000_0000h.

ecc logger info register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070118; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:28	IDX. Read-only. Reset: 0h. Description: This field indicates the index of the ECC error. 4'h1: 1st ECC Error 4'h2: 2nd ECC Error 4'h3: 3rd ECC Error ... 4'h8: 8th ECC Error Other values: undefined
27	DEDB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with DEDB
26	SECB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with SECB
25:24	RAM_NUM. Read-only. Reset: 0h. This field indicates which RAM this ECC error is related with.
23	WR_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram write access
22	RD_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram read access
21:13	Reserved.
12:0	RAM_ADDR. Read-only. Reset: 0000h. This field indicates the ECC error?s related write or read address.

USBCONTAINER[0...1]x0007011C (USB31::USBCONTAINERS0REGCNTR0::ECC_LOGGER_INFO_3)

Read-only. Reset: 0000_0000h.

ecc logger info register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x0007011C; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:28	IDX. Read-only. Reset: 0h. Description: This field indicates the index of the ECC error. 4'h1: 1st ECC Error 4'h2: 2nd ECC Error 4'h3: 3rd ECC Error ... 4'h8: 8th ECC Error Other values: undefined
27	DEDB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with DEDB
26	SECB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with SECB
25:24	RAM_NUM. Read-only. Reset: 0h. This field indicates which RAM this ECC error is related with.
23	WR_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram write access
22	RD_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram read access
21:13	Reserved.
12:0	RAM_ADDR. Read-only. Reset: 0000h. This field indicates the ECC error?s related write or read address.

USBCONTAINER[0...1]x00070120 (USB31::USBCONTAINERS0REGCNTR0::ECC_LOGGER_INFO_4)

Read-only. Reset: 0000_0000h.

ecc logger info register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070120; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:28	IDX. Read-only. Reset: 0h. Description: This field indicates the index of the ECC error. 4'h1: 1st ECC Error 4'h2: 2nd ECC Error 4'h3: 3rd ECC Error ... 4'h8: 8th ECC Error Other values: undefined
27	DEDB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with DEDB
26	SECB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with SECB
25:24	RAM_NUM. Read-only. Reset: 0h. This field indicates which RAM this ECC error is related with.
23	WR_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram write access
22	RD_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram read access
21:13	Reserved.
12:0	RAM_ADDR. Read-only. Reset: 0000h. This field indicates the ECC error?s related write or read address.

USBCONTAINER[0...1]x00070124 (USB31::USBCONTAINERS0REGCNTR0::ECC_LOGGER_INFO_5)

Read-only. Reset: 0000_0000h.

ecc logger info register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070124; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:28	IDX. Read-only. Reset: 0h. Description: This field indicates the index of the ECC error. 4'h1: 1st ECC Error 4'h2: 2nd ECC Error 4'h3: 3rd ECC Error ... 4'h8: 8th ECC Error Other values: undefined
27	DEDB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with DEDB
26	SECB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with SECB
25:24	RAM_NUM. Read-only. Reset: 0h. This field indicates which RAM this ECC error is related with.
23	WR_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram write access
22	RD_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram read access
21:13	Reserved.
12:0	RAM_ADDR. Read-only. Reset: 0000h. This field indicates the ECC error?s related write or read address.

USBCONTAINER[0...1]x00070128 (USB31::USBCONTAINERS0REGCNTR0::ECC_LOGGER_INFO_6)

Read-only. Reset: 0000_0000h.

ecc logger info register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x00070128; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:28	IDX. Read-only. Reset: 0h. Description: This field indicates the index of the ECC error. 4'h1: 1st ECC Error 4'h2: 2nd ECC Error 4'h3: 3rd ECC Error ... 4'h8: 8th ECC Error Other values: undefined
27	DEDB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with DEDB
26	SECB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with SECB
25:24	RAM_NUM. Read-only. Reset: 0h. This field indicates which RAM this ECC error is related with.
23	WR_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram write access
22	RD_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram read access
21:13	Reserved.
12:0	RAM_ADDR. Read-only. Reset: 0000h. This field indicates the ECC error?s related write or read address.

USBCONTAINER[0...1]x0007012C (USB31::USBCONTAINERS0REGCNTR0::ECC_LOGGER_INFO_7)

Read-only. Reset: 0000_0000h.

ecc logger info register

_inst[USB[1:0]]_aliasSMN; USBCONTAINER[1:0]x0007012C; USBCONTAINER[1:0]=16[F,D]8_0000h

Bits	Description
31:28	IDX. Read-only. Reset: 0h. Description: This field indicates the index of the ECC error. 4'h1: 1st ECC Error 4'h2: 2nd ECC Error 4'h3: 3rd ECC Error ... 4'h8: 8th ECC Error Other values: undefined
27	DEDB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with DEDB
26	SECB. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error is related with SECB
25:24	RAM_NUM. Read-only. Reset: 0h. This field indicates which RAM this ECC error is related with.
23	WR_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram write access
22	RD_EN. Read-only. Reset: 0. If this bit is 1, it indicates this ECC error occurs after ram read access
21:13	Reserved.
12:0	RAM_ADDR. Read-only. Reset: 0000h. This field indicates the ECC error?s related write or read address.